

**BACK TO
METAL**

WHY THIS BOOK EXISTS

YOU ARE ALLOWED TO RUN YOUR OWN COMPUTERS

A generation of engineers has now been trained to believe that owning a server is a kind of professional failure. It is not. It is a trade, and like every trade it has terms: you take on hardware, capacity and a pager, and in exchange you stop paying a margin on every byte you move and every hour you idle. For a great many companies that trade is worth making, and the arithmetic is not close.

What has been missing is not the argument. It is the **method**. Repatriation is written about as a decision and executed as a crisis, which is why so many attempts stall six months in, with half the estate moved, two platforms to run and nobody able to say whether it is going well. This book is the other thing: 122 Moves, each one a single job with a stated cutover, a stated risk and a rollback that works.

It is honest about the parts that do not pay. Several Moves here end by telling you to keep paying somebody else, because a CDN, a DDoS scrubbing centre and outbound email are three things you will not beat on your own, and a book that pretends otherwise is selling something. The point was never to own everything. It was to own the parts where ownership is cheaper, faster and yours.

Which is why the whole thing is **open source**. Every Move, the typesetter that turns them into this book, and the site that publishes them are yours: the text under Creative Commons, the software under the MIT licence, all of it at github.com/OneUptime/back-to-metal. Correct it, extend it, add the services your own estate actually runs. Infrastructure knowledge this practical should not sit behind a consulting invoice.

BACK TO METAL

Leaving the cloud with Kubernetes

Nawaz Dhanda

BY THE MAKERS OF ONEUPTIME • [ONEUPTIME.COM](https://oneuptime.com)

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Leaving the cloud with Kubernetes

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The right of Nawaz Dhandala to be identified as the author of this work has been asserted in accordance with the Copyright, Designs and Patents Act 1988.

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Every runbook in this book changes production infrastructure, and several of them move data that cannot be recreated. Read the Rollback section before the first step of any Move, take a backup you have restored at least once, and rehearse against a copy. Prices, service names and product behaviour are those the author observed while writing and will drift; the figures in each Move are illustrative of the shape of a saving, not a quotation. Nothing here is legal, financial, security or compliance advice. You remain responsible for your own systems, your own data and your own obligations to the people whose data it is.

Disclosure: the author founded OneUptime, which this book recommends for alerting, on-call, incidents and status pages. It is recommended because it is Apache-2.0 licensed, self-hostable, and does in one place what that Part would otherwise ask you to assemble from four separate tools. The alternatives named beside it are real ones, and a reader who prefers them loses nothing else in this book.

Set in Archivo, drawn by Omnibus-Type, and JetBrains Mono, drawn by Philipp Nurullin and Konstantin Bulenkov. Both are licensed under the SIL Open Font License.

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MANY MOVES, NOT ONE MIGRATION

The reason cloud repatriations fail is almost never that the technology did not work. It is that they were attempted as one decision, executed as one project, and abandoned somewhere in the middle with two platforms running and nobody able to say whether it was going well.

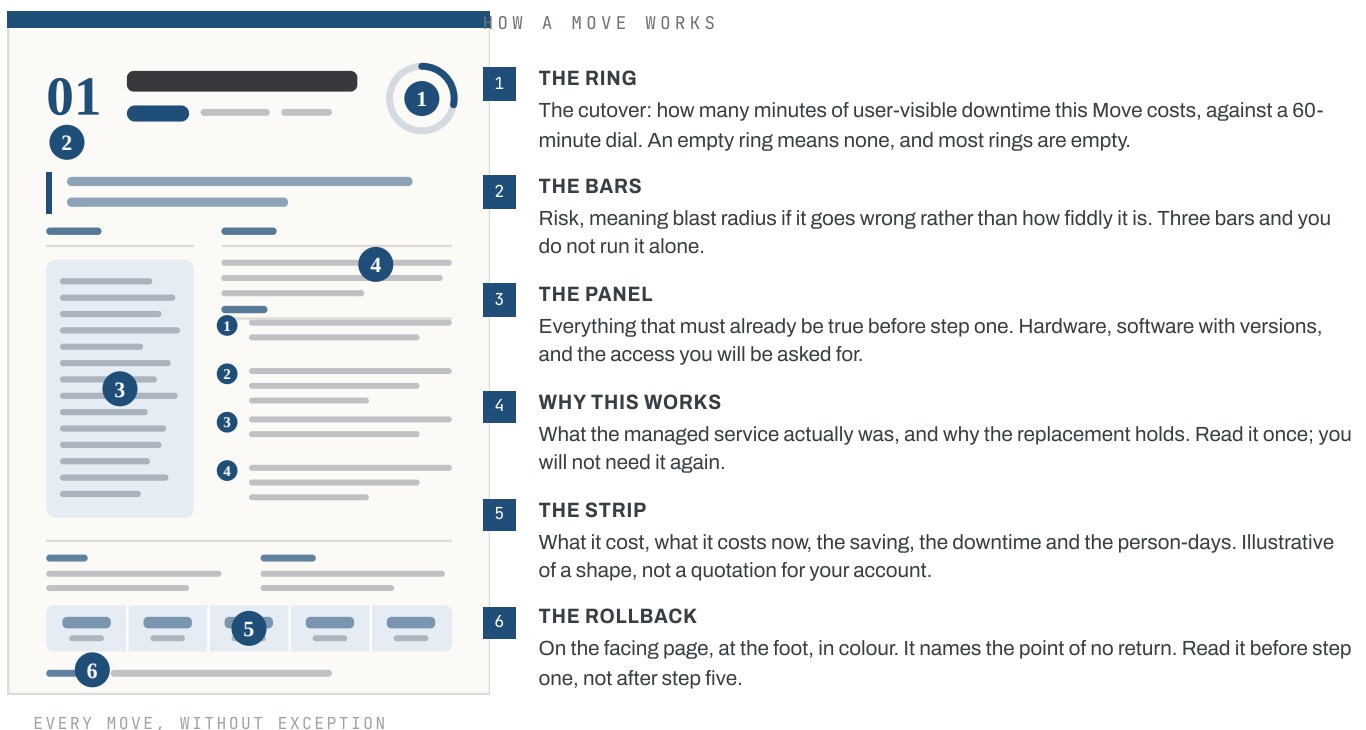
This book takes the opposite shape. It is 122 Moves. Each is one job with a stated cutover, a stated risk and a rollback that has been thought about. Each can be done on a Tuesday and undone on a Wednesday. You can stop after any of them and still be in a coherent place, which is the only property that matters in a project measured in quarters.

They are ordered so that a reader who starts at Move 01 and works forward has, by construction, met every prerequisite of the Move in front of them. The dependency at the foot of each page always points backwards. That is not a stylistic choice; the build refuses to compile a book where it does not hold.

Every Move carries all three clouds. The job is the same whether you are leaving AWS, Google Cloud or Azure — what differs is the extraction, so each Move opens with three lines naming the real service on each provider and the one thing that is different there. Writing three whole variants of every runbook would have tripled the book to say the same thing three times.

Three of the Moves conclude that you should keep paying somebody else. A global content network, scrubbing capacity at the edge and outbound email are businesses other people run better than you will, and each is cheap next to what it replaces. The aim was never to own everything.

Work in any order the dependencies allow. Rehearse everything. And read the Rollback section before the first step of every Move, including the ones that look like nothing.



THE REFERENCE BUILD

Every Move in this book is written against one cluster, so that a runbook can name a real thing rather than a category: 2 sites, 6 nodes each, 64 cores and 512 GB a node, 25 GbE to the top of the rack. Two sites because one site is not a platform, and 6 nodes because that is where losing one stops hurting. Scale the numbers; do not scale away the redundancy.

COMPUTE

- ☐ Six 1U or 2U dual-socket servers per site
- ☐ 64 physical cores a node, current-generation EPYC or Xeon
- ☐ 512 GB ECC RAM a node, all channels populated
- ☐ Redundant hot-swap power supplies, both fed
- ☐ BMC on every node - Redfish, not just a web console
- ☐ Two spare nodes on the shelf, racked and burned in

STORAGE

- ☐ Four NVMe drives a node, mixed-use endurance, 3.84 TB each
- ☐ Enterprise drives with power-loss protection - not consumer SSDs
- ☐ A separate boot pair, mirrored
- ☐ One spare drive of each size, on site, unopened
- ☐ A backup target in a third location you do not control

NETWORK

- ☐ Two 25 GbE ToR switches a site, MLAG or stacked
- ☐ Two uplinks a node, LACP to separate switches
- ☐ A separate 1 GbE out-of-band management switch
- ☐ Your own address space, or a /24 from the facility
- ☐ Two transit providers, or one transit and one IX
- ☐ A serial console server, reachable when the cluster is not

SITE AND POWER

- ☐ Two facilities far enough apart to fail separately
- ☐ A- and B-side power, on separate PDUs
- ☐ Remote hands with a response time in the contract
- ☐ Cross-connects ordered and tested before you need them
- ☐ Physical access for at least two named people
- ☐ A written escalation path with a phone number on it

THE SOFTWARE STACK

- ☐ Talos Linux on every node - no shell, no package manager, no drift
- ☐ Kubernetes, on a version at least one minor behind the newest
- ☐ Cilium for CNI, with kube-proxy replacement
- ☐ Rook, and the Ceph cluster it manages
- ☐ cert-manager, external-dns, and Envoy Gateway
- ☐ Argo CD, with the cluster state in Git
- ☐ Prometheus and Loki for the stores, OneUptime for what wakes someone
- ☐ Velero, and a restore you have actually performed

ACCESS AND IDENTITY

- ☐ An OIDC provider that is not the cluster
- ☐ Break-glass credentials in a safe, on paper
- ☐ A bastion or a mesh - not a public API server
- ☐ Hardware tokens for anyone who can delete a cluster
- ☐ An audit log shipped somewhere the cluster cannot reach
- ☐ A status page that is not hosted on the thing it reports on
- ☐ The domain registrar account locked, with recovery codes printed

ON THE LAPTOP

- ☐ kubectl
- ☐ helm
- ☐ talosctl
- ☐ k9s
- ☐ argocd
- ☐ velero
- ☐ psql and pg_dump
- ☐ mysql client
- ☐ redis-cli
- ☐ rclone
- ☐ aws CLI, still logged in
- ☐ dig and mtr
- ☐ openssl
- ☐ ipmitool
- ☐ a serial cable
- ☐ the runbook, printed

THE ON-RAMP

Almost nobody should sign a facility contract before they have run this stack once. A cluster of 3 refurbished machines with 12 cores and 64 GB each, on a managed switch under a desk, will run every Move in Parts II and III unchanged and most of Part IV besides. It costs about \$1,840 once and about \$30 a month in electricity, it saves nothing whatsoever, and it is the cheapest way to find out whether the rest of this book is for you.

WHY THREE, AND WHY SECOND-HAND

Three because a control plane needs three members to lose one and keep going, and everything you will learn about quorum, fencing and upgrades needs somewhere to go wrong. Second-hand because a machine two generations old runs Kubernetes exactly as well as a new one at roughly a tenth of the price, and because you want to be willing to break it.

WHAT IT CANNOT TEACH YOU

Every word of Part I. A homelab has one power feed, one switch, no cross-connect, no remote hands, no second site and nobody to escalate to at three in the morning. It cannot show you what a colocation contract is for, or what happens when the A feed goes away and the B feed was never tested. The building is a different discipline, which is why it has a Part of its own.

THE ONE PART TO BUY NEW

The drives. A consumer SSD without power-loss protection will corrupt etcd the first time the power flickers, and it will do it in a way that looks like a Kubernetes bug for a day and a half. Buy small enterprise NVMe with the protection, second-hand if you like, and put the money you saved on the chassis into the disks.

WHAT IT GENUINELY PROVES

The whole software estate. Talos, the control plane, Cilium with kube-proxy replaced, Rook and Ceph, the registry, secrets, the observability stack, and a Postgres cutover rehearsed end to end against a copy. If a Move works here it will work on the metal; the difference is scale, not shape.

WHEN TO STOP USING IT

Never. The on-ramp becomes the non-production cluster the rest of the book keeps asking for — the one you restore etcd onto, rehearse an upgrade on, and point a load generator at. Every estate needs a cluster it is allowed to destroy, and this is the cheapest one you will ever own.

IF THREE MACHINES AT HOME IS NOT POSSIBLE

Rent three of the cheapest dedicated hosts you can find, by the month, and run exactly the same experiment. It costs more than electricity and less than being wrong, it gives you real addresses and a real network, and you can hand it all back at the end of the month. What it will not give you is a machine you can physically pull a disk out of.

THE WHOLE SHOPPING LIST

- ☐ Three refurbished mini PCs or small-form-factor desktops
- ☐ 64 GB in each - memory is what runs out first, not cores
- ☐ Two NVMe drives a node: one to boot, one for data
- ☐ A managed switch with VLANs and a spare port for out-of-band
- ☐ A small UPS, mostly to survive the brownouts that corrupt etcd
- ☐ A label printer, because you will regret not having one
- ☐ A separate router or firewall you are willing to break
- ☐ A domain you own, for real certificates rather than self-signed ones

TEN RULES FOR LEAVING THE CLOUD

-
- 1 **Move the cheapest thing first.** Not the most expensive. The first Move exists to prove the platform and the team, and it should be something whose failure is a shrug. You are buying evidence, and evidence is worth more early than savings are.
 - 2 **Never run two platforms longer than you planned to.** The cost of a repatriation is not the hardware. It is the months in which both estates are live, both are on call, and every change has to be made twice. Pick the overlap window before you start, write it down, and treat overrunning it as the incident it is.
 - 3 **Data moves last and leaves last.** Every stateless thing can be cut over and cut back in minutes. State cannot. Move compute first, prove it against the cloud database across the link, and only then move the database - by which point you will have learned the things you would otherwise have learned during a data migration.
 - 4 **Keep the bill until the invoice proves you can stop.** A resource is not decommissioned when nothing points at it. It is decommissioned when a full month has passed, the line has gone from the bill, and nobody has filed a ticket. Turning things off too early is how a migration becomes an outage a fortnight later.
 - 5 **Buy the second of everything before you need the first.** Two switches, two power feeds, two transit providers, two people who can get into the building. Every single point of failure you accept at the start becomes an incident you will attend in person, at night, in a year.
 - 6 **Own your addresses.** An IP range you rent from a provider is a migration you have to do again. Your own address space and your own ASN cost a few hundred pounds and take a few weeks, and they are the difference between changing suppliers and changing everything.
 - 7 **Automate the rebuild, not the build.** Anyone can install a cluster once. What matters is whether you can lose a site and rebuild it from a repository and a backup, with no hands on the original. If the answer involves remembering something, you do not have a platform yet.
 - 8 **Count the salary.** A saving that costs one and a half engineers is a real saving only if you write the one and a half engineers into the comparison. Most repatriations still win by a wide margin once you do. The ones that do not, you want to know about in month one rather than month ten.
 - 9 **Keep paying for the three things you will not beat.** A global CDN, DDoS scrubbing at the edge, and outbound email deliverability. Each is a business somebody else is already running better than you will, and each is cheap next to what it replaces. Ownership is a means, not a principle.
 - 10 **The migration is finished when the account is closed.** Not when the traffic moves. An estate with a dormant cloud account still has the cost, the credentials, the compliance surface and the temptation. The last Move in this book is closing the account, and until it is done the project is still open.

BEFORE YOU TOUCH ANYTHING

Every Move in this book changes something that is running. Most are reversible for a stated window, a few are reversible only until a particular step, and 13 are not reversible at all. The difference between a migration and an incident is almost never the technique - it is whether somebody worked out the way back before they started, and whether anybody had ever tested it.

READ THE ROLLBACK FIRST

Not the runbook. The Rollback section of every Move names the point of no return: the step after which the old system can no longer serve traffic or no longer holds current data. Read it, decide whether you are willing to reach that step today, and only then start at step one.

KEEP THE OLD THING RUNNING, AND KEEP PAYING FOR IT

The instinct after a successful cutover is to delete the source and book the saving. Do not. The Reversible field on each Move is how long the old system must stay warm, powered and receiving its own backups. That overlap is the entire cost of being wrong, and it is cheap.

WRITE DOWN THE STATE YOU CANNOT RECREATE

Before Part II, list every store whose contents exist nowhere else: the primary database, the uploads bucket, the secrets, the certificate private keys, the TOTP seeds for the accounts that own the domain. Everything else is rebuildable from a repository. That list is short, and it is the only part of the estate where a mistake is permanent.

TWO PEOPLE, ONE KEYBOARD

Every High-risk Move in this book assumes a second person who is not typing, is reading the runbook aloud, and has the authority to stop. It is the cheapest safety control in infrastructure and the first one teams drop when they are behind schedule.

A BACKUP NOBODY RESTORED IS NOT A BACKUP

It is a file. Before any Move in Part II, restore the backup you are relying on into a scratch environment and count the rows against production. The number of organisations that discover their backups were empty during the incident that needed them is not small, and every one of them had a green dashboard.

CUT OVER WHEN YOU CAN UNDO IT, NOT WHEN YOU ARE CONFIDENT

Confidence is not evidence. The right time to move traffic is when the rollback has been rehearsed end to end, on the same day of the week, with the same people awake. A Tuesday morning with the team at their desks beats a Saturday night with one person and a laptop, every time.

VERIFY BY COUNTING, NOT BY LOOKING

A migration is finished when the destination answers the same questions as the source with the same answers. Row counts, checksums, a replayed hour of real traffic compared response by response. A page that loads is not evidence, and neither is an application that starts.

STOP WHEN YOU ARE SURPRISED

Not when something breaks - when something is merely unexpected. A row count that is off by eleven, a certificate with the wrong issuer, a replica that caught up faster than it should have. Surprise means the model in your head and the system in front of you have diverged, and continuing is guessing.

These are operational practices, not a warranty. The runbooks in this book were written against the services and versions named in each Move and will drift as those change. Rehearse every Move against a copy of your own estate before you run it against the real one, and treat any figure here as the shape of an answer rather than a quotation for your account.

WHAT IT ACTUALLY COSTS

The comparison that gets a repatriation approved and then regretted is the one that counts the hardware and forgets the people. This one counts both, at list price and with no commitment discount on either side, for the reference build opposite. Substitute your own effective rate; the shape does not change.

ON AWS, COMPUTE ALONE

1,536 vCPU of current-generation general purpose instances, on demand, is **\$56,513** a month before a single gigabyte of storage, a load balancer or a byte of egress. Egress is the line that ends most arguments: 100 TB a month is \$9,216, every month, for the privilege of your own traffic leaving.

RENTED BY THE MONTH

The same class of machine from a dedicated-host provider is \$5,760, and removes the racking, the spares and half a person: **\$23,260** all in. Most of the saving, none of the loading dock, and a supplier you can still leave in thirty days.

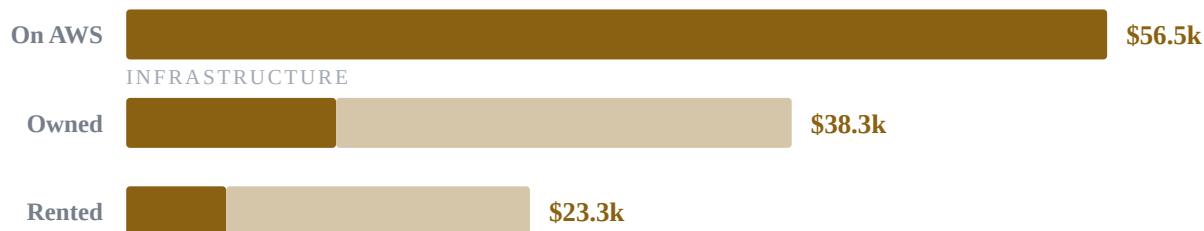
OWNED, IN TWO FACILITIES

\$12,087 of infrastructure - metal amortised over five years, power, racks, transit, cross-connects and remote hands - plus \$26,250 of additional salaried time. Total **\$38,337** a month for the same 1,536 vCPU, with the redundancy the cloud figure does not include.

WHAT THE DIFFERENCE BUYS

Roughly \$18,175 a month, or \$218,102 a year, against a capital outlay of \$288,000 that pays for itself in about 16 months. Every Move in this book states its own version of this table, and every one of them is a fraction of this.

THE SAME 1,536 VCPU, THREE WAYS, PER MONTH



Every AWS figure is a public list price for us-east-1 observed while writing, before any Savings Plan, private pricing agreement or credit. Every hardware figure is a mid-market street price for the specification opposite. Both will drift. The comparison is a method you can repeat against your own invoice, not a quotation - and if your effective rate is half of list, halve the left-hand column and read the conclusion again.

WHAT REPLACES WHAT

Every Move names the real service on all three clouds and the one thing that differs on each, so this table is the map rather than the content. Find the row you are paying for and then read the Move. The right-hand column is the point of the whole book, and where it says keep paying, that is a conclusion rather than a gap.

AWS	GOOGLE CLOUD	AZURE	WHAT YOU RUN INSTEAD
EC2	Compute Engine	Virtual Machines	Bare metal you own, or dedicated hosts by the month
EKS	GKE	AKS	Talos Linux, and Kubernetes you run yourself
Fargate	Cloud Run	Container Apps	Deployments and Jobs on your own nodes
Auto Scaling groups	Managed instance groups	Scale sets	Fixed capacity, priority classes, and a spare shelf
AMI	Machine images	Managed images	A Talos machine config, applied over the network
RDS for PostgreSQL	Cloud SQL for PostgreSQL	Database for PostgreSQL	PostgreSQL under CloudNativePG, on local NVMe
Aurora	AlloyDB	Cosmos DB for PostgreSQL	PostgreSQL - the storage layer does not come with you
RDS for MySQL	Cloud SQL for MySQL	Database for MySQL	MySQL 8.4 LTS or Percona Server, under an operator
ElastiCache	Memorystore	Azure Cache for Redis	Valkey for the half that is really a cache
S3	Cloud Storage	Blob Storage	Ceph RGW under Rook - archive stays where it is
EBS	Persistent Disk	Managed Disks	Ceph RBD under Rook, with local NVMe underneath it
EFS	Filestore	Azure Files	CephFS under Rook, or a filesystem the application stops needing
MSK	Managed Service for Apache Kafka	Event Hubs	Kafka under Strimzi - after deleting the ones that were never Kafka
SQS and SNS	Pub/Sub	Service Bus	NATS JetStream, or a table in the database you already have
OpenSearch Service	Elasticsearch on GKE	Azure AI Search	OpenSearch you run - the index is rebuildable
DynamoDB	Bigtable, Firestore	Cosmos DB	PostgreSQL, Cassandra, or keep paying below the threshold
Redshift	BigQuery	Synapse	Keep paying - and bring the lake substrate home
AWS Backup	Backup for GKE	Azure Backup for AKS	Velero, CSI snapshots and pgBackRest, to a third site
ALB and NLB	Cloud Load Balancing	Load Balancer	Envoy Gateway on addresses advertised over BGP
CloudFront	Cloud CDN	Azure Front Door	Keep paying - change vendor, not model
Route 53	Cloud DNS	Azure DNS	Two independent authoritative providers, never one

WHAT REPLACES WHAT II

Three rows here say keep paying, and they are set in red because they are the most important lines in the table. A global content network, scrubbing capacity at the edge and outbound mail deliverability are not technical problems you have not solved yet. They are businesses, and you are not in them.

AWS	GOOGLE CLOUD	AZURE	WHAT YOU RUN INSTEAD
ACM	Certificate Manager	App Service Certificates	cert-manager with ACME, plus an offline internal root
AWS WAF and Shield	Cloud Armor	Azure WAF and DDoS Protection	Keep paying for scrubbing - the capacity is upstream of you
Global Accelerator	Premium Tier network	Front Door	Your own ASN and address space, or the CDN you kept
NAT Gateway	Cloud NAT	NAT Gateway	Node addresses, and one SNAT hop for the few that need it
SES	No first-party equivalent	Communication Services	Keep paying for outbound - inbound comes home
ECR	Artifact Registry	Container Registry	A pull-through cache, then Harbor as the registry of record
CodeBuild and CodePipeline	Cloud Build	Azure Pipelines	Ephemeral runners on your own nodes, with a warm cache
Secrets Manager	Secret Manager	Key Vault	External Secrets as a dated bridge, then OpenBao or SOPS
KMS	Cloud KMS	Key Vault	Named custodians, an HSM or TPM-sealed keys - custody is the hard part
IAM roles for service accounts	Workload Identity	Workload Identity	Your own service-account issuer, federated back to each cloud
CloudWatch metrics	Cloud Monitoring	Azure Monitor	Prometheus, with VictoriaMetrics behind it and a cardinality budget
CloudWatch Logs	Cloud Logging	Log Analytics	Alloy or the OpenTelemetry Collector into Loki
X-Ray	Cloud Trace	Application Insights	OpenTelemetry into Tempo, once every service speaks traceparent
Lambda	Cloud Functions	Functions	CronJobs, Deployments and KEDA - in that order of preference
Step Functions	Workflows	Logic Apps	Argo Workflows or Temporal - and a dated exception for the rest
Cost Explorer	Cloud Billing reports	Cost Management	A model built from invoices, counting power, spares and the rota
CloudWatch alarms and SNS	Cloud Monitoring alerting	Azure Monitor alerts	OneUptime - alerting, on-call rotas and escalation
Systems Manager Incident Manager	No first-party equivalent	No first-party equivalent	OneUptime - incidents, timelines and postmortems
No first-party equivalent	No first-party equivalent	No first-party equivalent	OneUptime - a status page your customers can read
Synthetic Canaries	Uptime checks	Standard availability tests	OneUptime - synthetic and uptime monitoring, from outside the cluster

WORK OUT WHAT TO BUY

DONE WHEN You know what to order, and why.

What to buy. Cores, memory, disks, network cards, switches, optics and the topology they hang on — chosen against your own numbers rather than a vendor configurator, and tested before anything depends on it.

☐	01	The month of numbers you do not have	0 min	24	☐	11	Headroom, and the number of machines you order	0 min	44
☐	02	What a vCPU is actually worth	0 min	26	☐	12	Two switches, and the spine you do not need yet	0 min	46
☐	03	The estate nobody sizes	0 min	28	☐	13	The switch, and the software running on it	0 min	48
☐	04	Cores against clock, and the licence that decides it	0 min	30	☐	14	The card, the driver, and the kernel you will run	0 min	50
☐	05	SMT, NUMA, and the workload that notices	0 min	32	☐	15	Copper inside the rack, glass between them	0 min	52
☐	06	Memory is what runs out first	0 min	34	☐	16	The network that works when the network is down	0 min	54
☐	07	One rack unit or two, and the watts it draws	0 min	36	☐	17	Who you buy from, and what is safe to buy used	0 min	56
☐	08	The way in when the machine will not boot	0 min	38	☐	18	Nothing carries production until it has been made to fail	0 min	58
☐	09	The drive you are allowed to put etcd on	0 min	40	☐	19	The shelf, and the failure rates you stock against	0 min	60
☐	10	How much flash, and where it sits	0 min	42	☐	20	The bill of materials	0 min	62

FIND SOMEWHERE TO PUT IT

DONE WHEN The machines are racked, powered and reachable.

Where the machines live, who owns the room, and what you are signing. Nothing here is software, and getting it wrong is the only mistake in this book you cannot fix with a deploy.

☐	21	What the three machines under the desk cannot tell you	0 min	66
☐	22	The four places the machines could live	0 min	68
☐	23	The room in your own building	0 min	70
☐	24	The second site you are not building this year	0 min	72
☐	25	The flood plain, the fault line and the grid	0 min	74
☐	26	Carrier neutrality and the cross-connect price list	0 min	76
☐	27	What a tier rating promises, and what it does not	0 min	78
☐	28	Committed kilowatts, metered kilowatts and the derate	0 min	80
☐	29	What the service credit is actually worth	0 min	82

☐	30	The notice period and the door out	0 min	84
☐	31	The 80 per cent you are allowed to draw	0 min	86
☐	32	Two feeds, one of which you have tested	0 min	88
☐	33	The inlet temperature you are allowed	0 min	90
☐	34	The importer of record and the loading dock	0 min	92
☐	35	Who can get in at 03:00	0 min	94
☐	36	The first rack	0 min	96
☐	37	Labels, port maps and the as-built	0 min	98
☐	38	The drives that leave, and the snapshots that stay	0 min	100

BUILD THE CLUSTER

DONE WHEN A cluster that can take production traffic, and has never seen any.

Everything between a racked machine and a cluster that can take production traffic. Nothing here moves a workload; all of it decides how well the rest of the book goes.

☐	39	The address plan, and the three clocks you start today	0 min	104	☐	47	Local NVMe, and the default StorageClass nobody gave you	0 min	120
☐	40	The management VLAN, and the way in at 03:00	0 min	106	☐	48	Encryption at rest, and who holds the key	0 min	122
☐	41	Talos as the host OS, and where it cannot go	0 min	108	☐	49	Ceph under Rook: block, shared filesystem and the object store	0 min	124
☐	42	Netboot, node identity and failure-domain labels	0 min	110	☐	50	Humans on the API server: OIDC, after break-glass	0 min	126
☐	43	Three control-plane nodes, the API VIP and the cluster CA	0 min	112	☐	51	The second cluster, and what it shares	0 min	128
☐	44	Cilium, with kube-proxy off	0 min	114	☐	52	The etcd restore drill	0 min	130
☐	45	BGP to the top-of-rack, and the MTU that breaks five per cent	0 min	116	☐	53	The links you build first: to the cloud, to the office	0 min	132
☐	46	Cluster DNS, and the resolver the network used to be	0 min	118	☐	54	The object store you now own, and how it loses data	0 min	134

MAKE IT FIT TO RUN PRODUCTION

DONE WHEN Identity, secrets, images, policy and dashboards, before any data arrives.

What developers touch. Get this wrong and the migration succeeds while everybody who has to use it quietly hates you.

☐	55	Workload identity without the cloud's	0 min	138
☐	56	Secrets: External Secrets as the dated bridge	0 min	140
☐	57	The root of trust after managed key management	0 min	142
☐	58	The mirror, then Harbor as the registry of record	0 min	144
☐	59	cosign, Trivy and an admission policy	0 min	146
☐	60	East-west: security groups into NetworkPolicy	0 min	148
☐	61	East-west encryption, and an identity per workload	0 min	150
☐	62	Metrics off the managed monitor, and the cardinality budget	0 min	152

☐	63	Logs and traces, brought home	0 min	154
☐	64	Builds onto your own metal	0 min	156
☐	65	Argo CD, and the two infrastructure-as-code estates	0 min	158
☐	66	Infrastructure code that only compiles for one cloud	0 min	160
☐	67	The first stateless service, end to end	0 min	162
☐	68	The container service that was not Kubernetes	0 min	164
☐	69	The fixed fleet: priority, preemption, NUMA and the GPUs	0 min	166
☐	70	The workloads that must stay virtual machines	30 min	168

MOVE THE DATA

DONE WHEN Your state is on your disks, and the old copy is still warm.

The part that can lose a company its data, and therefore the part with the longest overlap windows, the most rehearsal and the fewest one-way steps.

☐	71	The Postgres pre-flight	15 min	172	☐	81	The queues come home, the fan-out does not	0 min	192
☐	72	The verification gate	0 min	174	☐	82	The event bus, and the control-plane events that stop arriving	0 min	194
☐	73	Postgres onto your own disks	20 min	176	☐	83	Kafka off the managed broker	0 min	196
☐	74	Point-in-time recovery with pgBackRest	0 min	178	☐	84	Application buckets, and the object API your code assumes	0 min	198
☐	75	Quorum commit and fencing, after the restore was proven	2 min	180	☐	85	The data that stays: deep archive and the warehouse	0 min	200
☐	76	The proprietary Postgres engines: the exit is a stream	20 min	182	☐	86	The batch estate: who schedules the jobs	0 min	202
☐	77	MySQL, wherever it is managed	20 min	184	☐	87	Search off the managed cluster	0 min	204
☐	78	The licensed databases, and the contract that can veto it	30 min	186	☐	88	Document and wide-column stores: the hardest exit	30 min	206
☐	79	Redis: which half of it is actually a cache	0 min	188	☐	89	The serverless estate in four piles	0 min	208
☐	80	Brokered messaging: JMS, durable subscriptions, your broker	10 min	190	☐	90	Orchestrated workflows: the dated exception	0 min	210

MOVE THE TRAFFIC

DONE WHEN Users are reaching your addresses, not somebody else's.

How traffic reaches you once the addresses are yours. Also the Part with the most honest advice to keep paying somebody else.

☐	91	Your own ASN, a /24 and the address space you already have	0 min	214	☐	100	The managed API front door: keys, quotas and WebSockets	0 min	232
☐	92	Turning on dual-stack, and the code that parses an address	0 min	216	☐	101	Edge authentication after the balancer did it	5 min	234
☐	93	Transit: two upstreams and the 95th percentile	0 min	218	☐	102	The shadow edge: two gateways and a weighted record	0 min	236
☐	94	Egress off the managed NAT, and the allowlists you renegotiate	0 min	220	☐	103	Go-live: the soak, the freeze, and how you abort	10 min	238
☐	95	The first VIP: L2, and failover when nobody will peer	0 min	222	☐	104	Authoritative DNS, and the registrar you forgot	0 min	240
☐	96	VIPs over BGP: ECMP, BFD and the rehash	0 min	224	☐	105	CDN: change vendor, not model	0 min	242
☐	97	TLS off the managed certificate service	0 min	226	☐	106	DDoS: the capacity you rent forever	0 min	244
☐	98	Envoy Gateway, after ingress-nginx	0 min	228	☐	107	Email: inbound comes home, outbound does not	0 min	246
☐	99	Balancer rules and proxy annotations into HTTPRoute	0 min	230					

RUN IT, AND CLOSE THE ACCOUNT

DONE WHEN You can carry it at 03:00, and the cloud bill is zero.

Running it once it is yours. Every repatriation article stops before this Part, and every repatriation that regrets itself failed inside it.

☐	108	The upgrade calendar and the control-plane upgrade	0 min	250	☐	116	Capacity planning, and the cloud footprint you keep	0 min	266
☐	109	The fleet roll with no thirteenth machine	10 min	252	☐	117	The patch SLA after the vulnerability feed stopped	0 min	268
☐	110	The inventory, the RMA loop and which rack unit	0 min	254	☐	118	Audit logs after the provider's	0 min	270
☐	111	The 03:00 disk	0 min	256	☐	119	The auditor's checklist, and the contracts nobody read	0 min	272
☐	112	The support contracts you buy for open source	0 min	258	☐	120	What it actually costs	0 min	274
☐	113	The rota	0 min	260	☐	121	Going back: how the programme is aborted	0 min	276
☐	114	Alerts that survive the cluster, and the pager you drop	0 min	262	☐	122	The last account	0 min	278
☐	115	Velero, the off-site copy and the rebuild drill	0 min	264					

 IRON

Servers, disks, switches and optics, chosen against your numbers and burned in before anything trusts them.

 SITE

The facility, the contract and the power. The Part where a mistake costs a lorry rather than a deploy.

 CLUSTER

The OS, the control plane, the network and the disks. Nothing here moves a workload, and everything here decides how the rest goes.

 PLATFORM

Builds, registries, secrets, identity and observability. What your own engineers actually touch.

 DATA

State, and the way back from moving it. The longest overlaps and the most rehearsal in the book.

 EDGE

Addresses, balancers, DNS and certificates - and the three things you keep paying somebody else to do.

 WATCH

Upgrades, on-call, capacity and compliance. The Part that decides whether this was a good idea.

Every Move states its cutover in minutes of user-visible downtime, its risk as blast radius rather than difficulty, and how long the thing it replaced must stay warm before you turn it off. 110 of the 122 Moves in this book cost no downtime at all; the whole book, executed end to end, costs 202 minutes.

WHICH MOVE DO I NEED I

Nobody opens a book like this at Move 01. They open it because something on the bill, or on the pager, has become intolerable. This is the index for the way the question actually arrives.

The bill went up forty per cent and nobody can say why

- [01](#) The month of numbers you do not have [03](#) The estate nobody sizes
- [120](#) What it actually costs
- [116](#) Capacity planning, and the cloud footprint you keep

We are being rate-limited by a database we pay a fortune for

- [71](#) The Postgres pre-flight [73](#) Postgres onto your own disks
- [47](#) Local NVMe, and the default StorageClass nobody gave you
- [09](#) The drive you are allowed to put etcd on

Something has to move this quarter and nothing may break

- [67](#) The first stateless service, end to end
- [102](#) The shadow edge: two gateways and a weighted record
- [103](#) Go-live: the soak, the freeze, and how you abort
- [72](#) The verification gate

A single availability zone took us down anyway

- [24](#) The second site you are not building this year
- [115](#) Velero, the off-site copy and the rebuild drill
- [42](#) Netboot, node identity and failure-domain labels
- [75](#) Quorum commit and fencing, after the restore was proven

Our secrets are in four places and one of them is a spreadsheet

- [56](#) Secrets: External Secrets as the dated bridge
- [57](#) The root of trust after managed key management
- [55](#) Workload identity without the cloud's
- [48](#) Encryption at rest, and who holds the key

The audit is in eight weeks and the evidence lives in a console

- [119](#) The auditor's checklist, and the contracts nobody read
 - [118](#) Audit logs after the provider's
 - [38](#) The drives that leave, and the snapshots that stay
 - [110](#) The inventory, the RMA loop and which rack unit
-

Egress is now the largest line on the invoice

- [93](#) Transit: two upstreams and the 95th percentile
- [53](#) The links you build first: to the cloud, to the office
- [94](#) Egress off the managed NAT, and the allowlists you renegotiate
- [105](#) CDN: change vendor, not model

The finance director has asked what happens if we just left

- [20](#) The bill of materials [120](#) What it actually costs
- [22](#) The four places the machines could live
- [121](#) Going back: how the programme is aborted

We cannot get a straight answer about an outage we did not cause

- [27](#) What a tier rating promises, and what it does not
- [29](#) What the service credit is actually worth
- [25](#) The flood plain, the fault line and the grid
- [114](#) Alerts that survive the cluster, and the pager you drop

The container registry is throttling our deploys

- [58](#) The mirror, then Harbor as the registry of record
- [59](#) cosign, Trivy and an admission policy [64](#) Builds onto your own metal
- [65](#) Argo CD, and the two infrastructure-as-code estates

We have no idea what would happen if the cluster were deleted

- [52](#) The etcd restore drill
- [115](#) Velero, the off-site copy and the rebuild drill
- [65](#) Argo CD, and the two infrastructure-as-code estates
- [74](#) Point-in-time recovery with pgBackRest

Nobody wants to be on call for hardware

- [113](#) The rota [110](#) The inventory, the RMA loop and which rack unit
 - [111](#) The 03:00 disk [22](#) The four places the machines could live
-

WHICH MOVE DO I NEED II

The rest of the same index. Every number points at a Move, and the build refuses to compile a reference that does not.

The monitoring bill is larger than the thing it monitors

- [62](#) Metrics off the managed monitor, and the cardinality budget
- [63](#) Logs and traces, brought home
- [114](#) Alerts that survive the cluster, and the pager you drop
- [120](#) What it actually costs

Latency between our services is worse than it should be

- [45](#) BGP to the top-of-rack, and the MTU that breaks five per cent
- [46](#) Cluster DNS, and the resolver the network used to be
- [05](#) SMT, NUMA, and the workload that notices
- [69](#) The fixed fleet: priority, preemption, NUMA and the GPUs

Half the estate is serverless and nobody knows what it costs

- [89](#) The serverless estate in four piles
- [66](#) Infrastructure code that only compiles for one cloud
- [68](#) The container service that was not Kubernetes
- [90](#) Orchestrated workflows: the dated exception

Someone has to hold an IP address we actually own

- [91](#) Your own ASN, a /24 and the address space you already have
- [96](#) VIPs over BGP: ECMP, BFD and the rehash
- [94](#) Egress off the managed NAT, and the allowlists you renegotiate
- [104](#) Authoritative DNS, and the registrar you forgot

We are locked into one database we cannot price-compare

- [76](#) The proprietary Postgres engines: the exit is a stream
- [78](#) The licensed databases, and the contract that can veto it
- [71](#) The Postgres pre-flight [73](#) Postgres onto your own disks

We were told we cannot leave because of compliance

- [119](#) The auditor's checklist, and the contracts nobody read
- [27](#) What a tier rating promises, and what it does not
- [35](#) Who can get in at 03:00 [118](#) Audit logs after the provider's

We want out but the CDN and the mail are non-negotiable

- [105](#) CDN: change vendor, not model
- [107](#) Email: inbound comes home, outbound does not
- [106](#) DDoS: the capacity you rent forever
- [116](#) Capacity planning, and the cloud footprint you keep

The migration stalled and we are now running two of everything

- [120](#) What it actually costs
- [121](#) Going back: how the programme is aborted
- [103](#) Go-live: the soak, the freeze, and how you abort
- [116](#) Capacity planning, and the cloud footprint you keep

I IRON

What to buy. Cores, memory, disks, network cards, switches, optics and the topology they hang on — chosen against your own numbers rather than a vendor configurator, and tested before anything depends on it.

20 MOVES	20 ZERO DOWNTIME	12 HIGH RISK	0 MINUTES, TOTAL
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IF YOU ONLY DO THREE

01

THE MONTH OF NUMBERS YOU DO NOT HAVE

Installs a memory-reporting agent in all three clouds and opens a thirty-day sampling window, because none of them …

20

THE BILL OF MATERIALS

Assembles every Iron decision into a document a supplier can quote against, and sets the three-year cost frame the …

02

WHAT A VCPU IS ACTUALLY WORTH

Converts a month of samples into a derated core requirement, and corrects the derate on the families where a vCPU is …

THE MONTH OF NUMBERS YOU DO NOT HAVE · WHAT A VCPU IS ACTUALLY WORTH · THE ESTATE NOBODY SIZES · CORES AGAINST CLOCK, AND THE LICENCE THAT DECIDES IT · SMT, NUMA, AND THE WORKLOAD THAT NOTICES · MEMORY IS WHAT RUNS OUT FIRST · ONE RACK UNIT OR TWO, AND THE WATTS IT DRAWS · THE WAY IN WHEN THE MACHINE WILL NOT BOOT · THE DRIVE YOU ARE ALLOWED TO PUT ETCD ON · HOW MUCH FLASH, AND WHERE IT SITS · HEADROOM, AND THE NUMBER OF MACHINES YOU ORDER · TWO SWITCHES, AND THE SPINE YOU DO NOT NEED YET · THE SWITCH, AND THE SOFTWARE RUNNING ON IT · THE CARD, THE DRIVER, AND THE KERNEL YOU WILL RUN · COPPER INSIDE THE RACK, GLASS BETWEEN THEM · THE NETWORK THAT WORKS WHEN THE NETWORK IS DOWN · WHO YOU BUY FROM, AND WHAT IS SAFE TO BUY USED · NOTHING CARRIES PRODUCTION UNTIL IT HAS BEEN MADE TO FAIL · THE SHELF, AND THE FAILURE RATES YOU STOCK AGAINST · THE BILL OF MATERIALS

■ 2 LOW RISK

■ 6 MEDIUM RISK

■ 12 HIGH RISK

01 THE MONTH OF NUMBERS YOU DO NOT HAVE

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
IRON	—	LOW	IMMEDIATELY

Installs a memory-reporting agent in all three clouds and opens a thirty-day sampling window, because none of them reports guest working set without one.

AWS	CloudWatch	the hypervisor publishes CPU, network and disk and says nothing about memory; the CloudWatch agent has to be installed and handed a configuration that names the memory metrics you want.
GOOGLE	Cloud Monitoring and the Ops Agent	the only one of the three that reports memory in its default configuration, so the work here is confirming coverage rather than writing configuration.
AZURE	Azure Monitor	memory arrives only through the Azure Monitor agent behind a data collection rule, an object you create per subscription and then associate with every machine you care about.

BEFORE YOU START

ACCESS

- Read access to the billing export in every cloud account you run in, including the ones nobody has opened this year
- Permission to install an agent on production instances, and a change window if that needs one

SOFTWARE

- A Prometheus or equivalent time-series store with at least forty days of retention, so the window survives its own analysis
- The three provider command-line tools, `aws`, `gcloud` and `az`, none older than six months
- `kubectl` against every managed cluster, with rights to read the metrics API

PEOPLE

- Whoever owns each application, to say which workloads are allowed to be slow and which are not

WHY THIS WORKS

You cannot buy a machine against a number you have never measured. The figure a purchase order needs is working set — the memory a process is actually touching — and no cloud reports it, because the hypervisor cannot see inside the guest. What the console shows you instead is allocation, which is the number you were billed for and not the number you need. Thirty days is the shortest window that contains a month end, a full weekly cycle and at least one deploy of every service, and it is the reason this is the first Move in the book: every purchasing decision in Part I waits on it, so the clock starts before anything else does.

SWAP

If an agent rollout to production needs a change advisory board that meets fortnightly, start with a representative fifth of the fleet and widen. A partial month of real data beats a full month you never got approval for.

DO IT FASTER

The agent configuration is three files. Ship them through whatever configuration tool already reaches those machines rather than building a new path for one job.

WATCH OUT

The agents themselves cost money — custom metrics are billed per metric per month on all three, and a careless configuration collecting every counter on every host has surprised people with a four-figure line. Name the metrics you want.

LEFTOVERS

The data collection rules, agent configurations and the extra metric ingestion stay billing until you remove them. Diary the removal for the day the purchase order is signed.

—	—	—	0 min	4 days	30 days
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. This Move only adds cost — a few hundred dollars of agent metrics for a month — and it is the cheapest money in the programme.

- 01 Install the memory agent everywhere it is missing. On AWS that is the CloudWatch agent with a configuration block naming `mem_used_percent` and `mem_available`; on Azure it is the Azure Monitor agent plus a data collection rule associated with each machine; on Google Cloud confirm the Ops Agent is running and reporting rather than assuming it.
- 02 Point your own time-series store at all three metric APIs and let it pull, rather than reading each console separately. One store means one query language at the end of the month instead of three.
- 03 Add per-container working set and per-node IOPS from the managed clusters you already run, using `kubectl` against the metrics API. Container memory is what sizes a node; node memory is what sizes a machine.
- 04 Export a full month of billing data per account and reconcile it against the running inventory. Anything on the invoice that is not in the inventory is the part of the estate nobody remembers, and it is usually five to fifteen per cent.
- 05 Write down what cannot be measured at all: burstable credit balances, page cache against anonymous memory, and the build minutes that never appeared as an instance. An honest gap list is worth more than an invented figure.
- 06 Leave it running for thirty days. Do not analyse it at day seven and do not shorten the window because the quarter is ending; a month with no month end in it is not a month.

ROLLBACK

Nothing here changes how anything runs, so backing out is uninstalling three agents and deleting a data collection rule. The point of no return does not exist in this Move; it exists in Move 20, which spends money against these numbers. The only real cost of abandoning it is the calendar: stop at day twelve and you have twelve days of data, and starting again means waiting another thirty. If the time-series store is lost, restore it from its own snapshot rather than re-running the window, because the window cannot be re-run retrospectively.

02 WHAT A VCPU IS ACTUALLY WORTH

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
IRON	THE VCPU AS A UNIT OF PURCHASE	HIGH	UNTIL THE ORDER IS SIGNED

Converts a month of samples into a derated core requirement, and corrects the derate on the families where a vCPU is a whole core rather than a thread.

AWS	EC2 instance families	a vCPU is one SMT thread on the x86 families and one whole physical core on every Graviton generation, which is the correction most people miss.
GOOGLE	Compute Engine machine families	a vCPU is a thread on the Intel and AMD families, and a full core on C4A, T2D, T2A, N4A and H3, all of which are sold in the same units.
AZURE	Virtual Machines series	a vCPU is a thread on the D, E and F series, and a physical core on the Cobalt 100 Dpsv6 and Epsv6 series, and the portal draws them identically.

BEFORE YOU START

ACCESS

- The completed thirty-day window from Move 01, in one queryable place
- The instance inventory per account, with family and size, exported rather than screenshotted

SOFTWARE

- A spreadsheet or notebook that other people can check, because this number will be argued with
- The provider documentation for each family you run, saved as PDF against the day you read it

PEOPLE

- Whoever signs the purchase order, present for the derate conversation rather than shown the answer

WHY THIS WORKS

A vCPU is a billing unit, not a quantity of computer. On most x86 families it is one hardware thread, so two of them share one physical core and a fleet of 400 vCPUs is 200 cores of silicon. On the Arm families it is a whole core, so 400 vCPUs is 400 cores. Applying the halving derate to an Arm fleet buys twice the hardware you need, and it is the single most expensive arithmetic error available in this Part. The other correction is statistical: a fleet peak is the sum of peaks that never happened at the same moment, so the requirement is built from per-workload p95 and p99 and the fleet number falls out of that, not the other way round.

SWAP

If you have no month of samples and cannot wait for one, size from the peak of the last quarter's billing and mark the whole result provisional. It will be high by a third and you will pay for that.

DO IT FASTER

Most estates have five workload shapes wearing forty names. Cluster them first and the arithmetic takes an afternoon instead of a week.

WATCH OUT

Rightsizing recommendations from the providers optimise for a smaller rented instance, which is a different question from how much silicon to own. They are a useful sanity check and a bad input.

LEFTOVERS

The exported inventory dates fast. Re-run the export the week the order goes in rather than reusing this one.

—	—	—	0 min	5 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet, though this is usually where somebody notices a family of oversized instances that can be shrunk this week, on the cloud, for free.

NEEDS 01 The month of numbers you do not have

- 01 | Group every instance by family and mark each group as thread-per-vCPU or core-per-vCPU from the vendor documentation. Do this before any arithmetic; the whole result turns on it.
- 02 | Pull out the burstable families — the t-series, Azure B-series, and e2-medium and smaller. Their utilisation is measured against a credit budget rather than against a core, so their samples say nothing about core requirement. Discard them and re-measure those workloads on a non-burstable shape for a week.
- 03 | For every remaining workload, take p95 and p99 of CPU over the window, per workload rather than per fleet. Size against p99 for anything with a latency commitment and p95 for everything else, and record which you chose for each.
- 04 | Convert to physical cores: halve for the thread-per-vCPU groups, take at face value for the core-per-vCPU groups. Sum by workload class, not by account.
- 05 | Apply a generational factor against the parts you are actually going to buy, taken from a benchmark that resembles your workload rather than from a marketing figure. A current-generation core is not a 2019 core, and pretending it is over-buys as reliably as the derate error under-buys.
- 06 | Write the requirement as a range with the assumptions attached, and have somebody who was not in the room reproduce it from the same data. If they cannot, the number is not ready to spend against.

ROLLBACK

This Move produces a number, and a number is undone by writing a better one. Redo the derate, redo the percentile, publish the correction to everyone who saw the first version. The point of no return is Move 20, where the requirement becomes a signed order and the difference between right and wrong stops being a spreadsheet revision and becomes machines in a loading bay. Keep the raw samples and the working, and restore the earlier version from source control if a later revision turns out to be the mistake.

03 THE ESTATE NOBODY SIZES

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
IRON	NON-PRODUCTION CAPACITY THAT COSTS NOTHING WHEN IT IS SWITCHED OFF, AND HOSTED CI RUNNERS	MEDIUM	UNTIL THE ORDER IS SIGNED

Sizes dev, staging and CI from the same month of samples, because on hardware you own idle non-production costs exactly what busy non-production costs.

AWS	CodeBuild and non-production accounts	build minutes are billed per minute of a build container and never appear in an instance inventory at all.
GOOGLE	Cloud Build and non-production projects	the same absence, plus a free tier that hides the real volume until you export the billing data.
AZURE	Azure Pipelines with Microsoft-hosted agents	billed per parallel job rather than per minute, so the invoice tells you concurrency and not consumption.

BEFORE YOU START

ACCESS

- Billing exports for every non-production account and project, including the sandbox nobody claims
- The build service's usage report for the same month, in minutes rather than in currency

SOFTWARE

- The same time-series store as Move 01, with the non-production accounts already flowing into it
- A list of every pipeline and its trigger, so concurrency can be reasoned about rather than guessed

PEOPLE

- Whoever runs the delivery pipeline, to say what queue time is currently tolerated

WHY THIS WORKS

Non-production is invisible in a cloud cost model because it is cheap when nobody is using it: scale it to zero at seven in the evening and the bill agrees. On a machine you own, that saving does not exist. The machine draws power whether or not a test is running, and the capital was spent whether or not anyone deploys on a Friday. So non-production has to be sized like production, from measurement, and it usually comes to between a third and the whole of production again. Build capacity is worse, because hosted runners never appeared as instances anywhere, and the only honest record of them is billed minutes.

SWAP

If the sharing question cannot be settled in the time available, size for separate machines. Merging two clusters later is a weekend; splitting a shared one after production is on it is a quarter.

DO IT FASTER

Build executors are the one place where a small number of fast cores beats a large number of slow ones. Size them against wall-clock build time and stop optimising.

WATCH OUT

Free tiers hide volume. A pipeline that looks like four hundred minutes a month on the invoice can be four thousand, with the difference absorbed by an included allowance that will not exist on your own hardware.

LEFTOVERS

The idle environments this Move surfaces can be deleted today, on the cloud, before anything moves. That is usually the first saving in the whole programme.

—	—	—	0 min	4 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The environments this Move finds and nobody claims. Delete them in the cloud, this week, and watch a full billing cycle before you count the saving.

NEEDS 02 What a vCPU is actually worth

- 01 Inventory the non-production estate the same way you inventoried production, and reconcile it against the billing export. Everything on the invoice and not in the inventory is either a test environment somebody forgot or a resource that has been idle for a year.
- 02 Take the same p95 and p99 from the window, per environment. Staging usually turns out to be production-shaped and dev turns out to be much smaller than its instance sizes suggest.
- 03 Convert build minutes into concurrent build capacity. Take the busiest hour of the busiest weekday in the window, divide the minutes by sixty, and add the queue depth you are willing to accept. This is the number of build executors you need, and it is normally between four and twelve for a team of thirty.
- 04 Add the storage that comes with it — artefacts, layer caches, package proxies — measured, not estimated. Build caches are the fastest-growing thing in most estates and they behave nothing like application storage.
- 05 Decide the sharing question now: does non-production run on the same machines as production with resource limits and priority, or on its own smaller ones? Both are defensible; only one of them is in the machine count, and Move 11 needs the answer.
- 06 Write the decision down with its consequence stated plainly. Sharing saves capital and puts your test suite one bad manifest away from production. Separating costs money and makes the blast radius a conversation you never have to have.

ROLLBACK

The output is a number and a decision, and both are revised by revising them. The point of no return is Move 20: once the machine count includes a separate non-production tier, unpicking that means either returning hardware inside a vendor window or living with capacity you did not want. Until then the cost of being wrong is an hour of arithmetic. Keep the environment inventory under source control so an earlier version can be restored and compared when somebody argues the estate was smaller than this.

04 CORES AGAINST CLOCK, AND THE LICENCE THAT DECIDES IT

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
IRON	—	HIGH	UNTIL THE ORDER IS SIGNED

Picks socket count, generation and vendor against the derated requirement, then re-checks the answer against per-core licensing, which frequently inverts it.

AWS	Dedicated Hosts and licence-included instances	the licence was bundled into the hourly rate, or brought under a host you did not have to count cores on yourself.
GOOGLE	Sole-tenant nodes	the same bundling, with node affinity labels doing the work that a physical machine boundary will do for you afterwards.
AZURE	Azure Hybrid Benefit and dedicated hosts	the one place where the licensing arithmetic was already visible to you, which makes this Move less of a surprise there.

BEFORE YOU START

ACCESS

- The derated core requirement from Move 02, and the non-production requirement from Move 03
- Your current licence entitlements, in writing, from whoever holds the contracts

SOFTWARE

- Benchmark results for the candidate parts on something that resembles your workload, not a synthetic score
- A three-year power price for the site you expect to use, even if the site is not chosen yet

PEOPLE

- Someone from procurement or legal who can get a licensing position confirmed in writing rather than assumed

WHY THIS WORKS

Silicon is priced per core and software is priced per core, and the two curves point in opposite directions. A dense dual-socket part gives you the cheapest core you will ever buy and, under a per-core licence, the most expensive server you will ever run. Oracle counts every physical core in the host at the 0.5 x86 factor unless the partitioning is one it has approved as hard; SQL Server licenses every core on the server with a four-core minimum per processor; Windows Server carries an eight-core minimum per processor and sixteen per server. So the shape of the fleet is decided by what is licensed, not by what is fastest, and the licensed workloads want few fast cores on a machine of their own.

SWAP

Where a licensed database is the only reason for an expensive shape, price migrating it off that engine instead. Sometimes the licence is the migration, and the hardware question disappears.

DO IT FASTER

Ask two vendors to quote the same requirement rather than the same configuration. They will each propose a different part and the gap between them is the information you wanted.

WATCH OUT

Core factors and minimums change between contract versions. The position that held for your last renewal may not hold for the estate you are about to build.

LEFTOVERS

Licences already bought against cloud instances may or may not carry to hardware you own. Establish that now, because it is worth more than the servers.

—	—	—	0 min	5 days	2 weeks
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet, but the licensing audit this Move forces has been known to find entitlements already paid for and never deployed.

NEEDS 02 What a vCPU is actually worth

- 01 Split the requirement into licensed and unlicensed workloads before choosing anything. These are two different fleets that happen to arrive on the same delivery.
- 02 For the unlicensed majority, pick the cheapest core that meets the clock floor your latency-sensitive services need. One socket is usually right for a Kubernetes node; two sockets earn their place only for memory capacity or PCIe lanes, not for core count.
- 03 For the licensed workloads, invert the question. Choose the smallest core count that clears the performance requirement and the highest clock available at that count, then check the per-server minimums, which frequently make a small machine cost the same as a medium one.
- 04 Price the generation behind the current one. It is worth buying when the discount exceeds the performance-per-watt gap over three years of electricity, and on a site where you now pay the power bill directly that calculation is different from the one the cloud made for you.
- 05 Get the licensing position in writing before the order goes anywhere. Ask the specific question — how many licences does this exact configuration require — and keep the answer. Verbal reassurance from a reseller is not a defence in an audit.
- 06 Publish the two fleet shapes with their cost per core and their cost per licensed core side by side. The second number is the one that changes minds.

ROLLBACK

A choice of part is reversed by choosing a different one, at no cost, right up until Move 20 turns it into an order. The point of no return is that signature: after it, a licensing error is a bill rather than a revision. Keep the benchmark data, the vendor quotes and the written licensing position in source control, so the reasoning can be restored and re-argued when somebody asks in eighteen months why the estate is shaped like this.

05 SMT, NUMA, AND THE WORKLOAD THAT NOTICES

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
IRON	—	MEDIUM	IMMEDIATELY, BY REBOOT

Settles SMT and the NUMA layout before latency-sensitive work arrives, and sets the kubelet policies without which two containers can share one physical core.

AWS	EC2 CPU options	you could disable threads per core at launch, and the underlying topology was otherwise hidden behind the instance shape.
GOOGLE	Compute Engine visible NUMA and threads-per-core settings	exposed on some families, absent on others, and never something you had to reason about per host.
AZURE	Virtual Machines constrained core sizes	the licensing-driven answer to the same question, which set core count without giving you the topology underneath.

BEFORE YOU START

ACCESS

- Firmware settings on a loan or evaluation machine of the part chosen in Move 04
- A latency-sensitive workload you can run in a loop and measure honestly

SOFTWARE

- A load generator that reports percentiles rather than averages
- `ipmitool` or the vendor equivalent, so firmware settings can be changed without a trip to the rack

PEOPLE

- The owner of the most latency-sensitive service, to agree what regression would be unacceptable

WHY THIS WORKS

An instance shape hid two decisions from you. The first is simultaneous multithreading: with it on you get more throughput and less predictable latency, because two threads contend for one core's L1 and L2. The second is the NUMA layout, which on a chiplet EPYC matters more than the socket count, since the NUMA-per-socket setting and the last-level-cache boundary decide whether a memory access crosses a link. Kubernetes honours none of this by default. Without the CPU manager on its static policy and the topology manager set to single-numa-node, the scheduler will happily place two containers on the two halves of one physical core and both will run slowly for reasons nothing reports.

SWAP

If nothing in the estate has a hard latency commitment, leave SMT on, leave the topology manager at its default, and revisit when something does. Complexity you cannot measure is complexity you will misconfigure.

DO IT FASTER

Decide once, per machine class, and put the answer in the machine configuration rather than in a runbook. A setting that has to be applied by hand will be wrong on the fourth machine.

WATCH OUT

Changing the CPU manager policy on a running node requires removing its state file and restarting the kubelet; the policy alone does not take effect and the failure is silent.

LEFTOVERS

Halving your logical core count by turning SMT off changes the derated requirement from Move 02. If you choose off, go back and redo that arithmetic before Move 11 counts machines.

—	—	—	0 min	3 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing. This Move changes what the hardware you have not yet bought will be asked to do.

NEEDS 04 Cores against clock, and the licence that decides it

- 01 Measure the workload with SMT on and again with it off, at the same offered load, and compare p99 rather than throughput. Most services prefer it on; a small, loud minority does not, and the minority is usually the one with a latency commitment.
- 02 Set the NUMA-per-socket option in firmware with `ipmitool` or the vendor tool, choosing the value that matches how you intend to schedule, and record the resulting node topology. On the current EPYC parts the useful settings are one node per socket for general work and a per-quadrant split for workloads that fit inside a cache boundary.
- 03 Configure the kubelet with the CPU manager static policy, the full-pcpus-only option, and the topology manager on single-numa-node. Without the second of those, a guaranteed pod can still be handed the two threads of one core.
- 04 Reserve cores explicitly for the system and the kubelet rather than leaving them to compete. Two cores per socket is a reasonable starting point and it should be measured, not inherited.
- 05 Re-run the load test with the policies in place and confirm the improvement is real. If it is not, keep the simpler configuration; these settings cost scheduling flexibility and should be paid for with evidence.
- 06 Write down the symptom of each wrong setting — the throughput cliff above a load threshold, the p99 that only degrades when a neighbouring container is busy — because these faults arrive looking like an application regression and are debugged for weeks by the wrong team.

ROLLBACK

Every setting in this Move is reversed by a firmware change and a reboot, or by a kubelet configuration change and a node drain. There is no point of no return here at all, which is precisely why it belongs before the purchase order rather than after it: the same experiment costs an afternoon now and a maintenance window later. Keep the firmware profiles exported to a file so a known good configuration can be restored on a machine that has been fiddled with.

06 MEMORY IS WHAT RUNS OUT FIRST

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
IRON	—	HIGH	UNTIL THE ORDER IS SIGNED

Sizes memory from measured working set plus reserves, rounds up to a population that fills every channel, and orders it first because it is the longest lead.

AWS	EC2 memory-optimised families	memory came in fixed ratios to vCPU, so you bought cores you did not want to reach a capacity you did.
GOOGLE	Compute Engine custom machine types	the one place you could set memory independently of cores, within a per-core band that still constrained the answer.
AZURE	Virtual Machines E-series and constrained-core sizes	memory was bought by series, and constrained cores existed mainly to make the licensing arithmetic survivable.

BEFORE YOU START

ACCESS

- The thirty-day window from Move 01, with per-container working set separated from page cache
- A live quote for the exact module part number, dated, because last quarter's price is not a price

SOFTWARE

- The vendor's memory population guide for the chosen socket, which is a real document and not a rule of thumb
- The bandwidth figures for one and two modules per channel on that part, at the speed you will actually run

PEOPLE

- Whoever approves capital spend, warned that this line moves between quote and order

WHY THIS WORKS

Nodes do not run out of CPU. They run out of memory, and when they do the kubelet starts evicting, which is a much worse failure than being slow. So memory is sized from the measured working set from Move 01, plus page cache, plus the kubelet and system reserved, plus the eviction threshold you intend to set — and then rounded up to a DIMM population that fills every memory channel, because a half-populated channel map throws away bandwidth out of all proportion to the capacity it saves. This is also the line with the longest lead time and the most volatile price in the whole bill of materials, which is why it is sized here and ordered first.

SWAP

Where capacity matters more than bandwidth — a large in-memory cache, a build farm — the second module per channel is worth its speed penalty. Measure the workload rather than applying a rule.

DO IT FASTER

Order one machine's worth of modules early and test them in a loan chassis. A population that the firmware will not train at the rated speed is much cheaper to discover on one machine than on twenty.

WATCH OUT

Registered and load-reduced modules are not interchangeable, ranks affect the achievable speed, and mixing part numbers within a channel map is how a machine ends up running a whole speed grade below its rating with nothing reporting it.

LEFTOVERS

Modules pulled from decommissioned machines are rarely worth carrying forward. Different rank, different speed, out of warranty, and the failure they cause is intermittent.

—	—	—	0 min	3 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. This Move is where the largest single line on the bill of materials gets its number.

NEEDS 02 What a vCPU is actually worth 04 Cores against clock, and the licence that decides it

- 01 Take the measured working set per node class and add the reserves that are not optional: kubelet reserved, system reserved, the eviction threshold, and page cache sized against how much of your read traffic currently comes from it. Fifteen to twenty-five per cent above raw working set is a defensible starting point, and it should be written as a calculation rather than a factor.
- 02 Round the result up to a population that fills every channel. Current AMD EPYC on the SP5 socket has twelve channels per socket and accepts two modules per channel, so a full first fill is twelve modules and the second fill is twenty-four. The Intel Xeon 6 6900P parts also have twelve channels but are one module per channel, so twelve is both the first fill and the last.
- 03 Check the speed penalty before choosing the second fill. Two modules per channel runs slower than one on the same part, by a documented step, and that step is worth reading rather than guessing: on a memory-bandwidth-bound workload it can cost more than the capacity gains.
- 04 Decide now whether memory is ever added later, because on a one-module-per-channel part the answer is no and on a two-module part it means replacing modules rather than adding them. Either way it is a decision made at purchase, not deferred by it.
- 05 Get a live quote and treat it as expiring. Server memory prices and lead times have moved a long way outside their historical range as high-bandwidth allocation has consumed fabrication capacity, and the figure that matters is the one your supplier will honour this week.
- 06 Order this line first, ahead of everything else on the bill of materials. A rack of machines waiting for modules is the most common way this programme slips a quarter.

ROLLBACK

Until the order is signed this is a spreadsheet and reversing it costs an afternoon. After that it is the hardest line to change, because modules are consumed by the population map and a smaller order cannot be topped up without replacing what arrived. The point of no return is the purchase order in Move 20. Keep the sizing calculation, the quote and the population guide in source control so the reasoning can be restored when a machine arrives with a different part number on the label than the one you priced.

07 ONE RACK UNIT OR TWO, AND THE WATTS IT DRAWS

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
IRON	—	MEDIUM	UNTIL THE ORDER IS SIGNED

Chooses chassis and supplies from the CPU and DIMM population, and produces the per-node and per-rack draw the facility contract in Part II is written against.

AWS	EC2 instance families	density, cooling and power were somebody else's problem, expressed to you only as a price per hour.
GOOGLE	Compute Engine machine families	the same abstraction, with the added detail that local SSD capacity was fixed per shape rather than chosen.
AZURE	Virtual Machines series	the L-series existed precisely because drive bays are a physical constraint, which is the constraint this Move makes visible again.

BEFORE YOU START

ACCESS

- The processor and memory selections from Moves 04 and 06, as part numbers rather than descriptions
- The drive count from Move 10 if it exists yet, or a working assumption you will revisit

SOFTWARE

- The vendor's chassis configuration tool, which will refuse combinations a datasheet suggests are fine
- A power calculator from the same vendor, used for shape rather than for precision

PEOPLE

- Whoever will negotiate the facility contract in Part II, because the draw you produce here is their opening number

WHY THIS WORKS

The chassis is where every earlier decision becomes a physical object with a height, a depth, an airflow direction and a power draw. A 1U machine is denser and its fans work harder, which costs watts and makes noise a real consideration in a room you own; a 2U machine gives you drive bays, PCIe slots and cooling headroom for less fan power. Neither is right in general. What is right in general is that the answer is derived from the CPU thermal design power and the module population you have just chosen, and that the output is a number in watts per node and per rack, because that number is what the facility contract in Part II is written against.

SWAP

Where the estate is small and the room is your own, 2U everywhere is a defensible default. Density is a constraint of expensive floor space, and if the floor is cheap, buy the cooling headroom.

DO IT FASTER

Ask the vendor for a configured power report for the exact build. It takes them an hour and it is more accurate than any calculator you will run.

WATCH OUT

Fan power rises steeply with inlet temperature. A 1U node validated in a cool lab can draw considerably more in a warm aisle, and that difference lands on your electricity bill every hour for three years.

LEFTOVERS

Rails, cable-management arms and bezels are usually quoted separately and forgotten. They are trivial money and they hold up an installation day.

—	—	—	0 min	3 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. The output of this Move is a number in watts, and Part II is where it starts costing money.

NEEDS 04 Cores against clock, and the licence that decides it 06 Memory is what runs out first

- 01 Set the height from what has to fit. Drive bays and PCIe slots drive 2U; nothing else usually does. Count the slots you need for network cards and any accelerator before choosing, because a slot you discover you need after delivery is a new chassis.
- 02 Fix rack depth and the rail kit against a cabinet you have not signed for yet. Deep chassis in a shallow cabinet is the most common delivery-day failure, and the cabinet is chosen in Part II, so record the constraint rather than assuming a standard.
- 03 Choose front or rear input and output so that cabling and airflow agree. Cold aisle at the front, hot aisle at the back, cables where the air is not. A machine that has to be cabled from the cold aisle will be cabled from the cold aisle for its whole life.
- 04 Size the supplies against measured draw rather than badge rating. Titanium-rated units are generally 200 to 240 V input only and peak in efficiency near half load, so a pair of oversized supplies running at fifteen per cent is both more expensive and less efficient than the right pair.
- 05 Confirm that the two supplies can be fed from two different sources. Two supplies on one busbar is one feed and two failure points, which is worse than one supply and an honest single point of failure, because it is believed to be redundant.
- 06 Refuse multi-node and blade chassis for this estate. A shared midplane, shared supplies and vendor-locked spares move the failure domain from the node to the enclosure, which is exactly the direction you are trying to travel.
- 07 Publish per-node and per-rack draw at idle, at your expected load and at the peak the machine can reach. Part II needs all three, and the third one is what a facility charges against.

ROLLBACK

Nothing is committed until Move 20 signs the order, and until then a chassis choice is reversed by changing a line in a configuration. The point of no return is that order, after which a machine that does not fit the cabinet is a logistics problem rather than an edit. Keep the vendor's configured build sheets and power reports in source control alongside the bill of materials so the original assumptions can be restored and compared when the delivered draw turns out to differ from the quoted one.

08 THE WAY IN WHEN THE MACHINE WILL NOT BOOT

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
IRON	THE PROVIDER'S SERIAL CONSOLE AND BOOT DIAGNOSTICS	HIGH	UNTIL THE ORDER IS SIGNED

Treats the management controller as a purchase-order line, tests Redfish on a loan unit, and finds which console features sit behind a licence before delivery.

AWS	EC2 Serial Console and EC2 Instance Connect	a console you could reach from an API call, with no licence, no firmware and no cable.
GOOGLE	Compute Engine interactive serial console	the same idea, enabled per project or per instance, and free.
AZURE	Serial console and boot diagnostics	boot screenshots included, which is the feature people miss most when it stops existing.

BEFORE YOU START

ACCESS

- A loan or evaluation unit of the exact chassis chosen in Move 07, with a management port you can reach
- The vendor's licence matrix for that generation, in writing rather than from a forum post

SOFTWARE

- A Redfish client, or `curl` and the specification, to test the API rather than the web interface
- `ipmitool` for the operations the vendor has not yet moved to Redfish, which is still several of them

PEOPLE

- Whoever writes the purchase order, so the licence lands on it rather than on next quarter's budget

WHY THIS WORKS

Every cloud gave you a way to watch a machine boot and to type at it when the network was not working. That was a product feature, and on your own hardware it is a baseboard management controller: a small computer inside the server with its own processor, its own network port and its own firmware. It is the difference between diagnosing a failed boot from your desk and driving to a building at three in the morning. It is also, on every major vendor, partly behind a licence — and the moment to discover that is before delivery, when it is a line item, rather than after, when it is an emergency purchase at list price.

SWAP

Where the vendor's licence is priced absurdly, a serial console server reaching every machine's serial port covers the same ground for boot diagnosis, though not for virtual media. Move 16 buys that hardware anyway.

DO IT FASTER

Ask the vendor for a temporary licence key on the loan unit. Most will issue one for thirty days and it turns this Move from a negotiation into a test.

WATCH OUT

Management controllers are a security boundary and a well-known target. They belong on their own network, they do not belong on the internet, and their default credentials belong changed before the machine leaves the loading bay.

LEFTOVERS

Some vendors bind licences to a service tag. A machine returned under warranty can come back with a controller that has forgotten it was licensed, and re-issuing the key is a support case rather than a click.

—	—	—	0 min	2 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. This is the Move that stops Part II being carried out by driving to a building.

NEEDS 07 One rack unit or two, and the watts it draws

- 01 Establish which console features are licensed on the exact model quoted. Remote console and virtual media sit behind iDRAC Enterprise on Dell, behind iLO Advanced on HPE, and behind the DCMS key on Supermicro, and the naming changes between generations.
- 02 Test Redfish against the loan unit rather than the datasheet. Ask it for the system inventory, the power state, the boot source override, the event log and a firmware version, and confirm each returns what the specification says it should. Vendors claim conformance and implement subsets.
- 03 Test the operations you will actually need under pressure: power cycle, boot to a one-time source, mount an image over virtual media, and read the log after a forced reset. Do them over the network from a laptop, which is how you will do them for real.
- 04 Confirm that `ipmitool` still works for anything Redfish does not cover, and record which is which. Running both is normal; pretending one is enough is how a runbook stops halfway through.
- 05 Record the firmware support horizon: how long this controller will receive updates, how those updates are delivered, and whether an update requires a host reboot. That horizon is the number that actually dates the platform, more than the processor does.
- 06 Put the licence, at the correct part number, on the purchase order for every machine, including the spare. A controller licence bought under pressure costs more and arrives slower than one bought with the server.

ROLLBACK

Until the order is signed, a controller decision is reversed by editing the quote. After that, adding a licence to twenty machines is a purchase at list price and a per-machine application step, which is annoying rather than fatal. The point of no return is the purchase order in Move 20. Keep every key, its service tag and the vendor's licence matrix in the same place as the bill of materials, so entitlement can be restored quickly when a controller is replaced and arrives blank.

09 THE DRIVE YOU ARE ALLOWED TO PUT ETCD ON

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
IRON	—	HIGH	PER DRIVE, AT THE COST OF A CEPH REBUILD

Fixes interface, form factor and endurance, and makes power-loss protection a hard requirement, because a drive without it is a quorum defect you have bought.

AWS	EBS io2 and gp3, and instance store	durability, endurance and power-loss behaviour were the provider's problem, expressed as an IOPS number you paid for.
GOOGLE	Persistent Disk and Local SSD	the same split, with local devices explicitly ephemeral so nobody was tempted to trust them.
AZURE	Premium SSD v2 and local temporary disks	again a provisioned IOPS figure standing in for every physical property of the device underneath.

BEFORE YOU START

ACCESS

- The chassis and bay configuration from Move 07, because the form factor question is decided by the backplane
- Datasheets for the candidate devices, with the endurance term stated rather than implied

SOFTWARE

- A test rig capable of measuring fsync latency, not just throughput
- The vendor's compatibility matrix for the exact backplane in the exact chassis

PEOPLE

- Whoever will hold the spares budget, since endurance class and spare count are the same conversation

WHY THIS WORKS

etcd is a consensus system that makes a durability promise on every write, so it calls fsync constantly and its latency is the fsync latency of the device under it. A drive with power-loss protection has a capacitor bank that lets it acknowledge a flush as soon as the data is in its own protected buffer. A drive without one must genuinely push the data to flash on every fsync, which turns a microsecond into a millisecond and turns a healthy cluster into one that loses leadership under load. That is why this is a hard requirement and not a preference: a consumer solid-state drive in a control-plane node is a quorum outage you have paid for in advance.

SWAP

Where a control-plane node cannot take two devices, put the consensus data on the fastest single protected drive and take the node loss seriously. Three nodes each with one good drive beats three nodes each with two bad ones.

DO IT FASTER

Buy one of each candidate and measure fsync latency at queue depth one. It is a ten-minute test and it settles arguments that datasheets cannot.

WATCH OUT

Firmware matters as much as the part number. Fleets have been bitten by a specific firmware revision that reports wear incorrectly or drops out under a particular queue pattern, and the fix is a controlled update rather than a return.

LEFTOVERS

Drives pulled from the old estate are almost never suitable here. Unknown wear, unknown firmware, no protection, and the failure mode is silent.

—	—	—	0 min	4 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. This Move decides whether Part III has a control plane that stays elected.

NEEDS 07 One rack unit or two, and the watts it draws

- 01 Fix the interface first. NVMe for anything that matters, SATA only for cold bulk where cost per terabyte dominates, and SAS only where an enclosure you already own forces the issue. There is no case in this book for putting the cluster's consensus data on anything but NVMe.
- 02 Make power-loss protection a stated requirement on the purchase order, checked per part number. It is the property most often absent from a drive that otherwise looks identical on a comparison table.
- 03 Settle the form factor against the backplane, because compatibility runs one way. A U.2 drive works in a U.3 bay; a U.3 drive does not work in a U.2 bay. A mixed order placed without checking can arrive entirely unusable.
- 04 Set the endurance class in drive writes per day against the measured write rate from Move 01, and read the warranty term it is quoted over. Two drives with different terms are not comparable on that number alone, and the shorter term flatters the figure.
- 05 Separate the control-plane devices from everything else at order time: a mirrored pair of small, high-endurance drives per control-plane node, sized for the consensus data and its own snapshot rather than for capacity.
- 06 Name the wear metrics you will alarm on, and confirm the candidate reports them: percentage used, available spare against its own threshold, media and data integrity errors, and unsafe shutdowns. A drive that does not report these is a drive you will replace after it has already caused an incident.

ROLLBACK

A drive choice is reversed one device at a time, and outside the control plane that is a Ceph operation rather than an outage: mark the device out, let the cluster restore its data by backfilling from the surviving copies, and replace it. The cost is rebuild time and the risk window while the pool is degraded. The point of no return is the purchase order in Move 20, since a bay full of the wrong form factor cannot be corrected by software. Keep the compatibility matrix and the measured latency results with the bill of materials.

10 HOW MUCH FLASH, AND WHERE IT SITS

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
IRON	BLOCK VOLUMES THAT GROW WITH AN API CALL	HIGH	UNTIL THE BAYS ARE FULL

Turns the measured capacity requirement into drives against Ceph's replication and its nearfull thresholds, and separates the etcd, local-NVMe and OSD tiers.

AWS	EBS	a volume grew with a modify call and the capacity behind it was somebody else's planning problem.
GOOGLE	Persistent Disk and Hyperdisk	the same elasticity, with performance provisioned separately from capacity on the newer types.
AZURE	Managed Disks	resized in place with the same one-way ratchet, since none of the three let you shrink a volume afterwards either.

BEFORE YOU START

ACCESS

- The measured capacity and write rate from Move 01, separated by workload rather than summed
- The bay count per chassis from Move 07, and the device choice from Move 09

SOFTWARE

- A capacity model you can show people, with the replication factor and the reserve as visible inputs
- The Ceph documentation for the version you intend to run, since the default thresholds are version-specific

PEOPLE

- Whoever owns the largest dataset, to say how fast it is actually growing rather than how fast it grew once

WHY THIS WORKS

Capacity used to be a request. Now it is a purchase with a lead time, and the arithmetic between the two is not one to one, because a replicated storage system consumes more raw space than it presents. Three-way replication triples raw consumption. Erasure coding does not, and pays for it in recovery cost when a device fails. Both need free space held back, because Ceph warns at eighty-five per cent, stops backfilling at ninety and stops accepting writes at ninety-five — so a cluster planned to sit at its nearfull mark passes every check on a healthy day and fails during precisely the recovery it was bought for.

SWAP

For cold, large, rarely-read data, erasure coding across more nodes saves real money. It costs recovery time and CPU, and it is a poor choice for anything latency-sensitive.

DO IT FASTER

Model the cluster at full, at one device down and at one node down in the same spreadsheet. Three columns settle most of the arguments this Move produces.

WATCH OUT

Mixed drive sizes in one pool skew placement, so the largest devices take proportionally more data and fill first. Keep a pool's devices the same size, or accept the imbalance deliberately.

LEFTOVERS

The old provider's volumes keep billing after the data has moved, and unattached ones bill silently. They are deleted in Part V, after the copy is proven, and not before.

—	—	—	0 min	3 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. The provider volumes stay until Part V has moved the data and proved the copy.

NEEDS 02 What a vCPU is actually worth 09 The drive you are allowed to put etcd on

- 01 Separate the three tiers before sizing any of them, because they have different endurance, different sizes and different failure consequences: a mirrored pair of small high-endurance drives per control-plane node for etcd, local NVMe for the single-writer databases that will not sit on a replicated pool, and the OSD devices that make up the cluster's own storage.
- 02 Size the OSD tier from usable capacity backwards. Decide the replication scheme per pool, multiply, then divide by the fraction of the cluster you are willing to fill. Planning to sixty-five per cent of raw is a reasonable default and anything above seventy-five is a decision that needs writing down.
- 03 Add the growth you actually measured plus the lead time you will actually face. Twelve to eighteen months of headroom is normal when new drives take six weeks to arrive, and it is cheaper than an emergency order.
- 04 Check the failure arithmetic. Losing one device must not push the cluster past its backfill threshold while it recovers, and losing a whole node must leave enough space for the data that node held to land somewhere. Both of those are capacity questions, and both are missed.
- 05 Buy chassis with bays deliberately left empty. Capacity that grew with an API call now grows only when somebody walks to the rack, and an empty bay is the cheapest form of that walk.
- 06 Write down the point at which you would add a node rather than more drives, and what triggers it. Without that line, a cluster grows by drive until it is unbalanced and nobody can say when that happened.

ROLLBACK

Until the bays are full, this is reversed by ordering more drives, which is the whole point of leaving them empty. After that it becomes a node purchase with a lead time. The point of no return is the moment the bays fill and the next increment is a chassis rather than a device. Under-buying is recoverable and over-buying is money spent early; both are survivable. Keep the capacity model in source control so an earlier version can be restored and compared against what the cluster actually consumed.

11 HEADROOM, AND THE NUMBER OF MACHINES YOU ORDER

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
IRON	ELASTIC NODE CAPACITY: CLUSTER AUTOSCALING AND NODE AUTO-PROVISIONING	HIGH	UNTIL THE ORDER IS SIGNED

Multiplies the node shape by the derated requirement, adds the losses per-hour pricing hid, and produces the machine count, rack-unit budget and total draw.

AWS	Cluster Autoscaler and Karpenter on EKS	capacity arrived in ninety seconds and left again, so nobody ever had to state a fleet size.
GOOGLE	GKE cluster autoscaling and node auto-provisioning	the same elasticity, with node pools created for you when a pod would not fit.
AZURE	AKS cluster autoscaler and node auto-provisioning	the same again, and the same reason the fleet size was never a number anyone wrote down.

BEFORE YOU START

ACCESS

- The derated requirement from Move 02 and the non-production decision from Move 03
- The measured largest single pod in the estate, by both cores and memory

SOFTWARE

- The node shape from Moves 04 through 07, with usable cores and usable memory calculated rather than assumed
- A bin-packing model, even a crude one, that can be shown to somebody who doubts the answer

PEOPLE

- Whoever will carry the on-call rota, to agree how much failure the fleet should absorb without a human

WHY THIS WORKS

Autoscaling let you avoid this arithmetic by paying a premium per hour to never do it. Owning the hardware means doing it once, properly, and then living with the answer for three years. The requirement from Move 02 is not the fleet size, because a node does not give a pod all of its cores or all of its memory: the kubelet and the system take some, the storage and networking daemons take some, bin packing wastes some, and the largest single pod you run sets a floor on how much of a node must stay free. Add the capacity to lose a node — or a rack — without evicting production, and the honest number is meaningfully larger than the raw requirement.

SWAP

Where the workload genuinely spikes by an order of magnitude a few times a year, keep a small cloud account for the spike and size the metal for the base. Move 116 makes that a deliberate design rather than an admission of defeat.

DO IT FASTER

Model two fleets — many small nodes and few large ones — against the same workload. The overhead figures usually make the decision for you within an hour.

WATCH OUT

A fleet sized exactly to the requirement has no room for the rolling upgrade in Move 109, which needs somewhere to put the workloads from the node it is draining. Headroom is not optional; it is the upgrade path.

LEFTOVERS

The autoscaling configuration, the launch templates and the node auto-provisioning policies stay in the cloud accounts until Part VII closes them. Leave them alone until then; they are the way back.

—	—	—	0 min	4 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. Autoscaling stays exactly as it is until Part VII, because it is what carries production while all of this is being built.

NEEDS 03 The estate nobody sizes 05 SMT, NUMA, and the workload that notices 06 Memory is what runs out first

07 One rack unit or two, and the watts it draws 10 How much flash, and where it sits

- 01 Compute usable capacity per node: total cores and memory, less kubelet and system reserved, less the daemonsets that run everywhere — the storage agents, the network agent, the log shipper, the metrics agent. On a modern estate that overhead is several cores and several gigabytes per machine, and it is the same on the smallest node as on the largest, which is an argument for larger nodes.
- 02 Model bin-packing waste against the largest single pod rather than the average one. If your biggest workload needs twelve cores, every node has to be able to hold twelve free cores when it is scheduled, and that constraint costs more than the pod does.
- 03 Add failure headroom explicitly and say what it protects against, in the terms the on-call rota will experience it. Tolerating one node loss is the minimum. Tolerating the loss of a whole rack is a different number and it is the one that matters once Part II puts more than one rack in the plan.
- 04 Fix the control-plane count at three, or five for an estate that expects to grow past a few hundred nodes, and state which failure domain each one sits in. Three control-plane nodes in one rack is three machines and one failure domain.
- 05 Sum the result into three outputs: the machine count, the rack-unit budget from Move 07, and the total power draw at expected and peak load. These three numbers are the entire input to the Part II facility conversation.
- 06 Show the fleet size to the people who have been asking for this programme, and be honest about what it means. Capacity is now bought in whole servers with a lead time. Nothing scales to meet a launch on Thursday unless it was bought in advance, and that is the real trade this book is asking them to make.

ROLLBACK

Until the order is signed the machine count is a number in a model, and it is revised by revising the model. The point of no return is Move 20. Ordering too few is recoverable at the cost of a lead time and a second delivery; ordering too many is capital spent early and racked slowly. Neither is fatal, which is worth saying plainly to whoever is nervous. Keep the model and its inputs in source control so a previous version can be restored when somebody asks why the estate is this size.

12 TWO SWITCHES, AND THE SPINE YOU DO NOT NEED YET

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
IRON	—	HIGH	UNTIL THE SECOND RACK IS CABLED

Chooses the fabric topology from measured east-west traffic, refuses a spine below four racks, and records whether the fabric runs MLAG or BGP to the host.

AWS	VPC networking	a flat, high-bandwidth fabric with no topology you could see and no oversubscription you could measure.
GOOGLE	VPC networks and subnets	the same absence of shape, with the added property that a network spans regions, which nothing you build will do.
AZURE	Virtual Networks	again a fabric with no visible shape, plus accelerated networking as the one knob that hinted there was hardware underneath.

BEFORE YOU START

ACCESS

- East-west traffic measurements from the existing estate, per availability zone and between zones
- The rack count and machine count from Move 11

SOFTWARE

- Flow data from the current cloud, exported for a representative week rather than a quiet Sunday
- A cabling diagram tool, or graph paper, because the mesh is easier to refuse once drawn

PEOPLE

- Whoever will be cabling the racks in Part II, since a topology nobody can cable is not a topology

WHY THIS WORKS

A leaf-spine diagram is the standard answer to a question most estates do not have. Below four racks, a pair of top-of-rack switches with a peer link between them carries everything, costs a fraction as much, and has fewer devices to fail and fewer to upgrade. A spine earns its place when the rack count or the cabling makes a direct mesh unmanageable, and buying it early spends ports, optics, power and rack units for years before anything uses them. What decides the shape is the east-west traffic you have actually measured, and the second output — whether the fabric runs a multi-chassis link aggregation or plain routing to the host — determines how Cilium is configured in Move 45 and how service addresses are advertised in Move 96.

SWAP

Where a spine is genuinely needed later, adding one to a well-cabled pair of leaves is a planned change with a maintenance window, not a rebuild. Design the addressing in Move 39 so that it stays true.

DO IT FASTER

Draw the cabling for the target rack count before choosing. If it fits on one page and a person can follow it, you do not need a spine.

WATCH OUT

A multi-chassis link aggregation is two switches pretending to be one, and the pretence has failure modes — split brain, unequal state, upgrade coupling — that plain routing does not. Choosing it because the diagram is simpler is choosing it for the wrong reason.

LEFTOVERS

Cloud flow logs cost money to keep and this Move is the last time you need them at this resolution. Turn the sampling back down when you are done.

—	—	—	0 min	4 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet, though the flow-log sampling rate this Move needed can go back down as soon as the measurement is finished.

NEEDS 11 Headroom, and the number of machines you order

- 01 Measure east-west traffic properly, per hour, across a full week. Most estates discover their real internal traffic is an order of magnitude smaller than they assumed, and a small number of pairs — the application to its database, the log shipper to its store — account for nearly all of it.
- 02 Compute the oversubscription ratio you would run at with a given uplink count, using those measurements. A ratio between three and one and five and one is unremarkable for general workloads; storage replication traffic is the thing that argues for less.
- 03 Decide the topology from that number and the rack count. One to three racks is a pair of top-of-rack switches with a peer link and no spine. Four or more, or a cabling plan that has become a mesh, is where leaf-spine starts paying for itself.
- 04 Settle the redundancy model, and settle it now rather than at installation. Either the two leaves present as one logical switch to the host through a multi-chassis link aggregation, or each leaf is an independent router and the host runs a routing protocol to both. Both work; they lead to different configurations in Part III.
- 05 Size uplinks and leave ports free. A fabric with no spare ports is a fabric that cannot absorb the next rack, and ports are cheapest when bought with the switch.
- 06 Write the topology decision down with the traffic data attached, because it will be challenged by somebody holding a reference architecture, and the traffic data is the only answer that ends that conversation.

ROLLBACK

A topology decision is free to change right up until the second rack is cabled, because a pair of leaves with spare ports can become the first two leaves of a spine design without recabling the first rack. The point of no return is the day the second rack's uplinks are terminated against a design, since undoing that means a fabric outage and an afternoon with a cable tester. Keep the traffic measurements and the cabling diagram in source control so the original design can be restored and compared when the third rack arrives.

13 THE SWITCH, AND THE SOFTWARE RUNNING ON IT

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
IRON	—	HIGH	UNTIL THE ORDER IS SIGNED

Buy tier-one merchant-silicon leaves running a supported network operating system, and checks the quoted licence tier for the routing features assumed.

AWS	VPC route tables and Transit Gateway	routing was a set of API objects and the forwarding plane was invisible.
GOOGLE	VPC routes and Cloud Router	the same, with a managed router that spoke BGP on your behalf without you configuring one.
AZURE	Route tables and Virtual WAN	again managed routing, and again nothing you could log into when a path went wrong.

BEFORE YOU START

ACCESS

- The topology and redundancy decision from Move 12, since it changes which features you need licensed
- Loan units of the exact model being quoted, not a similar one from the same family

SOFTWARE

- The vendor's licence matrix for the exact part number, and the release notes for the version you intend to run
- A traffic generator capable of producing the microbursts you measured, rather than a steady stream

PEOPLE

- Somebody in procurement who can get a licensing answer in writing before the order goes in

WHY THIS WORKS

The switch is the one device in this book that a small team should refuse to be clever about. This book buys tier-one merchant-silicon leaves running a vendor network operating system, and the recommendation is Arista EOS on Trident-class hardware, because its routing and EVPN behaviour is documented, supported and dull, and dull is the property you want from the thing every packet crosses. A commercially supported Enterprise SONiC distribution is the reasonable alternative where procurement requires an open operating system. Community SONiC and Cumulus Linux are both real, and both are set aside here — the latter because its current release supports only one silicon vendor, which removes the point of merchant silicon.

SWAP

For a single-rack estate that will never grow, a pair of well-supported one-rack-unit leaves from any tier-one vendor is sufficient, and the operating system argument matters much less than the support contract.

DO IT FASTER

Ask for the loan units at the same time as the quote. The evaluation and the negotiation then run in parallel rather than in series, which is worth two weeks.

WATCH OUT

The same model number can ship with different silicon in different regions or generations. Confirm the exact silicon on the quote, because the feature set and the buffer behaviour follow it, not the badge.

LEFTOVERS

Evaluation licences expire and take features with them. If a loan unit becomes a production unit, make the licence conversion a purchase-order line rather than a surprise.

—	—	—	0 min	2 weeks	2 weeks
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. This Move buys the thing every packet in Parts III to VI will cross.

NEEDS 12 Two switches, and the spine you do not need yet

- 01 Fix port count and speed against three years of growth from Move 11, not against today. Twenty-five gigabit to the host with hundred gigabit uplinks is the sensible floor, and the spare ports you leave are the cheapest capacity in the estate.
- 02 Check buffer depth against the microbursts you measured rather than against a rule of thumb. Shallow-buffer leaves are fine for most traffic and are not fine for a storage cluster that synchronises writes across racks.
- 03 Read the licence tier on the exact part number being quoted. Routing protocols, EVPN, telemetry streaming and even some monitoring features sit behind tiers, the tiers are renamed between generations, and the detail that catches people is that the base tier on a quote is rarely the tier the design assumes.
- 04 Run the loan units against your own design for a week. Configure the routing you intend to run, pull a cable, pull a power lead, upgrade the operating system, and watch what the fabric does. A switch is bought for its behaviour when something breaks.
- 05 Confirm the automation path you will actually use — a configuration API, a structured data model, and a way to diff intended against running state. Configuring switches by hand is how a two-switch fabric drifts into two different fabrics.
- 06 Get the licensing and support answer in writing, including what a hardware replacement looks like and how long the model will be supported. That horizon dates the fabric more surely than its speed does.

ROLLBACK

Until the order is signed, changing switch vendor costs a fortnight of evaluation and nothing else. The point of no return is that signature: after it, a fabric that cannot do what the design assumed is either a licence purchase at list price or a replacement, and the optics you bought in Move 15 are coded for a vendor. Keep the loan-unit test results, the licence matrix and the running configurations in source control, so a known good configuration can be restored on a switch that has been replaced under warranty.

14 THE CARD, THE DRIVER, AND THE KERNEL YOU WILL RUN

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
IRON	THE PROVIDER'S PARAVIRTUAL NETWORK INTERFACE AND ITS SINGLE DRIVER	MEDIUM	PER NODE, AT THE COST OF A CARD SWAP AND A REBOOT

Chooses the network card against the Talos kernel rather than a datasheet, because there is no DKMS and no third place a driver can come from.

AWS	Elastic Network Adapter	one paravirtual device, one driver, in every kernel you were ever offered, so the question never arose.
GOOGLE	virtio-net and the gVNIC adapter	two devices, both in-tree, and the choice between them was a performance setting rather than a procurement decision.
AZURE	the synthetic adapter with accelerated networking	a virtual function appearing beside a synthetic device, with the driver handled by the platform image.

BEFORE YOU START

ACCESS

- The Talos release notes and kernel configuration for the version you intend to run in Part III
- The slot budget from Move 07 and the fabric decision from Move 13

SOFTWARE

- The exact model and firmware level of each candidate card, from the vendor rather than from a reseller listing
- A loan machine you can boot Talos on and check that the interface appears

PEOPLE

- Whoever will own the node lifecycle, since a card change later means touching every machine

WHY THIS WORKS

Talos Linux is an immutable, API-driven operating system with no shell and no package manager, and that is the property this book wants. It also means a driver is either compiled into the kernel Talos ships or supplied as an official system extension, and there is no third option: no DKMS, no vendor installer, no compiling on the host. Kernel module signatures are enforced. So the network card is chosen against that kernel, by exact model and firmware requirement rather than by family, before the order goes out — because on the far side of delivery there is no recovery path except a different card.

SWAP

Where an existing card is unsupported and cannot be replaced, the honest answer is a different host operating system for those machines, not a patched Talos. Keep the exception small and write down why it exists.

DO IT FASTER

Buy one card of each candidate before the fleet order. The test is an afternoon and it removes the only failure in this Move that cannot be fixed remotely.

WATCH OUT

Card firmware and switch software have compatibility opinions, particularly around link training and forward error correction at twenty-five gigabit and above. Test the card against the switch you actually bought, on the cable you actually bought.

LEFTOVERS

Cards pulled from the old estate are usually the wrong generation and always the wrong firmware. Treat them as spares for a lab, not for production.

—	—	—	0 min	3 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. The provider's network adapter stays in service until the workloads leave it in Parts IV to VI.

NEEDS 07 One rack unit or two, and the watts it draws 13 The switch, and the software running on it

- 01 List the candidate cards by exact model, not family. Two cards with the same marketing name can use different silicon and different drivers, and the one you get is the one on the part number.
- 02 Check each against the Talos kernel: in-tree driver, official system extension, or neither. Neither means it does not go on the list, however good the card is, because there is nowhere for the driver to come from.
- 03 Check the firmware requirement as carefully as the driver. A card whose driver needs firmware that is not shipped will bind and then fail to bring the link up, which is a much more confusing failure than a missing driver.
- 04 Boot Talos on a loan machine with the card in it and confirm the interface appears, links at the expected speed, and that the node comes back with the same interface name after a reboot. Ten minutes of this is worth every datasheet.
- 05 Decide the redundancy honestly. Two ports on one card is one PCIe device and one failure domain; genuine card redundancy is two cards, and it costs a slot. Twenty-five gigabit to the host against hundred gigabit uplinks is the sensible floor today.
- 06 Refuse SR-IOV and RDMA for the general cluster. They bypass the eBPF datapath that Move 60 enforces east-west policy in, and outside high-performance computing and dedicated storage fabrics they buy nothing you can measure against that cost.

ROLLBACK

Changing card model after delivery is a per-node operation: drain the node, swap the card, reboot, return it to service. That is a maintenance window per machine rather than a disaster, which is why this Move is Medium risk rather than High. The point of no return is not a moment but a count: past about a dozen machines, swapping every card costs more than the difference between the cards ever will. Keep the tested model, firmware and Talos version recorded together so a known good combination can be restored on a replacement machine.

15 COPPER INSIDE THE RACK, GLASS BETWEEN THEM

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
IRON	—	LOW	IMMEDIATELY, AT THE COST OF RE-CODING OR RE-ORDERING

Orders cabling and optics once both ends exist, settles connector polish and vendor coding at order time, and buys one spare of every type in the same order.

AWS	VPC and Direct Connect	the physical layer existed and was billed to you as bandwidth, never as a connector type.
GOOGLE	VPC and Cloud Interconnect	the same, with the cross-connect being the one place a physical detail ever reached you.
AZURE	Virtual Networks and ExpressRoute	again bandwidth as a product, and again a physical handoff you specified once and forgot.

BEFORE YOU START

ACCESS

- The cabling diagram from Move 12 with both ends of every link identified
- The switch part numbers from Move 13 and the card models from Move 14

SOFTWARE

- The switch vendor's transceiver compatibility list and its position on third-party modules, in writing
- A link budget for anything longer than a rack, so the optic class is chosen rather than guessed

PEOPLE

- Whoever will terminate the cabling, to agree on lengths and routing before anything is cut

WHY THIS WORKS

Cabling is the cheapest part of the estate and the one that produces the most baffling faults, because a marginal optical link does not fail — it degrades, intermittently, under load, and presents as an application problem. The rules are short. Passive direct-attach copper inside a rack, out to about three metres, because it is cheap and it draws almost no power. Active optical or discrete transceivers beyond that. Single-mode fibre between racks rather than multimode, because the incremental cost is small and it does not need replacing at the next speed step. Everything else in this Move is about ordering the right variant of each, once, with the spares.

SWAP

Where a run is awkward and short, an active copper cable saves the transceiver cost and the power. Past five metres it stops being the cheaper answer.

DO IT FASTER

Order cables in the colours you will use for each purpose, and order them cut to length. Both cost almost nothing and both save an hour every time somebody works in the rack.

WATCH OUT

Bend radius is a real limit, not advice. A fibre pulled around a rack post at a tight angle will pass a link test on the day it is installed and fail six weeks later when the cabinet is tidied.

LEFTOVERS

Leftover optics are only useful if they are the coded variant your switch accepts. Keep them labelled with the vendor coding, not just the speed.

—	—	—	0 min	2 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. This Move is a small line on the bill of materials and an outsized share of the faults you will chase in Part III.

NEEDS 13 The switch, and the software running on it 14 The card, the driver, and the kernel you will run

- 01 | Order nothing until both ends of every link exist on paper. A cable with one end is a guess, and a box of guesses is how a rack build stalls on the day.
- 02 | Choose the cabling medium per link from the length: passive copper inside the rack, active optical or discrete transceivers past three metres, single-mode fibre between racks and out to the meet-me room.
- 03 | Record connector type and polish per link. Duplex LC for the common single-mode optics, and MPO-12 with an angled polish for parallel single-mode. An angled channel with a flat-polished patch lead in it presents as a marginal optic and gets diagnosed last, after the switch, the card and the driver have all been blamed.
- 04 | Settle the coding question at order time. Optics carry vendor identity in their memory, and switch vendors variously warn, refuse to link, or decline support on third-party modules. Confirm your choice against the support contract signed in Move 13, and buy accordingly.
- 05 | Configure breakout ports before expecting a link. A hundred gigabit port split into four twenty-five gigabit lanes needs the port mode set on the switch first, and until it is set the link stays down with no useful error.
- 06 | Buy one spare of every optic and cable type in the same order. They are cheap, they are the most common failure in the rack, and a spare that has to be ordered is a fault that lasts a week.

ROLLBACK

Everything in this Move is reversed by unplugging something. The cost is re-coding an optic where the vendor allows it and re-ordering where it does not, plus the delay of a second delivery. There is no point of no return in the usual sense; the nearest thing is the moment fibre is installed in a containment run and pulling it means a maintenance window. Keep the port map and the cable schedule in source control so the intended state can be restored when somebody has moved a cable and not said so.

16 THE NETWORK THAT WORKS WHEN THE NETWORK IS DOWN

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
IRON	—	HIGH	IMMEDIATELY

— Buys the out-of-band estate as hardware in its own right, on its own uplink and power, because the management controller rides the network you are fixing.

AWS	EC2 Instance Connect Endpoint	a control path that reached an instance without touching your own network at all.
GOOGLE	Identity-Aware Proxy TCP forwarding	the same property, reached through the provider's front door rather than yours.
AZURE	Azure Bastion	a managed jump host that stayed reachable when everything you had built was not.

BEFORE YOU START

ACCESS

- The switch and machine counts from Moves 11 and 13, to size the port count
- The facility's position on a second uplink, which is a Part II conversation you can start now

SOFTWARE

- A console server that speaks SSH and logs sessions, not one that needs a Java applet
- A cellular or fixed uplink option priced, even if the decision is deferred

PEOPLE

- Whoever will be woken up, since this is the equipment that decides whether they can help from home

WHY THIS WORKS

Every cloud gave you a control path that did not depend on your network working. On your own hardware that path has to be bought, and it is the cheapest line on the bill of materials and the one most often struck out on the grounds that the management controller from Move 08 covers it. It does not. The controller is reached over a network, and the fault you most need it for is the production switch pair itself. What is needed is a small management switch, a serial console server with ports for every switch and enough machines to matter, and an uplink and power that do not traverse the production fabric.

SWAP

Where the facility offers a managed out-of-band service with its own uplink, taking it is reasonable and cheaper than building one. Read what it depends on before believing it is independent.

DO IT FASTER

Buy the console server with more ports than you need. Adding a second one later means a second address, a second credential set and a second thing to forget.

WATCH OUT

An out-of-band network that reaches the internet directly is an out-of-band network that will be scanned. It gets its own firewall and its own authentication in Move 40, and it is never left open in the meantime.

LEFTOVERS

The cloud bastion and instance-connect endpoints keep billing quietly. They stay until Part VII, because they are how you reach the old estate while it still matters.

—	—	—	0 min	2 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF — Nothing yet. The provider's bastion and console endpoints are still the way into the old estate.

NEEDS 08 The way in when the machine will not boot 13 The switch, and the software running on it

- 01 Count the serial ports you need: every switch, every power distribution unit that has one, and enough machines that a console is available where the management controller is not. Add a quarter for growth and buy the next size up.
- 02 Buy a management switch that is not part of the production fabric, and a console server that logs every session to somewhere off the device. The log is what tells you what the last person did at three in the morning.
- 03 Give the out-of-band estate its own uplink. A small business broadband line or a cellular router is enough, and its whole purpose is that it does not share a fate with the transit you buy in Part VI.
- 04 Give it its own power. A separate small uninterruptible supply for the management switch and console server, so that the thing you use to diagnose a power fault is not on the circuit that failed.
- 05 Price the whole set. It is normally under two per cent of the hardware bill, and stating that percentage is usually enough to keep it on the purchase order.
- 06 Record what this equipment is for, in one paragraph, on the bill of materials. Six months later somebody will ask why there is a second switch, and the paragraph is what the engineer woken at three in the morning will be glad of.

ROLLBACK

Nothing here is committed to anything: the equipment is small, cheap and independent, and taking it out is unplugging it. The point of no return does not exist in this Move, which is why leaving it out is such a common and such an expensive decision — the cost of including it is small and known, and the cost of omitting it is a drive to a building. Keep the console server configuration exported so it can be restored onto a replacement unit without rebuilding the port map by hand.

17 WHO YOU BUY FROM, AND WHAT IS SAFE TO BUY USED

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
IRON	THE HARDWARE REFRESH SOMEONE ELSE DID FOR YOU	MEDIUM	14 TO 30 DAYS, PER THE VENDOR'S RETURN WINDOW

Prices the specification three ways, reads the warranty for response against restore, and treats lead time rather than unit price as the schedule risk.

AWS	EC2 instance generations	a refresh was a change of instance type, done by somebody else on a schedule you never saw.
GOOGLE	Compute Engine machine generations	the same, with live migration meaning you often did not notice hardware changing underneath you.
AZURE	Virtual Machines series refreshes	again somebody else's capital cycle, surfaced to you only as a new series in the portal.

BEFORE YOU START

ACCESS

- The complete specification from Moves 04 through 15, in enough detail that three suppliers quote the same thing
- Your organisation's position on refurbished equipment, since some procurement policies forbid it outright

SOFTWARE

- A comparison sheet with warranty terms as columns rather than a footnote
- A written inspection list for any used equipment, agreed before anything is shipped

PEOPLE

- Two reference customers per supplier, spoken to rather than read about

WHY THIS WORKS

The cloud did your hardware refresh for you, and the price of taking it back is that you now have a supplier relationship, a warranty to read and a lead time to plan around. The three ways to buy the same specification are a tier-one vendor with next-business-day on-site parts, a whitebox builder at a lower price with return-to-base service, and refurbished. All three are legitimate. What separates them is not the unit price but what happens in year four, when the platform reaches end of service life and spares stop being manufactured — and, in the meantime, how long a machine takes to arrive, which is the risk that actually moves this programme's dates.

SWAP

For a first rack that mainly needs to prove the approach, a whitebox builder with return-to-base is often the right economics. Move to a tier-one contract when the estate is carrying revenue and an outage costs more than the premium.

DO IT FASTER

Ask each supplier for their lead time before their price. Two of the three will usually disqualify themselves on the date, which shortens the exercise considerably.

WATCH OUT

A support entitlement follows the serial number, not the invoice. A machine sold on twice can arrive with a valid warranty that the vendor will not honour for you until it is transferred, and transfers can be refused.

LEFTOVERS

Suppliers will offer to hold stock. A holding agreement without a signed order is worth exactly nothing when somebody else pays first.

—	—	—	0 min	3 weeks	2 weeks
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. The old estate carries production until Part V and Part VI have moved off it.

NEEDS 11 Headroom, and the number of machines you order 13 The switch, and the software running on it

- 01
- Write one specification and send it to all three channels unchanged. Suppliers who are allowed to propose their own configuration will each propose a different one, and you will end up comparing three things instead of three prices.
- 02
- Read each warranty for the distinction that matters: response against restore. A four-hour response commitment means an engineer is dispatched, not that the machine works again. Parts-only against parts-and-labour is the second column, and end of service life is the third.
- 03
- Price the refurbished option honestly. It is legitimate for chassis, processors, switches and optics, and it is where a memory price spike will push you. It is never right for flash, batteries or supercapacitors, because those wear and their remaining life is the thing you cannot see.
- 04
- For anything used, apply the inspection list before payment: percentage used and power-on hours from every drive, the controller event log, the current firmware level and whether it can still be updated, and whether the serial number is still entitled to vendor support.
- 05
- Compare lead times as carefully as prices. Memory is the long pole, quoted build times move by months, and a quote is not a delivery date. Ask for a date, in writing, with a penalty or a cancellation right attached.
- 06
- Call the reference customers and ask one question: what happened the last time you had a failure. The answer separates suppliers more reliably than any specification sheet.

ROLLBACK

A purchase from a reputable supplier can be returned inside its window, which is fourteen to thirty days on most contracts and is the reason this Move sits before the acceptance testing in Move 18 rather than after it. Past that window the equipment is yours. The point of no return is the expiry of the return period, not the delivery date, and the two are frequently confused. Keep every quote, warranty document and entitlement check filed with the bill of materials so a support position can be restored when a vendor disputes it.

18 NOTHING CARRIES PRODUCTION UNTIL IT HAS BEEN MADE TO FAIL

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
IRON	—	HIGH	IMMEDIATELY; THIS IS THE LAST CHEAP MOMENT

Writes the acceptance suite before the order is placed and makes passing it a term of the purchase order, so a failed machine is a return rather than an argument.

AWS	EC2 instance retirement notices	a failing host was replaced by somebody else, usually before you knew there was a problem.
GOOGLE	Compute Engine live migration	the same protection, applied transparently, which is why nobody who grew up on it has ever burned in a server.
AZURE	Virtual Machines planned maintenance and health events	the same again, surfaced as a notification rather than as a hardware fault you had to diagnose.

BEFORE YOU START

ACCESS

- Power and cooling for the whole batch at full load, which is often the constraint on how many can be tested at once
- The supplier's return terms from Move 17, with the acceptance criteria referenced in them

SOFTWARE

- A memory test capable of multiple full passes, and a disk exerciser that can run your own access profile
- A way to log inlet and exhaust temperature and machine check events per machine, unattended

PEOPLE

- Whoever will sign off the delivery, so that pass and fail mean the same thing to them as to you

WHY THIS WORKS

A machine that has never been pushed hard has not been tested; it has been switched on. Infant mortality in server hardware is real and it concentrates in the first weeks, which means the cheapest moment to find a bad machine is before it has anything on it and while the supplier's return window is still open. That is only true if the acceptance criteria were written down before the order went in and made a term of it. Written afterwards, they become a negotiation. Written into the purchase order, a machine that fails them is a return.

SWAP

Where the batch is too large to soak at once, test in waves and accept the schedule cost. Testing half the machines properly beats testing all of them for an hour.

DO IT FASTER

Automate the suite so a machine is racked, powered and tested without anyone watching it. The suite will be run again on every replacement machine for the next five years.

WATCH OUT

A correctable memory error is not a passing grade. Correctable errors trend upward before a module fails, and the machine that reports a handful during acceptance is the one that reports thousands in month four.

LEFTOVERS

Test images, temporary credentials and the scratch network used for the soak all outlive the exercise unless someone removes them.

—	—	—	0 min	1 week	1 week
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. This Move spends a week to avoid spending a quarter chasing a machine that was never healthy.

NEEDS 17 Who you buy from, and what is safe to buy used

18 NOTHING CARRIES PRODUCTION UNTIL IT HAS BEEN MADE TO FAIL

6 STEPS · 0 MIN · HIGH RISK

THE RUNBOOK

IRON

- 01 Write the suite before the order goes in and attach it to the purchase order. Every machine runs a multi-pass memory test, a disk exerciser at the workload's own profile plus a full-surface pass per device, a sustained processor and thermal soak with temperatures logged, and both network ports at line rate on the cable that will actually be used.
- 02 Define failure in advance and in writing: any machine check exception, any correctable memory error trend rather than a single event, any storm of correctable errors on the bus, and any thermal throttle at the rated inlet temperature. A machine that throttles in a cool room will throttle in a warm aisle.
- 03 Run the suite for long enough to matter. Twenty-four hours of soak per machine is a reasonable floor and forty-eight is better, and the batch can run in parallel if the power is there.
- 04 Log everything per serial number, and keep the logs. They are the evidence in a return conversation and the baseline you will compare against in Move 111 when a machine starts misbehaving in production.
- 05 Return the failures immediately rather than setting them aside. A machine kept because it only failed once is a machine that will fail again on a Sunday.
- 06 Record the point this makes explicitly: the three second-hand machines under a desk that proved the software estate proved nothing about memory error behaviour, dual-feed supplies, bonded links across two switches, controller recovery or drive endurance under a replicated storage cluster. The homelab is an excellent on-ramp and it is not an acceptance test.

ROLLBACK

Everything here is reversed by putting a machine back in its box, which is exactly why it happens now. The point of no return is the expiry of the supplier's return window: after that a failed acceptance test is a warranty claim rather than a refund, and a machine you have to keep. Keep the per-serial logs somewhere durable and backed up, so the acceptance baseline can be restored and compared when the same machine misbehaves two years later.

IRON · 0 MIN · HIGH RISK

Nothing yet. This Move spends a week to avoid spending a quarter chasing a machine that was never healthy.

19 THE SHELF, AND THE FAILURE RATES YOU STOCK AGAINST

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
IRON	—	MEDIUM	IMMEDIATELY

Sets on-site spares from population and published failure rates, stocks a whole cold spare node, and counts the consumables nobody budgets for.

AWS	EC2 capacity pools	a failed host meant a new instance, and the spare parts inventory was a problem you never had.
GOOGLE	Compute Engine host maintenance	the same, with the fleet's spare capacity standing in for your shelf.
AZURE	Virtual Machines fault domains	again capacity as the spare, which is why nobody arriving from a cloud has a parts cupboard.

BEFORE YOU START

ACCESS

- The final population from Move 11: machines, drives, modules, supplies, fans, optics
- The support contract from Move 17, and specifically its promised turnaround

SOFTWARE

- A parts list keyed by the exact part numbers on the bill of materials, not by description
- Somewhere to record what was consumed, so next year's number is measured rather than repeated

PEOPLE

- Whoever can physically reach the site, since a spare nobody can fit is inventory rather than a spare

WHY THIS WORKS

Spares are the difference between a fault that costs an hour and a fault that costs a week, and the right number of them is arithmetic rather than instinct. Take the population, take the published annualised failure rate for each component class, and stock what a year is likely to consume plus a margin for the fact that failures cluster. The largest public storage fleet dataset has reported annualised drive failure in the region of one to one and a half per cent for several years running — a figure worth quoting and worth discounting, because it is a disk fleet rather than a flash one. Fans and power supplies are the commodity failures. Memory modules are the ones that hide for weeks as a correctable-error trend before anyone notices.

SWAP

Where a facility offers a bonded parts locker or a vendor holds local stock, that can substitute for some of the shelf. Confirm the access process at three in the morning before relying on it.

DO IT FASTER

Buy spares as part of the main order. They are cheaper, they arrive together, and the part numbers are guaranteed to match what was delivered.

WATCH OUT

Firmware drift between a spare and the fleet is the quiet failure here. A drive or supply that has sat on a shelf for a year needs updating before it goes in, and Move 110 needs to say so.

LEFTOVERS

Spares for a generation you no longer run are money on a shelf. Review the cupboard when the fleet changes, not five years later.

—	—	—	0 min	2 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. This is a line you are adding, and it is the line that makes Part VII survivable.

NEEDS 11 Headroom, and the number of machines you order 17 Who you buy from, and what is safe to buy used

- 01
- Compute the expected annual consumption per component class from the population and the published failure rates, then round up. For a small estate the rounding dominates the arithmetic, and that is fine: the point is to have a defensible number rather than a precise one.
- 02
- Stock a whole cold spare machine. It beats any return-and-repair cycle on time to service, and it is the only spare that covers a fault you have not yet diagnosed. For a fleet of a dozen it is the single best value line in this Move.
- 03
- Stock the commodity failures on site: supplies, fans, a small number of drives of each size in use, and modules of the exact part number. Modules in particular must match, because a replacement of a different rank or speed changes how the whole channel map runs.
- 04
- Count the consumables nobody budgets: rails, cage nuts, patch leads, optics of every type from Move 15, labels and a printer, and a bay map that says which drive is in which slot. These cost very little and their absence stops an installation day.
- 05
- Record the promised turnaround from the support contract next to the turnaround the vendor actually achieves. They are different numbers and only the second one belongs in the runbook that Move 110 will write.
- 06
- Put the spares on the bill of materials rather than buying them later out of a different budget. Later means the first failure arrives before the shelf does.

ROLLBACK

Spares are inventory, and inventory is reversed by not buying more of it. There is no commitment here beyond the cash. The point of no return is the moment the fleet's part numbers change and the shelf no longer matches, which is a slow, quiet failure rather than an event. Keep the parts list and the consumption record in source control alongside the bill of materials so the previous year's figures can be restored and compared when next year's number is set.

20 THE BILL OF MATERIALS

CUTOVER

0 MIN

LAYER	LEAVING	RISK	REVERSIBLE
IRON	COMMITTED-SPEND DISCOUNTS ON RENTED COMPUTE	HIGH	NO

Assembles every Iron decision into a document a supplier can quote against, and sets the three-year cost frame the rest of the book is judged against.

AWS	Savings Plans and Reserved Instances	a one or three-year commitment whose renewal date is the real deadline for this programme.
GOOGLE	Committed Use Discounts	the same commitment shape, per project or across an organisation, with the same renewal cliff.
AZURE	Reservations and savings plans	the same again, and the one where the licensing benefit is usually tangled into the commitment.

BEFORE YOU START

ACCESS

- Authority to commit capital, and the finance calendar it has to fit inside
- The renewal date and terms of every committed-spend agreement across all three providers

SOFTWARE

- Every decision from Moves 01 to 19 in one document, with the reasoning attached rather than summarised
- A three-year cost model that includes people, because a comparison that omits salary is the reason repatriations get approved and then regretted

PEOPLE

- Finance and procurement in the same room as whoever signs the order, once rather than four times

WHY THIS WORKS

Everything in Part I has been a decision. This Move turns them into a document a supplier can quote against, which is a different artefact: part numbers, quantities, unit prices, lead times, firmware baselines, per-rack draw, the rack-unit budget, the spares, the acceptance terms from Move 18, and an explicit exclusion list for the cabinet, the power distribution and the cross-connects, which belong to Part II. It also fixes the date. The renewal of your committed-spend discount is the deadline that actually governs this programme, because that is the moment the cloud bill either falls off a cliff or is re-committed for another three years.

SWAP

Renting dedicated machines by the month gets most of the saving with none of the capital, the lead time or the loading bay. It is the honest answer for a team that wants the economics without the facility, and it costs roughly half the saving.

DO IT FASTER

Ask for a quote on the whole document rather than line by line. Suppliers discount the bundle and it removes a fortnight of correspondence.

WATCH OUT

A committed-spend discount usually cannot be cancelled and sometimes cannot be transferred. Check whether yours can be sold on the marketplace before assuming the remaining term is a sunk cost.

LEFTOVERS

The cloud commitment keeps billing until its term ends whether or not anything runs on it. Plan the migration to finish near that date, not long before it.

\$74,013/mo	\$38,337/mo	48%	0 min	1 week	8 to 16 weeks
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet, and this is the point. The cloud bill does not move until Part V and Part VI have carried the workloads across; everything before then is spending twice.

NEEDS 14 The card, the driver, and the kernel you will run 15 Copper inside the rack, glass between them

16 The network that works when the network is down 18 Nothing carries production until it has been made to fail

19 The shelf, and the failure rates you stock against

- 01 Assemble the document. Part numbers and quantities, unit prices, lead times per line, firmware baselines, the per-rack draw from Move 07, the rack-unit budget from Move 11, the spares from Move 19 and the acceptance terms from Move 18 written in as a condition of payment.
- 02 State the exclusions explicitly. The cabinet, the power distribution units, the cross-connects and the facility itself are Part II and are not on this document. An exclusion list prevents the two most common surprises, which are buying a cabinet twice and buying none at all.
- 03 Set an order sequence rather than a single date. Memory and optics have the longest lead times and a switch without transceivers is a paperweight, so the sequence is memory first, then switches and optics, then chassis, then spares.
- 04 Build the three-year frame with both numbers in it. For the reference estate in this book, the same core count on rented compute is about \$56,500 a month before storage or egress, and the owned infrastructure is about \$12,100 a month all in. Add the salary honestly — the platform needs more of it than the cloud did — and the total-against-total comparison is closer to half rather than four-fifths. Publish both, and let nobody quote only the first.
- 05 Name the renewal date of the commitment you are choosing not to renew, and work backwards from it. That date, not the delivery date, is what the programme plan hangs on.
- 06 Sign it, and tell everyone the number. From this moment the programme has a budget, a delivery window and a deadline, and Part II can be negotiated against a real draw instead of an estimate.

ROLLBACK

This Move is irreversible: a signed purchase order is money committed and, past the supplier's return window from Move 17, hardware you own. The point of no return is the signature itself, which is why the whole of Part I exists to make it a well-evidenced one. What remains reversible is the programme, not the purchase — Move 121 describes how to abort and keep running on the cloud, and Move 20's hardware retains resale value. Keep every quote, model and decision in source control so the case can be restored and re-examined afterwards.

II SITE

Where the machines live, who owns the room, and what you are signing. Nothing here is software, and getting it wrong is the only mistake in this book you cannot fix with a deploy.

18	18	10	0
MOVES	ZERO DOWNTIME	HIGH RISK	MINUTES, TOTAL

IF YOU ONLY DO THREE

21

WHAT THE THREE MACHINES UNDER THE DESK CANNOT TELL YOU

Writes down on one page what a three-machine desk cluster proved and what it did not, and attaches each remaining gap

⋮

22

THE FOUR PLACES THE MACHINES COULD LIVE

Prices retail colocation, wholesale space, rented dedicated servers and a room you already occupy against the

⋮

WHAT THE THREE MACHINES UNDER THE DESK CANNOT TELL YOU · THE FOUR PLACES THE MACHINES COULD LIVE · THE ROOM IN YOUR OWN BUILDING · THE SECOND SITE YOU ARE NOT BUILDING THIS YEAR · THE FLOOD PLAIN, THE FAULT LINE AND THE GRID · CARRIER NEUTRALITY AND THE CROSS-CONNECT PRICE LIST · WHAT A TIER RATING PROMISES, AND WHAT IT DOES NOT · COMMITTED KILOWATTS, METERED KILOWATTS AND THE DERATE · WHAT THE SERVICE CREDIT IS ACTUALLY WORTH · THE NOTICE PERIOD AND THE DOOR OUT · THE 80 PER CENT YOU ARE ALLOWED TO DRAW · TWO FEEDS, ONE OF WHICH YOU HAVE TESTED · THE INLET TEMPERATURE YOU ARE ALLOWED · THE IMPORTER OF RECORD AND THE LOADING DOCK · WHO CAN GET IN AT 03:00 · THE FIRST RACK · LABELS, PORT MAPS AND THE AS-BUILT · THE DRIVES THAT LEAVE, AND THE SNAPSHOTS THAT STAY

■ ■ ■ 3 LOW RISK	■ ■ ■ 5 MEDIUM RISK	■ ■ ■ 10 HIGH RISK
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21 WHAT THE THREE MACHINES UNDER THE DESK CANNOT TELL YOU

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
SITE	—	LOW	IMMEDIATELY

Writes down on one page what a three-machine desk cluster proved and what it did not, and attaches each remaining gap by name to the Move that closes it.

AWS	the free tier and a personal account	where most people first ran a cluster, and where nothing about physical failure was ever visible.
GOOGLE	the free tier and a personal project	the same, with a generous allowance that made experimenting feel like operating.
AZURE	a free account and a personal subscription	again a place to learn the software and never the hardware underneath it.

BEFORE YOU START

ACCESS

- The desk cluster, left running at full load for a week rather than benchmarked for ten minutes
- Somewhere to publish one page that the whole team will actually read

SOFTWARE

- A thermal and power meter at the wall, so the numbers on the page are measured rather than asserted
- The wear and error counters from each device in the desk cluster, read rather than assumed

PEOPLE

- Whoever is sponsoring this programme, because this page is what stops the plan being sized off a desk

WHY THIS WORKS

The on-ramp in this book's front matter is three refurbished small machines, a managed switch and about eighteen hundred dollars. It is an excellent thing to have done and it proves a great deal: the software estate installs, the cluster forms, the storage layer works, the deployment pipeline reaches it. It also convinces people that the migration is de-risked, which it is not, because nothing about that desk cluster resembles the parts of the estate that fail at three in the morning. This Move writes the gap list down on one page and attaches each item to the Move that closes it, which turns a warning into a map.

SWAP

If there is no desk cluster, build one. Eighteen hundred dollars and a weekend is the cheapest possible way to discover that the software estate does not do what you assumed.

DO IT FASTER

The gap list is short and mostly the same for everyone. Start from the six items above and add your own rather than beginning with a blank page.

WATCH OUT

The most seductive false conclusion here is about storage. A replicated storage cluster on three consumer devices works beautifully for a week and tells you nothing about how it behaves after a year of writes and a device failure during recovery.

LEFTOVERS

The desk cluster stays useful for the whole programme as a place to break things. Do not dismantle it when the real hardware arrives.

—	—	—	0 min	3 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing. The desk cluster earns its electricity for the rest of the programme.

21 WHAT THE THREE MACHINES UNDER THE DESK CANNOT TELL YOU

6 STEPS · 0 MIN · LOW RISK

THE RUNBOOK

SITE

- 01 | Leave the desk cluster at genuine load for a week and read what happens. Sustained thermals, fan behaviour, the power draw at the wall, and whether anything throttles. A ten-minute benchmark tells you none of this.
- 02 | Read the wear and error counters off every device in it. A year of real writes against a consumer part is the number nobody has, and the desk cluster is the cheapest place to start collecting it.
- 03 | Write the gap list. A fabric wider than one switch, dual power supplies fed from two sources, out-of-band access from another city, drive endurance measured against a year of writes, and the four-week wait for a replacement part are all things the desk cluster cannot show you.
- 04 | Attach each gap to the Move that closes it, by number. The fabric is Moves 12 and 13, the supplies are Move 07 and Move 32, out-of-band is Moves 16 and 40, endurance is Move 09, and the wait for a part is Move 19.
- 05 | State plainly which of the gaps are the reason for Part II. The facility exists to supply power that does not fail, cooling that does not stop and physical access that is controlled, and none of those are things a desk can offer.
- 06 | Publish the page to everyone working on the programme and put a date on it. It gets revisited once, after the first rack is built in Move 36, to see which gaps turned out to matter.

ROLLBACK

A page of writing is undone by rewriting it, so there is nothing to reverse here. The point of no return is elsewhere entirely: it is Move 20, which has already been signed by the time most readers reach this page, and Move 28, which commits to a facility. This Move exists to make sure neither of those was signed on the strength of a desk. Keep the page in source control so the original gap list can be restored and compared against what actually went wrong.

22 THE FOUR PLACES THE MACHINES COULD LIVE

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
SITE	THE REGION-AND-ZONE ABSTRACTION	HIGH	IMMEDIATELY

Prices retail colocation, wholesale space, rented dedicated servers and a room you already occupy against the committed power figure, with staff time counted.

AWS	Regions and Availability Zones	a physical estate presented as two labels, with the buildings, the power and the people all abstracted away.
GOOGLE	regions and zones	the same abstraction, with the difference that a network spans regions, which is a property you will not be reproducing.
AZURE	regions and availability sets	the same again, with the set describing a fault domain inside a building you were never going to see.

BEFORE YOU START

ACCESS

- The committed power figure, rack-unit budget and machine count from Move 20
- Quotations from at least four suppliers, requested against the same specification

SOFTWARE

- A single comparison sheet with the five columns above, filled in for each option
- The staffing model from Part VII brought forward, because the rota changes which options are affordable

PEOPLE

- Whoever will approve headcount, since the difference between these options is measured in people as much as in money

WHY THIS WORKS

A region and a zone were labels. Underneath them were four things you are now choosing between: retail colocation, where you rent cabinets and power in somebody else's building; wholesale space, where you rent a room or a hall; rented dedicated machines, where somebody else owns the metal and you own everything above the operating system; and a room in a building you already occupy. All four are legitimate. What decides between them is not preference but arithmetic: capital cost, monthly cost at the committed power figure from Move 20, staff time consumed, lead time to the first boot, and what it costs to leave.

SWAP

For a team of fewer than about six engineers, rented dedicated machines get most of the economics with almost none of the physical work. It is the right answer more often than this book's title suggests.

DO IT FASTER

Ask each supplier for a price at your committed power and at twice it. The gap between those two numbers tells you what growth will cost, and it is where the surprises live.

WATCH OUT

Retail colocation quotes are per cabinet and per kilowatt, and cross-connects, remote hands and cabinet accessories are all extra. A quote that looks half the price of another is usually missing three lines.

LEFTOVERS

Nothing is decommissioned by this Move. Everything in the cloud keeps running, and will keep running until Part VI.

—	—	—	0 min	2 weeks	2 weeks
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. This Move only spends money on quotations, which are free, and on the time to read them, which is not.

NEEDS 20 The bill of materials 21 What the three machines under the desk cannot tell you

- 01 Request quotations against the same specification: the committed kilowatts, the rack-unit count, the cross-connects and the term. Suppliers left to propose their own shape will not be comparable.
- 02 Price all four options on one comparison sheet, on capital cost and monthly cost. For the reference estate in this book, a cabinet with power, transit, a cross-connect and remote hands lands near \$2,900 a month per site before the machines are amortised, and the same machines rented as dedicated servers land near \$480 each per month with the racking and the hands included.
- 03 Add the staff time each option consumes, in fractions of a person. This is where retail colocation and rented metal separate: the rented option removes the racking, the hands and most of the physical rota, and removes none of the platform work.
- 04 Carry the rota arithmetic forward from Move 113 rather than leaving it to Part VII. A rota that covers nights and weekends without burning people out needs a headcount, that headcount has a cost, and that cost changes which of these four options is affordable. This is the last point at which the reader can act on the answer.
- 05 Add lead time and exit cost. Retail colocation is weeks; wholesale is months; rented metal is days; a room in your own building is however long the plant works take, plus consent.
- 06 Treat a mixed landing as a real answer rather than indecision. Rented machines in another city are usually the cheapest disaster-recovery target you will ever price, and running the primary estate in colocation with a rented secondary is a defensible design rather than a compromise.

ROLLBACK

No commitment is made here: this Move produces a comparison and a recommendation, and both are revised by revising them. The point of no return is Move 28, where a power commitment is signed, and Move 30, where a term is. Until then, changing your mind costs the time spent gathering quotations. Keep the comparison sheet and every quotation in source control, so the reasoning can be restored when somebody asks in a year why the estate landed where it did.

23 THE ROOM IN YOUR OWN BUILDING

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
SITE	—	HIGH	IMMEDIATELY

Sets out the narrow case for a room in a building you occupy, then prices the suppression, detection, consent and insurance most estimates leave out.

AWS	Outposts	the closest the provider came to putting its estate in your building, and it came with the building requirements attached.
GOOGLE	Google Distributed Cloud	the same idea, with the same implication that a room has to meet a specification before hardware can go in it.
AZURE	Azure Stack HCI and Azure Local	again a supported on-premises footprint, and again the room, the power and the cooling were yours to provide.

BEFORE YOU START

ACCESS

- The lease or title, read for what it says about plant, alterations and hours of access
- The building's electrical single-line diagram, and the substation each supply comes from

SOFTWARE

- A cooling load calculation from the per-rack draw in Move 20, done by somebody qualified
- The applicable detection and suppression standards for your jurisdiction, obtained rather than paraphrased

PEOPLE

- A facilities manager, a mechanical and electrical consultant, and an insurance broker, all engaged before any number is published

WHY THIS WORKS

There is a real case for a room in a building you occupy, and it is narrow. It needs four things at once: you own the building or hold a long lease on it, there is a second electrical supply from a different substation, the plant room has the structural and cooling capacity for the load, and somebody's job title contains the word facilities. Where all four hold, the economics can be excellent. Where any one of them does not, the missing pieces are expensive, slow and regulated, and the honest conclusion for most companies is no. That conclusion should come from the arithmetic rather than from the tone of this page.

SWAP

A room in your own building is a much better idea for the second site than the first. Low stakes, real learning, and the failure mode is a restore rather than an outage.

DO IT FASTER

Ask a colocation provider for a tour and take the facilities manager. An hour in a real hall answers more design questions than a week of drawings.

WATCH OUT

Cooling is the constraint that surprises people. A room that is comfortable with four kilowatts of equipment is a different building problem at twenty, and the plant that solves it may not fit where you assumed.

LEFTOVERS

Where the answer is no, the work is not wasted: the cooling load, the draw and the fire requirements are all inputs to the colocation conversation in Move 28.

—	—	—	0 min	1 week	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing. This Move is a study, and its most common output is a decision not to spend.

NEEDS 22 The four places the machines could live

- 01
- Test the four conditions before pricing anything. Ownership or a long lease, a genuinely separate second supply, plant capacity for the cooling, and a facilities owner for the room. A no on any of them ends the exercise cheaply, which is the point of running it first.
- 02
- Price the fire strategy properly. Detection designed to the applicable standard, clean-agent or water-mist suppression, the annual test and the maintenance contract behind it. This is the line most estimates omit entirely and it is rarely small.
- 03
- Price the electrical works: distribution, the uninterruptible supply and its batteries, and the generator if there is one. A generator brings planning consent and fuel storage approval with it, and both have lead times measured in months rather than weeks.
- 04
- Get an insurance schedule from an underwriter who has seen the actual design. Cover for the equipment, for business interruption and for the building is priced against the fire strategy, and an underwriter's questions will find gaps the design review missed.
- 05
- Add maintenance contracts for every system: cooling, suppression, the uninterruptible supply and the generator. These recur annually for the life of the room and belong in the monthly comparison from Move 22, not in the capital line.
- 06
- Publish the total against the colocation quotation from Move 22, over the same three years, with staff time in both. Let the arithmetic reach the conclusion, and record it either way so the question is not reopened every quarter.

ROLLBACK

Nothing is committed until works are ordered or a contract is signed, so this Move is reversed by filing the study and choosing something else. The point of no return is the first irreversible building work — a penetration, a plant order, a consent application — after which the money is spent whether or not the room is finished. Keep the study, the quotations and the underwriter's questions in source control so the case can be restored and re-examined when somebody proposes the room again in two years.

24 THE SECOND SITE YOU ARE NOT BUILDING THIS YEAR

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
SITE	—	HIGH	IMMEDIATELY

Refuses a second site in year one, names the three conditions that change the answer, and buys a costed restore into rented metal in another city instead.

AWS	multi-Region architectures and cross-Region replication	a second site was a configuration option, priced per gigabyte rather than per building.
GOOGLE	multi-region resources and dual-region storage	the same, with some services spanning regions by default so nobody had to design it.
AZURE	paired regions and geo-redundant storage	again a second site as a checkbox, which is why its real cost is so widely underestimated.

BEFORE YOU START

ACCESS

- The comparison from Move 22, including the rented-metal option and its price per machine per month
- Whatever contractual or regulatory commitments exist about resilience, read rather than remembered

SOFTWARE

- A recovery objective for each service, expressed as time to restore and acceptable data loss
- A price for the rented standby estate at the size it would need to be, not at the size of production

PEOPLE

- Whoever signs customer commitments, since one of the three conditions is a promise already made

WHY THIS WORKS

A second site doubles the facility bill, the cabling, the spares pool and the change surface, and it does so in exchange for protection against a failure that a well-chosen facility already makes rare. In year one, when the team is learning to run one site, a second one usually makes things less reliable rather than more. The defensible answer for most estates is one good site plus a costed and rehearsed restore into rented machines in another city — which is cheap, which can be tested every quarter, and which does not require anyone to be woken up to fail over.

SWAP

Where a customer commitment is the trigger, price the commitment. Sometimes the contract can be renegotiated for less than the second site costs, and nobody has ever asked.

DO IT FASTER

Rented machines in a different city can be provisioned in a day. Buying a small standby estate now and drilling against it beats designing a second site for a year.

WATCH OUT

A second site that is never exercised is not a second site. If nothing runs there routinely, it will not work when it is needed, and the drill in Move 115 is the only thing that makes it real.

LEFTOVERS

The provider's cross-region replication keeps billing while you decide. Leave it until Part V has moved the data, then stop it deliberately.

—	—	—	0 min	3 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. Cross-region replication in the cloud stays until Part V has moved the data it protects.

NEEDS 22 The four places the machines could live

- 01 Name the three conditions under which a second site is built in year one: a regulator that requires it, a signed customer commitment that names it, or revenue per hour of downtime that exceeds the annual cost of the second site. If none holds, the answer is no and it is written down as a decision rather than a delay.
- 02 Dispose of the two false second sites explicitly. A second cabinet in the same building is not a second site, and a second building on the same substation and the same fibre route is not one either. Both are common and both are sold as resilience.
- 03 Cost the alternative properly: rented machines in another city, sized to carry a reduced service rather than full production, with the data arriving there continuously and a rehearsed restore procedure. Move 115 builds it and drills it.
- 04 Set the recovery objective per service, and be honest about which services genuinely need to come back in an hour and which can wait until the morning. Most estates find the list is shorter than expected.
- 05 Put the second site in the budget for the year in which one of the three conditions becomes true, with the trigger written next to it. That turns a refusal into a plan.
- 06 Review the decision annually, on a date, rather than whenever somebody raises it in an incident review.

ROLLBACK

This Move produces a decision and a trigger, both of which are revised by revising them. The point of no return is signing a second facility agreement in Move 30, which commits a term and a monthly charge for years. Until that signature the second site remains a budget line. Keep the decision, the conditions and the recovery objectives in source control so the reasoning can be restored when the question is reopened, as it will be after the first incident.

25 THE FLOOD PLAIN, THE FAULT LINE AND THE GRID

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
SITE	—	HIGH	IMMEDIATELY

Checks the address rather than the brochure: flood history, seismic category, the substation behind each feed, and the fibre routes drawn on a map.

AWS	Availability Zone selection	you chose a letter, and the physical risk behind it was assessed by somebody else and never shown to you.
GOOGLE	zone selection	the same, with the provider publishing rather less about the physical estate than its competitors.
AZURE	availability zone numbers	again a number standing in for a building, and again no way to ask where the fibre enters.

BEFORE YOU START

ACCESS

- The exact address and, ideally, the as-built utility drawings, which facilities are reluctant to send
- A site visit booked, because some of these questions are answered by looking at a wall

SOFTWARE

- The national flood map and the applicable seismic hazard data for the address
- A route-tracing tool or a carrier who will draw the fibre path for you rather than describing it as diverse

PEOPLE

- Somebody who has done this before, or a consultant for one day, because the questions are unfamiliar and the answers are consequential

WHY THIS WORKS

Everything in the facility contract is negotiable except where the building is. A site on a flood plain, on a single substation, or with two fibre entries that leave through the same duct cannot be repaired by better contract terms, and none of those facts appear in a brochure. So the address is checked before the contract is read: the national flood map and the building's own history, the seismic design category and whether cabinets need bracing that must be ordered with them, the substation feeding each supply, and the fibre routes drawn on an actual map. This is also where the two distances that matter get measured.

SWAP

Where a site is otherwise ideal but has a single substation, ask what the generator arrangement is and how often it is tested under load. A well-maintained generator estate can be a better answer than a second substation nobody checks.

DO IT FASTER

Ask three questions on the first phone call: which substations feed you, where does the fibre enter, and has the building ever flooded. Two of the four candidates will usually remove themselves.

WATCH OUT

Facilities will describe a route as diverse when it is diverse for the first two hundred metres. The question to ask is where the two routes are closest to each other, not whether they are separate.

LEFTOVERS

The latency measurements taken here are the baseline for Move 53 and Move 103. Record them properly rather than reading them off a screen once.

—	—	—	0 min	4 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing. This Move spends four days to avoid signing a three-year term against a building on a flood plain.

NEEDS 22 The four places the machines could live

- 01
- Check the flood position at the address rather than the postcode: the published flood map, the building's own history, and what is in the basement. Plant in a basement on a flood plain is a specific and recurring way that a well-run facility fails.
- 02
- Establish the seismic design category and whether the cabinets need bracing or anchoring. If they do, it is ordered with the cabinets rather than discovered on the installation day.
- 03
- Trace the electrical supply back to the substation. Ask which substation feeds each of the two supplies, and get the answer in writing. Two feeds from one substation are one feed with two cables, and this is the single most common misrepresentation in the industry.
- 04
- Get the fibre entries drawn on a map. Diverse entry means two physically separate routes leaving the building in different directions, not two ducts in the same trench. Ask where they converge, because they almost always converge somewhere.
- 05
- Measure the two distances that matter. Road time for a person carrying a spare part, honestly, at the worst time of day. And fibre distance to your users and to the cloud region you are leaving, at roughly one millisecond of round trip per hundred kilometres of glass, measured along the real path rather than the straight line.
- 06
- Score the site and write it down. A site that fails on flood, on substation diversity or on fibre diversity is refused here, before anybody has spent time negotiating a schedule for it.

ROLLBACK

Refusing a site costs nothing except the time spent assessing it, and that time is the cheapest in the whole Part. There is no point of no return here; the nearest one is Move 28, where power is committed to a specific building. Every finding in this Move should be dated and kept, because facilities change hands, and the substation answer that was true in March is worth re-checking at renewal. Keep the assessments in source control so an earlier survey can be restored and compared.

26 CARRIER NEUTRALITY AND THE CROSS-CONNECT PRICE LIST

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
SITE	—	MEDIUM	IMMEDIATELY

Counts the carriers that will sell at your volume and prices every cross-connect you will need across three years against the facility's own list.

AWS	Direct Connect	a port you ordered through a console, with the physical cross-connect ordered separately and forgotten about.
GOOGLE	Cloud Interconnect	the same shape, with a dedicated and a partner variant, and the partner one hiding the physical layer completely.
AZURE	ExpressRoute	again a logical circuit, and again a cross-connect underneath it that somebody had to order in a building.

BEFORE YOU START

ACCESS

- The carrier list for the specific building, not the operator's national list
- The facility's cross-connect price list, in writing, including installation and disconnection charges

SOFTWARE

- A three-year plan of every cross-connect the estate will need, drawn as a list rather than remembered
- The cloud provider's own documentation on how their authorisation and assignment paperwork expires

PEOPLE

- Whoever will negotiate the facility contract in Move 30, because cross-connects belong in that negotiation

WHY THIS WORKS

Carrier neutral is a marketing term that means the facility will let other people's fibre into the building. What matters is narrower: which carriers have live fibre in the meet-me room today, and which of those will sell to a customer of your size. The second list is always much shorter than the first. Underneath that sits a price list nobody quotes in the headline number — a cross-connect is commonly fifty to three hundred and fifty dollars a month each, plus installation, recurring forever, and outside the cabinet price. You will need more of them than you think.

SWAP

Where a building has few carriers, a wavelength or dark fibre to a nearby carrier hotel can restore the choice. It costs more per month and it is better than being captive to one provider.

DO IT FASTER

Ask the facility for a copy of a recent, redacted cross-connect invoice. It answers the per-connection question and several others in one document.

WATCH OUT

Some facilities charge to disconnect as well as to install, and some require notice. Both belong in the exit runbook that Move 30 writes.

LEFTOVERS

The cloud interconnect on the far side keeps billing until Part VI has moved the traffic and Part VII closes the account. Do not cancel it early to save a few hundred dollars.

—	—	—	0 min	1 week	2 to 4 weeks
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. The cloud interconnect stays until Part VI has moved the traffic across it.

NEEDS 22 The four places the machines could live

26 CARRIER NEUTRALITY AND THE CROSS-CONNECT PRICE LIST

6 STEPS • 0 MIN • MEDIUM RISK

THE RUNBOOK

SITE

- 01 Ask for the list of carriers with live fibre in the meet-me room. Then ask each of them, separately, whether they will sell at your committed volume. This is the list that matters and it is usually three or four names rather than thirty.
- 02 Count every cross-connect you will need over three years: two transit providers, a peering fabric port, the private interconnect back to the cloud you are leaving, and one per additional cabinet. Missing any of these turns a monthly number into a surprise.
- 03 Price them against the facility's own list, with installation and disconnection charges included. Ask specifically whether the price is per connection or per pair, because the two conventions differ by a factor of two.
- 04 Negotiate the cross-connect cost inside the facility agreement rather than afterwards. It is the line facilities are most willing to move, and once the contract is signed it is the line they are least willing to move.
- 05 Time the cloud on-ramp paperwork properly. The authorisation letter and connecting facility assignment that a cloud provider issues expires after ninety days if the cross-connect has not been completed, so it is ordered when the cabinet is ready and not when the contract is signed.
- 06 Record which carrier terminates in which position, because Move 37 will need it and Move 93 will order transit against it.

ROLLBACK

Nothing is committed until the facility contract is signed in Move 30 and the first cross-connect is ordered. Cancelling an ordered cross-connect before installation is usually free and after it is usually a disconnection fee. The point of no return, such as it is, is the installation itself, and it is small money. Keep the carrier list, the price list and the cross-connect plan in source control so the original commercial position can be restored when a renewal quotes different numbers.

27 WHAT A TIER RATING PROMISES, AND WHAT IT DOES NOT

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
SITE	—	MEDIUM	IMMEDIATELY

Separates tier topology from tier certification and from the similarly numbered parallel schemes, and retires the availability percentages quoted beside them.

AWS	the shared responsibility model and its compliance reports	the building's resilience arrived as an attestation you inherited rather than assessed.
GOOGLE	its compliance reports manager	the same inheritance, with the physical estate described and never inspected.
AZURE	the Service Trust Portal	again somebody else's audit, and again the reason nobody arriving from a cloud has read a tier standard.

BEFORE YOU START

ACCESS

- Whatever certifies the facility claims, requested as documents rather than logos on a website
- A tour booked, with the plant rooms included rather than the reception and the hall

SOFTWARE

- A written question list, prepared beforehand, because the useful questions do not occur to you on the day
- The relevant standard's scope statement, read once, so you know what the certificate covers

PEOPLE

- Somebody technical on the tour who will look at the plant, not only at the racks

WHY THIS WORKS

A sales deck runs three different things together. Tier topology describes redundancy and whether the facility can be maintained without being switched off, and nothing else. Tier certification is a separate exercise and comes in three distinct forms — one for the design documents, one for the constructed building, and one for how it is operated — of which only the second tells you the building was built the way it was drawn. And there are parallel schemes with confusingly similar numbering that are different standards with different scopes. Knowing which of these a facility actually holds is the difference between a tour and an assessment.

SWAP

An uncertified facility with an excellent operations record and open answers can be a better bet than a certified one that will not let you into the plant room. The certificate is evidence, not the thing itself.

DO IT FASTER

Ask two questions on the phone before booking a tour: which certificate, and for which building. That removes half the shortlist without leaving your desk.

WATCH OUT

Concurrently maintainable and fault tolerant are precise terms with different meanings, and they are used interchangeably in marketing. The distinction is whether the facility survives a failure or only a planned maintenance.

LEFTOVERS

The question list from this Move is reused at every renewal and at every new site. Keep it.

—	—	—	0 min	2 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing. This Move costs two days and changes what you are willing to sign in Move 28.

NEEDS 22 The four places the machines could live

- 01 Ask which certificate the facility holds and for which of the three forms. A design certificate and a constructed-facility certificate are different documents, and only the second says the building matches the drawings.
- 02 Ask for the certificate itself with its date and scope. Certificates apply to a specific building and sometimes to a specific phase of it, and a group-level claim is not a certificate for the hall you will occupy.
- 03 Note which scheme it is. The parallel schemes number their levels the same way and mean different things, and a facility rated under one will happily be described as though it were rated under another.
- 04 Retire the availability percentages. The figures still quoted beside each tier were removed from the standard in 2009 and never constituted a guarantee of anything. If a percentage appears in a proposal, ask what contractual commitment stands behind it, and Move 29 will tell you the answer is a capped credit.
- 05 Use the tour, and the question list, to ask what the topology means in practice: can the uninterruptible supply be maintained without dropping load, can a cooling unit be taken out in July, when was the generator last run under load, and for how long.
- 06 Write down what the facility actually holds, what it actually claims, and the gap between them. That gap is a negotiating position for Move 28 and Move 29.

ROLLBACK

Nothing is committed here, so this Move is undone by filing the notes. The point of no return is Move 28, which commits power in a specific building; everything before that is assessment. If a facility later turns out to have been misdescribed, these notes are the record, so keep them dated and in source control where an earlier version can be restored and compared against what the facility says at renewal.

28 COMMITTED KILOWATTS, METERED KILOWATTS AND THE DERATE

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
SITE	—	HIGH	NO

Reads the power schedule rather than the price, and fixes the billing model, the committed figure and the continuous-load derate in writing.

AWS	per-hour instance pricing	power was inside the rate, invisible, and never something you committed to in advance.
GOOGLE	per-second billing with sustained use discounts	the same invisibility, with the discount applied after the fact rather than committed up front.
AZURE	per-hour pricing with reservations	again power inside the rate, and again no schedule to read.

BEFORE YOU START

ACCESS

- The per-cabinet draw from Move 20 at expected and at peak load
- The facility's power schedule as a document, not as a paragraph in a proposal

SOFTWARE

- The three billing models written out with your own numbers substituted into each
- A note of what happens at the commitment boundary in each model: alarm, invoice, or breaker

PEOPLE

- Counsel to read the schedule with you, because the words that matter here are contractual rather than technical

WHY THIS WORKS

The headline price of a cabinet is not the number that governs your bill. The power schedule is. There are three billing models in common use — a flat charge against a committed figure, metered consumption with a committed floor, and a hybrid with an overage rate — and they behave completely differently when you grow. Underneath that sits the derate: the eighty per cent continuous-load convention is an artefact of one country's wiring code and has no direct equivalent in others, so one facility may sell against the full breaker rating where another will not. The only number that matters is the one the contract says you may draw continuously.

SWAP

Where the facility will only sell a flat commitment, size it against the year-two estate rather than the year-one one, and negotiate the increase right anyway. Paying for a little unused power is cheaper than an amendment under time pressure.

DO IT FASTER

Ask for a worked example invoice at your committed figure and at one hundred and twenty per cent of it. It settles the model question in an afternoon.

WATCH OUT

Power and cooling are frequently sold as one number and metered as two. Check whether cooling is included, charged as a factor on power, or billed separately against a facility efficiency figure that can change.

LEFTOVERS

Cloud commitments continue in parallel through the whole programme. You will pay both for a period, and Move 20 has already budgeted for it.

—	—	—	0 min	1 week	4 to 8 weeks
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. This Move adds a monthly bill that runs in parallel with the cloud one until Part VI.

NEEDS 20 The bill of materials 26 Carrier neutrality and the cross-connect price list

28 COMMITTED KILOWATTS, METERED KILOWATTS AND THE DERATE

6 STEPS · 0 MIN · HIGH RISK

THE RUNBOOK

SITE

- 01 Identify which billing model the schedule uses before discussing price. A flat charge against a commitment and a metered charge with a floor produce very different bills for the same estate, and the difference grows with you.
- 02 Establish what the facility does when you exceed the commitment. Some alarm and call you, some invoice at an overage rate, and some trip the breaker. Get the answer in writing, because the third one is an outage caused by a contract term.
- 03 Fix the derate in the contract. Ask directly what you are permitted to draw continuously per feed, as a number in kilowatts, and have it written into the schedule rather than inferred from a breaker rating.
- 04 Set the committed figure for the first cabinet from Move 20, with honest headroom. Committing too low makes growth an amendment; committing too high pays for power you never draw, and most schedules have no mechanism to reduce a commitment.
- 05 Negotiate a written right to increase power mid-term, with a stated notice period and a stated price. This is worth more than a discount on the initial rate and it is routinely omitted.
- 06 Get the metering point identified. Power measured at the cabinet and power measured at the distribution board are different numbers, and only one of them is yours.

ROLLBACK

This Move is irreversible in the way that matters: a signed power commitment is a monthly charge for the term, and most schedules have no mechanism to reduce it. The point of no return is the signature on the schedule, not the day the machines arrive. What is still reversible is where the machines go, since Move 22 kept a rented-metal option alive. Keep the schedule, the worked examples and the negotiation record in source control so the agreed position can be restored when the facility's billing disagrees with it.

29 WHAT THE SERVICE CREDIT IS ACTUALLY WORTH

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
SITE	CLOUD SERVICE-LEVEL AGREEMENTS AND THEIR CREDITS	LOW	UNTIL THE SCHEDULE IS SIGNED

Establishes what the facility owes when it fails, which is a capped credit, and redirects the negotiation onto the derate, the power increase and the exit.

AWS	the EC2 and EKS service level agreements	a monthly percentage and a credit, claimable by ticket, capped at a share of the bill.
GOOGLE	its compute service level agreements	the same structure, with the same claim window and the same cap.
AZURE	its virtual machine service level agreements	the same again, with credits expressed as a percentage of the monthly charge for the affected service.

BEFORE YOU START

ACCESS

- The draft service level schedule and the exclusions attached to it
- Your own cost of an hour of downtime, calculated rather than asserted

SOFTWARE

- The claim procedure written out as a runbook, including the deadline, because credits expire unclaimed
- A note of the measurement point for each commitment, since it decides whether a failure counts

PEOPLE

- Whoever will actually file a claim at eight in the morning after a bad night

WHY THIS WORKS

Everybody negotiates the service level and almost nobody reads what it pays. Facility agreements follow the same shape as the cloud ones you are leaving: a commitment on power and cooling, a set of exclusions, a window inside which a claim must be filed, and a credit that is the sole and exclusive remedy, capped at a fraction of one month's recurring charge. That credit will never approach the cost of an outage, and no amount of negotiation will make it. Establishing that early is useful, because it frees the negotiating hours for the three clauses that do change your position: the derate, the mid-term power increase and the exit.

SWAP

Where an outage genuinely would be catastrophic, the answer is the second site in Move 24 or the rehearsed restore in Move 115, not a better credit. Insurance and architecture are the tools here; contract remedies are not.

DO IT FASTER

Ask the facility directly what the largest credit it has ever paid was. The answer, or the refusal to answer, tells you what you need to know.

WATCH OUT

Some schedules require you to prove the outage with your own monitoring, at your own measurement point. If so, Move 114 needs to be collecting that data from the day the cabinet is live.

LEFTOVERS

The cloud service level agreements keep applying to whatever is still running there. Do not stop claiming against them while the migration is in flight.

—	—	—	0 min	3 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing. This Move saves negotiating hours rather than money, and spends them somewhere better.

NEEDS 28 Committed kilowatts, metered kilowatts and the derate

THE RUNBOOK

- 01 Read the commitment and find its measurement point. Power measured at the feed and power measured at the cabinet are different commitments, and cooling measured at the unit return is not the same as cooling measured at your inlet.
- 02 List the exclusions. Planned maintenance, force majeure and anything caused by your own equipment are standard, and together they remove most of the events you would want to claim for.
- 03 Find the cap and the claim window. The credit is almost always limited to a share of one month's recurring charge and the window is often as short as thirty days. Both are usually non-negotiable.
- 04 Calculate what the maximum possible credit would be for your estate, in currency, and put that number next to your cost of an hour of downtime. The comparison is the whole argument of this Move.
- 05 Redirect the negotiation. Spend the hours on the derate from Move 28, on the written right to increase power, and on the exit terms in Move 30. Those three change what the agreement is worth; the credit does not.
- 06 Write the claim procedure down anyway, with the deadline in the calendar. A credit you are entitled to and do not claim is a credit you have donated.

ROLLBACK

Nothing here binds you until the schedule is signed with the rest of the agreement in Move 30, and until then every term is a draft. The point of no return is that signature, after which the remedy for a failure is whatever the document says it is. Keep the schedule, the exclusions and your downtime cost calculation in source control, so the analysis can be restored and re-run at renewal rather than done again from nothing.

30 THE NOTICE PERIOD AND THE DOOR OUT

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
SITE	—	MEDIUM	UNTIL THE TERM COMMENCES

Fixes the term, the escalator, the auto-renewal notice and the relocation clause, and writes the exit as a runbook with its notice date in the calendar.

AWS	on-demand and Savings Plans	leaving was a matter of stopping instances, with the commitment as the only thing that outlived the decision.
GOOGLE	on-demand and Committed Use Discounts	the same, with the commitment attached to a project or an organisation rather than to a building.
AZURE	pay-as-you-go and reservations	again, exit by deletion, which is the property that makes a facility term feel so unfamiliar.

BEFORE YOU START

ACCESS

- The full draft agreement, including every schedule and any documents it incorporates by reference
- The depreciation schedule the bill of materials in Move 20 assumed

SOFTWARE

- A one-page summary of the commercial terms that non-lawyers on the team will actually read
- A calendar everybody uses, with reminders that survive the person who set them leaving

PEOPLE

- Counsel to draft the exit runbook with you, and whoever holds the budget for the term length

WHY THIS WORKS

The cloud let you leave by deleting things. A facility agreement does not, and the shape of the exit is decided at signature rather than at departure. Four terms carry almost all of it: the initial term and whether it matches the depreciation schedule the bill of materials assumed, the annual escalator, the auto-renewal and the notice required to stop it, and the conditions under which the facility may move you within the building or out of it. The fifth thing is not a clause at all but a date in a calendar, because an agreement that rolls into a further term because nobody diarised the notice is the commonest way this saving quietly disappears.

SWAP

Where a longer term buys a materially better rate, take it only if the exit and relocation terms are good. A cheap five-year agreement in a building that can move you is not cheap.

DO IT FASTER

Ask for the facility's standard agreement early, before pricing. Reading it takes a day and it tells you which battles are worth fighting.

WATCH OUT

Some agreements incorporate a facility handbook or rules by reference, and that document can be changed unilaterally. Ask for the current version and ask what happens when it changes.

LEFTOVERS

Disconnection fees for cross-connects are frequently forgotten in exit budgets. Move 26 priced them; carry the number into the exit runbook here.

—	—	—	0 min	3 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. This Move decides what turning it all off will cost when Part VII arrives.

NEEDS 28 Committed kilowatts, metered kilowatts and the derate 29 What the service credit is actually worth

THE RUNBOOK

- 01
- Match the initial term to the depreciation schedule from Move 20. A five-year hardware life against a three-year facility term means a renegotiation halfway through the asset's life, which is a weak position; a term longer than the hardware life is a commitment to a building you may not want.
- 02
- Fix the escalator. An uncapped index-linked increase is an open cheque; a fixed percentage is predictable. Ask for a cap either way and put the resulting figure into the three-year model.
- 03
- Find the auto-renewal and the notice period, which is often ninety or one hundred and eighty days. Put the notice date in the shared calendar the day the agreement is signed, with two reminders before it.
- 04
- Read the relocation clause. Facilities reserve the right to move customers within a building and occasionally between buildings. Ask what notice you get, who pays, and whether your cross-connects are re-provisioned at their cost.
- 05
- Write the exit as a runbook rather than as a clause: space returned to a bare cabinet, cross-connects formally disconnected and their fees paid, the final invoice reconciled, who does the de-racking and where the equipment goes. Move 38 handles the drives.
- 06
- Publish the one-page summary to the team. The people who will be affected by the notice date are not the people who signed, and that is exactly how a notice date gets missed.

ROLLBACK

An unsigned agreement is reversed by not signing it, and this Move deliberately happens before the term commences for that reason. The point of no return is the commencement date, after which the exit is governed by the notice period rather than by a decision. Keep the executed agreement, the one-page summary and the calendar entries in one place, so the commercial position can be restored quickly when a renewal notice arrives and nobody remembers what was agreed.

31 THE 80 PER CENT YOU ARE ALLOWED TO DRAW

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
SITE	—	HIGH	IMMEDIATELY

Turns the bill of materials into a per-cabinet power budget in the units the facility bills in, with outlets counted and the phases balanced.

AWS	instance families and their implied draw	power existed only as a component of the hourly rate you paid.
GOOGLE	machine families and their carbon reporting	the closest any of the three came to telling you what a workload drew, and it was a report rather than a budget.
AZURE	virtual machine series and their sustainability calculator	again a report after the fact, never a constraint you designed against.

BEFORE YOU START

ACCESS

- The per-node and per-rack draw from Move 07, and the machine count from Move 11
- The feed type the facility is offering, as volts, phases and amps rather than as a kilowatt figure

SOFTWARE

- The power distribution unit datasheets for the exact model you intend to order, with the outlet mix listed
- A cabinet elevation drawing, even a rough one, so outlets and rack units are counted together

PEOPLE

- Whoever will order the distribution units, because the outlet mix cannot be changed afterwards

WHY THIS WORKS

The facility bills in kilowatts against a committed figure, and your bill of materials is in machines. This Move does the conversion explicitly, because the arithmetic has two traps in it. The first is the derate: a 208 volt three-phase 30 amp feed is 10.8 kVA on the nameplate and about 8.6 kW once a continuous-load derate is applied, while a 400 volt three-phase 32 amp European feed is 22.2 kVA and is sold against a different convention entirely. The second is outlets, which nobody counts until the strips arrive: every machine takes two, every switch takes two, and the mix of socket types is fixed the day the distribution units are ordered.

SWAP

Where the facility offers metered distribution units, take them. Per-outlet measurement turns Move 120's cost arithmetic from an estimate into a reading, and it is worth the small premium.

DO IT FASTER

Ask the facility which distribution units their other customers use in that hall. Matching them means the remote hands technician in Move 35 already knows the equipment.

WATCH OUT

Inrush at power-on is much higher than steady draw. A cabinet that sits comfortably inside its budget can trip a breaker when everything starts at once after a supply failure, which is precisely when you least want it.

LEFTOVERS

Distribution units bought for one feed type do not work on another. If the facility choice is not final, do not order the strips.

—	—	—	0 min	4 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. This Move turns a bill of materials into the number the facility invoices against.

NEEDS 20 The bill of materials 28 Committed kilowatts, metered kilowatts and the derate

- 01
- Convert the feed to a usable figure. Multiply volts by amps by the phase factor for the nameplate, then apply the derate the contract in Move 28 actually states. Write both numbers down, because people will quote the first one at you.
- 02
- Sum the measured draw of everything going in the cabinet at expected load and at peak, using the figures from Move 07 rather than badge ratings.
- 03
- Count the outlets by type. Every machine takes two, every switch takes two, the console server takes one or two, and the mix of socket types is fixed when the strips are ordered. Running out of one type with spare capacity in the other is a common and avoidable delivery-day failure.
- 04
- Balance the phases. A three-phase strip distributes outlets across phases in a fixed pattern, so which slot a machine plugs into decides which phase it loads. An unbalanced cabinet trips a breaker at a total draw well below its rating.
- 05
- Leave headroom for year two. A cabinet planned to its committed figure on day one has nowhere to put the machines Move 11 said you would add, and increasing the commitment is an amendment under Move 28's terms.
- 06
- Publish the cabinet plan: position, draw, outlet, phase. Move 36 racks against it and Move 37 documents it.

ROLLBACK

Nothing here is committed until the distribution units are ordered, and a cabinet plan is revised by revising it. The point of no return is the strip order, because the outlet mix and the feed type are fixed at that moment and a wrong choice is a re-purchase rather than a reconfiguration. Keep the cabinet plan and the power arithmetic in source control so the intended layout can be restored when somebody has moved a machine to a different outlet without updating the drawing.

32 TWO FEEDS, ONE OF WHICH YOU HAVE TESTED

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
SITE	—	HIGH	IMMEDIATELY

Establishes that a dual-fed cabinet is worth one feed at the derate, puts single-supply equipment behind a transfer switch, and pulls the A feed to prove it.

AWS	Availability Zones	the redundancy was inside the zone and you never saw it, tested it, or paid for it as a line.
GOOGLE	zones	the same, with the provider's own testing invisible and the failure domains described rather than demonstrated.
AZURE	availability zones and fault domains	again resilience as a property of the platform, which is why an untested B feed feels acceptable to people arriving from it.

BEFORE YOU START

ACCESS

- A commissioning window with the cabinet loaded and no production on it
- Permission from the facility to open a breaker on the A feed, agreed in advance

SOFTWARE

- Per-outlet or per-feed metering readings, recorded before and during the test
- A rack transfer switch for the single-supply equipment, ordered with the distribution units

PEOPLE

- A second person on site during the test, because the failure mode is a dark cabinet

WHY THIS WORKS

A cabinet with two feeds looks like twice the power and is not. The usable budget is one feed at the derate, because the whole point is that either feed can carry the load alone, so a pair of 8.6 kilowatt feeds is an 8.6 kilowatt cabinet and each distribution unit runs at around forty per cent in normal operation. That arithmetic is routinely got wrong in the direction that produces an outage. The second half of this Move is the part almost nobody does: pulling the A feed with the cabinet fully loaded and nothing yet depending on it, and measuring what the B feed actually carried and what tripped.

SWAP

Where the facility will not permit a live feed test, ask them to schedule it during their own maintenance. Failing that, test at the distribution unit level rather than not at all.

DO IT FASTER

Do this on the day the first cabinet is loaded and before anything is installed on the machines. It is a two-hour test that becomes a three-week negotiation once production depends on the cabinet.

WATCH OUT

Some power supplies draw noticeably more from one feed when the other is absent, so the B feed reading under test can exceed half the normal total by more than expected. That is the number the budget has to survive.

LEFTOVERS

Test readings are evidence. Keep them, because they are what you compare against when a feed reading drifts in year two.

—	—	—	0 min	3 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. This Move spends a day proving that the resilience you are paying for exists.

NEEDS 31 The 80 per cent you are allowed to draw

THE RUNBOOK

- 01
- Recompute the cabinet budget as one feed at the derate rather than the sum of the two. If the plan from Move 31 exceeded that figure, fix it now, before anything is racked against it.
- 02
- Wire every dual-supply machine to both feeds, with one supply on each distribution unit, and record which outlet each cable occupies in the cabinet plan.
- 03
- Deal with the single-supply equipment properly. A console server, a small access switch or an out-of-band unit with one supply goes behind a rack transfer switch fed from both strips, rather than being assigned to a feed and hoped about.
- 04
- Load the cabinet fully and take a baseline reading per feed. This number is the one you will compare against for the rest of the cabinet's life.
- 05
- Pull the A feed. Open the breaker with the cabinet loaded and watch what happens: the reading on B, anything that restarted, anything that did not come back, and whether the transfer switch carried its load without interruption. Record the numbers rather than the impression.
- 06
- Restore the A feed, confirm both strips return to their baseline, and write up what tripped. Every fault this test finds is a fault you would otherwise have found during a real supply failure, at three in the morning, with production on the cabinet.

ROLLBACK

The test is reversed by closing the breaker, and the wiring decisions are reversed by moving cables. Nothing here is permanent. The point of no return is the day production depends on the cabinet, after which this test stops being a commissioning step and becomes a change with a risk assessment. Keep the baseline readings and the cabinet plan in source control so the commissioned state can be restored and compared when something is rewired later.

33 THE INLET TEMPERATURE YOU ARE ALLOWED

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
SITE	—	HIGH	IMMEDIATELY

Commits the inlet temperature at the top of the cabinet inside containment, then fixes the blanking, brush strips and aisle orientation that make it achievable.

AWS	EC2 host management	thermal behaviour was the provider's problem and surfaced to you, at most, as a retirement notice.
GOOGLE	Compute Engine host maintenance	the same, with live migration hiding thermal events entirely.
AZURE	Virtual Machines health events	again somebody else's cooling, and the reason nobody arriving from a cloud owns a temperature probe.

BEFORE YOU START

ACCESS

- The facility's cooling commitment as drafted, with its measurement point stated
- The density limit per cabinet in kilowatts, which is often lower than the power commitment implies

SOFTWARE

- A temperature probe or a set of them, because a thermal survey done by hand is not a survey
- Blanking panels and brush strips ordered with the cabinet rather than after the first hot spot

PEOPLE

- The facility's engineer, walked around the cabinet with you rather than consulted by email

WHY THIS WORKS

A cooling commitment measured as a room average at the cooling unit's return tells you nothing about the machine at the top of your cabinet, which is where the air is hottest and where the failures start. So the commitment is written at the point that matters: inlet air at the top of the cabinet, inside containment. The envelope itself is well established — the recommended band is eighteen to twenty-seven degrees and mainstream equipment is allowable considerably higher — and in practice the binding constraint is rarely temperature at all. It is the facility's density limit per cabinet, which is a different number and one that is frequently discovered late.

SWAP

Where containment is not available, a cabinet with good blanking in a well-run hall is usually acceptable. Where neither is available, reduce the density rather than accepting the temperature.

DO IT FASTER

Buy a handful of small logging probes and leave them in the cabinet permanently. Move 114 will alert on them and you will never have to run a survey again.

WATCH OUT

Running at the top of the allowable envelope saves the facility money and costs you fan power, which you pay for. Warmer is not automatically cheaper once the draw is yours.

LEFTOVERS

Blanking panels are the cheapest thermal improvement available and the one most often left in a box. Fit them all, including the ones for gaps you intend to fill next year.

—	—	—	0 min	3 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. Getting this wrong shortens the life of everything Move 20 bought.

NEEDS 31 The 80 per cent you are allowed to draw

- 01
- Get the cooling commitment written at your inlet, at the top of the cabinet, inside containment. If the facility will only commit to a room average, understand that you are buying a number that does not describe your equipment.
- 02
- Set the envelope explicitly rather than by folklore. Write down the recommended band you intend to run in and the allowable maximum for your class of equipment, so that a warm day is a known condition rather than an incident.
- 03
- Confirm the density limit per cabinet and check it against Move 31's budget. A hall rated at five kilowatts a cabinet will not take a cabinet designed for eight, whatever the power schedule says you may draw.
- 04
- Fix the airflow discipline before the thermal survey, not after. Consistent hot and cold aisle orientation, blanking panels in every unused rack unit, brush strips on every cable cut-out, and no cabinet drawing its air from the hot aisle.
- 05
- Run a thermal survey with a probe, at the bottom, middle and top of the cabinet, front and rear, at full load. Record the numbers against the commitment.
- 06
- Repeat the survey after Move 36 has racked the cabinet properly and again in the warmest month. A cabinet that is comfortable in February is not evidence of anything.

ROLLBACK

Every mitigation here is physical and reversible: panels come out, probes move, the aisle orientation is a decision about which way cabinets face. The point of no return is the cabinet's orientation once it is bolted down and cabled, which is an afternoon's work to change on day one and a serious disruption once populated. Keep the survey readings in source control so the commissioned thermal baseline can be restored and compared when a machine starts throttling next summer.

34 THE IMPORTER OF RECORD AND THE LOADING DOCK

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
 SITE	—	 MEDIUM	UNTIL THE SHIPMENT IS BOOKED

Names the importer of record, settles the customs and import-VAT position, and books the dock, the lift and the window against a cabinet approaching a tonne.

AWS	EC2 capacity	hardware logistics were entirely invisible, and the only thing that ever crossed a border was a packet.
GOOGLE	Compute Engine capacity	the same, with regional expansion happening without any involvement from you at all.
AZURE	Virtual Machines capacity	again a service, and again the reason a customs entry feels like an unfamiliar kind of problem.

BEFORE YOU START

ACCESS

- The customs registration for the importing entity in the destination country, obtained rather than assumed
- The facility's delivery procedure, dock hours and lift capacity, in writing

SOFTWARE

- A commercial invoice reviewed by the broker before shipping, with correct classification and values
- A delivery plan naming who receives, who signs, and where the packaging goes afterwards

PEOPLE

- A customs broker engaged early, and somebody at the facility expecting the pallet on the day

WHY THIS WORKS

Getting the hardware into the building has a legal half and a physical half, and both stop a delivery cold. The legal half is that a colocation provider will not act as importer of record, because that would put its name on a customs debt for equipment it does not own. You need an entity with a customs registration in the destination country and a broker who has read the commercial invoice before the pallet flies. The physical half is that a populated cabinet approaches a tonne, delivery windows are booked, docks have hours, and lifts have limits — and a pallet that will not fit a passenger lift is a delivery that goes back on the truck.

SWAP

Where the importing entity does not exist in the destination country, a delivered-duty-paid arrangement with the supplier can work, at a premium, and it removes the recovery of import tax. Price both.

DO IT FASTER

Ask the supplier to ship to a local staging warehouse first. It converts an international shipment with a border in it into a domestic delivery you can reschedule.

WATCH OUT

Lithium batteries in uninterruptible supplies have their own shipping rules, and a mixed pallet can be held for one item. Ask the supplier to ship batteries separately if there is any doubt.

LEFTOVERS

Keep the customs entry documents. They are needed to recover import tax and they will be asked for at the audit in Move 119.

—	—	—	0 min	5 days	3 to 6 weeks
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. This Move is pure overhead, and skipping it is how hardware sits at a border for a fortnight.

NEEDS 20 The bill of materials 28 Committed kilowatts, metered kilowatts and the derate

- 01 Name the importer of record and confirm it holds a customs registration in the destination country. The facility will decline this role, and discovering that when the shipment is at the border is expensive in both fees and days.
- 02 Settle the classification and the tax position with the broker. Servers generally fall under a heading that attracts no duty under the international agreement covering information technology equipment, but import tax falls due on the landed value, only the named importer can recover it, and origin-based tariffs can override the headline treatment.
- 03 Have the broker review the commercial invoice before anything ships. Wrong values, wrong classification or a missing importer reference are the three things that hold a pallet, and all three are fixed in an email beforehand.
- 04 Book the delivery window with the facility, in writing, with dock hours and an on-site contact. Ask specifically about lift capacity and dimensions, because a populated cabinet approaching a tonne does not go in a passenger lift.
- 05 Decide whether cabinets arrive populated or are built in place. Populated saves days of work and needs a goods lift and a dock; built in place is slower and fits through more doors.
- 06 Plan where the packaging goes. A pallet of cardboard in a hall is a fire-load problem and most facilities will charge you to remove it or refuse to let it stay.

ROLLBACK

Until a shipment is booked this is paperwork, and paperwork is revised. Once goods are in transit the importer of record is named on a customs entry and changing it means an amended entry, delay and fees. The point of no return is the booking, not the border. Keep every entry document, invoice and delivery note filed with the bill of materials so the import position can be restored and evidenced months later when finance or an auditor asks.

35 WHO CAN GET IN AT 03:00

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
SITE	THE SHARED-RESPONSIBILITY SPLIT FOR PHYSICAL SECURITY	HIGH	IMMEDIATELY

Settles the access list, the escort rules and a remote-hands agreement measured in ticket-to-touch, and ends with a ticket template a stranger can follow.

AWS	the shared responsibility model	physical security was entirely theirs, evidenced by a report you downloaded and filed.
GOOGLE	its shared responsibility documentation	the same split, with the physical layer described in a whitepaper rather than inspected.
AZURE	its shared responsibility guidance	again the provider's half, and again the reason nobody has an access list to maintain.

BEFORE YOU START

ACCESS

- The facility's access request procedure and how long it takes to add or remove a name
- The remote hands agreement as a document, with its billing increments and escalation path

SOFTWARE

- A ticket template with photographs, cabinet, rack unit and port number, written before it is needed
- A record of who is on the access list, kept somewhere the leaver process actually reaches

PEOPLE

- Whoever runs joiners and leavers, because an access list that is not part of that process rots within a quarter

WHY THIS WORKS

Physical access is now yours to manage in both directions. Inward: who is on the named list, how quickly a leaver comes off it, whether contractors need an escort, whether the cabinet is locked and who holds the second key, and how long camera footage is retained. Outward: what the facility's own staff will do for you when nobody can drive there, which is the remote hands agreement, and it is measured in ticket-to-touch rather than in response time. Underneath both sits a fact no contract changes — facility staff can open your cabinet — which is why Move 48 puts encryption and key custody beneath everything that follows.

SWAP

Where remote hands is slow or expensive, the cold spare machine from Move 19 covers more faults than any support contract, because it converts a repair into a replacement.

DO IT FASTER

Raise one trivial remote hands ticket in the first week — reseal a cable, read a serial number — purely to learn the process while it does not matter.

WATCH OUT

Camera retention is often shorter than you assume, sometimes only a fortnight. If evidence matters for the audit in Move 119, ask for the retention in writing.

LEFTOVERS

Access lists accumulate contractors. Review the list quarterly, on a date, rather than after an incident.

—	—	—	0 min	3 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. This Move is what makes the facility usable by a small team that does not live next to it.

NEEDS 28 Committed kilowatts, metered kilowatts and the derate 29 What the service credit is actually worth

- 01 | Fix the named access list and how a leaver is removed from it. Bind the removal to the same process that revokes everything else, or the list will contain former employees within a year.
- 02 | Settle the escort rules for contractors and suppliers, and who is permitted to authorise a visit at three in the morning. That authority needs a name and a deputy, not a role nobody occupies out of hours.
- 03 | Decide whether the cabinet is locked, who holds the keys and whether the facility holds a second one. Locked cabinets slow down remote hands; unlocked ones speed up everybody, including people you have not met.
- 04 | Read the remote hands agreement for the number that matters. Ticket-to-touch is how long until somebody is standing at your cabinet; response time is how long until somebody reads the ticket. They differ by hours. Note the billing increment, which is usually fifteen minutes with a minimum.
- 05 | Build the escalation ladder with a name and a telephone number at the top, and test it once during working hours by raising a trivial ticket and following it up.
- 06 | Write the ticket template. The technician does exactly what the ticket says and no more, so it carries photographs of the front and rear, the cabinet identifier, the rack unit, the port number and the expected outcome. A stranger should be able to follow it without ringing you.

ROLLBACK

Every arrangement here is reversed by changing a list or a document, and none of it is structural. The point of no return does not exist in this Move, which is why it is worth doing carefully now rather than under pressure later. Keep the access list, the ticket template and the escalation ladder in source control so a known good version can be restored when somebody edits the template and removes the photographs.

36 THE FIRST RACK

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
SITE	—	MEDIUM	IMMEDIATELY

Racks the first cabinet in the order that survives contact with the second: strips first, weight low, gaps blanked, and photographs before any data cable.

AWS	launching an instance	capacity appeared in ninety seconds and the physical work behind it was somebody else's shift.
GOOGLE	creating an instance	the same, with the machine's location expressed as a zone label.
AZURE	deploying a virtual machine	again minutes rather than days, which is the expectation this Move gently corrects.

BEFORE YOU START

ACCESS

- The cabinet, delivered and positioned, with the feeds live and the cross-connects at least ordered
- A booked day on site with the facility notified, because this takes longer than anyone estimates

SOFTWARE

- The cabinet plan from Move 31 and the cabling diagram from Move 12, printed and taken with you
- The acceptance results from Move 18, so you know which machines are known good

PEOPLE

- Two people for the whole day, because a two-rack-unit chassis at height is not a one-person job

WHY THIS WORKS

The first cabinet teaches you how to build the second, and the order of operations is what decides whether that lesson is cheap. Power distribution and cable management go in before anything heavy, because retrofitting them means unracking machines. Switches go where the cabling plan wants them rather than at the top out of habit. The heaviest chassis go low. Rails are checked against the actual cabinet depth rather than the specification, and the rack-unit budget has already counted the strips, the patch panels and the console server. Deliberate gaps are left for year two and every one of them is blanked.

SWAP

Where the supplier offers to rack and cable on delivery, take it for the first cabinet and watch them do it. It is the fastest way to learn what your own second cabinet should look like.

DO IT FASTER

Build the cabinet in the order above and resist the urge to cable as you go. Batch the cabling at the end and it takes half as long and looks like it was planned.

WATCH OUT

Cable slack is the thing that makes a cabinet maintainable. Too little and a machine cannot slide out on its rails; too much and the rear becomes impassable. Dress to a length rather than to a bundle.

LEFTOVERS

Packaging, rails you did not use and cables of the wrong length all accumulate. Take them away the same day, because the facility will charge you or refuse to store them.

—	—	—	0 min	1 week	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. There is now a cabinet drawing power in one building and a full cloud bill in another.

NEEDS 31 The 80 per cent you are allowed to draw 32 Two feeds, one of which you have tested 33 The inlet temperature you are allowed 34 The importer of record and the loading dock 35 Who can get in at 03:00

- 01
- Fit the power distribution units and the cable management first, then the patch panels. Anything that has to be retrofitted behind a populated cabinet costs an evening.
- 02
- Place the switches where the cabling plan wants them. Top of the cabinet is a habit rather than a rule, and middle placement halves the average cable run in a full cabinet.
- 03
- Rack the heaviest chassis low. Check every rail kit against the actual cabinet depth before lifting anything, because a rail that does not fit is discovered at the worst possible moment.
- 04
- Leave deliberate gaps for the machines Move 11 said you would add in year two, and blank every gap. An unblanked gap recirculates hot air straight back to the inlet above it.
- 05
- Power up, confirm the out-of-band network from Move 16 is reachable from outside the building, and confirm every management controller answers. Do this before touching a data cable, so that a machine that will not come up is a machine you can still reach.
- 06
- Photograph the front and the rear before any data cabling. These photographs go in the ticket template from Move 35 and they are what a remote technician will be looking at.
- 07
- Cable the data network last, to the plan, labelling both ends as you go. Move 37 turns the labels into a document.

ROLLBACK

Everything in a cabinet can be unracked, and on day one that is a day's work rather than a change window. The point of no return is the moment the cabinet carries production, after which every physical change is a maintenance window. This is the argument for leaving the gaps and blanking them now. Keep the cabinet plan and the photographs somewhere durable and backed up, so the as-built state can be restored and compared after somebody has worked in the cabinet without saying so.

37 LABELS, PORT MAPS AND THE AS-BUILT

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
SITE	—	LOW	IMMEDIATELY

Builds the label scheme, port map and as-built that let somebody who has never entered the room replace the correct part at three in the morning.

AWS	instance identifiers and tags	an identifier that followed the resource, with no physical position to reconcile it against.
GOOGLE	instance names and labels	the same, with a naming convention that never had to survive a machine being moved.
AZURE	resource names and tags	again logical identity only, which is why physical labelling is an unfamiliar discipline.

BEFORE YOU START

ACCESS

- The racked cabinet from Move 36, with the photographs taken
- The switch running configuration, exported, so port descriptions can be reconciled against reality

SOFTWARE

- A label printer and durable labels, because paper and tape do not survive a warm aisle
- A single source for the as-built, in source control rather than in a drawing on somebody's laptop

PEOPLE

- Whoever will be the remote technician's contact, since they are the one who will use this document

WHY THIS WORKS

The document this Move produces has one test: hand it to a technician in another city who has never seen your cabinet, and see whether they can find and replace the correct part without ringing you. That requires a label scheme that names the position rather than the machine, so that moving a machine does not invalidate the label; both ends of every cable labelled, because a cable with one label is half a label; a port map tying switch port to cabinet, rack unit and interface; and serial numbers recorded against positions, so the drive that fails in Move 111 can be found by its bay rather than by guesswork.

SWAP

Where a full asset system is available, use it, but keep the as-built as a flat document as well. The document is what somebody reads on a phone in a hall with no signal.

DO IT FASTER

Print the labels before going to site, from the cabinet plan. Labelling during the build takes minutes; labelling after it takes an afternoon and a torch.

WATCH OUT

Serial numbers on drives are frequently unreadable once installed. Record them during the build in Move 36, not afterwards, and photograph the labels while you can still see them.

LEFTOVERS

Old labels from a previous occupant of a cabinet are a genuine hazard. Remove them rather than labelling over them.

—	—	—	0 min	4 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. This is the cheapest four days in this Part and the one that pays for itself in Part VII.

NEEDS 36 The first rack

SITE

THE RUNBOOK

- 01 Choose a label scheme that names the position, not the equipment. Cabinet, rack unit, and where relevant the bay or port. A label that says which machine is in a slot is wrong the first time a machine moves; a label that says which slot it is stays true forever.
- 02 Label both ends of every cable with the same identifier, and put port descriptions into the switch configuration that match. A port map that disagrees with the switch is worse than no port map.
- 03 Record serial numbers against positions for every machine, drive bay and distribution unit. This is the table Move 110 will maintain and Move 111 will read at three in the morning.
- 04 Write the as-built: cabinet elevation, power plan with outlet and phase, network port map, cross-connect terminations, and the photographs from Move 36. One document, in source control, with a date.
- 05 Test it the only way that means anything. Send it to a supplier or technician in another city and ask them to describe how they would replace a specific drive in a specific machine. If they have to ask you a question, the document is not finished.
- 06 Put a review date on it and bind updates to the change process, because an as-built that is not updated when a cable moves becomes actively dangerous within about two months.

ROLLBACK

Documentation is corrected by correcting it, and there is nothing here to undo. The point of no return is the moment the document stops matching the cabinet, which happens silently and is the whole reason for the review date. Keep the as-built in source control with the photographs, so a previous version can be restored and compared when the cabinet and the document disagree and nobody remembers which changed.

38 THE DRIVES THAT LEAVE, AND THE SNAPSHOTS THAT STAY

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
SITE	PROVIDER-MANAGED SNAPSHOT, IMAGE AND BACKUP RETENTION	HIGH	NO

Sanitises the drives that leave against serial-level certificates and destroys the key material behind the snapshots, images and vaults that stay.

AWS	EBS snapshots, AMIs and Backup vaults	deletion is an API call, and vault lock and recycle bin settings can prevent it entirely.
GOOGLE	Persistent Disk snapshots, machine images and Backup vaults	the same, with retention policies that outlive the resources they protected.
AZURE	Managed Disks snapshots, images and Recovery Services vaults	the same again, with soft delete on by default and purge protection that delays completion by days.

BEFORE YOU START

ACCESS

- Administrative rights to delete snapshots, images and vaults in every account, including the locked ones
- The key management permissions needed to schedule destruction of the keys that wrapped them

SOFTWARE

- An inventory of every snapshot, image, object version and vault, per account, exported rather than browsed
- A sanitisation tool appropriate to each media type, and the certificate template you will require from a processor

PEOPLE

- Whoever owns data classification, because destruction is their decision and their sign-off

WHY THIS WORKS

Decommissioning has two halves and most programmes do only one. On hardware you hold, sanitisation follows the current standard, which now defers technique to the media-specific specification: cryptographic erase for self-encrypting media, the drive's own sanitise command for NVMe, and physical destruction only where a data classification or a contract demands it — because multi-pass overwriting a solid-state device is theatre when wear levelling keeps cells out of reach of the write path. On the cloud side you will never hold the platter, so the only real lever is the key. Destroy the key material first, then delete the snapshots, images, object versions and vaults, and expect deletion to complete days after you asked.

SWAP

Where a vault lock prevents deletion until a retention period expires, the honest answer is to diary the date and pay the storage until then. Locks exist precisely so that they cannot be worked around.

DO IT FASTER

Destroy the key first. On every provider, that renders the data unreadable immediately even where the objects themselves take days to disappear.

WATCH OUT

Cross-region and cross-account copies are the ones that survive a cleanup. Search by account and by region rather than trusting a single console view.

LEFTOVERS

Object storage keeps non-current versions and delete markers after a delete. A lifecycle rule that expires them is required, and it takes at least a day to run.

—	—	—	0 min	1 week	2 weeks
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Snapshots, machine images, object versions and backup vaults in the accounts that have been emptied — after the data owner has signed, and never before.

NEEDS 37 Labels, port maps and the as-built

- 01
- Build the inventory first, across every account and region. Snapshots outlive the resources they protected, and copies made into other regions do not appear in the region you are looking at.
- 02
- Confirm what can still be restored from each item and for how long, and get written agreement from the data owner that it is no longer needed. Nothing in this Move is undone, so the check happens before, not after.
- 03
- On hardware you hold, sanitise per media type: cryptographic erase where the device supports it, the device's own sanitise command for NVMe, physical destruction only where policy requires it. Verify each unit and record the result against its serial number.
- 04
- Require a certificate listing serial numbers from any processor you use, under a chain of custody, and check their certification before the equipment leaves the building. Keep the disposal note that satisfies your waste-electricals obligation.
- 05
- On the cloud side, schedule destruction of the key material that wrapped the data before deleting anything, and confirm the destruction has completed rather than merely been scheduled.
- 06
- Delete the snapshots, images, object versions and vaults, then verify. Allow for soft-delete windows, recycle bins and purge protection, all of which mean the deletion you requested completes days later, and some of which will refuse until a lock expires.
- 07
- Re-run the inventory afterwards and confirm it is empty. An inventory that still lists items after a deletion pass is the normal outcome of the first attempt.

ROLLBACK

This Move is irreversible by design and there is no rollback: once the key material is destroyed the data cannot be restored, and a sanitised drive cannot be read back. The point of no return is the key destruction, which precedes the deletions and is the moment everything downstream becomes unrecoverable. That is why step two requires written agreement from the data owner first. Keep the inventory, the certificates and the destruction records for the audit in Move 119.

III

CLUSTER

Everything between a racked machine and a cluster that can take production traffic. Nothing here moves a workload; all of it decides how well the rest of the book goes.

16	16	10	0
MOVES	ZERO DOWNTIME	HIGH RISK	MINUTES, TOTAL

IF YOU ONLY DO THREE

51

THE SECOND CLUSTER, AND WHAT IT SHARES

Builds the non-production cluster the drills and rehearsals in this book assume, and states plainly what it shares …

54

THE OBJECT STORE YOU NOW OWN, AND HOW IT LOSES DATA

States the replication factor, the off-site copy and the restore drill for the store that now holds backup …

39

THE ADDRESS PLAN, AND THE THREE CLOCKS YOU START TODAY

Fixes the pod, service, node and management ranges before a machine is racked, and starts the RIR, partner-allowlist …

THE ADDRESS PLAN, AND THE THREE CLOCKS YOU START TODAY · THE MANAGEMENT VLAN, AND THE WAY IN AT 03:00 · TALOS AS THE HOST OS, AND WHERE IT CANNOT GO · NETBOOT, NODE IDENTITY AND FAILURE-DOMAIN LABELS · THREE CONTROL-PLANE NODES, THE API VIP AND THE CLUSTER CA · CILIUM, WITH KUBE-PROXY OFF · BGP TO THE TOP-OF-RACK, AND THE MTU THAT BREAKS FIVE PER CENT · CLUSTER DNS, AND THE RESOLVER THE NETWORK USED TO BE · LOCAL NVME, AND THE DEFAULT STORAGECLASS NOBODY GAVE YOU · ENCRYPTION AT REST, AND WHO HOLDS THE KEY · CEPH UNDER ROOK: BLOCK, SHARED FILESYSTEM AND THE OBJECT STORE · HUMANS ON THE API SERVER: OIDC, AFTER BREAK-GLASS · THE SECOND CLUSTER, AND WHAT IT SHARES · THE ETCD RESTORE DRILL · THE LINKS YOU BUILD FIRST: TO THE CLOUD, TO THE OFFICE · THE OBJECT STORE YOU NOW OWN, AND HOW IT LOSES DATA

■ 2 LOW RISK	■ 4 MEDIUM RISK	■ 10 HIGH RISK
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39 THE ADDRESS PLAN, AND THE THREE CLOCKS YOU START TODAY

CUTOVER

0 MIN



LAYER CLUSTER	LEAVING CLOUD-MANAGED PRIVATE ADDRESSING AND INTER-NETWORK TRANSIT	RISK HIGH	REVERSIBLE IMMEDIATELY
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Fixes the pod, service, node and management ranges before a machine is racked, and starts the RIR, partner-allowlist and hiring clocks that nothing compresses.

AWS	VPC address ranges with Transit Gateway	the ranges were yours to pick and the routing between them was a managed service.
GOOGLE	VPC subnets with secondary ranges	pod and service ranges belong to the network rather than to the cluster, so the plan has to match ranges that already exist.
AZURE	Virtual Network address space with peering or Virtual WAN	the same picking exercise, with peering rather than a hub doing the transit.

BEFORE YOU START

ACCESS

- The current address ranges in every cloud account, exported, including the ones nobody uses
- The office ranges and every partner range you already connect to

SOFTWARE

- A single address plan document, in source control, that is the only place ranges are recorded
- The registry membership application for your region, opened rather than bookmarked

PEOPLE

- Whoever can sign a registry membership and whoever owns hiring, because both clocks start today

WHY THIS WORKS

Addressing is the one design decision that is free to change today and extremely expensive to change in eighteen months, because by then it is in firewall rules at partners you do not control. So the whole plan is allocated at once: pod, service, node, management, provisioning and a reserved range for a second site, none of which may overlap each other, the offices, any partner you will connect to, or the private space an acquisition will bring with it. The second half of this Move is not technical at all. Three clocks — registry membership, partner allowlisting and hiring — take months and nothing compresses them, so they are started this afternoon.

SWAP

Where an overlap with an office range is unavoidable, translation at the boundary works and is unpleasant. It is a reason to renumber the office rather than the cluster, because the office has fewer dependencies.

DO IT FASTER

Allocate on a clean boundary with obvious arithmetic, so that a person reading a firewall rule can tell what it is for without looking anything up.

WATCH OUT

Cloud-managed transit hubs did route summarisation for you. On your own fabric, an address plan that does not summarise produces a routing table that is hard to read and slow to converge.

LEFTOVERS

The old cloud ranges stay live throughout the migration and are only reclaimed in Part VII. Do not reuse them in the new plan.

—	—	—	0 min	5 days	6 to 12 weeks
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. The cloud networks carry production for the whole of Parts III and IV.

NEEDS 11 Headroom, and the number of machines you order 24 The second site you are not building this year

- 01 | Inventory every private range already in use across the clouds, the offices and the partners. Overlaps discovered later cannot be routed around; they have to be renumbered, and renumbering a production network is a project rather than a change.
- 02 | Allocate the ranges in one pass, with room to grow: pods, services, nodes, management, provisioning, and a reserved block for the second site that Move 24 declined to build this year. Write down why each is the size it is.
- 03 | Keep the pod range generous. Pod addresses are consumed per node and the number of nodes grows; a pod range sized to today's cluster is the most common address plan mistake and it surfaces as pods that will not schedule.
- 04 | Reconcile with the Google case explicitly if you are leaving it. There, pod and service ranges are secondary ranges owned by the network rather than by the cluster, so the plan has to align with ranges that already exist rather than being invented.
- 05 | Start the registry clock. Membership and an address allocation take weeks to months depending on region, and Move 91 cannot begin until it is done. Apply today, before the first machine is racked.
- 06 | Start the other two clocks the same afternoon: allowlist requests with every partner whose firewall will need your new addresses, and the hiring you concluded you needed in Move 22.

ROLLBACK

On paper this Move is free to redo, and it should be redone as often as needed until it is right. The point of no return is Move 42, which addresses the first node from this plan, and every day after that adds firewall rules at other people's organisations. The registry application can be withdrawn without penalty. Keep the plan in source control as the single source of truth, so an earlier allocation can be restored and compared when somebody claims a range was always meant for something else.

40 THE MANAGEMENT VLAN, AND THE WAY IN AT 03:00

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
CLUSTER	CLOUD SERIAL CONSOLE AND AGENT-BASED REMOTE SHELL	HIGH	IMMEDIATELY

Builds the management VLAN, rotates the BMC credentials off the sticker, and drills the break-glass path before Move 50 can lock everyone out at once.

AWS	EC2 Serial Console with Session Manager	a console and a shell, both reached through the provider's control plane rather than your network.
GOOGLE	the serial console with Identity-Aware Proxy	the same, except that serial access is a permission an organisation policy commonly disables, so many readers never used it.
AZURE	Serial Console with Bastion and Run Command	the same again, with three separate services covering what is here one VLAN and one jump host.

BEFORE YOU START

ACCESS

- The management range from Move 39 and the out-of-band hardware from Move 16, installed
- The default controller credentials from the build, which are about to be replaced

SOFTWARE

- A credential store that the break-glass path does not depend on, and a sealed copy of the emergency credentials
- `ipmitool` and a Redfish client for the smoke test, one per vendor if you have mixed hardware

PEOPLE

- Two people who will drill the break-glass path, because a path one person knows is not a path

WHY THIS WORKS

Every remote path you had went through somebody else's control plane, which meant it kept working when your own network did not. Reproducing that property is the whole content of this Move: a management network with no default route, reachable only through a jump host, carrying the management controllers bought in Move 08 and the console server bought in Move 16. The second half is break-glass — a path that works when the API server, the identity provider and internal name resolution are all down at the same time — and it is drilled here, because Move 50 puts identity in front of the cluster and can otherwise lock everyone out at once.

SWAP

Where a cellular router is the out-of-band uplink, test it with the primary link down rather than assuming. Some cellular services fail closed on a captive portal that nobody notices until it matters.

DO IT FASTER

Do the credential rotation as part of the acceptance testing in Move 18, before the machines are racked. It is much faster with a keyboard in front of the machine.

WATCH OUT

A jump host that authenticates against the same identity provider as everything else is not a break-glass path. Its whole value is being independent, and that independence has to be deliberate.

LEFTOVERS

The cloud bastion, session manager and serial console stay in place until Part VII. They are how you reach the old estate while it still carries production.

—	—	—	0 min	4 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. The provider's console and shell services remain the way into the old estate.

NEEDS 08 The way in when the machine will not boot 35 Who can get in at 03:00 39 The address plan, and the three clocks you start today

THE RUNBOOK

- 01 Build the management VLAN with no default route. Nothing on it reaches the internet and nothing on the internet reaches it; the only way in is the jump host, and the only way out is the out-of-band uplink from Move 16.
- 02 Rotate every controller credential off the sticker, and record the firmware baseline for each machine at the same time. Default credentials on a management controller are the single most exploited weakness in a private estate.
- 03 Run a Redfish smoke test per vendor: inventory, power state, boot override, and one virtual media mount. Confirm `ipmitool` covers whatever Redfish does not, and write down which tool does which.
- 04 Build the jump host outside the cluster's failure domain. It does not run on the cluster, it does not authenticate against the cluster's identity provider, and it does not resolve names using the cluster's name service.
- 05 Seal the break-glass credentials somewhere physical and auditable, and record who may open them and what happens afterwards. Rotation after use is part of the procedure, not an afterthought.
- 06 Drill it. Disable the identity provider for an hour, on purpose, in daylight, and have two people reach a machine console and recover it. Move 50 is gated on this drill having been done, and the drill is the reason it is gated.

ROLLBACK

Everything here is configuration and is reversed by reverting it. There is no point of no return in this Move, which is exactly why the drill happens now rather than after Move 50 has put identity in front of the API. If the break-glass credentials are lost, the controller passwords can be restored through a physical reset at the machine, which is the reason the whole path is worth documenting. Keep the procedure in source control and the credentials nowhere near it.

41 TALOS AS THE HOST OS, AND WHERE IT CANNOT GO

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
CLUSTER	MANAGED NODE IMAGES AND THEIR PATCH AND TIME-SYNC AGENTS	MEDIUM	IMMEDIATELY

Commits to Talos as the host OS, names the two cases it cannot take, and sets the time-sync, conntrack, hugepage and interrupt defaults the managed image provided.

AWS	Bottlerocket, patched by Systems Manager	an image somebody else built, patched on a schedule you did not set.
GOOGLE	Container-Optimized OS with automatic node upgrades	already immutable and shell-restricted, which makes this the shortest agent audit of the three.
AZURE	the AKS Ubuntu or Azure Linux node image	usually a general-purpose distribution carrying a full agent estate, which makes this the longest.

BEFORE YOU START

ACCESS

- The Talos version you intend to run, pinned, with its release notes read
- An inventory of every agent currently running on your managed nodes, and what each one is for

SOFTWARE

- talosctl at the same minor version as the cluster you will build
- The machine configuration held in source control from the first line, because it is the whole operating system

PEOPLE

- Whoever owns each agent on the inventory, to say what happens if it stops existing

WHY THIS WORKS

Talos Linux is an operating system with no shell, no package manager and no remote login. It is configured entirely by a machine configuration file applied over an API, and the absence of a shell is the feature: there is nothing to log into, nothing to drift, and nothing to patch by hand. The cost is that a handful of things genuinely cannot run on it — an agent that needs a shell or a host package, and any driver that is not in the kernel or an official extension — and those exceptions are written into the design here rather than discovered during a migration. The machine configuration also carries the host tuning the managed image used to supply.

SWAP

Where a conventional distribution is required by policy, the rest of this book still works, and the cost is that node configuration becomes something you maintain rather than something you declare.

DO IT FASTER

Start from the upstream example configuration and change only what you can justify. Most estates need a dozen changes, not a hundred.

WATCH OUT

Time synchronisation is the setting people skip, and clock drift breaks certificate validation, distributed consensus and log correlation all at once, in ways that look like three unrelated faults.

LEFTOVERS

Patch management, image build pipelines and the agents that fed them can be retired once the fleet is on Talos, but not before Part VII closes the old estate.

—	—	—	0 min	1 week	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. The managed node images keep running production until Part IV moves the workloads.

NEEDS 14 The card, the driver, and the kernel you will run 40 The management VLAN, and the way in at 03:00

- 01 Audit the agents. Go down the list from your managed nodes and sort each into one of three piles: replaced by something in the cluster, genuinely unnecessary, or requires a host shell. Only the third pile is a problem.
- 02 Deal with the third pile explicitly. Anything requiring a shell or a host package runs on the hypervisor tier built in Move 70, or on a small labelled pool running a conventional distribution. Write the exception into the design with its owner and its review date.
- 03 Check every driver you need against the Talos kernel, using the answer already obtained in Move 14. There is no out-of-tree module path, so a driver is in the kernel, in an official extension, or the hardware does not go in this cluster.
- 04 Write the machine configuration with the host tuning the managed image used to provide: time synchronisation with real servers, connection-tracking table sizing, neighbour table sizing, file-watch limits, transparent hugepages, processor idle states and interrupt affinity.
- 05 Record the symptom of each setting being wrong, next to the setting. A small connection-tracking table presents as random connection failures under load; undersized file watches present as a controller that silently stops reconciling. These are debugged for weeks by people who do not know the setting exists.
- 06 Build one machine from the configuration, then rebuild it twice from nothing, first confirming no volume is bound to it, and check that the same configuration produces the same machine every time. That reproducibility is the entire argument for this operating system.

ROLLBACK

A machine configuration is reverted and reapplied, and a node is rebuilt from nothing in minutes, which is what makes this Move low-risk in practice. The point of no return is not in this Move at all; it is Move 47, where a volume binds to a specific machine. Keep every machine configuration version in source control so a previous one can be restored exactly, and keep the agent audit with it so the exceptions can be reviewed rather than rediscovered.

42 NETBOOT, NODE IDENTITY AND FAILURE-DOMAIN LABELS

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
CLUSTER	GOLDEN MACHINE IMAGES, LAUNCH TEMPLATES AND THE INSTANCE METADATA SERVICE	MEDIUM	IMMEDIATELY

Netboots nodes from a per-MAC record that also carries node identity and the rack, PDU and switch labels every later spread constraint reads.

AWS	AMIs, launch templates and the instance metadata service	an image, a template and an identity endpoint that every workload assumed was there.
GOOGLE	machine images, instance templates and the metadata server	the same three, with the metadata server also serving identity tokens.
AZURE	managed images and the instance metadata service	again the same idea, and again nothing to replace them with unless you build it.

BEFORE YOU START

ACCESS

- The as-built from Move 37, which is where the rack, feed and switch facts come from
- The provisioning range from Move 39, and address assignment you control on it

SOFTWARE

- A boot chain: address assignment with a next-server pointer, a chain loader and a place to serve machine configurations
- The Talos boot assets for your pinned version, built or fetched from the upstream image service

PEOPLE

- Whoever will add the next machine, because this is the process they will follow and it should be written for them

WHY THIS WORKS

A managed node arrived complete: an image, a template that configured it, and a metadata service that told it who it was. On your own hardware all three have to exist somewhere, and the cheapest place to put them is one record per network address. Address assignment points the machine at a boot chain, the boot chain hands it a machine configuration, and the same record carries the labels that everything later depends on: which rack, which power feed, which switch. Without those labels an inherited spread constraint is a rule that matches everything and therefore protects nothing.

SWAP

For a small fleet, machine configurations on removable media work and remove the boot server entirely. It stops being reasonable somewhere around a dozen machines.

DO IT FASTER

Keep the per-machine record to the minimum that differs — address, identity, labels — and put everything shared in one patch applied to all of them.

WATCH OUT

Firmware boot order is the quiet failure. A machine that boots from a stale local disk instead of the network comes back as a version you retired, and it will pass every health check.

LEFTOVERS

Image build pipelines, launch templates and their versions stay in the cloud accounts until Part VII. They are the way back if the programme is aborted.

—	—	—	0 min	2 weeks	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. The managed node groups keep carrying production until Part IV.

NEEDS 37 Labels, port maps and the as-built 39 The address plan, and the three clocks you start today

41 Talos as the host OS, and where it cannot go

- 01 Choose the provisioning components with the support position stated rather than implied. The Talos boot assets and the upstream image service are maintained by the project; one common provisioning server is minimally maintained; one is an early-stage project; and the vendor's own management product is licensed under terms that require a commercial licence for production use. Pick deliberately and record why.
- 02 Build the boot chain: address assignment with a next-server pointer, a chain loader, and machine configurations served per hardware address. Set the firmware boot order so that a machine which has been power-cycled comes back the same way every time.
- 03 Put node identity in the same record. No metadata service is issuing it any more, so the record that boots the machine is also the record that names it.
- 04 Carry the failure-domain labels in that record too: rack, power feed and switch, taken from the as-built rather than typed from memory. Every spread constraint and anti-affinity rule later in the book reads these labels.
- 05 Check inherited zone rules against reality. A rule that spread pods across cloud zones will match nothing here unless the labels exist, and it will fail open rather than failing loudly. This is worth checking hardest for readers leaving Azure, where zone numbers are mapped per subscription and the same number means different buildings to different subscriptions.
- 06 Time it with a stopwatch, from power button to a node reporting Ready. Then do it again on a machine that has already been in service, because that is the path Move 109 will use for every upgrade.

ROLLBACK

A machine is reprovisioned from nothing in minutes, so this Move is reversed by changing a record and rebooting. The point of no return is Move 47, after which a machine holds a bound volume and cannot be rebuilt casually. Keep the per-machine records and the boot configuration in source control so a known good provisioning state can be restored, and so that the fleet can be rebuilt from the repository rather than from a server nobody backed up.

43 THREE CONTROL-PLANE NODES, THE API VIP AND THE CLUSTER CA

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
CLUSTER	THE MANAGED KUBERNETES CONTROL PLANE AND ITS CLUSTER-CREATION TOOLING	HIGH	IMMEDIATELY

Places three etcd members across separate power feeds and switches, raises the API VIP, and takes ownership of the cluster CA the provider used to rotate.

AWS	the EKS managed control plane	a per-hour charge, an endpoint, and a certificate authority somebody else rotated on your behalf.
GOOGLE	the GKE managed control plane	the same, with control-plane upgrades applied on a release channel you chose.
AZURE	the AKS managed control plane	the same again, with the availability guarantee sold as a paid tier, so a reader on the free tier has no commitment to lose.

BEFORE YOU START

ACCESS

- Three machines provisioned by Move 42, on separate feeds, switches and racks
- The power-loss-protected devices from Move 09, confirmed present in each of them

SOFTWARE

- talosctl and kubect1, both pinned to the cluster version you are building
- etcdctl for the health checks, and a way to measure fsync latency on each control-plane device

PEOPLE

- Whoever will hold the certificate authority material, because that custody starts today

WHY THIS WORKS

The control plane is three machines and a consensus store, and the consensus store is the part that decides whether you have a cluster. etcd is an fsync benchmark wearing a database's clothing: its write path is a durability promise on every commit, so it goes on the power-loss-protected devices specified in Move 09 and it never goes on replicated network storage. The three members are placed across the power feeds, switches and racks labelled in Move 42, so that losing one of anything loses one member and not two. The API address is raised by the operating system's own virtual-address support rather than by adding another component to the critical path.

SWAP

Five members are worth it for an estate expected to pass a few hundred nodes, and are unnecessary below that. Five members write to five devices on every commit, which is not free.

DO IT FASTER

Build the cluster twice from the same configuration before you trust it. The second build takes twenty minutes and proves the first was not luck.

WATCH OUT

Placing the consensus store on replicated network storage looks reasonable and destroys performance and availability together. It goes on local, protected devices, always.

LEFTOVERS

The managed cluster keeps running and keeps charging until Part VII. It is the way back for the whole of Parts IV, V and VI.

\$73/mo	\$0/mo	100%	0 min	2 weeks	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet, although the per-hour control-plane charge on the old cluster is the first line of the cloud bill that will go to zero in Part VII.

NEEDS 09 The drive you are allowed to put etcd on 32 Two feeds, one of which you have tested

42 Netboot, node identity and failure-domain labels

43 THREE CONTROL-PLANE NODES, THE API VIP AND THE CLUSTER CA

6 STEPS · 0 MIN · HIGH RISK

THE RUNBOOK

- 01 Confirm the placement before installing anything. Three control-plane machines on three feeds, three switches and, where you have them, three racks. Three members in one cabinet is one failure domain wearing a quorum's clothing.
- 02 Measure fsync latency on each control-plane device and record it. This is the number that predicts leadership stability under load, and a device that fails here fails the cluster later rather than now.
- 03 Bring up the three members and raise the API address using the operating system's own virtual-address support. Adding a separate load balancer in front of the API introduces a component that has to be up for the cluster to be reachable.
- 04 Take ownership of the certificate authority hierarchy. Record where the key material lives, who can use it, how kubelet certificates rotate, and when every certificate in the cluster expires. The at-rest encryption key is a separate concern and belongs to Move 48.
- 05 Build the annual expiry inventory now, while the numbers are in front of you. A certificate that expires unnoticed is the classic way a healthy cluster stops working on a Sunday.
- 06 Kill a control-plane member deliberately and watch the cluster stay up, then bring it back and confirm it rejoins. Do it again with the machine that currently holds leadership. This drill costs an hour and it is the difference between believing you have three members and knowing it.

ROLLBACK

Until workloads arrive, this cluster can be deleted and rebuilt in an afternoon, which is the argument for building it long before anything needs it. The point of no return is Move 47, where the first volume binds. Keep the certificate authority material somewhere sealed and backed up, and rehearse the consensus restore in Move 52 before trusting any of it; a control plane that cannot be restored from its own backup is three machines and a hope.

44 CILIUM, WITH KUBE-PROXY OFF

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
 CLUSTER	THE CLOUD-PROVIDER CNI PLUG-IN AND KUBE-PROXY	 HIGH	IMMEDIATELY

Installs Cilium with kube-proxy replaced, which ends the address accounting behind pod density and turns service resolution into a hash-map lookup.

AWS	the Amazon VPC CNI	pods took real network addresses from the subnet, so pod density was an address-accounting problem rather than a capacity one.
GOOGLE	GKE Dataplane V2	already Cilium, so this reader is taking over a control loop rather than changing a datapath.
AZURE	Azure CNI Powered by Cilium	also already Cilium, with the same consequence and the same short journey.

BEFORE YOU START

ACCESS

- The cluster from Move 43, healthy, with the API address reachable from every machine
- The pod and service ranges from Move 39, agreed and not overlapping anything

SOFTWARE

- `helm`, and the Cilium chart pinned to an exact chart version rather than to a range
- `cilium` for the connectivity test, and `kubectl` for the verification steps

PEOPLE

- Whoever will own the network datapath, because this is the component that carries every packet

WHY THIS WORKS

Two things change here and they are worth separating. The first is the address model: on one of the three clouds, pods took addresses from the network itself, which is why pod density per machine was governed by how many addresses an instance could hold. On your own fabric, pods get their own range from Move 39 and that accounting disappears. The second is service resolution. Replacing kube-proxy turns a linear walk through a rule chain into a hash-map lookup in the kernel, which stops mattering somewhere in the low thousands of services and then matters a great deal. The pod network is left deliberately flat here, because policy is Move 60.

SWAP

Where an older kernel rules out kube-proxy replacement, run Cilium with kube-proxy in place first and remove it after the kernel is updated. The two changes do not have to happen together.

DO IT FASTER

Do this on the second cluster from Move 51 first if it exists yet, or on a three-node scratch cluster if it does not. The whole cycle is twenty minutes.

WATCH OUT

Pinning the API host to a name rather than an address reintroduces a name-resolution dependency in the path that brings up name resolution. Pin the address.

LEFTOVERS

The cloud network plug-in and its address reservations stay in the old cluster until Part VII. Nothing here touches them.

—	—	—	0 min	1 week	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. The old cluster's network plug-in stays until Part VII closes the account.

NEEDS 43 Three control-plane nodes, the API VIP and the cluster CA

- 01 Install Cilium with `helm install` at a pinned chart version, with kube-proxy replacement enabled and the API host and port pinned at the virtual address raised in Move 43. Without those two values the agent cannot reach the API once kube-proxy is gone, which is a memorable way to break a cluster.
- 02 Confirm no kube-proxy rules survive. Check that the rule chains it created are absent rather than merely unused, because a half-removed kube-proxy produces packet behaviour that is extremely hard to reason about.
- 03 Run the `cilium` connectivity test and read the whole output. It exercises paths you would not think to test and it is the cheapest verification in this Part.
- 04 Check the connection-tracking table sizing set in Move 41 against real load, and watch for table exhaustion under a load test rather than in production.
- 05 Leave the pod network flat and unpoliced for now. Introducing policy at the same time as the datapath means that when something is unreachable, there are two candidate causes instead of one. Move 60 does policy.
- 06 Record the fallback explicitly: for the non-Talos exception from Move 41 on an older kernel, the standard rule-based dataplane of another network plug-in remains available, and it is the answer where the kernel is too old rather than too new.

ROLLBACK

Cilium is uninstalled and kube-proxy restored by reverting the machine configuration and reapplying it, which takes minutes on a cluster with no workloads on it. That is the argument for doing it now. The point of no return is Move 47, where the first volume binds and machines stop being disposable. Keep the chart version and every value in source control so a known good datapath configuration can be restored exactly rather than reconstructed from memory.

45 BGP TO THE TOP-OF-RACK, AND THE MTU THAT BREAKS FIVE PER CENT

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
CLUSTER	CLOUD NETWORK ROUTING AND MANAGED TRANSIT HUBS	HIGH	IMMEDIATELY

Advertises pod prefixes to the leaf switches over BGP and settles the MTU, including the path-MTU black hole that hangs large responses rather than failing them.

AWS	VPC route tables with Transit Gateway	routes were API objects and a managed hub computed the paths between networks.
GOOGLE	VPC routes with Network Connectivity Center	the same, and the shortest journey of the three, because an alias-address pod range was already routable inside the network.
AZURE	user-defined routes with Virtual WAN	again managed routing, with the hub doing the work your leaf switches are about to do.

BEFORE YOU START

ACCESS

- Switch configuration access on both leaves, and an agreed maintenance approach for changing routing on them
- The address plan from Move 39, so the advertised prefixes are the ones everyone expects

SOFTWARE

- The routing configuration for the network plug-in, using its current resources rather than the deprecated ones
- A path-discovery test that sends full-size packets, because a small ping proves nothing about this

PEOPLE

- Whoever administers the switches, since the peering has two ends and one of them is theirs

WHY THIS WORKS

With real pod prefixes advertised to the switches, there is no overlay: a packet from one pod to another is routed by the fabric, with no encapsulation, no extra header and nothing to debug in the middle. The routing protocol is configured through the network plug-in's own resources rather than by hand on each machine, and how the two uplinks behave follows the redundancy decision taken back in Move 12. The other half of this Move is the maximum transmit unit, which is worth its own attention because getting it wrong does not produce a clean failure. It produces a black hole where small requests succeed, large responses hang, and about five per cent of traffic is mysteriously broken.

SWAP

Where the switches will not run a routing protocol at all, a routed overlay works and costs a header, some throughput and a debugging tool. It is the right answer where the fabric is not yours.

DO IT FASTER

Agree the peering parameters with the network administrator once, in a document, and apply the same pattern to every rack. Per-rack improvisation is how a fabric becomes two fabrics.

WATCH OUT

Large frames must be consistent along the whole path, including the switch uplinks and anything in between. One device at the default size in the middle of a large-frame path is exactly the black hole this Move exists to prevent.

LEFTOVERS

The cloud transit hub keeps billing per attachment and per gigabyte until Part VII. It is also the path Move 53 rides on, so it stays.

—	—	—	0 min	1 week	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. The managed transit hub carries the migration traffic and stays until Part VII.

NEEDS 12 Two switches, and the spine you do not need yet 13 The switch, and the software running on it 44 Cilium, with kube-proxy off

45 BGP TO THE TOP-OF-RACK, AND THE MTU THAT BREAKS FIVE PER CENT

6 STEPS · 0 MIN · HIGH RISK

THE RUNBOOK

- 01 | Configure the routing peers on both leaf switches and on the cluster side, using the current resources rather than the deprecated peering policy. Peer each machine with both leaves so the loss of one switch removes one path and not the machine.
- 02 | Advertise the pod prefixes and confirm they appear in the switch routing tables with the next hops you expect. A prefix advertised but not installed is the most common first fault and it is usually a filter on the switch.
- 03 | Route the office prefix through the fabric so that the link built in Move 53 has somewhere to land. Doing it now means the office link is a routing change rather than a design change later.
- 04 | Settle the maximum transmit unit across every segment, and write down which segments are large-frame and which are not. Mixed segments are normal; undocumented mixed segments are how this fails.
- 05 | Test with full-size packets across every path, including the ones through the office link and the cloud circuit. A black hole is invisible to a small ping and obvious to a large one, and it presents to application teams as intermittent slowness rather than as a network fault.
- 06 | Fail one leaf switch deliberately, with no workloads running, and confirm the routes converge and traffic continues. Then fail the other. This is the drill that proves the fabric decision from Move 12 was implemented rather than merely designed.

ROLLBACK

Routing changes are reverted by withdrawing the advertisements and restoring the previous switch configuration, which is a minutes-long operation with no workloads on the cluster. The point of no return does not exist here; the risk is entirely about doing it later, with production on the fabric, when a routing mistake is an outage rather than an experiment. Keep both switch configurations and the cluster routing resources in source control so the working state can be restored quickly.

46 CLUSTER DNS, AND THE RESOLVER THE NETWORK USED TO BE

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
 CLUSTER	THE MANAGED CLUSTER DNS ADD-ON AND THE PROVIDER'S PRIVATE RESOLVER	 MEDIUM	IMMEDIATELY

Sizes CoreDNS against the real racks and replaces the provider resolver for everything outside the cluster, with the replica count proved by a load test.

AWS	the CoreDNS add-on with the Route 53 Resolver	a resolver at the second address in every subnet, which was always just there.
GOOGLE	Cloud DNS for GKE	where cluster name resolution was moved into the managed service, so this reader has no in-cluster deployment to carry across and is standing one up from nothing.
AZURE	the CoreDNS add-on with the Azure DNS Private Resolver	the same pattern, reached at a fixed platform address nobody configured.

BEFORE YOU START

ACCESS

- The cluster from Move 44 with its network working, and the failure-domain labels from Move 42
- The current resolver configuration in the cloud: forwarders, private views and stub domains, exported

SOFTWARE

- A name-resolution load generator, so the replica count is measured rather than copied from an article
- `kubect'l` to place, scale and observe the deployment

PEOPLE

- Whoever owns the internal name zones, because split views are their design and not yours

WHY THIS WORKS

Two separate things are being replaced and it helps to name them apart. Inside the cluster, name resolution is a deployment you now size, place and upgrade yourself, against the real failure domains labelled in Move 42 rather than against cloud zones. Outside it, every cloud provided a resolver at a fixed address that every machine used without anyone configuring it, and that resolver did split views, conditional forwarding and stub domains. Both have to exist here. The sizing is the part people get wrong, because the default replica count is a starting point that was never right for anybody, and a load test settles it in an hour.

SWAP

Where an existing internal name service already exists and is well run, keep it and forward to it. Building a second authority for the same zones is how two answers to one question appear.

DO IT FASTER

Copy the exported forwarders and views wholesale and prune afterwards. Reconstructing them from what people remember takes a fortnight and misses two.

WATCH OUT

Forcing upstream lookups over a connection-oriented transport avoids one class of truncation and failure, and costs latency. Decide deliberately rather than inheriting whichever default you land on.

LEFTOVERS

The provider's resolver endpoints and private zones bill per endpoint and per query, and they stay until Part VI has moved the traffic.

\$190/mo	\$0/mo	100%	0 min	4 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. The provider's resolver endpoints stay until the workloads that use them have moved.

NEEDS 42 Netboot, node identity and failure-domain labels 44 Cilium, with kube-proxy off

46 CLUSTER DNS, AND THE RESOLVER THE NETWORK USED TO BE

6 STEPS · 0 MIN · MEDIUM RISK

THE RUNBOOK

- 01 | Deploy the cluster resolver with anti-affinity across the racks and switches labelled in Move 42. Two replicas on one machine is one replica with extra billing.
- 02 | Turn on the node-local cache. It removes most of the load, removes the connection-tracking race that produces intermittent lookup failures on parallel queries, and it is the single highest-value change in this Move.
- 03 | Deal with the search-path amplification. The default search list turns one external lookup into several, and services that resolve external names in a hot path pay for it repeatedly. Set fully qualified names where it matters and measure the difference.
- 04 | Load test the deployment and set the replica count from the result. Include a failure case: kill a replica during the test and confirm the remaining ones absorb it without the error rate moving.
- 05 | Build the site resolver that replaces the provider's fixed address. Forward internal zones to it, define the split views and stub domains you exported, and give it two instances that do not share a machine.
- 06 | Point everything outside the cluster at the new resolver and confirm resolution from a machine, from a pod, and from the office link. Three vantage points, because they take three different paths.

ROLLBACK

Name resolution is reverted by scaling the old path back up and repointing, and with no production traffic on the cluster this is a minutes-long change. The point of no return does not exist in this Move; the danger is entirely that it is done badly and then relied upon. Keep the exported forwarder and view configuration in source control so the provider's resolver behaviour can be restored and compared when something resolves differently in two places.



CLUSTER · 0 MIN · MEDIUM RISK

Nothing yet. The provider's resolver endpoints stay until the workloads that use them have moved.

47 LOCAL NVME, AND THE DEFAULT STORAGECLASS NOBODY GAVE YOU

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
CLUSTER	MANAGED NETWORK BLOCK STORAGE AND ITS DEFAULT STORAGECLASS	HIGH	NO

Puts databases on local NVMe with power-loss protection, keeps replication in the application, and marks a default StorageClass so PVCs stop hanging Pending.

AWS	EBS with the EBS CSI driver	recent EKS versions ship no default StorageClass at all, so a hanging claim is already a familiar surprise here.
GOOGLE	Persistent Disk with a default StorageClass	provided, so the missing default is a new experience for this reader.
AZURE	Managed Disks with a default StorageClass	also provided, with the same consequence.

BEFORE YOU START

ACCESS

- The devices specified in Move 09, with power-loss protection confirmed rather than assumed
- The cluster from Move 43 with machines provisioned by Move 42

SOFTWARE

- The Talos user-volume configuration and a local volume provisioner, both pinned to versions
- kubectl to mark the default StorageClass and to watch binding behaviour

PEOPLE

- Whoever will own the databases in Part V, because this is the storage they will be running on

WHY THIS WORKS

A database on network storage pays a round trip on every write and inherits a second system's failure modes. A database on local flash does not, and the durability that the network volume was providing is moved into the application, where it belongs: replicas on other machines, and a backup repository somewhere else. This is why local volumes come before Ceph rather than after — Ceph sits on top of these devices in Move 49, while the consensus store and the single-writer databases stay underneath. The cost is real and specific: a local volume pins its pod to one machine, which is what makes the drain arithmetic in Move 109 and the failed device in Move 111 interesting.

SWAP

Where a workload genuinely needs to move between machines freely and cannot replicate itself, put it on Ceph in Move 49 and accept the latency. The choice is between where the replication lives, not whether it exists.

DO IT FASTER

Name the storage classes after what they are for rather than after the technology. A developer choosing between two classes should not need to know what either is made of.

WATCH OUT

A local volume that is deleted takes its data with it and there is no undelete. The reclaim policy is a decision, and the wrong one is discovered exactly once.

LEFTOVERS

The managed block volumes and their snapshots keep billing in the cloud until Part V has moved the data and Move 38 has removed them deliberately.

\$1,480/mo	\$0/mo	100%	0 min	1 week	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. The managed volumes carry production until Part V, and their snapshots stay for thirty days after that.

NEEDS 09 The drive you are allowed to put etcd on 10 How much flash, and where it sits

43 Three control-plane nodes, the API VIP and the cluster CA

47 LOCAL NVME, AND THE DEFAULT STORAGECLASS NOBODY GAVE YOU

6 STEPS · 0 MIN · HIGH RISK

THE RUNBOOK

- 01 | Declare the user volumes in the machine configuration with the filesystem and mount options you intend to keep. Changing them later means draining the machine and recreating the volume.
- 02 | Install the local volume provisioner at a pinned version and confirm it discovers the devices on every machine, with the right sizes and the right names.
- 03 | Set binding to wait for the first consumer. Without it the scheduler binds a claim to a machine before it knows where the pod can run, and the pod then cannot be scheduled anywhere else.
- 04 | Mark a default StorageClass. Recent managed clusters on one of the three clouds ship without one, and the symptom is a claim that stays pending forever with no error that says why.
- 05 | Verify power-loss protection is present on every device that will hold a database, and record the result per serial number. This is the property the whole of Part V rests on.
- 06 | Write down the consequence in the runbook, plainly: a pod with a local volume runs on one machine and only that machine. Draining it means moving the data or accepting the outage, and Move 109 and Move 111 are where that bill arrives.

ROLLBACK

The default StorageClass is flipped back in a minute, but this Move is irreversible in the way that matters: once a claim is bound to a local device there is no way back except a data migration with a write stop, which is not a configuration change. The point of no return is the first bound claim. Everything before that is free. Keep the volume configuration in source control, and make sure a backup exists and has been restored before anything real binds, because a local device has no snapshot behind it.

48 ENCRYPTION AT REST, AND WHO HOLDS THE KEY

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
CLUSTER	DEFAULT ENCRYPTION AT REST ON MANAGED VOLUMES	HIGH	NO

Encrypts the state and ephemeral partitions and the Ceph OSDs, and settles who holds the key when the only copy of it lives on the cluster it opens.

AWS	EBS default encryption with a managed key	a setting you had to switch on per account and per region, and the outlier of the three.
GOOGLE	Persistent Disk encryption	every disk encrypted with a provider-managed key whether or not anyone asked, so this reader has a default to replace rather than a setting to port.
AZURE	Storage Service Encryption with disk encryption sets	also on by default, with customer-managed keys as the option on top.

BEFORE YOU START

ACCESS

- Machines not yet carrying anything, because this cannot be applied to a populated device
- The physical security position from Move 35, which is what this Move is answering

SOFTWARE

- The chosen key provider configured and tested, including its recovery path
- The cluster encryption configuration, with a named key and a documented rotation procedure

PEOPLE

- A named custodian for each key, with a deputy, recorded somewhere that outlives both of them

WHY THIS WORKS

Move 35 already established the fact that makes this necessary: facility staff can open your cabinet. Encryption at rest is what makes that fact survivable, and it has to be turned on before there is anything on the devices, because there is no in-place conversion. The Talos state and ephemeral partitions are encrypted with a key provider chosen deliberately — a node identifier for convenience, a hardware module to bind a device to the machine it was installed in, or an external key server — and the Ceph devices are created encrypted in Move 49, because encrypting them afterwards is a rebuild rather than a setting. The cluster's own secret store gets a key with a named custodian.

SWAP

For an estate where physical access is genuinely controlled and the threat model is honest about it, a node-identifier key is a reasonable choice that keeps recovery simple. Write down the reasoning rather than the conclusion.

DO IT FASTER

Do this as part of the first install rather than as a second pass. Reinstalling twelve machines because encryption was forgotten costs a day that nobody planned.

WATCH OUT

Key rotation for the cluster secret store requires rewriting every secret. It is a documented procedure, it is not automatic, and it should be rehearsed once on the second cluster before it is needed.

LEFTOVERS

The provider's key management keys keep billing per key per month, and they are destroyed in Move 38 rather than here.

—	—	—	0 min	1 week	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. The provider's key management stays until Move 38 destroys the key material deliberately.

NEEDS 09 The drive you are allowed to put etcd on 35 Who can get in at 03:00 43 Three control-plane nodes, the API VIP and the cluster CA 47 Local NVMe, and the default StorageClass nobody gave you

- 01 Choose the key provider deliberately, and write down what each choice means when a machine fails. A node identifier is convenient and offers little protection against a stolen device. A hardware module binds the device to the machine, which is stronger and means a device moved to another chassis will not open. An external key server is strongest and adds a dependency.
- 02 If you choose an external key server, understand the ordering problem: a state partition sealed by a network service cannot take its network configuration from the configuration stored on that partition. The network has to come up before the partition opens, and that has to be designed rather than discovered.
- 03 Enable encryption on the state and ephemeral partitions at install time, on every machine, before anything is bound to them.
- 04 Set the cluster's own at-rest encryption for secrets with a named key, and record the custodian, the deputy and the rotation procedure. This key protects the secrets that every workload depends on.
- 05 Solve the recursion before it bites: the only copy of a key must not live solely on the cluster that key opens. Put a sealed copy somewhere physical and record who may open it, exactly as Move 40 did for break-glass.
- 06 Test the recovery path. Take a machine, move its device to another chassis, and confirm the behaviour matches what you chose in step one. This is the test that tells you what your decision actually means.

ROLLBACK

There is no in-place decryption, so this Move is irreversible on any machine it has been applied to: reversing it means reinstalling the machine. The point of no return is the first node installed with encryption enabled. That is acceptable precisely because it happens before anything is bound. If a key is lost, the data is not recoverable and the only path is a rebuild and a restore from the backups Move 54 and Move 74 create, which is the reason the sealed copy exists.

49 CEPH UNDER ROOK: BLOCK, SHARED FILESYSTEM AND THE OBJECT STORE

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
CLUSTER	MANAGED NETWORK BLOCK STORAGE, MANAGED SHARED FILESYSTEMS AND MANAGED OBJECT STORAGE	HIGH	NO

Installs Rook and Ceph for block and shared filesystem, and stands up the object gateway that Harbor, Loki and Velero will need before Part V begins.

AWS	EBS, EFS and S3	three products, three billing models, and three sets of assumptions baked into your applications.
GOOGLE	Persistent Disk, Filestore and Cloud Storage	the same three, with the shared filesystem sold in fixed capacity tiers.
AZURE	Managed Disks, Azure Files and Blob Storage	the same again, and the only one of the three offering a first-class Windows file-sharing protocol.

BEFORE YOU START

ACCESS

- The device count and capacity plan from Move 10, reconciled against what was actually delivered
- Machines installed with encryption from Move 48, since the devices are created encrypted

SOFTWARE

- `helm`, with Rook at a pinned chart version and the Ceph version it supports rather than the newest one
- `ceph` for the health checks, and `kubectl` for everything the operator does not do for you

PEOPLE

- A named owner for Ceph. This is the one condition this Move refuses to waive

WHY THIS WORKS

Three managed products are replaced by one system with three interfaces. Ceph provides block volumes, a shared filesystem and an object gateway from the same pool of devices, and Rook runs it as a set of Kubernetes resources rather than as a separate estate to administer. The failure domains come from the rack labels in Move 42, so the cluster spreads copies across racks rather than across machines that share a switch. The object gateway is stood up here rather than in Part V, because the registry, the log store, the backup tool and the database repository all need an object endpoint from Move 58 onwards.

SWAP

Where the estate needs a Windows file-sharing protocol as a first-class citizen, Ceph does not provide it directly and a separate file-sharing tier or an application change is required. Only one of the three clouds offered it, and that reader has real work here.

DO IT FASTER

Build it on the second cluster from Move 51 first and break it deliberately several times. Every hour spent breaking it there is a day saved later.

WATCH OUT

The protection scheme on a data pool cannot be changed in place once the pool holds data. Choose replication or erasure coding deliberately, with the recovery cost understood, because changing your mind means a new pool and a copy.

LEFTOVERS

The managed volumes, filesystems and buckets keep billing until Part V has moved the data and Move 38 has removed them against a signed data-owner approval.

\$3,940/mo	\$0/mo	100%	0 min	3 weeks	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. The managed block, file and object services carry production until Part V moves the data off them.

NEEDS 10 How much flash, and where it sits 42 Netboot, node identity and failure-domain labels

47 Local NVMe, and the default StorageClass nobody gave you 48 Encryption at rest, and who holds the key

49 CEPH UNDER ROOK: BLOCK, SHARED FILESYSTEM AND THE OBJECT STORE

7 STEPS · 0 MIN · HIGH RISK

THE RUNBOOK

- 01 Name the owner first. Ceph is a distributed storage system with its own failure modes, its own vocabulary and its own upgrade cadence, and an estate that runs it without somebody who understands it will have an outage nobody can diagnose.
- 02 Install Rook with `helm install` at a pinned chart version and let it create the storage devices encrypted, per Move 48. Encrypting them afterwards is a rebuild of every device, one at a time.
- 03 Build the placement rules on the rack labels from Move 42 rather than on machine names. A three-copy pool that puts all three copies in one cabinet is a pool that loses data when the cabinet loses power.
- 04 Set the replication scheme and the fullness thresholds against the drive count from Move 10. Reconcile them explicitly: a pool that will sit above the backfill threshold when one machine is lost is a pool that stops accepting writes during the recovery it exists for.
- 05 Create the block storage class and the shared filesystem, and confirm a claim binds, mounts, survives a machine reboot and can be restored from a snapshot before anything real uses either.
- 06 Stand up the object gateway with its own pool and its own placement, and create the first bucket. Move 58 onwards depends on this endpoint existing, so it is built now even though nothing uses it yet.
- 07 Pull a device out of a running machine and watch the cluster recover. Time it. That number is what Move 111 will be working against at three in the morning.

ROLLBACK

Uninstalling Rook before anything binds is clean. After that it is not: this Move is irreversible once a volume or a bucket holds data, because moving off Ceph is a data migration rather than a configuration change. The point of no return is the first bound volume or the first object written, not the installation. Prove the restore path before that point — take a snapshot, delete a claim, restore it — because a storage system nobody has restored from is a storage system nobody should be trusting.

50 HUMANS ON THE API SERVER: OIDC, AFTER BREAK-GLASS

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
CLUSTER	CLOUD IAM AUTHENTICATION TO THE MANAGED CLUSTER	HIGH	IMMEDIATELY

Points the API server at the identity provider you already run, maps groups to RBAC, and deletes the static admin kubeconfig from every laptop.

AWS	IAM with the authentication ConfigMap or access entries	a mapping from cloud principals to cluster identities that disappears with the account.
GOOGLE	Cloud IAM with the authentication plug-in	the same, and it also disappears when the project does.
AZURE	Microsoft Entra ID with Azure role-based access	the shortest path of the three, because that directory is a general-purpose identity provider you keep.

BEFORE YOU START

ACCESS

- The identity provider's configuration, and permission to register a new client against it
- The drilled break-glass path from Move 40, because this Move can otherwise lock everyone out at once

SOFTWARE

- The API server configuration for the identity provider, applied through the machine configuration
- kubectl with an authentication plug-in that fetches short-lived credentials rather than storing long ones

PEOPLE

- Whoever owns groups in the directory, because the mapping is theirs to maintain, not yours

WHY THIS WORKS

Authentication to your old cluster was a property of the cloud account, and it goes away when the account does. What replaces it is the identity provider the organisation already runs and intends to keep, with groups mapped to cluster roles and credentials that expire. The important part of this Move is its ordering: it runs after the break-glass path in Move 40 has been drilled, and it blocks go-live. Without that ordering, Parts III to VII get operated from a copied administrator credential on somebody's laptop, which then becomes permanent, and the audit in Move 119 finds it.

SWAP

Where no identity provider exists yet, this Move waits. Building one specifically for the cluster is a much larger project than it appears and it will be the wrong one.

DO IT FASTER

Test the whole flow on the second cluster from Move 51. The failure modes are confusing and they are much cheaper to meet there.

WATCH OUT

Group membership changes in the directory do not propagate until the credential is renewed. A removed engineer keeps access until their token expires, which is an argument for short lifetimes rather than long ones.

LEFTOVERS

The cloud identity mappings stay on the old cluster until Part VII. Do not remove them while that cluster is still carrying production.

—	—	—	0 min	1 week	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet, though the static administrator credentials on laptops go today, and that is a real reduction in risk.

NEEDS 40 The management VLAN, and the way in at 03:00 43 Three control-plane nodes, the API VIP and the cluster CA

THE RUNBOOK

- 01 | Confirm the break-glass drill from Move 40 has actually been done, by two people, within the last month. This is a gate rather than a suggestion.
- 02 | Register the cluster with the identity provider and configure the API server through the machine configuration. Get the claim names right first time by testing against a scratch cluster, because a wrong claim name looks exactly like a permissions problem.
- 03 | Map directory groups to cluster roles. Keep the mapping small and boring: one group with administrative rights, one per team with namespace rights, and nothing bespoke per person.
- 04 | Switch everyone to short-lived credentials through the authentication plug-in, and confirm that a credential actually expires by waiting for it to.
- 05 | Delete the static administrator credential from every laptop, and rotate it. This is the whole point of the Move and it is the step most often deferred until it is forgotten.
- 06 | Send the access log somewhere that survives the cluster. A record of who did what that lives only on the thing they did it to is not a record.

ROLLBACK

The API server configuration is reverted through the machine configuration and reapplied, and the static credential can be regenerated, so this is fully reversible within minutes. The point of no return does not exist here, which is exactly why the break-glass gate matters: the risk is not permanence but simultaneity, since a mistake locks out everybody at once. Keep the configuration and group mapping in source control so a working state can be restored without guesswork at the moment nobody can log in.

51 THE SECOND CLUSTER, AND WHAT IT SHARES

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
CLUSTER	THE SECOND MANAGED CLUSTER AND ITS PER-BRANCH EPHEMERAL VARIANTS	LOW	IMMEDIATELY

Builds the non-production cluster the drills and rehearsals in this book assume, and states plainly what it shares with production and what it must not.

AWS	a second EKS cluster	a per-hour control-plane charge and a node group, duplicated for staging.
GOOGLE	a GKE Autopilot cluster	the one that genuinely differs, because it removed node management from staging entirely and a self-hosted second cluster puts that cost straight back.
AZURE	a second AKS cluster	the same as the first, usually on the free control-plane tier.

BEFORE YOU START

ACCESS

- The machines allocated for non-production in Move 03, provisioned by Move 42
- The identity provider configuration from Move 50, ready to serve a second cluster

SOFTWARE

- The same pinned versions as production, because a rehearsal on different versions rehearses the wrong thing
- Separate storage pool configuration, so a mistake in staging cannot consume production capacity

PEOPLE

- Whoever will run the drills, since this cluster exists mainly for them

WHY THIS WORKS

Most of the drills in this book need somewhere to fail. The restore drill in Move 52, the upgrade rehearsal in Move 108, the verification gate in Move 72 and every per-branch clone all need a cluster that can be broken without a conversation. That cluster was sized back in Move 03 and is built here from the same boot assets as production, so that what it proves is transferable. What it shares is the interesting question, and the answers are deliberate: its own certificate authority and its own storage pool, but the same identity provider and, from Move 58, the same registry of record.

SWAP

Where hardware is tight, a smaller cluster of three machines still supports every drill in this book. What it cannot do is rehearse capacity behaviour, and that limitation should be written down.

DO IT FASTER

Build it before production, not after. Every mistake you are going to make is cheaper here, and the production build then takes a day instead of a fortnight.

WATCH OUT

Staging clusters accumulate exceptions — a version pinned back, a policy disabled for a demonstration — until they no longer resemble production. Reconcile the two on a schedule.

LEFTOVERS

The second managed cluster keeps its own control-plane charge and its own node group until Part VII. Autopilot readers should note that this line disappears entirely rather than shrinking.

\$890/mo	\$0/mo	100%	0 min	1 week	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet, though the per-branch ephemeral clusters can move here as soon as the pipeline in Move 64 reaches it.

NEEDS 03 The estate nobody sizes 42 Netboot, node identity and failure-domain labels

43 Three control-plane nodes, the API VIP and the cluster CA 47 Local NVMe, and the default StorageClass nobody gave you

50 Humans on the API server: OIDC, after break-glass

THE RUNBOOK

- 01 Build it from the same boot assets and the same pinned versions as production. A staging cluster that drifts is a staging cluster that stops predicting anything.
- 02 Give it its own certificate authority. Sharing one means a credential minted for staging is valid against production, which is the kind of mistake that is found during an incident review.
- 03 Give it its own storage pool. Sharing devices means a runaway test can fill the pool that production writes to.
- 04 Share the identity provider from Move 50 deliberately. One place to grant and revoke access is worth more than the isolation of a second directory, and the roles can still differ per cluster.
- 05 Share the registry of record from Move 58 once it exists. A second registry is how staging starts running images that production has never seen, which defeats the purpose of having staging.
- 06 State plainly, in writing and in the runbook, that this is not a disaster-recovery site. It has no copy of production data, it is not sized for production load, and describing it as a fallback is how a bad night becomes a worse one.

ROLLBACK

This cluster is deleted and rebuilt from the same configuration in an afternoon, which is what makes it useful. There is no point of no return here at all; it holds no production data by construction and it is meant to be destroyed regularly. Keep its configuration in source control next to production's, and diff them periodically, so the rehearsal cluster can be restored to a state that genuinely matches what it is rehearsing for.

52 THE ETCD RESTORE DRILL

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
CLUSTER	MANAGED CONTROL-PLANE BACKUP AND RESTORE	LOW	IMMEDIATELY

Destroys a quorum on purpose on the second cluster and restores it from a snapshot, so the recovery time is a measured number rather than an assumption.

AWS	EKS control-plane recovery	the consensus store was never exposed, and recovery meant recreating a cluster.
GOOGLE	GKE control-plane recovery with Backup for GKE	the only one of the three selling a backup product for cluster workloads, and it does not restore the consensus store either.
AZURE	AKS control-plane recovery	the same as the first, with the same answer: recreate the cluster.

BEFORE YOU START

ACCESS

- The second cluster from Move 51, with nothing on it anyone cares about
- The object store from Move 49, holding a snapshot that was taken rather than assumed

SOFTWARE

- `etcdctl` and `talosctl`, both pinned, and the snapshot verification step actually run
- A stopwatch, and somewhere to publish the number afterwards

PEOPLE

- Two people, one driving and one writing down what is missing, because the gaps are the output

WHY THIS WORKS

None of the three managed offerings ever let you touch the consensus store, so there is nothing to migrate here. What there is instead is a number you do not have: how long it takes you to get a cluster back after losing quorum. That number is only obtainable by doing it, and the second cluster from Move 51 exists precisely so that it can be done without a conversation. The finding is the wall-clock time, published, along with the discovery of which pieces you did not have to hand — usually the certificate authority material, and usually at the worst moment.

SWAP

Where the second cluster does not exist yet, a three-machine scratch cluster is enough for this drill. What it cannot rehearse is the scale of the restore, which matters less than the procedure.

DO IT FASTER

Script the restore rather than documenting it. A script that has been run is worth more than a document that has been read.

WATCH OUT

Restoring a snapshot creates a new cluster identity. Members from the old cluster will refuse to join, and that refusal looks like a networking fault to anyone who does not know.

LEFTOVERS

The managed control-plane backup product on one of the three clouds keeps billing until Part VII, and it never covered this anyway.

—	—	—	0 min	4 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. The managed backup product, where you bought one, stays until Part VII.

NEEDS 43 Three control-plane nodes, the API VIP and the cluster CA 49 Ceph under Rook: block, shared filesystem and the object store

51 The second cluster, and what it shares

THE RUNBOOK

- 01 | Take a snapshot and verify it. A snapshot that has not been verified is a file, and the verification step takes seconds.
- 02 | Confirm where the off-site copies live: the snapshot, the machine configurations and the certificate authority material. All three are needed and only the first is usually thought about. They go to the object store from Move 49, whose own durability is Move 54's problem.
- 03 | Destroy the quorum on the second cluster deliberately, having first confirmed nothing on it matters. Stop two of the three members and leave the cluster genuinely unable to elect.
- 04 | Restore through a single-member bootstrap with a new cluster identity, then bring the other two members back and confirm they join. Start the stopwatch at step three and do not stop it until the cluster is serving requests.
- 05 | Repoint the kubelets and confirm the workloads come back. A restored control plane with machines that cannot reach it is a half-finished restore.
- 06 | Publish the wall-clock number and the list of things you had to look up. Both go into the runbook, and the second list is the more valuable of the two.
- 07 | While you are here, run a defragmentation and check the backend size against its quota. A consensus store that hits its quota goes read-only, which presents as a cluster that will not accept any change at all.

ROLLBACK

There is nothing to reverse: the drill happens on a cluster built to be destroyed. The point of no return does not exist here, and that is the entire value of the exercise — it converts an unknown into a measured number at no risk. Repeat it quarterly, and repeat it after every upgrade in Move 108, because a restore procedure that worked on the previous version is not evidence about this one.

53 THE LINKS YOU BUILD FIRST: TO THE CLOUD, TO THE OFFICE

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
CLUSTER	PHYSICAL DATA-TRANSFER APPLIANCES AND MANAGED BULK-COPY SERVICES	MEDIUM	IMMEDIATELY

Builds the private circuit and the office link every later replication stream and mirrored comparison rides on, and prices the migration egress honestly.

AWS	Direct Connect or Site-to-Site VPN	a virtual interface over a cross-connect, or a tunnel, with data transfer priced per gigabyte.
GOOGLE	Dedicated or Partner Interconnect, or Cloud VPN	the same shapes, with the least conditional exit terms of the three.
AZURE	ExpressRoute or VPN Gateway	the same again, with the strictest exit terms, requiring the subscription to be terminated.

BEFORE YOU START

ACCESS

- The cross-connect ordered in Move 26, with the provider's authorisation paperwork still inside its validity window
- The address plan from Move 39 and the routed fabric from Move 45

SOFTWARE

- The provider's circuit configuration, and a tunnel configuration as the interim path while the circuit is provisioned
- A throughput test that runs for hours rather than seconds, because the interesting failures are sustained ones

PEOPLE

- Whoever negotiated the cloud contract, because the exit terms need to be obtained in writing before the first byte moves

WHY THIS WORKS

Every stream in Parts V and VI rides on this link: the database replication, the object copy, the mirrored comparison in Move 72, the shadow edge in Move 102. Building it early and sizing it properly is what turns those Moves into waiting rather than into crises. Sizing is the part people get wrong, because the number that matters is not steady-state throughput but catch-up throughput — three to five times steady state, so that a stream which falls behind can recover inside a working day rather than inside a fortnight. The lead time is what belongs on the schedule, not the work.

SWAP

Where the volume is small enough, the tunnel is the whole answer and no circuit is needed. Measure before ordering: a tunnel over good transit carries more than people expect.

DO IT FASTER

Run the free-egress request in parallel with the circuit order. Both take weeks and neither blocks the other.

WATCH OUT

A physical transfer appliance is still the right answer for very large one-off copies, and it is slower to arrange than people assume. Decide early, because it has its own lead time.

LEFTOVERS

The circuit and its port charges continue until Part VII. Do not cancel it when the last workload moves; Move 121's abort path needs it.

—	—	—	0 min	1 week	6 to 12 weeks
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. This Move adds a circuit charge that runs alongside the cloud bill until Part VII.

NEEDS 26 Carrier neutrality and the cross-connect price list 39 The address plan, and the three clocks you start today

45 BGP to the top-of-rack, and the MTU that breaks five per cent

53 THE LINKS YOU BUILD FIRST: TO THE CLOUD, TO THE OFFICE

7 STEPS • 0 MIN • MEDIUM RISK

THE RUNBOOK

- 01 | Order the circuit early. Six to twelve weeks is normal, this Move sits inside that wait rather than beside it, and every Part V date depends on it.
- 02 | Build the tunnel first as the interim path. It is available in an afternoon, it carries the low-volume traffic, and it means Parts IV and V are not blocked on a physical order.
- 03 | Address and route both links against the plan from Move 39 and the fabric from Move 45, so that the office prefix and the cloud prefix are reachable from the cluster without exceptions.
- 04 | Size for catch-up rather than for average. Three to five times steady state is the working rule, and the test that matters is a sustained transfer, not a burst.
- 05 | Price the migration egress honestly, including the per-gigabyte charge a managed address-translation gateway adds on top of the transfer itself. That second charge is routinely missed and it is frequently the larger of the two.
- 06 | Get the exit terms in writing before the first byte moves. All three providers now waive egress charges for a customer who is leaving, on materially different terms: one is close to unconditional, one reviews each request and does not require the account to close, and one requires the subscription to be terminated and the move completed inside sixty days. Regional law is also moving on this, and none of it helps unless you asked first.
- 07 | Test the whole path end to end with a sustained transfer and record the achieved throughput. That number, not the circuit's badge speed, is what Part V plans against.

ROLLBACK

Both links are removed by withdrawing routes and cancelling the circuit, and neither leaves anything behind. The point of no return does not exist in this Move, though the circuit's minimum term does have a cost attached. Keep the routing configuration and the negotiated exit terms in source control, so the commercial position can be restored and quoted back when a provider's support desk disagrees with what was agreed.

54 THE OBJECT STORE YOU NOW OWN, AND HOW IT LOSES DATA

CUTOVER

0 MIN



LAYER CLUSTER	LEAVING MANAGED OBJECT-STORAGE DURABILITY, VERSIONING AND OBJECT LOCK	RISK HIGH	REVERSIBLE NO
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States the replication factor, the off-site copy and the restore drill for the store that now holds backup repositories, log chunks and registry layers.

AWS	S3 with cross-region replication and Object Lock	durability quoted in elevens, replication as a bucket setting, and immutability as a mode.
GOOGLE	Cloud Storage with dual-region buckets and bucket lock	the only one selling a single bucket name spanning two regions, so this reader loses a geographic property rather than a feature.
AZURE	Blob Storage with geo-redundant storage	the same durability story, with immutability policies and a different redundancy vocabulary.

BEFORE YOU START

ACCESS

- The object gateway from Move 49, and the link from Move 53 for the off-site copy
- A second credential domain for the off-site target, so one compromised credential cannot reach both copies

SOFTWARE

- The pool protection scheme decided before any data is written, because it cannot be changed in place afterwards
- rc1one or mc for the off-site copy, pinned, with a schedule and an alert on failure

PEOPLE

- Whoever owns the recovery objectives from Move 24, because this Move is what makes those objectives real

WHY THIS WORKS

The object store built in Move 49 has quietly become the durability floor under everything: the database backup repositories from Move 74, the log chunks from Move 63, the cluster backups from Move 115 and the registry layers from Move 58. None of those is a backup until this Move is done, because a backup that lives in the same failure domain as the thing it protects is a copy rather than a backup. So this Move fixes the protection scheme, the failure domain it spreads across, versioning, immutability, an off-site copy in a second credential domain, and a restore drill with a published number.

SWAP

Where an off-site copy in your own second location does not exist yet, a small amount of cloud object storage is a perfectly good target and costs very little for backups. Keeping it is not a failure of the programme.

DO IT FASTER

Do the restore drill on the smallest real repository you have rather than the largest. The procedure is what you are testing, and it transfers.

WATCH OUT

A cheap off-site copy in the same credential domain as the primary is not protection against the failure that matters most, which is a credential doing something destructive everywhere at once.

LEFTOVERS

Cross-region replication and versioning in the cloud keep billing throughout the migration. They are removed in Move 38, deliberately and last.

\$2,310/mo	\$180/mo	92%	0 min	2 weeks	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. Cross-region replication and immutability policies stay until Move 38 removes them against a signed approval.

NEEDS 49 Ceph under Rook: block, shared filesystem and the object store 53 The links you build first: to the cloud, to the office

54 THE OBJECT STORE YOU NOW OWN, AND HOW IT LOSES DATA

6 STEPS · 0 MIN · HIGH RISK

THE RUNBOOK

- 01 Fix the data pool's protection scheme now: replication for latency-sensitive pools, erasure coding for large cold ones. This cannot be changed in place once the pool holds data, so it is decided before the first object is written.
- 02 Set the failure domain the pool spreads across to the rack labels from Move 42, and verify by asking the cluster where the copies of a test object actually are rather than assuming.
- 03 Turn on versioning where the workload benefits from it, and set object immutability where a retention requirement exists. Note that on some releases immutability can only be enabled when a bucket is created, so it is a creation-time decision rather than a later one.
- 04 Build the off-site copy over the link from Move 53, into a second credential domain. Schedule it, alert on failure, and check the alert works by breaking the copy deliberately once.
- 05 Run a restore drill: take a backup repository, delete it from the primary store, and restore it from the off-site copy. Time it and publish the number.
- 06 Write the failure modes down in plain language, because everyone downstream now depends on this store: what happens when a machine is lost, when a rack is lost, when the pool passes its fullness threshold, and when the off-site copy has been silently failing for a week.

ROLLBACK

The protection scheme is irreversible once the pool holds data, so there is no way back from that decision except a new pool and a copy. The point of no return is the first object written into a pool created with the wrong scheme. Everything else here — versioning, the off-site copy, the schedule — is reversible. Keep the pool configuration and the drill results in source control so the tested state can be restored and compared when the store behaves differently a year from now.

IV PLATFORM

What developers touch. Get this wrong and the migration succeeds while everybody who has to use it quietly hates you.

16	15	6	30
MOVES	ZERO DOWNTIME	HIGH RISK	MINUTES, TOTAL

IF YOU ONLY DO THREE

56

SECRETS: EXTERNAL SECRETS AS THE DATED BRIDGE

External Secrets Operator syncs the cloud secret stores into the cluster under a written end date, so workloads can …

63

LOGS AND TRACES, BROUGHT HOME

Grafana Alloy or an OpenTelemetry Collector replaces the retired log agent and the provider's trace daemon, which …

55

WORKLOAD IDENTITY WITHOUT THE CLOUD'S

Publishes the cluster's own service-account issuer and federates it with each cloud's IAM, so pods on your metal hold …

WORKLOAD IDENTITY WITHOUT THE CLOUD'S · SECRETS: EXTERNAL SECRETS AS THE DATED BRIDGE · THE ROOT OF TRUST AFTER MANAGED KEY MANAGEMENT · THE MIRROR, THEN HARBOR AS THE REGISTRY OF RECORD · COSIGN, TRIVY AND AN ADMISSION POLICY · EAST-WEST: SECURITY GROUPS INTO NETWORKPOLICY · EAST-WEST ENCRYPTION, AND AN IDENTITY PER WORKLOAD · METRICS OFF THE MANAGED MONITOR, AND THE CARDINALITY BUDGET · LOGS AND TRACES, BROUGHT HOME · BUILDS ONTO YOUR OWN METAL · ARGO CD, AND THE TWO INFRASTRUCTURE-AS-CODE ESTATES · INFRASTRUCTURE CODE THAT ONLY COMPILES FOR ONE CLOUD · THE FIRST STATELESS SERVICE, END TO END · THE CONTAINER SERVICE THAT WAS NOT KUBERNETES · THE FIXED FLEET: PRIORITY, PREEMPTION, NUMA AND THE GPUS · THE WORKLOADS THAT MUST STAY VIRTUAL MACHINES

■ 1 LOW RISK

■ 9 MEDIUM RISK

■ 6 HIGH RISK

55 WORKLOAD IDENTITY WITHOUT THE CLOUD'S

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
PLATFORM	CLOUD IAM WORKLOAD IDENTITY: CLUSTER OIDC FEDERATION, POD-IDENTITY AGENTS AND INSTANCE METADATA CREDENTIALS	MEDIUM	IMMEDIATELY

Publishes the cluster's own service-account issuer and federates it with each cloud's IAM, so pods on your metal hold the roles they held before, with no static key.

AWS	IAM roles for service accounts	an estate already on the newer pod-identity agent has to convert back to this shape first, because that agent exists only inside the managed cluster.
GOOGLE	workload identity federation	the one that genuinely differs, because the provider accepts an uploaded key set, so your issuer need not be reachable from the internet at all.
AZURE	federated credentials on a managed identity	capped at twenty per identity, which forces a naming scheme before the first one is created.

BEFORE YOU START

ACCESS

- Permission to create an identity provider in each cloud account that still holds resources
- A stable, durable place to publish the discovery document and key set, outside the cluster

SOFTWARE

- The API server issuer setting, applied through the machine configuration from Move 41
- An inventory of every role each workload currently assumes, which is the bulk of the work

PEOPLE

- Whoever owns cloud identity policy, because federating a new issuer is a decision they should make knowingly

WHY THIS WORKS

Your workloads currently get cloud credentials because the managed cluster's identity issuer is trusted by the cloud's identity service. Your own cluster can be trusted the same way: publish the discovery document and the signing keys at a stable address, tell the API server to issue tokens naming that address, and register it as a federated identity provider in each cloud that still holds resources you need. The result is that a pod running on your metal holds exactly the role it held before, with no static key anywhere. This has to work during the migration, because for months the workloads are here and the data is still there.

SWAP

Where a workload needs one cloud credential and nothing else, a short-lived static credential with a rotation job is defensible for a few weeks. Write the end date down when you create it.

DO IT FASTER

Do the inventory with the cloud's own access analyser rather than by reading policy documents. It finds roles that are granted and never used, which are the ones to delete rather than federate.

WATCH OUT

The provider that caps federated credentials per identity will let you create the twenty-first and fail, at the worst moment. Decide the naming scheme and the identity granularity before creating the first.

LEFTOVERS

The old cluster's identity provider stays registered until Part VII. Removing it early breaks the way back.

—	—	—	0 min	5 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. The cloud roles stay exactly as they are, and this Move is what lets your own machines keep using them.

NEEDS 43 Three control-plane nodes, the API VIP and the cluster CA 53 The links you build first: to the cloud, to the office

THE RUNBOOK

- 01 Build the per-role inventory first. Every workload, every role it assumes, and what that role can do. This is four days of the five and it is the part that finds the roles nobody remembers granting.
- 02 Publish the discovery document and the signing key set at a stable address that does not depend on the cluster being up. An object bucket with a fixed name is the usual answer; the address is effectively permanent, so choose it carefully.
- 03 Set the issuer on the API server through the machine configuration and confirm a freshly issued token names it. Applying this changes every token in the cluster, so do it before workloads arrive.
- 04 Register the issuer in each cloud. The audience value differs per provider and getting it wrong produces an authentication error that says nothing useful, so take it from the provider's own documentation rather than from a copied example.
- 05 Convert any estate on a newer agent-based identity mechanism back to the federated shape first. That agent only exists inside the managed cluster and there is nothing to carry across.
- 06 Write down the signing-key rotation order — publish the new key, wait for caches, start signing with it, retire the old one — and the revocation path for a role that has to be withdrawn in a hurry. Both are needed before anything depends on this.

ROLLBACK

The issuer setting is reverted through the machine configuration and the federated providers are deleted in each cloud, both in minutes. There is no point of no return here, and the risk is confined to the moment the issuer changes, which invalidates existing tokens. Keep the issuer address, the key set and the federation configuration in source control so a working identity path can be restored quickly, and keep the previous signing key until every consumer has stopped using it.



56 SECRETS: EXTERNAL SECRETS AS THE DATED BRIDGE

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
PLATFORM	—	LOW	IMMEDIATELY

External Secrets Operator syncs the cloud secret stores into the cluster under a written end date, so workloads can land before the root of trust has moved.

AWS	Secrets Manager and Parameter Store	two services, one billed per secret and per API call, the other largely free at the standard tier.
GOOGLE	Secret Manager	one service, versioned, billed per secret version and per access.
AZURE	Key Vault	one service holding secrets, certificates and keys as three first-class object types, which makes the eventual migration in Move 57 carry three shapes rather than one.

BEFORE YOU START

ACCESS

- The federated identity from Move 55, working, so the operator authenticates without a static credential
- Read access to the secret stores in every account that still holds one

SOFTWARE

- The External Secrets Operator at a pinned chart version, installed with `helm`
- An inventory of secrets by store, with a note of which are actually read at runtime

PEOPLE

- Whoever will own the eventual migration in Move 57, because the end date is their commitment

WHY THIS WORKS

Workloads have to land on your cluster months before the root of trust moves, and they need their secrets on the first day. This Move builds a bridge rather than a destination: an operator that reads the cloud secret stores, authenticated by the federated identity from Move 55 rather than by a key in a file, and writes the values into the cluster as secrets. Nothing is chosen as the store of record here; the cloud store stays authoritative until Move 57 replaces it. The one thing that must happen today is writing the end date down, because bridges that have no end date become architecture.

SWAP

For a small estate, a manual copy of a dozen secrets into the cluster is faster and honest, provided it is written down and dated the same way.

DO IT FASTER

Start with one namespace and one store. The shapes are all the same once the first one works, and the first one takes most of the day.

WATCH OUT

Syncing a secret into the cluster puts a plaintext copy in the consensus store unless at-rest encryption from Move 48 is on. It is on, if Move 48 was done, and it is worth confirming rather than assuming.

LEFTOVERS

The cloud secret stores keep billing per secret and per read for as long as this bridge exists, which is another reason for the end date.

\$210/mo	\$210/mo	0%	0 min	3 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing. This Move deliberately keeps paying the cloud secret bill so that workloads can move before the root of trust does.

NEEDS 55 Workload identity without the cloud's

THE RUNBOOK

- 01 Install the operator with `helm install` at a pinned chart version, authenticated through the federated identity from Move 55. If it needs a static key to start, something in Move 55 is not finished.
- 02 Define one store resource per cloud secret service and confirm the operator can read from each. Three services across three clouds is the common case and each has its own authentication shape.
- 03 Set the refresh interval as a cost decision rather than a preference. Two of the three providers bill per read, and a five-minute refresh across a few thousand secrets is a bill that shows up a month later with no obvious cause.
- 04 Sync only the secrets that are actually read at runtime. An inventory that lists four hundred secrets usually contains fifty that matter and three hundred and fifty that were created once by a pipeline.
- 05 Confirm rotation propagates: change a value at the source, and time how long until a pod sees it. That number is what Move 57 will have to match or beat.
- 06 Write the end date into the exit plan on the day the operator is installed, with a named owner. This is a bridge, and the date is what stops it becoming the design.

ROLLBACK

Removing the operator stops the synchronisation and leaves the existing cluster secrets in place, so nothing breaks at the moment of removal. There is no point of no return in this Move; the cloud store remains authoritative throughout, which is the entire design. Keep the store and secret definitions in source control so the sync configuration can be restored, and confirm the cloud copies are intact before deleting anything on either side.



57 THE ROOT OF TRUST AFTER MANAGED KEY MANAGEMENT

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
 PLATFORM	MANAGED KEY MANAGEMENT AND CLOUD HSM SERVICES, CLOUD-BACKED AUTO-UNSEAL, ENVELOPE ENCRYPTION IN CI, AND THE CLOUD SECRET STORE AS THE RECORD	 HIGH	30 DAYS

The managed key could be neither exfiltrated nor lost, so replacing it means naming the humans, the HSM or the TPMs that hold it now, with a rehearsed recovery.

AWS	KMS and CloudHSM	a key that could not be exported and could not be lost, scheduled for deletion with a waiting period you could cancel.
GOOGLE	Cloud KMS and Cloud HSM	the same properties and the same cancellable destruction window.
AZURE	Key Vault and Managed HSM	the same again, except that with purge protection enabled the old key cannot be destroyed on your timetable at all, which is a constraint Move 122 inherits.

BEFORE YOU START

ACCESS

- The physical security position from Move 35 and the encryption decisions from Move 48
- The secret inventory from Move 56, including the certificates and keys that one provider holds as separate object types

SOFTWARE

- The chosen secret store, licensed deliberately: an openly licensed fork rather than the source-available original where that matters to you
- A recovery procedure written before the system is depended upon, not after

PEOPLE

- Custodians with named deputies, and a rehearsal in the calendar before anything depends on this

WHY THIS WORKS

The managed key had two properties that are hard to reproduce together: nobody could take a copy of it, and nobody could lose it. Replacing it means choosing which of those you are willing to work for, and there are three honest custody models. Shares of the key held by named people, which is strong and depends on those people being reachable. A hardware module in the cabinet, which is strong and is a single physical thing in a building. Or a key sealed to the control-plane machines themselves, which is convenient and ties recovery to hardware that can fail. None of the three is best; each has a failure mode, and this Move is about naming yours.

SWAP

For fewer than about twenty secrets, file-level encryption committed to source control with a key held in two places is genuinely better than a secret manager nobody has rehearsed recovering.

DO IT FASTER

Rehearse the recovery with the actual custodians, in a room, once. It takes an afternoon and it converts a document into a capability.

WATCH OUT

A secret store that seals itself on restart and can only be opened by people who are asleep is an outage waiting for a power cut. Whatever model you choose, the recovery has to be possible at three in the morning.

LEFTOVERS

The cloud keys stay for thirty days after the migration, because that is the window in which you find the workload nobody knew was using one.

\$640/mo	\$40/mo	94%	0 min	2 weeks	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The cloud key management keys and the secret stores behind them — thirty days after the migration, and after a billing cycle has shown nothing still reading them.

NEEDS 48 Encryption at rest, and who holds the key 56 Secrets: External Secrets as the dated bridge

- 01 Choose the custody model and write down its failure mode in one sentence. Shares held by people fail when the people are unreachable; a module in the cabinet fails when the building does; a key sealed to machines fails when the machines are replaced.
- 02 Pick the software with the licence read rather than assumed. The widely deployed secret manager moved to a source-available licence; an openly licensed fork exists and is the book's default. For an estate small enough that a share-based ceremony is theatre, file-level encryption with a simple key tool is a legitimate and much simpler answer.
- 03 Stand it up, seal it, and rehearse the recovery with the custodians before migrating anything. Two days of ceremony rehearsal is the recommended budget and it is the step that gets cut and then regretted.
- 04 Migrate the secrets from the cloud stores, remembering that one provider holds certificates and keys as first-class objects alongside secrets, so three shapes travel rather than one.
- 05 End the Move 56 bridge on its written date. Turn off the synchronisation, confirm every workload still starts, and remove the federated read access afterwards rather than before.
- 06 Plan the destruction of the old keys with the provider difference in mind, only after a full billing cycle has confirmed nothing still reads them. Two providers schedule destruction with a cancellable waiting period; one, with purge protection on, will not let you destroy the key on your schedule at all, and Move 122 has to allow for that.

ROLLBACK

For thirty days the old keys still exist and the bridge can be restarted, so a failed migration is recoverable by repointing workloads at the cloud store. The point of no return is the destruction of the old key material, which is deliberately deferred to Move 38 and Move 122. Rehearse the recovery of the new store before that date, because after it the only way back from a lost key is a restore of data encrypted under a key that no longer exists, which is to say no way back at all.

58 THE MIRROR, THEN HARBOR AS THE REGISTRY OF RECORD

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
PLATFORM	MANAGED CONTAINER REGISTRIES: PRIVATE AND PUBLIC REPOSITORIES, GEO-REPLICATION, LIFECYCLE RULES AND PULL-THROUGH CACHE RULES	HIGH	30 DAYS

A proxy cache is the cheap half and Harbor becomes the registry of record, the component that stops anything scheduling or recovering when it is down.

AWS	ECR with pull-through cache rules	an account-level rule with an authorisation token that expires every twelve hours.
GOOGLE	Artifact Registry with remote and virtual repositories	the closest managed equivalent to what you are about to build yourself.
AZURE	Container Registry	with geo-replication gated behind the premium tier, which is the line that disappears here.

BEFORE YOU START

ACCESS

- The object store from Move 49 for the layers, and a local volume from Move 47 for the database
- The identity provider from Move 50, so people authenticate the same way here as everywhere

SOFTWARE

- A proxy cache at a pinned version, and skopeo for copying pinned digests rather than tags
- Harbor at a pinned chart version, installed with helm, with its database and cache configured deliberately

PEOPLE

- Whoever owns the build pipeline, because Move 64 will be pushing here within a fortnight

WHY THIS WORKS

This Move has a cheap half and an expensive half. The cheap half is a proxy cache in front of the public registries, which removes the rate limits and the outages you do not control, and takes an afternoon. The expensive half is making your own registry the record, because from that moment nothing schedules and nothing recovers while it is down. That is worth doing anyway, and it is worth understanding before you do it. The registry then needs a database and a cache of its own, which is this book's one deliberate exception to keeping databases out of Part IV.

SWAP

For a small estate, a simple registry with object storage behind it is enough, and it removes the database and cache entirely. You lose projects, quotas and scanning integration, which Move 59 then has to solve differently.

DO IT FASTER

Do the proxy cache today and defer the registry of record by a fortnight. The first half delivers most of the operational benefit on its own.

WATCH OUT

Garbage collection in a registry is a real operation with real risk, and it needs the images that are actually in use to be identifiable. Do not enable aggressive retention before Move 65 makes deployments declarative.

LEFTOVERS

The cloud registry keeps billing for storage and data transfer for thirty days after the switch, and that overlap is deliberate.

\$780/mo	\$60/mo	92%	0 min	2 weeks	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The managed registry's geo-replication today, and the registry itself after thirty days of the mirror running clean.

NEEDS 47 Local NVMe, and the default StorageClass nobody gave you 49 Ceph under Rook: block, shared filesystem and the object store

54 The object store you now own, and how it loses data 55 Workload identity without the cloud's

- 01 Stand up the proxy cache first, in front of the public registries you actually pull from. Point the cluster at it and confirm pulls succeed. This half is cheap and immediately valuable.
- 02 Understand what a cache is not. A cache holds what has been pulled; it does not hold what you have never pulled, and the difference only becomes visible during a recovery when nothing is warm. Copy the digests you depend on deliberately with skopeo, pinned by digest rather than by tag.
- 03 Install Harbor with `helm install` at a pinned chart version, with its database on a local volume from Move 47 and its layers in the object store from Move 49. This is the deliberate exception to keeping stateful services out of this Part, and it is made because the registry cannot depend on anything that depends on the registry.
- 04 Set up projects, robot accounts, quotas and garbage collection before anyone pushes. Retention rules written after a year of images are written against a full disk.
- 05 Move the pipeline to push here, and mirror from the cloud registry rather than cutting over. Both registries hold the images for thirty days, which is the reversibility window this Move claims.
- 06 Power the rack off and back on, deliberately, and watch what happens. The registry needs the cluster to run and the cluster needs the registry to start, and the only way to know how that resolves in your estate is to do it once on purpose.

ROLLBACK

For thirty days both registries hold the images, so reverting is repointing the pipeline and the cluster back at the cloud registry. The point of no return is the day the cloud registry's images are deleted in Move 38, after which a restore means rebuilding images from source. Keep the mirror running for the whole window, and verify that a representative image can be pulled from the cloud registry on the last day before it goes.

59 COSIGN, TRIVY AND AN ADMISSION POLICY

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
 PLATFORM	MANAGED IMAGE SCANNING, MANAGED CODE SIGNING, TAG IMMUTABILITY, SCAN-ON-PUSH GATES AND CLUSTER-LEVEL BINARY AUTHORISATION	 HIGH	IMMEDIATELY

Leaving the managed registry deletes signing, scanning and the admission gate at once, so all three are rebuilt with cosign, Trivy and a break-glass bypass.

AWS	ECR enhanced scanning with Signer	scan-on-push you can replace with an open scanner more or less directly.
GOOGLE	Artifact Analysis with Binary Authorization	an attester model that does not map one to one onto a policy rule, so this reader re-derives the policy rather than translating it.
AZURE	Microsoft Defender for Containers	again scan-on-push at the registry, with the findings landing in a security portal rather than in the registry.

BEFORE YOU START

ACCESS

- The registry of record from Move 58, holding images the pipeline already pushes
- The root of trust from Move 57, so signatures chain to something rather than to a key in a file

SOFTWARE

- cosign for signatures and attestations, pinned, with its key material held by Move 57
- A scanner in the pipeline and in the registry, and an admission controller at a pinned chart version

PEOPLE

- Whoever is on call, because they are the ones who will meet the bypass under pressure

WHY THIS WORKS

Moving off the managed registry removes three things at once, and only one of them is obvious. Scanning goes, signing goes, and the admission gate that refused unsigned or vulnerable images goes with them. All three are rebuilt here: signatures and attestations chained to the root of trust from Move 57, scanning in the pipeline and again in the registry so that a base image which rots after the build is still caught, and enforcement at admission. The interesting part is the failure mode. A policy that blocks the fix at three in the morning is a policy that gets deleted at five past, so the bypass is designed, written down and alerted on rather than improvised.

SWAP

Where an admission controller is too much for the team, registry-side scanning with a hard failure in the pipeline covers most of the ground. You lose the guarantee that only signed images run, and you should say so.

DO IT FASTER

Start with one policy — images must come from your registry — and enforce only that. It is the highest-value rule and it is uncontroversial.

WATCH OUT

Blocking on vulnerability severity alone produces a policy that fails builds for findings nobody can fix. Gate on signature and provenance, and handle vulnerabilities through the patch commitment in Move 117.

LEFTOVERS

The managed scanning subscriptions bill per image or per node and can be turned off once your own scanning is producing findings people act on.

\$340/mo	\$0/mo	100%	0 min	1 week	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The managed scanning subscription, once your own findings are being triaged by a person under Move 117.

NEEDS 57 The root of trust after managed key management 58 The mirror, then Harbor as the registry of record

THE RUNBOOK

- 01 Sign images in the pipeline with `cosign`, chained to the root of trust from Move 57, and attach a software bill of materials as an attestation at the same time. Signing without an inventory answers half the question Move 117 will ask.
- 02 Scan in the pipeline and again in the registry. The first catches what you built; the second catches a base image that developed a vulnerability three weeks after the build, which is the majority of real findings.
- 03 Install the admission controller at a pinned chart version and run every policy in audit mode first. Read a week of what it would have blocked before blocking anything, because the first run always finds something surprising and important.
- 04 Turn on enforcement one namespace at a time, starting with the least critical. A cluster-wide switch is how an admission policy takes down a production deploy on its first evening.
- 05 Design the break-glass bypass explicitly: who can use it, how it is invoked, how long it lasts, and what alert fires when it is used. An undesigned bypass is a deleted policy.
- 06 Re-derive rather than translate if you are leaving a cluster-level authorisation model. The attestor structure there does not map cleanly onto a policy rule, and a mechanical translation produces a policy that passes tests and permits things it should not.
- 07 Prove it works from both directions: push a signed, clean image and watch it admit; push an unsigned one and watch it refuse, with a message a developer can act on.

ROLLBACK

Every part of this is reverted by removing the policies, and the images remain signed and scanned regardless, so nothing is lost by backing out. There is no point of no return here. The real risk is the opposite failure — a policy enforced too widely too early, which is why the audit week and the namespace-by-namespace rollout exist. Keep the policies in source control so a known good set can be restored the moment a bypass is used at three in the morning.

60 EAST-WEST: SECURITY GROUPS INTO NETWORKPOLICY

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
PLATFORM	INSTANCE-LEVEL FIREWALLS: SECURITY GROUPS AND THEIR GROUP-TO-GROUP REFERENCES, VPC FIREWALL RULES AND NETWORK ACLS	HIGH	IMMEDIATELY

The pod network is flat until you say otherwise, so the existing firewall rules become the policy set, rolled out per namespace in audit mode before default-deny.

AWS	security groups with group-to-group references	a construct with no equivalent here, so every rule is re-expressed as a label selector rather than translated.
GOOGLE	VPC firewall rules with network tags	closer, because tags behave more like labels, and the dataplane was already Cilium-derived.
AZURE	network security groups with application security groups	the same shape as the first, with the same re-expression required.

BEFORE YOU START

ACCESS

- The exported firewall rules from every cloud account, including the ones attached to things nobody claims
- The cluster with its network observability enabled, so flows can be read rather than guessed

SOFTWARE

- Flow logging turned on, with somewhere to query it that is not the terminal you are standing in
- The policy resources chosen deliberately: the portable upstream shape, or the richer vendor one with its portability cost stated

PEOPLE

- One owner per namespace, because a default-deny rollout needs somebody who can say what that namespace legitimately talks to

WHY THIS WORKS

Move 44 left the pod network deliberately flat, which means that today anything can talk to anything. That is fine while nothing real is running and unacceptable the moment Part V arrives, so this is the Move that closes it. The policy set is derived from the firewall rules you already have rather than invented from a diagram, because the existing rules encode a decade of decisions nobody remembers making. The rollout is per namespace and in audit mode first, with flow logs answering what actually talks to what, and only then default-deny. North-south policy is Part VI.

SWAP

Where a namespace is genuinely too tangled to police, leave it flat and say so in writing, with a date. An honest exception beats a policy full of allow-everything rules.

DO IT FASTER

Start with the namespaces that have one or two dependencies. They take an hour each and they build the pattern the difficult ones will follow.

WATCH OUT

Policies are additive and there is no explicit deny, so a permissive rule somewhere else in the namespace silently defeats a restrictive one. Review the whole namespace rather than the rule you just wrote.

LEFTOVERS

The cloud security groups keep protecting the old estate and stay until Part VII. Do not tidy them while anything still runs there.

—	—	—	0 min	3 weeks	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. The cloud firewall rules protect the old estate for as long as it carries production.

NEEDS 44 Cilium, with kube-proxy off 46 Cluster DNS, and the resolver the network used to be

THE RUNBOOK

- 01 | Export the existing firewall rules and turn them into an inventory rather than a policy. Most of the three weeks this Move takes is here, and none of it is writing configuration.
- 02 | Re-express group-to-group references as label selectors. Two of the three clouds have no equivalent construct, so this is a rewrite, and a mechanical translation produces rules that are both too broad and subtly wrong.
- 03 | Turn on flow logging and watch a full week, including a deploy, a batch window and a weekend. What actually talks to what is never quite what the diagram says.
- 04 | Roll out per namespace in audit mode. Read the would-be denials and fix the policy rather than the workload, until a week passes with nothing surprising in the log.
- 05 | Handle name resolution and outbound traffic explicitly and first, because they are what break and they break quietly. A default-deny policy that forgets name resolution produces an application that fails in ways nobody attributes to the network.
- 06 | Switch to default-deny one namespace at a time, least critical first, with the previous policy one command away.
- 07 | Decide the resource shape deliberately. The upstream policy resource is portable and limited; the vendor's own is more expressive and ties you to this network plug-in. Write down which you chose and what it costs.

ROLLBACK

Policies are deleted and the network returns to flat within seconds, which makes this one of the safest High-risk Moves in the book. There is no point of no return. The danger is entirely in the direction of doing it late, after Part V has put data behind an unpoliced network. Keep every policy in source control so a working set can be restored instantly, and keep the audit-mode logs so a denial can be explained rather than argued about.



61 EAST-WEST ENCRYPTION, AND AN IDENTITY PER WORKLOAD

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
PLATFORM	MANAGED SERVICE MESH ADD-ONS AND THE IN-MESH MTLS THEY PROVIDED BY DEFAULT	MEDIUM	IMMEDIATELY

Gives every workload a cryptographic name and encrypts pod-to-pod traffic, which is the destination Move 55 named and what an auditor asks for before data lands.

AWS	App Mesh	retired, so this population is migrating regardless of anything in this book.
GOOGLE	Anthos Service Mesh and Cloud Service Mesh	Istio underneath, which makes this a control-plane swap rather than a rewrite.
AZURE	the Open Service Mesh add-on	also retired, which leaves this reader in the same position as the first.

BEFORE YOU START

ACCESS

- The namespace policy from Move 60 in place, because encryption without policy protects the wire and not the access
- A load test that can measure throughput before and after, honestly

SOFTWARE

- Node-to-node encryption available in the network plug-in already installed in Move 44
- If per-workload identity is in scope, a mesh chosen with its licence read rather than assumed

PEOPLE

- Whoever will answer the auditor's question in Move 119, so that what is built matches what will be asked

WHY THIS WORKS

There are two different things people mean by encrypting east-west traffic, and conflating them is why this Move takes either a week or three. The floor is node-to-node encryption, which is close to one flag, protects traffic on the wire between machines, and costs measurable throughput that this Move measures rather than guesses. The thing an auditor usually means is different: a cryptographic identity per workload, so that a service can prove which service it is, and mutual authentication between named services. That is the destination Move 55 pointed at, and it is worth reaching before Part V puts real data on the network.

SWAP

For most estates, node-to-node encryption plus the policy from Move 60 is a defensible position, and per-workload identity can wait for a specific requirement. Say which one you have.

DO IT FASTER

Do the encryption now and the identity later. They are independent and the first one is a genuine improvement on its own.

WATCH OUT

A mesh adds a component to every request path. Whatever is chosen must be something the team can debug at three in the morning, and that is a stronger constraint than any feature comparison.

LEFTOVERS

Two of the three managed mesh add-ons are retired or retiring, so there is often nothing left to turn off — the bill went away before you did.

\$290/mo	\$0/mo	100%	0 min	3 weeks	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The managed mesh add-on, where it still exists. On two of the three clouds it has already been retired for you.

NEEDS 55 Workload identity without the cloud's 60 East-west: security groups into NetworkPolicy

61 EAST-WEST ENCRYPTION, AND AN IDENTITY PER WORKLOAD

6 STEPS · 0 MIN · MEDIUM RISK

THE RUNBOOK

- 01 | Decide which of the two things you are doing, and write it down. Node-to-node encryption is a week. Per-workload mutual authentication is three, and it changes how services are named.
- 02 | Turn on node-to-node encryption in the network plug-in and measure throughput before and after with a real load test. The cost is measurable and it is better to know the number than to discover it during an incident.
- 03 | Confirm the encryption is actually applied by capturing traffic between two machines and looking at it. A flag that is set and not working is the failure mode here, and it is silent.
- 04 | If per-workload identity is in scope, choose the mesh with the licence read. One popular option now requires a paid subscription for stable builds above a modest organisation size, leaving only edge releases freely available; the book's default is the mesh that runs without sidecars in its lighter mode.
- 05 | Roll the mesh out per namespace and give each service a name that means something. The identity is the point; the encryption is a consequence.
- 06 | Prove it from the auditor's direction: show that service A can prove it is service A, that service B refuses an unauthenticated caller, and that both facts are logged somewhere durable.

ROLLBACK

Node-to-node encryption is a flag and is reverted in minutes with a brief interruption to cross-machine traffic. A mesh is removed per namespace and the services keep working, because the mesh sits beside them rather than inside them. There is no point of no return. Keep the throughput measurements, taken before and after, so that a later performance question can be answered with data rather than suspicion, and so the previous configuration can be restored knowingly.



PLATFORM · 0 MIN · MEDIUM RISK

The managed mesh add-on, where it still exists. On two of the three clouds it has already been retired for you.

62 METRICS OFF THE MANAGED MONITOR, AND THE CARDINALITY BUDGET

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
PLATFORM	MANAGED METRIC STORES, CONTAINER INSIGHT AGENTS, CUSTOM METRICS AND THE PROVIDER'S METRIC QUERY LANGUAGE	MEDIUM	14 DAYS

Prometheus scrapes into storage you sized yourself, where cardinality stops being a smooth monthly charge and becomes an OOMKill during an incident.

AWS	CloudWatch with Container Insights	a native metric service with its own expression language, so every dashboard and alert is rewritten by hand.
GOOGLE	Cloud Monitoring	the same, with its own query language and its own custom metric pricing.
AZURE	Azure Monitor with Log Analytics	the same again, and the one where metrics and logs share a query language and a bill.

BEFORE YOU START

ACCESS

- The object store from Move 49 and the durability terms from Move 54, because the metric store wants object storage rather than local disk
- An export of every dashboard and alert expression currently in use, which is the real inventory

SOFTWARE

- A scraper and a long-term store, both at pinned versions, with the store's licence read
- A series budget per machine, written down before the first scrape rather than discovered afterwards

PEOPLE

- Whoever owns each dashboard, because the rewriting is the work and it is theirs to verify

WHY THIS WORKS

On a managed metric service, a careless label was a slightly larger invoice. On storage you run, the same careless label is a memory limit exceeded during the incident you added the label to debug. That change — from a smooth monthly charge to a hard operational limit — is the single most important thing to understand about this Move. Everything else follows from it: a series budget per machine, drop rules applied at scrape time rather than at query time, and a fortnight of running both systems side by side so that the panels are proved before anybody trusts them.

SWAP

For a small estate, a single scraper with local storage and a modest retention is genuinely enough, and the object-store tier can wait until retention becomes a requirement.

DO IT FASTER

Rewrite the twenty dashboards people actually open, and delete the rest. Most estates have a long tail of panels nobody has looked at in a year.

WATCH OUT

The first bad label will arrive from an application team who added a request identifier to a metric. Drop rules at scrape time, plus an alert on series growth, are what stop that being an outage.

LEFTOVERS

Custom metrics, insight agents and the log ingestion that came with them keep billing until the managed service is turned off, which is fourteen days after the dual-run proves clean.

\$2,180/mo	\$140/mo	94%	0 min	3 weeks	2 weeks
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The managed metric service, its insight agents and its custom metrics — fourteen days after the dual-run shows the same incident on both.

NEEDS 49 Ceph under Rook: block, shared filesystem and the object store 54 The object store you now own, and how it loses data

62 METRICS OFF THE MANAGED MONITOR, AND THE CARDINALITY BUDGET

6 STEPS · 0 MIN · MEDIUM RISK

THE RUNBOOK

- 01 Stand up the scraper and the long-term store, with the store's data in the object store from Move 49 rather than on local disk. One common option is permissively licensed; another is under a copyleft licence with a replication factor of three by default, which is a capacity decision as much as a legal one.
- 02 Set a series budget per machine and enforce it with drop rules at scrape time. A rule that drops a label at query time saves nothing; the memory was consumed when the series was created.
- 03 Scrape everything and dual-run against the managed service for a fortnight. This is the reversibility window this Move claims, and it is what turns a dashboard from a translation into a verified one.
- 04 Rewrite the dashboards and alert expressions. All three clouds now sell a managed service speaking the same query language, and a reader on one of those adds a target and is nearly finished; the larger population is on a native expression language and every panel is rewritten by hand. Budget it as engineering rather than as configuration.
- 05 Compare the two systems on the same incident. Take a real event from the fortnight and confirm both would have shown it, at the same time, with the same shape.
- 06 Alert on the metric store itself — its ingestion rate, its series count against the budget, and its memory headroom — before turning the managed service off. A monitoring system with no monitoring is the classic second failure during a first one.

ROLLBACK

For fourteen days both systems are running and reverting is repointing the dashboards, which is why the dual-run is not optional. The point of no return is turning the managed service off, after which historical data before that date exists only where you exported it. Export what you need to keep first, and confirm it can be restored and queried, because a metric history that cannot be read is not a history.

63 LOGS AND TRACES, BROUGHT HOME

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
PLATFORM	MANAGED LOG INGESTION AND QUERY, MANAGED DISTRIBUTED TRACING, THE PROVIDER'S TRACE AGENT AND ITS PROPAGATION HEADER	MEDIUM	30 DAYS

Grafana Alloy or an OpenTelemetry Collector replaces the retired log agent and the provider's trace daemon, which makes trace propagation a rolling code change.

AWS	CloudWatch Logs with X-Ray	a proprietary trace context header, so services need a translation shim through the mixed weeks.
GOOGLE	Cloud Logging with Cloud Trace	the same shape, with its own propagation header and its own retention pricing.
AZURE	Log Analytics with Application Insights	again a proprietary context, and again a shim until every service is speaking the standard one.

BEFORE YOU START

ACCESS

- The object store from Move 49 for the log chunks, under the durability terms Move 54 set
- An export of every saved query in the old console, because each one is rewritten by hand

SOFTWARE

- A collector at a pinned version, and a log store and a trace store, each with its licence read
- Query limits set before any developer is given the address, not after the first unbounded query

PEOPLE

- The owner of each service that emits traces, because the header change lands in their code

WHY THIS WORKS

Logs and traces move together because they are collected by the same agent and correlated by the same identifier. The collector question is settled rather than left open, since the long-standing log agent reached end of life, and what replaces it also carries traces. The part that surprises people is the propagation header: two of the three clouds propagate their own trace context, so during the mixed weeks a service that has moved and a service that has not need a shim between the standard header and the proprietary one. That makes this a rolling code change across the estate rather than an infrastructure swap.

SWAP

Where traces are not actually used by anyone, say so and do the logs only. Half this Move delivered well beats all of it delivered badly.

DO IT FASTER

Ship logs first and traces second. They have different consumers and the log half is what an incident needs.

WATCH OUT

Log volume is almost always larger than the estimate, because the estimate came from the managed service's billing after its own sampling and filtering. Size for two to three times what the invoice implies.

LEFTOVERS

Managed log retention keeps billing after ingestion stops, sometimes for months. Set the retention down explicitly rather than assuming it lapses.

\$3,620/mo	\$220/mo	94%	0 min	4 weeks	30 days
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Managed log ingestion and tracing after thirty days of dual-running, and the retention behind them once the export is proved.

NEEDS 49 Ceph under Rook: block, shared filesystem and the object store 54 The object store you now own, and how it loses data

62 Metrics off the managed monitor, and the cardinality budget

THE RUNBOOK

- 01 Choose the collector deliberately and pin it. The old log agent is end of life, so the choice is between the vendor's own collector and the vendor-neutral one, and the second is the safer default for an estate that may change vendors again.
- 02 Stand up the log store with its chunks in the object store from Move 49, and state its licence in the same breath as its features, because one of the common choices is under a copyleft licence that some organisations will not accept.
- 03 Put high-cardinality identifiers — request identifiers, user identifiers, trace identifiers — into structured metadata rather than into labels. A label is an index and an index of unique values is a memory leak with a schema.
- 04 Set query limits before anybody is given the address. An unbounded query over a month of logs will take the store down, and it will be run within a week of the address being shared.
- 05 Stand up the trace store and roll out the standard propagation header service by service. Where the estate still speaks a proprietary header, run a translation shim at the boundary until the rollout completes.
- 06 Rewrite the saved queries. This is the larger half of the four weeks, and there is no tool that does it. Rewrite the ones people use, delete the rest, and have the owners confirm each one returns what they expect.
- 07 Dual-run for thirty days, then turn the managed services off. That window is the reversibility this Move claims.

ROLLBACK

For thirty days both pipelines run, so reverting is repointing the collector. The point of no return is turning the managed ingestion off, after which older logs live only in the export you took. Export what compliance requires first and confirm it can be restored and searched, because an archive that has never been read back is an assumption rather than a record.

64 BUILDS ONTO YOUR OWN METAL

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
PLATFORM	MANAGED CI BUILD SERVICES, THEIR HOSTED RUNNER POOLS AND SELF-MANAGED RUNNER AUTOSCALING GROUPS	MEDIUM	IMMEDIATELY

The control plane stays with the forge and only execution moves to ephemeral runners on your nodes, so one label change routes every build back if it goes wrong.

AWS	CodeBuild	billed per build minute, with private pools already able to run inside your network, so this is close to a pool swap.
GOOGLE	Cloud Build	a free allowance of concurrent builds that a self-hosted pool has to beat on latency rather than on price.
AZURE	Azure Pipelines	build stages and deploy stages tangled in one pipeline object, so here and only here this Move and Move 65 are cut together.

BEFORE YOU START

ACCESS

- The forge's runner registration flow, and permission to register a self-hosted pool
- The object store from Move 49 for the shared build cache

SOFTWARE

- An ephemeral runner controller at a pinned chart version, so a job never runs on a dirty machine
- A build cache backend on the object store, and a warm layer cache measured before and after

PEOPLE

- Whoever owns the pipeline, because build latency is the thing developers will notice within a day

WHY THIS WORKS

Only execution moves. The forge that holds the code, the pull requests and the pipeline definitions stays exactly where it is, and what changes is where the work runs: ephemeral runners on your own machines, registered to a pool, torn down after each job. That split is what makes this Move safe, because the way back is changing one label on a job definition. It is also where the endurance arithmetic from Move 09 gets tested for real, since a build fleet writes more, harder, than anything else in the estate, and a cheap device in a build node will wear out inside a year.

SWAP

Where builds are infrequent, keeping them hosted is entirely defensible. This Move earns its place on estates where build minutes are a real line on the invoice.

DO IT FASTER

Move the slowest, most cache-dependent pipeline first. It shows the biggest improvement and it exercises the cache design hardest.

WATCH OUT

Build nodes tainted for builds will still attract system daemons and log shippers. Account for that overhead or the machines will be slower than the arithmetic promised.

LEFTOVERS

The managed build service keeps billing per minute until the last job moves, and the hosted parallel job allowance disappears rather than being refunded.

\$1,940/mo	\$180/mo	91%	0 min	2 weeks	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The managed build service and its hosted pools, once a week of parallel running shows the same failure rate and better wall-clock times.

NEEDS 56 Secrets: External Secrets as the dated bridge 58 The mirror, then Harbor as the registry of record

59 cosign, Trivy and an admission policy

THE RUNBOOK

- 01 | Install the ephemeral runner controller at a pinned version and register one pool. Ephemeral is the requirement rather than the preference: a runner that survives a job carries state into the next one, and that is how a build passes on a machine and fails everywhere else.
- 02 | Measure build wall-clock time before and after, with the cache cold and warm. The managed service had a warm cache you did not have to think about, and reproducing it is most of the value of this Move.
- 03 | Put the layer cache and the package cache on the object store from Move 49, and confirm cache hits rather than assuming them. A cache that is being written and never read is a cost with no benefit.
- 04 | Do the endurance arithmetic against the device class from Move 09. Builds are the hardest write workload you will run, and a build fleet on the wrong drive class is a replacement programme in month nine.
- 05 | Rebuild the isolation the managed builder was quietly selling: rootless building, dedicated and tainted nodes so a build cannot land beside production, and a separate cache namespace per tenant.
- 06 | Route a fraction of jobs to the new pool by label, run both for a week, and compare failure rates as well as times. A pool that is faster and flakier is not faster.
- 07 | Leave the hosted macOS runners bought. There is no answer to them on your own hardware that is worth the trouble.

ROLLBACK

Changing one label on a job definition sends every build back to the hosted pool, which is why this Move is reversible immediately and why it is safe to try on a Tuesday. There is no point of no return. Keep the pipeline definitions and the runner configuration in source control so a working pool can be restored, and keep the cache buckets until a full week has passed with no fallback, because a cold cache on a rollback is a very slow morning.

65 ARGO CD, AND THE TWO INFRASTRUCTURE-AS-CODE ESTATES

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
PLATFORM	MANAGED DEPLOYMENT SERVICES, PIPELINE DEPLOY STAGES, SOURCE-REPOSITORY CONNECTIONS AND CLOUD-HOSTED REMOTE STATE BACKENDS	HIGH	30 DAYS

Argo CD reconciles the cluster from Git while the shrinking cloud estate stays described in its own modules, with one owner and one review path across both.

AWS	CodeDeploy and CodePipeline with an S3 state backend	a deploy service, a pipeline, and a state file with a lock table.
GOOGLE	Cloud Deploy with a Cloud Storage state backend	the same shape, with delivery pipelines as first-class objects.
AZURE	Azure Pipelines release stages	the same again, with a Blob Storage state backend, and the one where build and deploy are the same pipeline object.

BEFORE YOU START

ACCESS

- A repository the cluster can read, with a deploy credential that does not depend on the cluster
- The root of trust from Move 57, because the controller needs secrets before the platform exists

SOFTWARE

- Argo CD at a pinned chart version, with an application-of-applications structure prepared rather than improvised
- The state backend moved deliberately, with the licence of the tool decided before the state file moves

PEOPLE

- One owner for both estates, because two owners with two review paths is how they diverge

WHY THIS WORKS

There are two estates here and they never merge. The cluster is reconciled from a repository by a controller that runs inside it, and everything about the cluster is described that way. The shrinking cloud footprint — the accounts, the identities, the buckets that stay — is described in its own modules with its own state. What matters is that both have one owner and one review path, so that a change to either is proposed, reviewed and recorded the same way. The other thing that matters is the bootstrap: the controller that deploys the platform cannot itself depend on the platform it deploys.

SWAP

For a small estate, a pipeline that applies manifests directly is simpler and honest. What it loses is drift detection, which is most of the value once more than one person is deploying.

DO IT FASTER

Onboard the platform components first and the applications second. The platform is where the ordering problems live and they are better found without a product team waiting.

WATCH OUT

A reconciler with cluster-wide write access is the most powerful credential in the estate. Its access is scoped, its repository is protected, and Move 118 audits what it did.

LEFTOVERS

The managed deploy service, its pipelines and its source-repository connections keep billing until they are removed, and the connections in particular hold credentials that Move 122 must revoke.

\$410/mo	\$0/mo	100%	0 min	3 weeks	30 days
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The managed deploy service and its pipelines, thirty days after the reconciler has been the only thing deploying.

NEEDS 56 Secrets: External Secrets as the dated bridge 58 The mirror, then Harbor as the registry of record 64 Builds onto your own metal

- 01 Choose the reconciler and say why. This book uses Argo CD, for its interface and its explicit synchronisation model, and the alternative is a perfectly good project that suits teams who prefer a smaller footprint. Do not run both.
- 02 Structure the repository as an application of applications, so that the whole platform can be recreated by pointing a fresh controller at one path. Test that by doing it on the second cluster from Move 51.
- 03 Solve the bootstrap explicitly. The controller needs its own credentials before the secret platform it deploys is running, so those come from the root of trust in Move 57 or from a sealed file, and the answer is written down rather than assumed.
- 04 Move the state backend for the cloud estate. Take the licence decision at the same time: the widely used tool moved to a source-available licence at version 1.6 and an openly licensed fork exists, and this is the moment to choose rather than after the state file has moved.
- 05 Fix the deprecated locking arrangement if you have one. A separate lock table is a superseded pattern and native state locking replaces it; migrate to the current shape rather than carrying the old one across.
- 06 Write the recovery as an ordered procedure: from an empty cluster and a repository, in what order does the platform come back. Move 115 drills it for real, and this Move is where the order is decided.
- 07 Run both delivery paths for thirty days. The managed deploy service stays configured and unused, which is the reversibility this Move claims.

ROLLBACK

For thirty days the managed deploy path stays configured, so a failed adoption is reverted by using it again. The point of no return is deleting those pipelines, which is a Part VII activity. Because everything is in a repository, a bad state is corrected by reverting a commit and letting the controller reconcile, and a lost cluster is restored by pointing a fresh controller at the same path — which is the property that makes this Move worth its three weeks.

66 INFRASTRUCTURE CODE THAT ONLY COMPILES FOR ONE CLOUD

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
PLATFORM	PROVIDER-NATIVE STACK TEMPLATES, THEIR PROGRAMMING-LANGUAGE SYNTHESISERS AND THE SERVERLESS APPLICATION FRAMEWORKS BUILT ON THEM	MEDIUM	IMMEDIATELY

Provider-native templates have no target off the cloud that defines them, so the estate they describe is re-expressed once rather than translated stack by stack.

AWS	CloudFormation with the Cloud Development Kit	the hardest of the three, because the source is a program rather than a document, and the serverless framework sits on top of it.
GOOGLE	Deployment Manager	already past end of support and scheduled for shutdown, with Config Connector and a managed service as successors, so this reader is moving regardless.
AZURE	ARM templates with Bicep	a template language with a friendlier front end, which makes the source at least readable as a description.

BEFORE YOU START

ACCESS

- Every template, stack and synthesiser project in the estate, listed rather than remembered
- The state backend and repository from Move 65, ready to receive what survives

SOFTWARE

- A stack-by-stack inventory marked as stays, moves to manifests, or is deleted
- The successor tooling for any provider whose native service is being retired underneath you

PEOPLE

- The author of each synthesiser project, because reading somebody else's generated infrastructure is slower than rewriting it

WHY THIS WORKS

A provider-native template has no target other than the cloud that defines it. There is no compiler flag that emits your cluster. So the honest approach is to re-express the estate once rather than to translate stack by stack, and the split is clean: identity, networking and buckets — the shrinking cloud footprint that stays — are described in the tool chosen in Move 65, while everything describing compute becomes Kubernetes manifests under that same reconciler and is not translated at all. The synthesiser estates are the harder half, because their higher-level constructs have no equivalent to compile into, and this Move decides how much of the function estate in Move 89 can move at all.

SWAP

Where a stack describes something genuinely staying on the cloud forever, leave it alone. Rewriting working infrastructure code for its own sake is the least valuable month available.

DO IT FASTER

Start with the stacks that only create identity and networking. They convert almost mechanically and they unblock everything else.

WATCH OUT

Deleting a stack can delete the resources it created, including buckets with data in them. Set the retention behaviour deliberately on every resource before removing any stack.

LEFTOVERS

Orphaned resources left behind by half-deleted stacks are extremely common and Move 120 will find them in the billing data.

—	—	—	0 min	4 weeks	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Old stacks, once their resources are managed elsewhere — and the orphaned resources they left behind, once Move 120 has found them.

NEEDS 65 Argo CD, and the two infrastructure-as-code estates

66 INFRASTRUCTURE CODE THAT ONLY COMPILES FOR ONE CLOUD

7 STEPS · 0 MIN · MEDIUM RISK

THE RUNBOOK

PLATFORM

- 01 | Inventory every stack and mark each one. Stays in the cloud, becomes a Kubernetes manifest, or is deleted because nothing has used it in two years. The third category is usually larger than expected.
- 02 | Re-express what stays — identity, networking, buckets — in the tool chosen in Move 65, against the shrinking cloud footprint. This is ordinary work and it goes quickly.
- 03 | Do not translate the compute stacks. A description of an instance group has no meaning here; the workload is described as a manifest and the stack is deleted. Attempting a translation produces something that resembles the old estate and works like neither.
- 04 | Take the synthesiser estates one project at a time, and read what they actually create rather than what they appear to say. A single high-level construct can create a dozen resources, several of which nobody knows about.
- 05 | Where a provider's own template service is being retired, move to its successor only if you are staying on that cloud for that resource. Otherwise move straight to the target and skip the intermediate step entirely.
- 06 | Record which parts of the function estate cannot move because their infrastructure only exists as a synthesiser construct. Move 89 needs that list, and this is the Move that produces it.
- 07 | Delete each old stack only after its resources are managed elsewhere, and confirm the deletion does not remove anything still in use, which is the failure mode that ends this Move badly.

ROLLBACK

Nothing is destroyed while the estate is being re-expressed, because the new description and the old stack can both exist against the same resources for a while. There is no point of no return until a stack is deleted. Keep the original templates in source control after they stop being used, so the previous definition can be restored and compared when a resource turns out to have had a setting nobody carried across.



PLATFORM · 0 MIN · MEDIUM RISK

Old stacks, once their resources are managed elsewhere — and the orphaned resources they left behind, once Move 120 has found them.

67 THE FIRST STATELESS SERVICE, END TO END

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
 PLATFORM	ONE STATELESS SERVICE: ITS WORKLOAD IDENTITY ROLE, ITS LOAD-BALANCER TARGET GROUP AND ITS MANAGED LOG GROUP	 MEDIUM	IMMEDIATELY

One service goes from inventory to production traffic with every step written down, because the next eighty follow it and the rollback is one weighted DNS record.

AWS	an EKS Deployment behind an Application Load Balancer	with an IAM role for the pod and a CloudWatch log group behind it.
GOOGLE	a GKE Deployment behind a load balancer	with a service account and a Cloud Logging sink doing the same two jobs.
AZURE	an AKS Deployment behind Application Gateway	with a managed identity and a Log Analytics workspace behind it.

BEFORE YOU START

ACCESS

- A service whose owner is willing, whose dependencies are known, and which is not the most important thing you run
- The cloud load balancer in front of it, with the ability to add a second target and weight it

SOFTWARE

- Manifests and chart values, config and secrets from Move 56, image and signature from Moves 58 and 59
- Dashboards and log pipeline from Moves 62 and 63, already collecting from the new cluster

PEOPLE

- The service's owner, in the room, because the handover is the point and not the deployment

WHY THIS WORKS

This is the Move where everything built in Parts III and IV is used at once, on one real service, with every step written down — because the next eighty services follow this path and the difference between a good runbook and a vague one is measured in months. The traffic comes through the load balancer still running in the cloud, across the interconnect from Move 53, onto the routable pod addresses from Move 45. None of Part VI exists yet and none of it is needed. The rollback is one weighted record in the cloud's name service, which is why this can be done on a Tuesday afternoon.

SWAP

Where no service is small enough to be a good first candidate, build a trivial one that does something real — an internal tool, a status endpoint — and move that first. The point is the path, not the service.

DO IT FASTER

Do the second service the week after the first, with the same person, following the document. That is the run that turns a narrative into a runbook.

WATCH OUT

The appliances that stay rented — a managed identity gateway, a third-party service behind a private link — are dependencies too, and they need to be reachable from the new cluster before the first request arrives.

LEFTOVERS

The old target group, task definition and log group stay until the weight has been at one hundred per cent for a fortnight.

\$640/mo	\$0/mo	100%	0 min	2 weeks	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The old service's compute and its log group, a fortnight after the weight reaches one hundred per cent and stays there.

NEEDS 53 The links you build first: to the cloud, to the office 55 Workload identity without the cloud's

56 Secrets: External Secrets as the dated bridge 58 The mirror, then Harbor as the registry of record 59 cosign, Trivy and an admission policy

60 East-west: security groups into NetworkPolicy 62 Metrics off the managed monitor, and the cardinality budget

63 Logs and traces, brought home 65 Argo CD, and the two infrastructure-as-code estates

THE RUNBOOK

- 01 Map the dependencies before writing anything. What it calls, what calls it, what data it touches, and what it assumes about its environment. This map is the template for every service that follows.
- 02 Run the compatibility audit. The new cluster is on a newer Kubernetes minor than the managed one, so deprecated resource versions, removed flags and changed defaults all need checking against the manifests rather than discovered at admission.
- 03 Assemble the whole package: manifests, chart values, configuration and secrets, a signed image, an identity from Move 55 and a network policy from Move 60. Write down each step as you do it; this document is the deliverable.
- 04 Deploy with no traffic and let it run for a day. Watch its dashboards, read its logs, and confirm both are as usable as the ones it had before.
- 05 Add the new cluster as a second target behind the existing cloud load balancer, weighted at a small percentage, and watch error rates and latency at the same resolution as the old target.
- 06 Increase the weight in steps over a few days rather than in one change, and stop at any step where the two targets do not behave the same.
- 07 Hand the namespace over as a product: quotas, roles, a dashboard, a way to deploy and a way to roll back. A service that has moved but that its team cannot operate has not moved.

ROLLBACK

Set the weight back to zero on the cloud load balancer and the traffic returns instantly to where it was, with the old deployment still running and still warm. There is no point of no return in this Move at all, which is what makes it the right place to spend two weeks. Keep the runbook, the manifests and the dependency map in source control so the exact configuration can be restored, and leave the old deployment in place for a fortnight rather than tidying it up on the day.



68 THE CONTAINER SERVICE THAT WAS NOT KUBERNETES

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
 PLATFORM	MANAGED CONTAINER RUNTIMES THAT ARE NOT KUBERNETES: TASK-DEFINITION SERVICES, REQUEST-BILLED SERVERLESS CONTAINERS AND THEIR BUILT-IN SERVICE DISCOVERY	 MEDIUM	IMMEDIATELY

A task definition is not a Deployment, so the sidecar model, the exec path, the per-task network interface and scale-to-zero billing are each converted separately.

AWS	ECS with Fargate	a task definition with its own service discovery, exec access, placement constraints and a network interface per task, which is a security boundary you lose.
GOOGLE	Cloud Run	request-billed, with a concurrency model, revision traffic splitting and scale to zero that have no default equivalent on a fixed fleet.
AZURE	Container Apps	already KEDA, Dapr and Envoy over Kubernetes underneath, so it converts the most directly of the three.

BEFORE YOU START

ACCESS

- The task or service definitions exported, including the placement rules nobody remembers writing
- Traffic profiles per service, because the ones that are genuinely bursty may not move at all

SOFTWARE

- Manifests derived from each definition, with the sidecars converted rather than dropped
- An autoscaler configuration for the request-billed services, since concurrency was doing that job before

PEOPLE

- The owner of each service, to say whether idle cost or peak latency is the thing that matters to them

WHY THIS WORKS

These three are not one thing and converting them as though they were is why this Move exists separately. A task definition becomes a Deployment, a Service and a network policy, and loses its per-task network boundary, which has to be re-expressed as policy rather than assumed away. A request-billed container has a concurrency model and a scale-to-zero behaviour that a fixed fleet does not reproduce for free. And one of the three is already Kubernetes underneath, so it converts almost mechanically. The honest note is about billing: a workload that cost nothing while idle now occupies capacity you bought.

SWAP

Where a service is already on the platform that is Kubernetes underneath, convert it first. It is the fastest win and it builds the pattern for the harder two.

DO IT FASTER

Convert the definitions with a script and then read every output by hand. The mechanical part is quick and the review is where the sidecars and the placement rules are caught.

WATCH OUT

Scale to zero is a billing property, not a design one. A service that has never handled a cold start on a fixed fleet may have a first-request latency nobody has measured.

LEFTOVERS

The old service definitions, their service-discovery namespaces and their per-task interfaces stay until Part VII, and the discovery namespaces in particular are easy to forget.

\$1,260/mo	\$0/mo	100%	0 min	2 weeks	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The task definitions, service-discovery namespaces and request-billed revisions, once a week of split traffic shows the same behaviour on both sides.

NEEDS 60 East-west: security groups into NetworkPolicy 67 The first stateless service, end to end

THE RUNBOOK

- 01 | Sort the estate by runtime first. Three different services with three different conversion paths, and treating them as one is how a migration produces a fleet of half-converted workloads.
- 02 | Convert task definitions into a Deployment, a Service and a network policy. Take the sidecars seriously: a logging or proxy sidecar in the old definition has an equivalent here, and dropping it silently removes a function somebody depends on.
- 03 | Re-express the per-task network boundary as policy. Each task had its own interface and its own firewall membership; here that becomes a label selector in Move 60's policy set, and it must be written rather than assumed.
- 04 | Replace the exec path. Debugging access to a running task was a platform feature; here it is a role, a policy and an audit trail, and it should be granted deliberately.
- 05 | For the request-billed services, map concurrency onto replicas and requests, and add an autoscaler that responds to the same signal the platform used. Revision traffic splitting becomes two deployments and a weight.
- 06 | Be honest about the ones that should not move. A service that runs for four minutes a day and costs nothing while idle now occupies reserved capacity, and the right answer may be that it stays rented until Move 89 gives it a scale-from-zero design.
- 07 | Run both for a week per service, with traffic split, and compare cost as well as latency.

ROLLBACK

Each service is reverted by shifting its traffic weight back, with the old task still defined and still able to run, so this is reversible per service and within minutes. There is no point of no return. Keep the exported definitions in source control so the original placement rules and sidecars can be restored and compared when a converted service behaves differently under load.

69 THE FIXED FLEET: PRIORITY, PREEMPTION, NUMA AND THE GPUS

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
PLATFORM	NODE AUTOSCALING GROUPS, CLUSTER AUTOSCALERS, JUST-IN-TIME NODE PROVISIONERS, MANAGED NODE POOLS AND SERVERLESS POD RUNTIMES	HIGH	IMMEDIATELY

Node count is fixed, so scheduling becomes the scarce resource: accurate requests, tiered priority classes and deliberately wasted headroom replace the autoscaler.

AWS	Karpenter and managed node groups	a just-in-time provisioner meant this estate never had to express a node class at all, which makes it the most work here.
GOOGLE	node auto-provisioning and Autopilot	the same absence of node classes, with the platform choosing shapes on your behalf.
AZURE	cluster autoscaler with node pools and node auto-provisioning	closer to explicit node classes, so this reader starts from a shorter list.

BEFORE YOU START

ACCESS

- Real usage data per workload from Move 62, at percentiles rather than averages
- The SMT and topology decisions from Move 05, which this Move inherits rather than revisits

SOFTWARE

- Priority classes defined as a tier list rather than as ad-hoc numbers
- The device plugin and driver for any accelerators, delivered as a system extension in the image rather than built on the host

PEOPLE

- The owner of every batch workload, because batch is what will be preempted and they need to know

WHY THIS WORKS

When a scheduler could conjure a machine, inaccurate resource requests were a rounding error. On a fixed fleet they are the whole game, because a request that is twice the real usage wastes half a machine and a request that is half the real usage produces evictions under load. So requests get measured against actual usage; priority classes decide what yields to what when the fleet is full; low-priority placeholder workloads reserve headroom deliberately, so that it exists rather than being consumed by whatever asked last; and a documented overcommit position per machine class replaces the autoscaler's willingness to spend.

SWAP

Where an estate genuinely cannot fit, the answer is Move 116 and more capacity, not a cleverer scheduler. Say that out loud rather than tuning for a month.

DO IT FASTER

Fix the ten largest workloads' requests and stop. The distribution is always long-tailed and the top ten are most of the waste.

WATCH OUT

Preemption is invisible to the workload being preempted, which experiences it as a restart. Batch owners need to know their jobs must be resumable, and this Move is when they find out.

LEFTOVERS

The node groups, autoscalers and provisioner configurations stay in the cloud until Part VII, because they are the burst capacity Move 116 may deliberately keep.

—	—	—	0 min	3 weeks	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. The autoscaling configuration stays in the cloud, deliberately, because Move 116 may keep it as tested overflow.

NEEDS 05 SMT, NUMA, and the workload that notices 47 Local NVMe, and the default StorageClass nobody gave you

67 The first stateless service, end to end

THE RUNBOOK

- 01 | Correct the requests first, against measured usage rather than against what the manifest has always said. This single step recovers more capacity than everything else in this Move combined.
- 02 | Define priority classes as a small tier list: serving above batch, batch above best-effort, and a placeholder tier below everything. Three or four tiers, not fifteen.
- 03 | Deploy low-priority placeholder workloads to reserve headroom. This is deliberately wasted capacity that exists so that a serving workload can always be scheduled, and it is the fixed-fleet replacement for an autoscaler's spare machine.
- 04 | Write down the overcommit position per machine class. How much more may be requested than exists, and why. An undocumented overcommit ratio is a decision somebody made once and nobody can defend.
- 05 | Set the topology and processor policies from Move 05 and confirm a guaranteed workload actually receives whole physical cores rather than two halves of one.
- 06 | Install the accelerator driver and device plugin as a system extension baked into the machine image. This is the one place in the Part where the immutable host changes the procedure rather than only the tooling, and it means a driver change is an image change and a reboot.
- 07 | Run the descheduler to correct drift, and watch what it moves for a week before letting it act automatically.

ROLLBACK

Every change here is a scheduler setting and is reverted by reverting it, with the workloads rescheduling within minutes. There is no point of no return. The risk is operational rather than permanent: a priority tier applied wrongly can preempt something important, which is why the tiers are few and why the descheduler is watched before it is trusted. Keep the priority classes and requests in source control so a working scheduling configuration can be restored quickly.

70 THE WORKLOADS THAT MUST STAY VIRTUAL MACHINES

CUTOVER

30 MIN



LAYER	LEAVING	RISK	REVERSIBLE
PLATFORM	RENTED VIRTUAL MACHINES OUTSIDE THE CLUSTER, MANAGED HYPERVISOR ESTATES AND THE APPLIANCE IMAGES SOLD THROUGH CLOUD MARKETPLACES	MEDIUM	14 DAYS

Some workloads never containerised, so KubeVirt runs them as virtual machines on the same nodes, drawing storage, addresses, identity and backups from the cluster.

AWS	EC2 instances and Marketplace appliance images	exported as raw or VMDK, which is the friendliest of the three formats to import.
GOOGLE	Compute Engine instances and Marketplace images	exported as a disk image through a bucket, which adds a copy step but is otherwise straightforward.
AZURE	Virtual Machines and Marketplace images	exported as a VHD that must be fixed-size and correctly aligned, or the import fails after the copy has finished.

BEFORE YOU START

ACCESS

- The list of workloads that genuinely cannot containerise, with a reason next to each one
- Export permissions in the source cloud, and somewhere to stage the disk images

SOFTWARE

- KubeVirt at a pinned version, with the virtualisation and guest-agent system extensions in the machine image
- Storage classes that support the access mode live migration needs, which is not the default one

PEOPLE

- The owner of each appliance, because a vendor support position may forbid this and it is better to know first

WHY THIS WORKS

Some things never containerised and never will: a licensed database engine whose vendor supports one deployment shape, a Windows workload, an appliance sold only as a disk image. Running a hypervisor tier inside the cluster is how those workloads get the same storage, the same addresses, the same identity, the same policy and the same backups as everything else, instead of becoming a second estate with its own everything. The virtual machines run on the same nodes, their disks live on the storage from Move 49, and their identity and policy are inherited from Moves 55 and 60 rather than reinvented.

SWAP

Where a vendor will not support their appliance on your hypervisor, keep renting it. That is a legitimate outcome and it belongs on the list in Move 116 of things you deliberately keep.

DO IT FASTER

Convert the disk images in advance and keep them current with a periodic delta copy, so the cutover is a short final synchronisation rather than a full transfer.

WATCH OUT

Live migration is not free and not always possible. A virtual machine with a local disk or an attached device does not migrate, which means node maintenance in Move 109 has to plan around it.

LEFTOVERS

The source instances, their disks and their snapshots stay for fourteen days, and the marketplace subscriptions behind the appliances are cancelled in Move 122 rather than here.

\$2,870/mo	\$0/mo	100%	30 min	3 weeks	14 days
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The source instances and their disks after fourteen days, and their snapshots when Move 38 removes them against a signed approval.

NEEDS 47 Local NVMe, and the default StorageClass nobody gave you 49 Ceph under Rook: block, shared filesystem and the object store

55 Workload identity without the cloud's 60 East-west: security groups into NetworkPolicy

69 The fixed fleet: priority, preemption, NUMA and the GPUs

- 01 Justify each entry on the list. A workload goes here because it cannot containerise, not because containerising it is work nobody has scheduled. The list should be short and each line should have a reason.
- 02 Add the virtualisation and guest-agent extensions to the machine image and roll the nodes. On an immutable host this is an image change and a reboot, not an installation.
- 03 Install KubeVirt at a pinned version and run one throwaway virtual machine end to end before touching anything real.
- 04 Set the storage up properly. Disks live on the block storage from Move 49, and live migration requires a shared access mode that the default class does not provide — so that is a deliberate storage class rather than an oversight discovered during the first migration.
- 05 Export, convert and import each disk. One provider needs the image to be a fixed-size, correctly aligned disk, and the failure appears only after the copy has finished, so validate the format before transferring gigabytes.
- 06 Give each virtual machine an address from the plan in Move 39, an identity from Move 55 and a policy from Move 60. The hypervisor tier is not an exception to the platform; that is the whole point of putting it here.
- 07 Cut over one machine at a time: stop it, copy the final delta, start it here. Thirty minutes is the honest figure per machine, and the source stays in place for a fortnight.

ROLLBACK

The source instance is stopped rather than deleted, so for fourteen days reverting is starting it again and moving the address back. The point of no return is the deletion of the source disk, which is deliberately deferred. Because the cutover involves a stop and a copy, any data written on the new side after the switch has to be restored to the source if you go back — so the fortnight is a window for deciding, not for running both.



V DATA

The part that can lose a company its data, and therefore the part with the longest overlap windows, the most rehearsal and the fewest one-way steps.

20	12	12	147
MOVES	ZERO DOWNTIME	HIGH RISK	MINUTES, TOTAL

IF YOU ONLY DO THREE

72

THE VERIFICATION GATE

Builds the chunked hash-and-checksum comparison that every later data cutover has to pass before anyone is allowed to …

76

THE PROPRIETARY POSTGRES ENGINES: THE EXIT IS A STREAM

Leaves the proprietary PostgreSQL engines by logical replication, which is the only door, and settles what replaces …

71

THE POSTGRES PRE-FLIGHT

Audits roles, extensions, replica identity and collation weeks before any cutover, because these failures are silent …

THE POSTGRES PRE-FLIGHT · THE VERIFICATION GATE · POSTGRES ONTO YOUR OWN DISKS · POINT-IN-TIME RECOVERY WITH PGBACKREST · QUORUM COMMIT AND FENCING, AFTER THE RESTORE WAS PROVEN · THE PROPRIETARY POSTGRES ENGINES: THE EXIT IS A STREAM · MYSQL, WHEREVER IT IS MANAGED · THE LICENSED DATABASES, AND THE CONTRACT THAT CAN VETO IT · REDIS: WHICH HALF OF IT IS ACTUALLY A CACHE · BROKERED MESSAGING: JMS, DURABLE SUBSCRIPTIONS, YOUR BROKER · THE QUEUES COME HOME, THE FAN-OUT DOES NOT · THE EVENT BUS, AND THE CONTROL-PLANE EVENTS THAT STOP ARRIVING · KAFKA OFF THE MANAGED BROKER · APPLICATION BUCKETS, AND THE OBJECT API YOUR CODE ASSUMES · THE DATA THAT STAYS: DEEP ARCHIVE AND THE WAREHOUSE · THE BATCH ESTATE: WHO SCHEDULES THE JOBS · SEARCH OFF THE MANAGED CLUSTER · DOCUMENT AND WIDE-COLUMN STORES: THE HARDEST EXIT · THE SERVERLESS ESTATE IN FOUR PILES · ORCHESTRATED WORKFLOWS: THE DATED EXCEPTION

■ 3 LOW RISK

■ 5 MEDIUM RISK

■ 12 HIGH RISK

71 THE POSTGRES PRE-FLIGHT

CUTOVER

15 MIN



LAYER	LEAVING	RISK	REVERSIBLE
DATA	MANAGED POSTGRESQL	MEDIUM	IMMEDIATELY

Audits roles, extensions, replica identity and collation weeks before any cutover, because these failures are silent until the far side is already serving reads.

AWS	RDS and Aurora for PostgreSQL	provider-only extensions and a role graph built under a superuser you never had.
GOOGLE	Cloud SQL for PostgreSQL	its own provider-only extensions and a flag set that has to be enabled before logical decoding works.
AZURE	Database for PostgreSQL Flexible Server	an allowlist parameter that governs which extensions may load at all.

BEFORE YOU START

ACCESS

- Read access to the catalogues on the managed instance, and a maintenance window for one parameter change
- The second cluster from Move 51 to rehearse the audit against

SOFTWARE

- `psql` at a version at least as new as the server, and `pg_dump` for the globals-only role export
- A written comparison of the collation provider and version on both sides

PEOPLE

- Whoever owns the application, to schedule the reboot the one production change requires

WHY THIS WORKS

Every failure this Move looks for is silent. A missing role does not fail at migration time; it fails when an application connects. A provider-only extension does not fail until a query uses it. A table with no primary key replicates its inserts and silently discards its updates. And a collation difference between the source and the destination does not fail at all — it produces text indexes that are subtly, permanently wrong. All four are cheap to find weeks ahead and expensive to find on the far side of a cutover, which is why this Move is separate from Move 73 and happens long before it.

SWAP

Where an extension has no equivalent and the application depends on it, the honest answer may be that this database moves last, or moves as a dump and restore with a longer window.

DO IT FASTER

Run the audit as a set of queries checked into source control rather than as a session. It will be run again for every database in the estate.

WATCH OUT

The reboot is the whole cutover figure in this Move, and it is the step people schedule casually because the rest of the Move is read-only.

LEFTOVERS

Logical decoding retains write-ahead log for any slot that exists. A slot created during the audit and forgotten will fill the source's disk, which is an outage on the system you are still depending on.

—	—	—	15 min	2 weeks	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing. This Move only adds a parameter and a list of things to fix before Move 73 can safely run.

NEEDS 51 The second cluster, and what it shares 53 The links you build first: to the cloud, to the office

THE RUNBOOK

- 01 Export the role graph with a globals-only dump and read it. Managed instances build roles under a superuser you never had, so the graph frequently contains grants that cannot be reproduced literally and have to be re-expressed.
- 02 Inventory the extensions and mark the provider-only ones. Each cloud has its own — an object-store integration extension, a machine-learning integration, a storage extension behind an allowlist — and each is either replaced, dropped or blocks the migration. Decide which, in writing, now.
- 03 Find every table with no primary key or no replica identity. Logical replication carries inserts for these and silently discards updates and deletes, which produces a destination that diverges quietly and passes a row count.
- 04 Compare the collation provider and version on both sides. A text index built under one collation and used under another can return wrong results, and this is the failure that is hardest to detect and worst to discover late.
- 05 Rehearse the whole audit against a restored copy on the second cluster from Move 51, so the findings are proved rather than theorised.
- 06 Make the one production change: enable logical decoding. It is a static parameter on all three providers and it requires a reboot. On a single-instance deployment that reboot is routinely longer than five minutes; on a multi-zone one it is a failover. Schedule it weeks ahead of anything anyone has described as zero-downtime.

ROLLBACK

The audit changes nothing and the parameter change is reverted by another parameter change and another reboot, so this Move is reversible immediately at the cost of one more window. The point of no return does not exist here; it is in Move 73. Keep the audit output in source control, because it is the baseline the verification in Move 72 will be compared against, and because a restore of the source later needs to be checked against the same findings.



72 THE VERIFICATION GATE

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
DATA	MANAGED DATABASE-MIGRATION VALIDATION	LOW	IMMEDIATELY

Builds the chunked hash-and-checksum comparison that every later data cutover has to pass before anyone is allowed to proceed.

AWS	Database Migration Service data validation	a validation task that stops working the day the account does.
GOOGLE	Database Migration Service verification	the same, tied to the same lifetime.
AZURE	Database Migration Service assessment	the same again, and the same reason to own this rather than rent it.

BEFORE YOU START

ACCESS

- A physical replica on the source side to run against, so the comparison costs read load on a machine nothing is serving from
- The destination database from Move 73's rehearsal, or the second cluster from Move 51

SOFTWARE

- `psql` for the aggregates, and somewhere to store per-chunk results so a run can resume
- A written rule for which mismatches are benign and which mean stop

PEOPLE

- Whoever will make the go decision on cutover night, because they have to trust this before they rely on it

WHY THIS WORKS

Every cutover in Part V ends with somebody deciding whether the two sides agree. That decision should be a measurement rather than a judgement, and it should be the same measurement every time so that it can be trusted without re-reading it. What this Move builds is a deterministic aggregate over every table on both sides — a hash over the primary key and a checksum over the columns, chunked by key range so that it runs concurrently and can be restarted — plus an assertion that replication lag is zero. It is deliberately a gate and not a repair tool: it answers go or no-go and nothing else.

SWAP

For small tables a straight row-by-row comparison is simpler and just as good. The chunked aggregate earns its complexity above a few million rows.

DO IT FASTER

Run it continuously during the catch-up phase rather than only at the end. Differences found a week early are cheap; the same differences found in the window are not.

WATCH OUT

Floating-point columns and timestamps with time zones are the two types most likely to produce spurious mismatches through formatting rather than value. Normalise them explicitly in the aggregate.

LEFTOVERS

The managed validation task keeps billing while its migration task exists. It is turned off with the rest of the migration service in Part VII.

\$180/mo	\$0/mo	100%	0 min	1 week	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. The managed validation tooling goes with the migration service in Part VII.

NEEDS 51 The second cluster, and what it shares 53 The links you build first: to the cloud, to the office 71 The Postgres pre-flight

THE RUNBOOK

- 01 Write the aggregate as a hash over the primary key combined with a checksum over the remaining columns, computed per chunk of the key range. Deterministic ordering matters; an aggregate that depends on row order will disagree for no reason.
- 02 Chunk by primary key range so the comparison runs concurrently and restarts from where it stopped. On a large table a single unchunked query is both slow and unrestartable, which is how verification gets skipped.
- 03 Run it against a physical replica rather than the primary. The blast radius is then read load on a machine nothing serves from, which is what makes it safe to run as often as you like.
- 04 Assert replication lag is zero before comparing. A comparison taken while the stream is behind reports differences that are not differences, and the resulting arguments waste an hour of a maintenance window.
- 05 Write down which mismatches are benign — a sequence ahead by design, a monitoring table that writes on both sides — and which mean stop. Do this in advance and in daylight, not at two in the morning.
- 06 Run it against the rehearsal from Move 73 and confirm it detects a difference you introduce on purpose. A gate that has never said no is a gate nobody has tested.

ROLLBACK

This Move only reads, so there is nothing to reverse and no point of no return anywhere in it. That is what makes it the right thing to build before any data moves. Keep the queries and the benign-mismatch rules in source control, so the same gate is applied to every database in Part V and so that a previous run can be restored and compared when somebody questions a result.



73 POSTGRES ONTO YOUR OWN DISKS

CUTOVER

20 MIN



LAYER	LEAVING	RISK	REVERSIBLE
DATA	MANAGED POSTGRESQL	HIGH	7 DAYS

Moves Postgres onto CloudNativePG by logical replication, with writes stopped at PgBouncer for the sequence advance and the verification gate.

AWS	RDS for PostgreSQL	replicates from the writer endpoint, with the logical decoding parameter already enabled in Move 71.
GOOGLE	Cloud SQL for PostgreSQL	the same, with an additional role grant required before a publication can be created.
AZURE	Database for PostgreSQL Flexible Server	the same again; the older single-server product has no logical decoding at all and has to be dumped and restored.

BEFORE YOU START

ACCESS

- The logical decoding parameter enabled and the audit findings from Move 71 resolved rather than noted
- The link from Move 53, sized for catch-up rather than for steady state

SOFTWARE

- CloudNativePG at a pinned version, already rehearsed and failed over on the second cluster
- PgBouncer in front of the application, so writes can be paused without restarting anything

PEOPLE

- The application owner present for the window, because the pause at PgBouncer is visible to users

WHY THIS WORKS

Logical replication moves the data while both databases are live, so the downtime is not the copy — it is the few minutes it takes to stop writing to one side, advance the sequences, run the gate and start writing to the other. This is also the first production data to land on your own hardware, which means it lands on the local volumes from Move 47, under the encryption from Move 48, from the registry in Move 58, inside the policy from Move 60 and on the dashboards from Move 62. None of that existed by accident; the whole of Parts III and IV was ordered so that this Move would have somewhere safe to land.

SWAP

Under about fifty gigabytes with a tolerant window, a dump and restore is simpler and finishes inside an hour. Logical replication earns its complexity above roughly that size.

DO IT FASTER

Run the initial copy from a read replica rather than the primary. It costs one extra instance for a day and takes the load off the database serving traffic.

WATCH OUT

Logical replication does not carry sequences, large objects or schema changes. A migration that forgets the sequences works perfectly until the first insert collides, which is usually about ninety seconds later.

LEFTOVERS

The publication and subscription persist until dropped. Drop them deliberately after the week, or the source keeps retaining write-ahead log for a subscriber nobody is reading.

\$4,180/mo	\$410/mo	90%	20 min	3 weeks	7 days
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The managed instance, its read replicas and its snapshots — after seven days, after Move 75 has high availability working, and after a billing cycle has shown the line gone.

NEEDS 47 Local NVMe, and the default StorageClass nobody gave you 48 Encryption at rest, and who holds the key

53 The links you build first: to the cloud, to the office 56 Secrets: External Secrets as the dated bridge

58 The mirror, then Harbor as the registry of record 60 East-west: security groups into NetworkPolicy

62 Metrics off the managed monitor, and the cardinality budget 63 Logs and traces, brought home 71 The Postgres pre-flight

72 The verification gate

THE RUNBOOK

- 01 Rehearse the whole thing twice on the second cluster from Move 51 before touching production. Three weeks is the budget for this Move and two rehearsals are most of it.
- 02 Copy the schema only, apply it to the CloudNativePG cluster on the destination, and defer the indexes on large tables. They slow the initial copy and are faster to build afterwards.
- 03 Create the publication and the subscription and watch the catch-up in the subscription statistics. Set the write-ahead log retention limit on the source as a safety valve, so a stalled subscriber cannot fill the source's disk and take down the database you are still depending on.
- 04 Build the reverse subscription back to the managed instance before the cutover, and leave it running. This is what makes the Move reversible for a week rather than only until the first write.
- 05 Build the deferred indexes, then run the Move 72 gate and resolve every mismatch that is not on the benign list.
- 06 Open the window. Pause writes at PgBouncer, wait for lag to reach zero, advance every sequence with a margin, run the gate one final time, and resume against the new primary. Twenty minutes is the honest figure and most of it is the gate.
- 07 Watch error rates and latency for an hour, then leave the managed instance running with its automated backups on for seven days.

ROLLBACK

Until writes are resumed on the new side, backing out is releasing the pause and doing nothing else. After that, the reverse subscription is what makes the week real: repoint PgBouncer, let the reverse stream drain, and the managed instance is authoritative again. The point of no return is the day the source is deleted, which is Move 38 and not this Move. Keep the managed instance and its automated backups for seven days so a restore to a known good point stays available.

74 POINT-IN-TIME RECOVERY WITH PGBACKREST

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
DATA	MANAGED AUTOMATED BACKUPS AND SNAPSHOTS	HIGH	IMMEDIATELY

Rebuilds point-in-time recovery with pgBackRest, because every claim of reversibility later in the book rests on a restore that has actually been run.

AWS	RDS automated backups and snapshots	a retention window and a restore-to-time button that you never had to operate yourself.
GOOGLE	Cloud SQL automated backups and point-in-time recovery	the same, with the write-ahead log retention configured separately.
AZURE	Flexible Server backups with geo-redundant options	the same again, with the redundancy sold as a tier.

BEFORE YOU START

ACCESS

- The object store from Move 54, with its durability, off-site copy and restore drill already proved
- A second credential domain for the repository, so one compromised credential cannot reach both copies

SOFTWARE

- pgbackrest at a pinned version, configured with asynchronous archiving rather than the default
- A scratch cluster to restore into, because a restore that has never been performed is a hope

PEOPLE

- Whoever owns the recovery objective, so the number this Move produces is the number they expected

WHY THIS WORKS

Every claim of reversibility in this book, and every rollback paragraph in Part V, rests on a restore that has actually been run. The managed service gave you that for a monthly charge and you never had to think about the repository, the archive path or the recovery objective. Owning it means all three become explicit. The repository goes on the object store from Move 49, in a different failure domain and a different credential domain, under immutability — and because immutability on that store has to be set when a bucket is created rather than added later, the bucket is made correctly before the first backup rather than hardened afterwards.

SWAP

For a small database, a nightly dump to the object store plus write-ahead log archiving is simpler and adequate. It gives a worse recovery point and it is a great deal better than nothing.

DO IT FASTER

Restore into the second cluster from Move 51 rather than building a scratch environment. It exists for exactly this.

WATCH OUT

A backup that has been running successfully for weeks can still be unrestoreable if the archive has a gap. Verify the archive continuously rather than checking that the backup job exited zero.

LEFTOVERS

Managed automated backups keep billing for their retention window after the instance is deleted, which is a line that surprises people in Part VII.

\$620/mo	\$70/mo	89%	0 min	2 weeks	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Managed automated backups, once a restore from your own repository has been performed, verified by the gate and timed.

NEEDS 49 Ceph under Rook: block, shared filesystem and the object store 54 The object store you now own, and how it loses data

55 Workload identity without the cloud's 72 The verification gate 73 Postgres onto your own disks

THE RUNBOOK

- 01 Create the repository bucket with immutability enabled at creation. On the object store you now own, this cannot be added afterwards on most releases, and a repository that can be deleted by the credential that writes it is not a backup.
- 02 Configure pgbackrest with a full and differential schedule and asynchronous write-ahead log archiving. Synchronous archiving on a busy database becomes the bottleneck and then becomes an incident.
- 03 State the recovery point objective honestly. The archive timeout bounds it only on a quiet database; a busy one is bounded by how fast segments fill plus how long the push takes, and that is the case that matters.
- 04 Put the repository in a different failure domain and a different credential domain from the cluster. A repository the cluster can delete with its own identity is a copy, not a backup.
- 05 Take a full backup and then restore it to a point in time into a scratch cluster. Time the restore. That number is the recovery time objective, and it is usually longer than anyone assumed.
- 06 Run the Move 72 gate against the restored copy. A restore that completes and produces different data has not succeeded; it has failed quietly, which is worse.
- 07 Put the restore drill on a schedule and treat a failed drill as an incident, because a repository nobody has restored from is exactly as useful as no repository at all.

ROLLBACK

Nothing in this Move changes the running database, so it is reversible immediately and has no point of no return. What it changes is what the rest of the book is allowed to claim: no Move after this one may describe itself as reversible unless a restore from this repository has been performed and verified. Keep the repository configuration and the timed drill results in source control, so the recovery objective is a published number rather than an assumption.



75 QUORUM COMMIT AND FENCING, AFTER THE RESTORE WAS PROVEN

CUTOVER

2 MIN



LAYER	LEAVING	RISK	REVERSIBLE
DATA	MANAGED MULTI-ZONE DATABASE FAILOVER	HIGH	IMMEDIATELY

Replaces the managed failover checkbox with quorum commit, and says plainly that only fencing gives safety.

AWS	RDS Multi-AZ	a checkbox that doubled the price and handled failover without ever telling you how.
GOOGLE	Cloud SQL high availability	the same checkbox with the same opacity and the same doubling.
AZURE	Flexible Server zone-redundant high availability	the same again, with the zone mapping being per-subscription and therefore worth checking.

BEFORE YOU START

ACCESS

- A proven restore from Move 74, timed and verified, because that is the gate on this Move
- The failure-domain labels from Move 42, so the replicas are placed across real racks

SOFTWARE

- The operator's quorum configuration, written explicitly rather than left at a default
- A fencing mechanism configured and tested, not merely available

PEOPLE

- Somebody from the on-call rota watching the deliberate failover, because this is the behaviour they will meet at night

WHY THIS WORKS

The managed checkbox did two different things and never distinguished them: it kept a second copy current, and it decided when to promote it. Quorum commit gives you the first — a write is not acknowledged until enough replicas hold it — and it is durability, not consensus. It cannot by itself stop a partitioned old primary from continuing to accept writes. That is what fencing does, and it is the part almost everyone omits, which is why it is named in this Move's title. This runs only after Move 74 has proven a restore, because building automatic failover before the recovery path works inverts the principle the previous Move exists to establish.

SWAP

Where two minutes of unavailability is genuinely unacceptable, the answer is connection-level retry and a queue in front, not a cleverer failover. Applications that survive a failover are applications that expected one.

DO IT FASTER

Test the failover on the second cluster first, several times, including the partition case. It is the only way to see the fencing work.

WATCH OUT

Synchronous replication to a replica that is slow makes every write slow. Quorum with more replicas than required is what stops one degraded machine from becoming a performance incident.

LEFTOVERS

The managed high-availability tier can be turned off on the source, halving that bill immediately, once Move 73 has cut over and the week has passed.

\$2,090/mo	\$0/mo	100%	2 min	1 week	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The managed high-availability tier on the source instance, which is usually half its cost, once the cutover week from Move 73 has passed.

NEEDS 42 Netboot, node identity and failure-domain labels 73 Postgres onto your own disks 74 Point-in-time recovery with pgBackRest

75 QUORUM COMMIT AND FENCING, AFTER THE RESTORE WAS PROVEN

7 STEPS · 2 MIN · HIGH RISK

THE RUNBOOK

- 01 Confirm Move 74's restore has been performed and timed. This is a gate: automatic failover in front of an unproven recovery path is a mechanism for turning one bad night into two.
- 02 Place the members across the failure domains labelled in Move 42. Three replicas on three machines that share a switch is one failure domain with extra replication traffic.
- 03 Configure quorum commit explicitly, naming how many replicas must acknowledge, and set the commit behaviour honestly. A configuration that acknowledges before any replica has the data is faster and is not what you think you bought.
- 04 Configure fencing and test it. Quorum commit is durability; fencing is what stops a partitioned old primary from accepting writes that will be lost. Without it, the configuration is safe against machine loss and unsafe against network partition, which is the more common failure.
- 05 Kill the primary deliberately, during business hours, with people watching. Measure the promotion time, watch the application reconnect, and confirm the old primary does not accept a single write after it is fenced.
- 06 Write down what the two minutes consist of — detection, promotion, reconnection — so that the number is understood rather than quoted.
- 07 Do not attempt quorum across sites. Move 24 already declined the second site, and Move 115 settles the recovery question against a measured rebuild time instead.

ROLLBACK

The configuration is reverted to a single primary in a minute, so this Move is reversible immediately, and the data is untouched because nothing here moves it. There is no point of no return, and if a promotion does go wrong the way back is the proven restore from Move 74 rather than anything in this Move. The real risk is the opposite: enabling automatic promotion without fencing, which is safe until the first partition and then loses writes. Keep the configuration and the failover drill results in source control, and re-run the drill after every upgrade in Move 108, because failover behaviour is exactly the kind of thing a minor version changes quietly.

76 THE PROPRIETARY POSTGRES ENGINES: THE EXIT IS A STREAM

CUTOVER

20 MIN



LAYER	LEAVING	RISK	REVERSIBLE
DATA	PROPRIETARY MANAGED POSTGRESQL-COMPATIBLE ENGINES	HIGH	7 DAYS

Leaves the proprietary PostgreSQL engines by logical replication, which is the only door, and settles what replaces fast cloning and near-instant crash recovery.

AWS	Aurora PostgreSQL	a decoupled storage layer that gives near-instant crash recovery and fast cloning, neither of which comes with you.
GOOGLE	AlloyDB	the same shape, with a columnar engine on top that has no landing-side equivalent at all.
AZURE	Cosmos DB for PostgreSQL	distributed by an extension that is openly licensed, so this reader can self-host the same distribution rather than facing a closed engine.

BEFORE YOU START

ACCESS

- Logical decoding enabled per Move 71, on the writer endpoint rather than a reader
- A measured initial copy rate from a rehearsal, because that is what sets the schedule

SOFTWARE

- The same CloudNativePG destination and the same gate from Move 72 as Move 73 used
- A written list of features in use that have no equivalent, with a decision beside each

PEOPLE

- Whoever uses fast cloning, usually the delivery team, because their workflow changes

WHY THIS WORKS

These engines are PostgreSQL at the protocol and a different system underneath, and that difference is the whole content of this Move. There is no file-level backup you can take away, no base backup that restores anywhere but the same engine, and no snapshot that means anything outside it. Logical replication is the only door. That makes the migration bounded by two things: how fast the initial copy runs, and how long the source will hold a replication slot open while it does. Everything else here is deciding what replaces the features that do not travel, before anyone touches production.

SWAP

Where the columnar engine is load-bearing, the honest landing is a separate analytical store rather than an attempt to reproduce it in PostgreSQL. Move 85 has the substrate.

DO IT FASTER

Measure the copy rate on a rehearsal against a full-sized dataset. Every schedule in this Move derives from that one number and estimating it is how the window overruns.

WATCH OUT

Near-instant crash recovery is a property you are giving up. On your own hardware, recovery time after an unclean stop is proportional to the work in flight, and Move 75's failover figure reflects that.

LEFTOVERS

The engine's storage is billed separately from its compute on some variants and continues after the instance stops. Check both lines before declaring the saving.

\$6,240/mo	\$580/mo	91%	20 min	3 weeks	7 days
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The proprietary engine's instances and its storage line — after seven days, and after checking that the storage line is billed separately and has actually stopped.

NEEDS 71 The Postgres pre-flight 72 The verification gate 73 Postgres onto your own disks 74 Point-in-time recovery with pgBackRest

76 THE PROPRIETARY POSTGRES ENGINES: THE EXIT IS A STREAM

7 STEPS • 20 MIN • HIGH RISK

THE RUNBOOK

- 01 List the engine-specific features actually in use and decide each one before the migration starts. Fast cloning for per-branch test databases, restore granularity finer than a backup repository gives, near-instant crash recovery, and scale to zero on the serverless variants. Each is consciously replaced or consciously dropped.
- 02 Replace fast cloning explicitly. On your own hardware the equivalent is a restore from the repository built in Move 74 into the second cluster, which is slower and adequate, or a filesystem-level snapshot where the storage supports it.
- 03 Follow Move 73's runbook for the stream itself: schema first, deferred indexes, publication and subscription, catch-up watched, reverse subscription built before the cutover.
- 04 Watch the slot rather than the lag on these engines. The constraint is how long the source will hold the replication slot open while the initial copy runs, and a slot that is dropped mid-copy means starting again.
- 05 Run the Move 72 gate, then cut over with writes paused, sequences advanced and the gate run once more. Twenty minutes, and the same shape as Move 73.
- 06 For the distributed variant on one cloud, decide between a rewrite and self-hosting the same openly licensed distribution extension. That is a genuine choice rather than a workaround, and it is usually the better answer.
- 07 Keep the source running for seven days with the reverse stream in place, exactly as in Move 73.

ROLLBACK

The reverse subscription makes the week real: repoint the connection, let the reverse stream drain, and the managed engine is authoritative again. Until writes resume on the new side, backing out is releasing the pause. The point of no return is the deletion of the source, which is Move 38. Keep the source and its automated backups for seven days so a restore to a known good point remains possible, and confirm the reverse stream is actually flowing rather than merely configured.



DATA • 20 MIN • HIGH RISK

The proprietary engine's instances and its storage line — after seven days, and after checking that the storage line is billed separately and has actually stopped.

77 MYSQL, WHEREVER IT IS MANAGED

CUTOVER

20 MIN



LAYER	LEAVING	RISK	REVERSIBLE
DATA	MANAGED MYSQL	HIGH	7 DAYS

Moves MySQL on GTIDs rather than replication slots, cut over behind read_only, a GTID position check and the verification gate.

AWS	RDS for MySQL and Aurora MySQL	the latter carrying a rewind feature whose replacement on your own hardware is an ordinary physical backup plus binary logs.
GOOGLE	Cloud SQL for MySQL	the same shape, with its own external replica flow and its own parameter names.
AZURE	Database for MySQL Flexible Server	again the same, with the parameter change requiring a restart as it does everywhere.

BEFORE YOU START

ACCESS

- Row-based binary logging with full row images enabled, which costs a restart on every managed variant
- The link from Move 53 and the gate from Move 72

SOFTWARE

- A parallel dump and load tool for the initial copy, pinned, and mysql for the position checks
- A written comparison of the source's mode settings against the version you are landing on

PEOPLE

- The application owner for the read-only window, because it is visible as failed writes rather than as slowness

WHY THIS WORKS

MySQL replicates by global transaction identifiers rather than by replication slots, which changes the shape of the cutover: what you check before switching is a position rather than a lag, and what makes the switch safe is putting the source into a read-only state so that no write can land after the position was taken. The transfer itself is a parallel dump and load followed by positioned replication. What is not available is the physical backup tool everybody reaches for first, because it needs access to the data directory and no managed provider gives you that — where a provider mentions it at all, it is a way in rather than a way out.

SWAP

Where the provider's own migration service is available and the database is small, it is a reasonable path for the initial copy. Do not let it own the cutover decision; the gate does that.

DO IT FASTER

Run the parallel dump from a read replica rather than the primary, with enough threads to saturate the link from Move 53 rather than the default.

WATCH OUT

The character set problem is the one that ruins a migration retrospectively. Fix the columns before the copy, not after, because after the copy the truncation has already been carried across.

LEFTOVERS

Binary log retention on the source continues to consume storage after the cutover. Reduce it deliberately once the seven days have passed.

\$2,760/mo	\$310/mo	89%	20 min	3 weeks	7 days
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The managed instance, its read replicas and its binary log retention — after seven days, and after the gate has been run once more against the source before it goes.

NEEDS 47 Local NVMe, and the default StorageClass nobody gave you 48 Encryption at rest, and who holds the key

53 The links you build first: to the cloud, to the office 58 The mirror, then Harbor as the registry of record

60 East-west: security groups into NetworkPolicy 62 Metrics off the managed monitor, and the cardinality budget 72 The verification gate

THE RUNBOOK

- 01 Enable row-based binary logging with full row images. This is a prerequisite and it costs a restart on every managed variant, so it is scheduled well before the cutover in the same way Move 71 schedules its reboot.
- 02 Audit the drift nobody logged. Mode settings that differ from upstream, and columns using the three-byte character set that have been silently truncating four-byte characters for years. The second one is a data problem, not a migration problem, and it is found here or never.
- 03 Take the initial copy with a parallel dump and load, recording the position it corresponds to. A copy without a recorded position cannot be followed by replication and has to be taken again.
- 04 Start positioned replication from that point and watch it catch up. The check that matters is the executed transaction set on both sides, not a lag figure in seconds.
- 05 Run the Move 72 gate against the destination while replication is live, and resolve every mismatch not on the benign list.
- 06 Cut over: set the source read-only, confirm the positions match exactly, run the gate once more, and repoint the application. Twenty minutes, most of it the final gate run.
- 07 Land on a long-term-support release or the openly licensed alternative distribution, under an operator, and keep the source for seven days.

ROLLBACK

Until the source is taken out of read-only and the application is repointed, backing out is a single setting. For the week afterwards, replication is reversed by streaming from the new primary back to the source, and the source's own backups remain available for a restore to a known good point. The point of no return here is the deletion of the source instance in Move 38 and not a moment in this runbook. Keep the recorded positions, because a rollback without them is a full recopy.



78 THE LICENSED DATABASES, AND THE CONTRACT THAT CAN VETO IT

CUTOVER

30 MIN



LAYER	LEAVING	RISK	REVERSIBLE
DATA	MANAGED LICENSED DATABASE ENGINES	HIGH	NO

Prices core-based licensing, mobility rules and partitioning policy on hardware you own, because the contract can veto the migration before any data moves.

AWS	RDS for Oracle and RDS for SQL Server	licence-included or bring-your-own, with the cloud counting cores differently from the way your own hardware will be counted.
GOOGLE	Cloud SQL for SQL Server and Bare Metal Solution for Oracle	the second existing precisely because the licensing made the first impossible.
AZURE	SQL Managed Instance and Oracle Database at Azure	again the licensing shaping the product rather than the technology.

BEFORE YOU START

ACCESS

- Your actual licence agreements, read rather than remembered, including any support-assurance terms
- The core counts and partitioning capability of the machines from Move 04

SOFTWARE

- A licence position model for both the current estate and the target, side by side
- The technical path: a backup and restore, or a replica onto the hypervisor tier from Move 70

PEOPLE

- Counsel and a licensing specialist, because a wrong position here is discovered in an audit rather than in a test

WHY THIS WORKS

The arithmetic here is not the one readers expect, and it can go either way. One major vendor's core factor table applies to hardware you own and does not apply in an authorised cloud, so the same silicon frequently needs half as many processor licences on your own metal. Against that, the same vendor recognises only a named list of partitioning technologies as hard, general-purpose virtualisation is not on it, and a small guest on a large uncapped host is licensed for the whole host. The other major vendor carries a per-processor core minimum and a reassignment rule that decides whether the hypervisor tier from Move 70 may move a guest at all. The technical work is the small half.

SWAP

Where the licence makes ownership impossible, one provider sells dedicated hardware in its own facility specifically for this engine. It is not repatriation and it is frequently the right answer.

DO IT FASTER

Model the licence position before doing any technical work at all. It takes a fortnight and it can end the Move before anyone builds anything.

WATCH OUT

Licence terms change between contract versions. A position that was true at your last renewal may not hold for a new deployment, and the vendor's audit team reads the current terms.

LEFTOVERS

Licence-included managed instances stop charging when deleted; bring-your-own licences you have already bought may or may not transfer, and that is worth more than the hardware.

\$9,400/mo	\$4,100/mo	56%	30 min	4 weeks	4 weeks
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The managed licensed instances, once the licence position is confirmed in writing and the data has been verified on the far side.

NEEDS 47 Local NVMe, and the default StorageClass nobody gave you 48 Encryption at rest, and who holds the key

53 The links you build first: to the cloud, to the office 70 The workloads that must stay virtual machines 72 The verification gate

78 THE LICENSED DATABASES, AND THE CONTRACT THAT CAN VETO IT

7 STEPS · 30 MIN · HIGH RISK

THE RUNBOOK

- 01 Model the licence position for the target before anything else. Core factor, per-processor minimums, and whether your partitioning approach is recognised. This is three of the four weeks and it is the part that decides whether the Move happens.
- 02 Check the mobility rule. One vendor restricts how often a licence may be reassigned between physical machines unless a support-assurance agreement is in place, and that rule governs whether live migration in Move 70 is permitted for these guests.
- 03 Get the position in writing from the vendor or a reseller with authority, on the specific configuration. A verbal reassurance about core factors is not a defence.
- 04 Pin the guests if the licensing requires it. Where general-purpose virtualisation is not recognised as hard partitioning, the guest is licensed against the whole host, and the answer is a dedicated host with a matching core count rather than a clever configuration.
- 05 Do the technical work: a backup and restore, or an availability replica onto the hypervisor tier from Move 70, plus Windows nodes for the workloads that stay on Windows. Thirty minutes of cutover per database and it is the easy part.
- 06 Run the Move 72 gate in whatever form the engine supports, and keep the source available until the licence position has survived one audit cycle.
- 07 Accept the honest outcome. Sometimes this estate does not move, and that conclusion, written down with its arithmetic, is a successful result for this Move.

ROLLBACK

This Move is irreversible in the way that matters: a licence position taken on your own hardware cannot be undone by moving back, because the migration itself may have triggered a reassignment the agreement counts. There is no way back from a licence audit. The point of no return is the day the workload starts on your hardware under your own licences. Keep the source database and its backups so the data can be restored, and keep the written licence position permanently, because it is the evidence.



DATA · 30 MIN · HIGH RISK

The managed licensed instances, once the licence position is confirmed in writing and the data has been verified on the far side.

79 REDIS: WHICH HALF OF IT IS ACTUALLY A CACHE

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
DATA	MANAGED REDIS, VALKEY AND MEMCACHED	MEDIUM	IMMEDIATELY

Classifies the key space before moving anything, sends the genuinely ephemeral half to Valkey and hands the durable half to the next Move.

AWS	ElastiCache for Redis, Valkey and Memcached	three engines, one service, and a default eviction policy that is not the stock one.
GOOGLE	Memorystore	the same idea with its own defaults, and its own cluster mode behaviour.
AZURE	Azure Cache for Redis	the same again, with tiers that change which features exist.

BEFORE YOU START

ACCESS

- Read access to the managed instance for a full key-space scan, run in a way that does not block it
- The eviction policy currently configured, which is not the stock default on any of the three

SOFTWARE

- A scanning script that classifies by key pattern and records the time-to-live distribution per pattern
- `valkey-cli` for the destination checks, at a version matching what you will run

PEOPLE

- The owner of each key pattern, because only they can say whether losing those keys is acceptable

WHY THIS WORKS

Most Redis is not a cache. It holds sessions, rate-limit counters, distributed locks and a job queue, and every one of those is durable state wearing a cache's clothing. Moving all of it as though it were ephemeral is how a migration loses everybody's shopping basket. So this Move classifies the key space first, by pattern and by time-to-live distribution, and only the half that is genuinely disposable moves here. The other half — sessions, locks, queues — is handed to Move 81, which treats it as the durable state it is.

SWAP

Where a cache is small and its miss cost is low, moving it with no warm-up is fine. Where a cold cache means a stampede on the database, warm it before switching or switch during a quiet hour.

DO IT FASTER

Do the classification once and keep it. It is the input to Move 81 and it is the document that stops the same argument being had twice.

WATCH OUT

A client library configured for cluster mode against a single node, or the reverse, fails in ways that look like a network problem. Confirm the topology the client expects matches what you built.

LEFTOVERS

The managed cache keeps billing per node-hour until the last client moves, and reserved nodes keep billing past that.

\$1,180/mo	\$60/mo	95%	0 min	1 week	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The cache nodes serving the ephemeral half, once the new endpoint has been serving for a week — but not the instance itself until Move 81 is done.

NEEDS 47 Local NVMe, and the default StorageClass nobody gave you 58 The mirror, then Harbor as the registry of record

60 East-west: security groups into NetworkPolicy 62 Metrics off the managed monitor, and the cardinality budget

THE RUNBOOK

- 01 | Scan the key space incrementally rather than listing it, and classify by pattern. Every estate has between five and twenty real patterns and a long tail of one-offs.
- 02 | Record the time-to-live distribution per pattern. Keys with no expiry are almost never a cache, and that single check finds the durable half faster than any conversation.
- 03 | Get an owner to confirm, per pattern, whether losing those keys during a cutover is acceptable. Where the answer is no, the pattern goes to Move 81.
- 04 | Note the two behaviour changes that are silent at the boundary. Cluster mode routes by key hash, so commands touching multiple keys that worked against a single node will be refused; and the managed eviction default differs from the stock one, so a cache that never evicted may start evicting, or the reverse.
- 05 | Stand up the destination on the openly licensed fork, and say why: the original project is now under a set of licences of which only one is approved as open source, and the fork is under a permissive licence with multi-vendor governance.
- 06 | Move the ephemeral half by pointing the application at the new endpoint and letting the cache fill. There is nothing to copy, which is what makes this Move three days rather than three weeks.
- 07 | Land the memcached estate on stock memcached at the same time; it belongs with the caching layer and it converts trivially.

ROLLBACK

Pointing the application back at the managed endpoint returns everything to where it was, and because only disposable keys moved, nothing needs to be restored. There is no point of no return in this Move. Keep the classification in source control, and keep the managed instance running until Move 81 has dealt with the durable half, because the two halves frequently share a client configuration.



80 BROKERED MESSAGING: JMS, DURABLE SUBSCRIPTIONS, YOUR BROKER

CUTOVER

10 MIN



LAYER	LEAVING	RISK	REVERSIBLE
DATA	MANAGED MESSAGE BROKERS	HIGH	7 DAYS

Rebuilds JMS semantics on a broker you run, because durable subscriptions, message selectors and transacted sessions are what neither a stream nor a table replaces.

AWS	Amazon MQ for ActiveMQ and RabbitMQ	the same brokers you are landing on, which makes this the most direct exit of the three.
GOOGLE	no first-party managed broker	so these readers are already self-hosting or buying from a marketplace vendor, and have the shortest path.
AZURE	Service Bus	JMS 2.0 over AMQP only on the premium tier, with the lower tier restricted to an older profile and a reduced feature set.

BEFORE YOU START

ACCESS

- The broker configuration exported: queues, topics, subscriptions, selectors and dead-letter settings
- A list of durable subscribers and who owns each, because they cannot be double-consumed

SOFTWARE

- The chosen broker under its cluster operator, at a pinned version, with quorum queues rather than mirrored ones
- The shovel or federation mechanism configured, so the drain is gradual rather than a reposit

PEOPLE

- The owner of every durable subscriber, present for the quiescence window

WHY THIS WORKS

A broker is not a stream and it is not a table, and the difference is the set of guarantees the application was written against: durable topic subscriptions that accumulate while a consumer is away, message selectors that filter server-side, transacted sessions that commit a receive and a send together. None of those survives a translation into a log-based system without rewriting the application, so this Move rebuilds them on a broker you run. Which broker follows from what is load-bearing: a message-queue broker where the estate is protocol-driven, and the enterprise messaging broker where the older messaging semantics genuinely matter.

SWAP

Where the estate is entirely queues with no durable topics or selectors, Move 81's substrate is simpler and cheaper, and this Move can be skipped entirely.

DO IT FASTER

Move one queue family at a time, with the shovel running, rather than the whole broker. Each family is an afternoon and the risk is contained.

WATCH OUT

Message ordering guarantees differ between brokers and between queue types on the same broker. An application that depends on ordering needs the guarantee checked rather than assumed.

LEFTOVERS

The managed broker bills per instance-hour and per storage, and a broker with no consumers still bills. Delete it deliberately after the week.

\$1,640/mo	\$120/mo	93%	10 min	2 weeks	7 days
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The managed broker instances and their storage, seven days after the last durable subscriber has moved and stayed moved.

NEEDS 47 Local NVMe, and the default StorageClass nobody gave you 53 The links you build first: to the cloud, to the office

58 The mirror, then Harbor as the registry of record 60 East-west: security groups into NetworkPolicy

62 Metrics off the managed monitor, and the cardinality budget

THE RUNBOOK

- 01 Establish which semantics are actually load-bearing. Durable topic subscriptions, selectors and transacted sessions each decide which broker you land on, and most estates use fewer of them than the configuration suggests.
- 02 Stand up the destination under its operator at a pinned version, with quorum queues in place of the older mirrored ones. The mirrored form is superseded and behaves worse under partition, so it is not carried across.
- 03 Recreate the topology: queues, topics, durable subscriptions, selectors, dead-letter destinations and their retry counts. Export first and diff afterwards, because a subscription that is silently missing produces messages that vanish.
- 04 Set up the shovel or federation link between old and new so that messages drain across rather than being cut over in one step. This is what turns a big-bang repoint into something you can pause.
- 05 Move producers first and let consumers drain the old broker. In-flight messages are never migrated by hand; they are consumed where they are.
- 06 Quiesce the durable subscribers for the window. This is the ten minutes: a durable subscription cannot be double-consumed the way a queue can, so its consumer stops on one side before it starts on the other.
- 07 Keep the source broker for seven days with its topology intact, so a subscriber that turns out to have been missed has somewhere to go back to.

ROLLBACK

For seven days the source broker still has its topology, so reverting is repointing producers back and letting the new side drain. Messages already consumed on the new side are not restored to the old one, which is why producers move before consumers rather than after. The point of no return is the deletion of the source broker with its durable subscriptions, which is Part VII. Keep the exported topology so it can be restored onto a rebuilt broker if the migration is aborted.

81 THE QUEUES COME HOME, THE FAN-OUT DOES NOT

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
DATA	MANAGED QUEUES AND TOPIC FAN-OUT	HIGH	14 DAYS

Brings managed queues and internal topic fan-out home to NATS JetStream or Postgres by drain-and-switch, and keeps renting mobile push and SMS.

AWS	SQS and SNS	separate queue and topic products, with a visibility timeout and a redrive policy that are load-bearing and undocumented in most codebases.
GOOGLE	Pub/Sub	one subscription model with acknowledgement deadlines, so the semantics map differently rather than directly.
AZURE	Storage queues and Service Bus queues	two products with different durability, and applications that frequently use both without anyone noticing.

BEFORE YOU START

ACCESS

- Every queue and topic listed with its settings, including the redrive policies nobody configured deliberately
- The durable half of the key space handed over by Move 79

SOFTWARE

- The chosen substrate: a permissively licensed streaming server for high volume, or the database for low
- Consumers made idempotent, because at-least-once means at-least-once on both sides

PEOPLE

- The owner of each producer and each consumer, because the switch is a code change on both

WHY THIS WORKS

The semantics are the migration. Visibility timeout and its extension, at-least-once delivery, the retention ceiling, message grouping and the redrive policy are all load-bearing in real applications and documented in almost none of them, and the three clouds do not agree on any of them. So the audit comes first. After that the mechanics are simple: producers switch to the new substrate and consumers drain the old queue until it is empty. In-flight messages are never migrated, which is exactly why the reversibility window is the source queue's own retention period rather than a number this book chose.

SWAP

For a queue doing fewer than a few messages a second, a table in the database with a locking select is genuinely the right answer and removes a whole system from the estate.

DO IT FASTER

Move the queues with a single producer and a single consumer first. They are an hour each and they build the pattern.

WATCH OUT

A dead-letter queue on the old side will keep receiving failures from consumers that have not moved. Check it is empty before deleting anything, because it is where the messages you lost are sitting.

LEFTOVERS

Queue services bill per request, so a queue nobody uses costs almost nothing and is easy to forget. Move 120 will find them in the billing data.

\$840/mo	\$50/mo	94%	0 min	2 weeks	14 days
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The queue and topic resources, once both dead-letter queues are empty and the source retention window has passed. Push and SMS stay bought.

NEEDS 58 The mirror, then Harbor as the registry of record 60 East-west: security groups into NetworkPolicy

62 Metrics off the managed monitor, and the cardinality budget 73 Postgres onto your own disks 79 Redis: which half of it is actually a cache

80 Brokered messaging: JMS, durable subscriptions, your broker

THE RUNBOOK

- 01 Audit the semantics per queue: visibility timeout and whether the consumer extends it, retention, ordering or grouping, and the redrive policy. Write down which of these the application actually depends on rather than which are configured.
- 02 Pick the substrate per workload. High-volume, high-fan-out work goes to the streaming server; low-volume work goes into the database you already run, which is a queue with transactions and one fewer system to operate.
- 03 Take the durable half of the key space from Move 79 — sessions, locks, counters, the job queue — and place each one deliberately. A job queue in a cache was always a queue in the wrong place.
- 04 Make every consumer idempotent before anything moves. Both sides deliver at least once and a migration doubles the opportunities to prove it.
- 05 Switch producers to the new substrate and leave consumers running against both. The old queue drains and then stays empty, which is the signal that the switch is complete.
- 06 Keep mobile push and SMS bought. Repatriating those means becoming a messaging provider badly, and the certification and deliverability work is a business rather than a migration.
- 07 Wait out the source's retention period before deleting anything, because that period is the reversibility window this Move claims.

ROLLBACK

For the length of the source's retention, producers are switched back and consumers drain the new substrate instead. Messages already consumed are not restored to the old queue, so the order — producers first, consumers last — is what makes the rollback clean. The point of no return is the expiry of that retention window, after which an unconsumed message on the old side is gone. Verify the dead-letter queues on both sides are empty before calling the Move finished.



82 THE EVENT BUS, AND THE CONTROL-PLANE EVENTS THAT STOP ARRIVING

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
DATA	MANAGED EVENT BUSES AND PROVIDER CONTROL-PLANE EVENTS	MEDIUM	7 DAYS

Rebuilds custom buses, schema registry, archive and replay on a broker you own, and names the control-plane events that stop arriving with the account.

AWS	EventBridge	custom buses, cross-account rules, content-based filtering, a schema registry and an event archive with replay.
GOOGLE	Eventarc with Pub/Sub	the same idea assembled from two services, with the filtering expressed differently.
AZURE	Event Grid	the same again, with system topics carrying the provider's own resource events.

BEFORE YOU START

ACCESS

- Every rule, bus and subscription exported, including the cross-account ones that nobody in your account can see
- The substrate from Move 81, running and carrying real traffic

SOFTWARE

- A self-hosted schema registry, and a retained stream configured for the archive and replay use
- Bucket notifications on your own object store, which cover the storage half of the control-plane events

PEOPLE

- The owner of each rule, because a rule with no owner is either critical or dead and only they can say which

WHY THIS WORKS

There are two halves here and only one of them has a replacement. The custom half — your own buses, your own rules, your own schemas — moves onto the substrate from Move 81: rules become subjects, the registry becomes a self-hosted one, and archive and replay become a retained stream rather than a product feature. The other half is the provider's own control-plane events: object-created notifications, database events, resource-lifecycle events, audit routing. Those stop arriving because the resources that emit them are the ones being deleted, and pretending otherwise is how a migration silently loses an automation nobody remembered.

SWAP

For a handful of rules, a direct call from the producer to the consumer is simpler than a bus and removes a system. Buses earn their place when the number of consumers per event is genuinely unknown.

DO IT FASTER

Convert the rules with the most consumers first. They prove the substrate and they are the ones where the bus is actually earning its keep.

WATCH OUT

Cross-account rules are invisible from the receiving side. Ask the other account's owner rather than reading your own configuration, or the rule will be discovered when it stops firing.

LEFTOVERS

Event archives keep billing for their retention after the bus stops receiving. Reduce the retention explicitly rather than waiting for it to lapse.

\$460/mo	\$30/mo	93%	0 min	2 weeks	7 days
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Custom buses and their rules after seven days. The provider's system events stop on their own when the resources behind them go.

NEEDS 49 Ceph under Rook: block, shared filesystem and the object store 58 The mirror, then Harbor as the registry of record

60 East-west: security groups into NetworkPolicy 62 Metrics off the managed monitor, and the cardinality budget

81 The queues come home, the fan-out does not

82 THE EVENT BUS, AND THE CONTROL-PLANE EVENTS THAT STOP ARRIVING

7 STEPS · 0 MIN · MEDIUM RISK

THE RUNBOOK

- 01 | Separate the two halves before doing anything. Custom application events move; provider control-plane events do not, and the second list is the one that needs decisions rather than work.
- 02 | Convert each rule into a subject or topic on the substrate from Move 81, keeping the content-based filtering where the substrate supports it and moving it into the consumer where it does not.
- 03 | Stand up a schema registry you run, and migrate the schemas. A registry is cheap to run and expensive to be without once producers and consumers are versioned independently.
- 04 | Replace archive and replay with a retained stream. This is a capacity decision rather than a feature: retention costs storage, and the retention you choose is the replay window you have.
- 05 | Wire your own object store's bucket notifications for the storage half of the control-plane events. That covers object-created and object-removed, which is most of what people actually used.
- 06 | Use cluster events plus the alerting path from Move 62 for the resource-lifecycle half. It is not the same thing and it covers the same need for anything running on your own hardware.
- 07 | Write down what is genuinely lost, with the automation it fed. An honest list of three lost event types beats an assumption that everything was covered.

ROLLBACK

For seven days both paths exist, so reverting is repointing producers back at the managed bus. Events consumed on the new side are not restored to the old archive, which is why the retained stream is configured before producers move. The point of no return is the deletion of the archive, which is Part VII. Keep the exported rules in source control so the topology can be restored, and keep the lost-events list where Move 119's auditor can read it.

83 KAFKA OFF THE MANAGED BROKER

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
DATA	MANAGED KAFKA	HIGH	7 DAYS

Deletes the Kafka clusters that were never Kafka, then mirrors the rest onto Strimzi with a written per-topic decision to replay or to skip.

AWS	MSK	an identity-backed authentication mechanism on every client, so every client configuration changes.
GOOGLE	Managed Service for Apache Kafka	a standard authentication profile, which is the closest of the three to what you will run.
AZURE	Event Hubs with its Kafka endpoint	a protocol endpoint on a product that is not Kafka underneath and does not support log compaction, which frequently decides the answer for this reader.

BEFORE YOU START

ACCESS

- Topic-level metrics: partitions, retention, throughput and consumer group lag over a representative fortnight
- The link from Move 53, sized for the mirroring throughput rather than the steady state

SOFTWARE

- The Kafka operator at a pinned version, on the current coordination architecture rather than the removed one
- The mirroring tool configured per topic, with a written decision to replay or to skip for each

PEOPLE

- The owner of every consumer group, because offsets are the part that does not travel cleanly

WHY THIS WORKS

A large share of managed Kafka clusters are not doing anything Kafka is for. So this Move opens with a test rather than a migration: partition count, retention, real throughput, whether anybody replays, and whether ordering is load-bearing. Clusters that fail that test become a queue on the substrate from Move 81 and are deleted, which is the cheapest outcome available. For what genuinely stays Kafka, the landing is the operator on the current architecture without the old coordination service, and the mirroring tool carries the topics — with one honest caveat that governs everything else.

SWAP

Where the source is a protocol endpoint on a product that is not Kafka, log compaction does not exist there, so any compacted topic must be rebuilt rather than mirrored. That usually decides the whole migration for that reader.

DO IT FASTER

Move the queue-shaped topics first with no history. They are quick, they build confidence, and they shrink the cluster you have to mirror.

WATCH OUT

Partition counts are not freely changeable after the fact for keyed topics, because changing them changes which partition a key lands in. Choose the destination partition count deliberately.

LEFTOVERS

The managed cluster bills per broker-hour and per gigabyte of storage, and storage keeps billing for the retention period after production stops.

\$3,180/mo	\$240/mo	92%	0 min	3 weeks	7 days
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The managed brokers and their storage, seven days after the last consumer group moved and a replay from your own cluster has been verified.

NEEDS 47 Local NVMe, and the default StorageClass nobody gave you 49 Ceph under Rook: block, shared filesystem and the object store

58 The mirror, then Harbor as the registry of record 60 East-west: security groups into NetworkPolicy

62 Metrics off the managed monitor, and the cardinality budget 81 The queues come home, the fan-out does not

82 The event bus, and the control-plane events that stop arriving

THE RUNBOOK

- 01 Run the test first. Partitions, retention, throughput, replay, ordering. Write the answer per cluster and delete the ones that were never Kafka, moving them to the substrate from Move 81.
- 02 Stand up the destination under its operator at a pinned version, on the current coordination architecture. The separate coordination service was removed from Kafka in the 4.0 release and there is no reason to reproduce it.
- 03 State the caveat before configuring anything: consumer offsets are cluster-local and the mirroring tool's translation of them is approximate. Consumers are made idempotent before a single topic is mirrored, and transactional producers do not survive the boundary at all.
- 04 Decide per topic whether history is replayed or skipped. A topic used as a queue does not need its history; a topic used as a log does, and mirroring a month of it is a schedule item rather than a footnote.
- 05 Change every client's authentication configuration. This differs on each cloud — an identity-backed mechanism on one, a standard profile on another, and a protocol endpoint on the third — and it is a change to every producer and consumer.
- 06 Mirror, then move consumers group by group, watching lag on both sides. Producers move last for a log-shaped topic and first for a queue-shaped one, which is the opposite of Move 81 and worth stating explicitly.
- 07 Keep the source for seven days with the mirror still running in case a consumer group was missed.

ROLLBACK

For seven days the mirror runs and the source retains its data, so a consumer group can be pointed back and will find its messages. Offsets will not be exact, which is why idempotent consumers are a prerequisite rather than a recommendation. The point of no return is the expiry of the source's retention, after which the history exists only on your side. Verify a replay from the new cluster before that date rather than assuming the mirrored data is complete, because after it the only way to restore a topic is from whatever produced it in the first place.

84 APPLICATION BUCKETS, AND THE OBJECT API YOUR CODE ASSUMES

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
DATA	MANAGED OBJECT STORAGE (APPLICATION BUCKETS)	HIGH	30 DAYS

Mirrors application buckets onto the cluster's own object store behind a dual-write window, then audits every client for the object-store behaviour it assumes.

AWS	S3	multipart entity tags that are not a plain digest, so verification needs matching part sizes to mean anything.
GOOGLE	Cloud Storage	a separate checksum alongside the digest, with the digest absent on composed objects.
AZURE	Blob Storage	a block composition model, so the checksum you compare is not the one you uploaded.

BEFORE YOU START

ACCESS

- Read access to every bucket, including the ones only a pipeline uses, and write access to the destination
- The object store from Move 49, made durable by Move 54, with capacity checked against the real total

SOFTWARE

- rc1one or mc at a pinned version, with the checksum mode chosen per source cloud rather than defaulted
- A comparison script written once with three modes, or three times, because the verification differs per provider

PEOPLE

- The owner of every application that reads or writes a bucket, for the client audit

WHY THIS WORKS

Copying objects is the easy half and it still has three traps: the entity tag on a multipart object is not a digest of the content, so a naive comparison reports every large object as different; non-current versions and delete markers are not copied by any sync tool; and storage class, access policy, immutability, lifecycle, cross-origin rules and event wiring are all configuration that has to be rebuilt by hand. The larger half is the client audit, because application code accumulates assumptions about the object store it was written against, and those assumptions are invisible until they are wrong.

SWAP

Where a bucket is only read by one application and is small, a straight copy during a short outage is far simpler than a dual-write window and finishes in an hour.

DO IT FASTER

Copy the cold data weeks in advance and the hot data at the end. Most buckets are mostly cold and the final delta is then measured in gigabytes rather than terabytes.

WATCH OUT

Presigned links generated before the switch keep pointing at the old store until they expire. Their lifetime is the minimum length of your dual-read window, and somebody has always set one to a year.

LEFTOVERS

Non-current versions and delete markers keep billing after the current objects are deleted. A lifecycle rule that expires them is required and takes at least a day to run.

\$2,940/mo	\$210/mo	93%	0 min	4 weeks	30 days
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Application buckets and their versions after thirty days, and their lifecycle and replication rules once the comparison passes on the final day. Archive buckets stay, per Move 85.

NEEDS 49 Ceph under Rook: block, shared filesystem and the object store 54 The object store you now own, and how it loses data

55 Workload identity without the cloud's 60 East-west: security groups into NetworkPolicy

62 Metrics off the managed monitor, and the cardinality budget

84 APPLICATION BUCKETS, AND THE OBJECT API YOUR CODE ASSUMES

7 STEPS · 0 MIN · HIGH RISK

THE RUNBOOK

- 01 | Inventory the buckets and their real sizes, including non-current versions, which are frequently larger than the current data and are not copied by any tool.
- 02 | Write the comparison properly. A multipart entity tag is a digest of digests, so it only matches when the part size matches; one provider gives a separate checksum that is absent on composed objects; and one composes blocks so that the value you compare is not the value you uploaded. Choose the mode per source.
- 03 | Rebuild the configuration by hand: storage class, access policy, immutability, lifecycle rules, cross-origin rules and event wiring. None of it is copied by a sync tool and all of it is load-bearing somewhere.
- 04 | Do the client audit, which is most of the four weeks. Addressing style, signature clock skew, presigned link lifetimes, assumptions about part sizes, and conditional writes. Each of these is a code change or a configuration change, and each is silent until it fails.
- 05 | Dual-write for a period: the application writes to both stores and reads from the old one. This is what makes the thirty-day window real rather than nominal.
- 06 | Run the initial copy, then repeated incremental copies until the delta is small, then switch reads. Verify with the comparison script rather than with a file count.
- 07 | Keep both stores for thirty days, with the dual-write running, before deleting anything on either side.

ROLLBACK

For thirty days the old bucket still holds everything and the application can be pointed back at it, with anything written in the interim restored by copying the delta in the other direction. That reverse copy is why the dual-write exists. The point of no return is the deletion of the source bucket in Move 38, against a signed data-owner approval. Verify the comparison script passes on the last day before that deletion rather than trusting a copy that finished weeks earlier.

 DATA · 0 MIN · HIGH RISK

Application buckets and their versions after thirty days, and their lifecycle and replication rules once the comparison passes on the final day. Archive buckets stay, per Move 85.

85 THE DATA THAT STAYS: DEEP ARCHIVE AND THE WAREHOUSE

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
DATA	—	LOW	IMMEDIATELY

Keeps deep archive and the warehouse on the provider deliberately, and hardens the one account that holds the archive and the third backup copy.

AWS	S3 Glacier Deep Archive and Redshift	a one-hundred-and-eighty-day minimum storage term on the deepest tier, which changes the retrieval arithmetic.
GOOGLE	Archive storage and BigQuery	a three-hundred-and-sixty-five-day minimum on the archive class, the longest of the three.
AZURE	Archive tier and Synapse	a one-hundred-and-eighty-day minimum, with rehydration priorities that change both the time and the price.

BEFORE YOU START

ACCESS

- The archive inventory with its storage classes and the minimum-term charges that apply to each
- The warehouse's real query cost, so the decision to keep it is priced rather than assumed

SOFTWARE

- A separate credential set for the retained account, outside the identity path built in Move 55
- Immutability, multi-factor deletion and a budget alarm on the retained account

PEOPLE

- Whoever owns retention policy, because this account is now the long-term record

WHY THIS WORKS

This is an honest no with a runbook attached. Deep archive is cheap to hold and expensive to move, and repatriating it means a one-off egress bill plus a media-refresh obligation you do not want. The managed warehouse means somebody else owns a query engine, a scheduler and a storage tier, and owning those three is a team rather than a Move. So both stay, deliberately, and the work here is different: hardening the one account that will outlive the migration, because it now holds the archive and the third copy of your backups, and it is no longer protected by everything else you were running.

SWAP

Where the archive is small and cold enough to fit on your own store under immutability, bringing it home is fine and removes an account. Price the one-off egress before deciding.

DO IT FASTER

Split the archive by whether it must be retrievable in hours or in days. The second half can go one tier deeper and is usually most of the volume.

WATCH OUT

A retained account with no active workloads is a security blind spot. It stops appearing in dashboards, its credentials stop being rotated, and its budget alarm is the only thing anybody looks at.

LEFTOVERS

The warehouse's scheduled queries and its ingestion pipelines keep billing whether or not anyone reads the results. Review them before deciding what the warehouse actually costs.

\$1,860/mo	\$1,860/mo	0%	0 min	1 week	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing here, by design. What you can turn off is the warehouse's scheduled queries that nobody reads, which is usually a third of its cost.

NEEDS 84 Application buckets, and the object API your code assumes

85 THE DATA THAT STAYS: DEEP ARCHIVE AND THE WAREHOUSE

6 STEPS · 0 MIN · LOW RISK

THE RUNBOOK

- 01 Price the retrieval properly, with early-deletion charges rather than list retrieval rates. The deepest tiers carry a minimum storage term of six months on two clouds and a full year on the third, and deleting early bills the remainder anyway.
- 02 Decide, and write the decision down: the archive stays, the warehouse stays, and the reason for each is arithmetic rather than inertia. Move 122 needs this in writing so it does not try to close the account.
- 03 Bring the lake substrate home. Open table formats on the object store from Move 84, with a query engine in front, is the part that does move, and it is where most of the ongoing query cost actually lives.
- 04 Harden the retained account. Separate credentials outside the identity path from Move 55, immutability on the archive buckets, multi-factor deletion, and a budget alarm that alerts a human rather than a mailbox.
- 05 Rehearse a retrieval end to end and time it, including the rehydration wait. A retrieval nobody has performed is a retention policy rather than a capability.
- 06 Put the account into Move 119's evidence pack and Move 120's cost model as a permanent line, so it is never a surprise on a future invoice.

ROLLBACK

Nothing here is a one-way door: the archive can be repatriated later at the price of the egress bill, and the warehouse can be migrated when someone is willing to own a query engine. The point of no return does not exist in this Move. Keep the retrieval drill timings and the hardening configuration in source control, so the account's intended state can be restored and audited when nobody has looked at it for a year.

 DATA · 0 MIN · LOW RISK

Nothing here, by design. What you can turn off is the warehouse's scheduled queries that nobody reads, which is usually a third of its cost.

86 THE BATCH ESTATE: WHO SCHEDULES THE JOBS

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
 DATA	MANAGED WORKFLOW ORCHESTRATION AND MANAGED SPARK	 MEDIUM	30 DAYS

Moves the DAGs and the Spark jobs onto schedulers you run, because the lake substrate came home and nothing yet writes to it on a schedule.

AWS	Managed Workflows for Apache Airflow with EMR	the same open-source scheduler on both sides, so the work is the hooks rather than the graphs.
GOOGLE	Cloud Composer with Dataproc	the same again, with the same consequence.
AZURE	Data Factory with Synapse Spark	a low-code designer with no open-source equivalent, so this estate is a rewrite priced as one rather than a migration.

BEFORE YOU START

ACCESS

- The scheduler's metadata database, exported, and the graphs in source control rather than only in the console
- The lake substrate from Move 85 and the object store from Move 84

SOFTWARE

- The scheduler at a pinned version with the cluster executor, and its metadata on a database cluster of its own
- The Spark operator reading and writing the object store, at a pinned version

PEOPLE

- The owner of every pipeline, because provider-specific operators are code changes in their repositories

WHY THIS WORKS

The scheduler repatriates well on two of the three clouds because it is literally the same software on both sides, which means the graphs themselves are not the work. The work is everything around them: the provider-specific hooks and operators embedded in the tasks, the metadata database, the executor, and the secrets backend. Each of those has a clean landing — a database cluster of its own, the cluster executor on the fixed fleet from Move 69, and the secrets path from Move 56. The third cloud is genuinely different, because a designer-based pipeline has nothing to port, and calling that a migration rather than a rewrite is how the estimate goes wrong by a factor of three.

SWAP

For a handful of jobs, cluster cron jobs are simpler than a scheduler and remove a component. A scheduler earns its place when dependencies between jobs are real.

DO IT FASTER

Move the graphs that use only generic operators first. They are usually half the estate and they move in a day.

WATCH OUT

The metadata database grows and is rarely pruned. Import a cleaned copy rather than a full one, or the first thing your new scheduler does is inherit five years of task instances.

LEFTOVERS

The managed scheduler bills per environment-hour whether or not anything runs, and the managed Spark service bills per cluster. Both keep running through the dual period.

\$3,410/mo	\$290/mo	91%	0 min	4 weeks	30 days
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The managed scheduler environment and the managed Spark clusters, thirty days after both sides produced the same outputs.

NEEDS 56 Secrets: External Secrets as the dated bridge 58 The mirror, then Harbor as the registry of record

60 East-west: security groups into NetworkPolicy 62 Metrics off the managed monitor, and the cardinality budget

65 Argo CD, and the two infrastructure-as-code estates 69 The fixed fleet: priority, preemption, NUMA and the GPUs

73 Postgres onto your own disks 84 Application buckets, and the object API your code assumes

85 The data that stays: deep archive and the warehouse

THE RUNBOOK

- 01 | Inventory the tasks by the operators they use. Anything using a provider-specific hook or operator is a code change; anything using a generic one moves untouched. That split is the whole estimate.
- 02 | Stand up the scheduler at a pinned version with its metadata on its own database cluster, and the cluster executor pointed at the fixed fleet from Move 69. Batch runs at the lower priority tier defined there, which is the point of having tiers.
- 03 | Move the secrets backend to Move 56's path rather than carrying the provider's. A scheduler holding credentials in its own database is a finding waiting for Move 118.
- 04 | Rewrite the provider-specific operators. This is ordinary work and it goes at a predictable rate once the first few are done.
- 05 | Land the Spark jobs on the operator, reading and writing the object store from Move 84. Sizing is now yours: executor count, memory and shuffle space are a capacity decision rather than a cluster type.
- 06 | Where the source was a designer rather than code, price a rewrite and schedule it as one. Do not attempt to translate a visual pipeline into a graph automatically.
- 07 | Run both schedulers for thirty days with the new one writing to a separate output path, and compare results before switching consumers.

ROLLBACK

For thirty days the managed scheduler still has the graphs and can be re-enabled, and because the new side writes to a separate output path nothing has been overwritten. Reverting is disabling the new schedules and re-enabling the old ones. The point of no return is deleting the managed environment, which is Part VII. Keep the exported metadata so a scheduler can be restored with its history rather than starting blank.



87 SEARCH OFF THE MANAGED CLUSTER

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
DATA	MANAGED SEARCH CLUSTERS	MEDIUM	7 DAYS

Restores a snapshot into a self-hosted OpenSearch cluster, dual-writes across the gap and cuts the read alias, because the index is rebuildable.

AWS	OpenSearch Service	a provisioned domain exits cleanly through a manual snapshot repository; the serverless collection type has no snapshot API at all.
GOOGLE	Elasticsearch through the marketplace or self-managed	usually already yours, so the exit is a network change rather than a migration.
AZURE	Azure AI Search	no backup or restore of any kind, so this estate exits by reindexing from the source of truth.

BEFORE YOU START

ACCESS

- Permission to register a manual snapshot repository on the source domain, and a bucket both sides can read
- The source of truth for every index, because reindexing is the fallback and sometimes the only path

SOFTWARE

- OpenSearch at a pinned version, sized for your hardware rather than for the source's instance types
- A dual-write path in the application, or an indexer that can write to both clusters

PEOPLE

- The owner of each index, to say whether a rebuild is acceptable and how long it takes

WHY THIS WORKS

Search indexes are derived data, which makes this the cleanest exit in the Part: the worst case is a rebuild from the source of truth, and the best case is a restore. For a provisioned domain the path is short — register a manual snapshot repository on the source, restore into a self-hosted cluster reading the same repository, dual-write across the gap, then move the read alias. The automated snapshots cannot be restored off the domain at all, so registering a manual repository is a mandatory first step rather than a convenience. Two service variants have no snapshot path and exit by reindexing instead.

SWAP

Where the index is small and rebuildable in under an hour, skip the snapshot entirely and reindex. It is simpler and it proves the rebuild path works, which you want anyway.

DO IT FASTER

Restore into the second cluster from Move 51 first to validate the version lineage. A blocked restore discovered there costs an hour rather than a window.

WATCH OUT

Heap above the compressed-pointer threshold makes a node slower rather than faster, and it is the single most common sizing error on a self-hosted search cluster.

LEFTOVERS

The managed domain bills per instance-hour and per gigabyte of storage, and a domain with no traffic still bills both.

\$2,240/mo	\$180/mo	92%	0 min	2 weeks	7 days
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The managed domain and its storage, seven days after the alias moved and the dual-write stopped.

NEEDS 49 Ceph under Rook: block, shared filesystem and the object store 54 The object store you now own, and how it loses data

58 The mirror, then Harbor as the registry of record 60 East-west: security groups into NetworkPolicy

62 Metrics off the managed monitor, and the cardinality budget 84 Application buckets, and the object API your code assumes

THE RUNBOOK

- 01 Establish which variant you have. A provisioned domain restores from a manual repository; a serverless collection and one provider's search service have no restore path and must be reindexed from the source of truth.
- 02 Register a manual snapshot repository on the source and take a snapshot. The automated ones are not restorable outside the domain, which is the single most important fact in this Move.
- 03 Size the destination for your hardware rather than copying the source's shape. Shard count and shard size chosen against your machines, dedicated master nodes, and heap at half of memory and below the threshold where compressed pointers stop working.
- 04 Restore into the self-hosted cluster from the same repository, remembering that restores are constrained by the source's version lineage and that a version gap can block the restore entirely.
- 05 Dual-write from the application or the indexer, so both clusters stay current while you compare them.
- 06 Compare result sets on real queries, not document counts. Relevance can differ across versions and configurations, and a search that returns the same number of different documents is a regression nobody will attribute to this Move.
- 07 Move the read alias, watch for a week, then stop the dual-write.

ROLLBACK

For seven days the alias is moved back and the managed domain is still current because the dual-write is still running. That is the whole rollback and it takes seconds. The point of no return is stopping the dual-write, after which the managed side goes stale — and even then the index can be restored by rebuilding from the source of truth, which is why this Move is Medium risk rather than High.



88 DOCUMENT AND WIDE-COLUMN STORES: THE HARDEST EXIT

CUTOVER

30 MIN



LAYER	LEAVING	RISK	REVERSIBLE
DATA	MANAGED DOCUMENT, KEY-VALUE AND WIDE-COLUMN STORES	HIGH	30 DAYS

Decides per store whether the table moves to Cassandra, collapses into Postgres, or stays on the bill below a stated monthly threshold.

AWS	DynamoDB	export to object storage and import back round-trips the items and builds secondary indexes from a specification, but only into a new table, and carries no expiry, stream, capacity or tag configuration.
GOOGLE	Firestore and Bigtable	the document store exports in a format importable only back into itself and carrying no index definitions; the wide-column store exits through its compatibility interface.
AZURE	Cosmos DB	multiple APIs, each with its own export path and its own set of things that do not travel.

BEFORE YOU START

ACCESS

- Export permissions, and the throughput data that tells you what shape the workload actually is
- A stated monthly threshold below which a store stays, agreed with whoever owns the budget

SOFTWARE

- A repository interface in front of every store that stays, so the decision can be revisited without a rewrite
- The destination chosen deliberately: the permissively licensed wide-column database rather than the source-available alternative

PEOPLE

- The owner of each store, because collapsing into a relational schema is a data-model decision

WHY THIS WORKS

This is the hardest exit in the book and the honest answer is a decision procedure with a threshold rather than a slogan. Three outcomes are legitimate: the table moves to a wide-column database you run, the table collapses into the relational database you already operate, or the table stays on the bill because it is below a monthly figure where migrating it costs more than paying for it for three years. The export limitations are real and specific, and the harder problem is not the data at all — it is that on-demand capacity meant nobody ever had to know the workload shape you must now size hardware against.

SWAP

Keeping a store on the bill is a legitimate outcome and should be written down as a decision with its threshold, not as an unfinished task.

DO IT FASTER

Do the threshold arithmetic before any technical work. It frequently removes two thirds of the stores from the programme in an afternoon.

WATCH OUT

A single hot partition that the managed service absorbed invisibly becomes a hot node on your own hardware. Find it in the measurement step or find it in production.

LEFTOVERS

Streams, expiry jobs and on-demand backups on the source all keep billing after the table stops being read.

\$4,620/mo	\$740/mo	84%	30 min	6 weeks	30 days
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The tables that moved, their streams and their on-demand backups, thirty days after the dual-write stops. The ones below the threshold stay, deliberately.

NEEDS 72 The verification gate 73 Postgres onto your own disks 84 Application buckets, and the object API your code assumes

THE RUNBOOK

- 01 Measure the workload shape first: read and write rates, item sizes, access patterns and how much of the traffic is a single hot partition. On-demand capacity hid all of this and hardware sizing needs all of it.
- 02 Apply the threshold. Below an agreed monthly figure, the store stays and a repository interface goes in front of it, so the decision can be revisited later without touching application code. Record which stores fall this side of the line.
- 03 For stores that move, read the export limitations rather than assuming a round trip. One export can only create a new table and carries no expiry, stream, capacity or tag configuration; another is importable only back into its own service and carries no index definitions; the wide-column case exits through a compatibility interface.
- 04 Rebuild everything the export does not carry: secondary indexes, expiry settings, stream configuration and capacity settings. Each is a separate step and each is silently absent otherwise.
- 05 Choose the landing deliberately. Where a wide-column store is genuinely required, the destination is the permissively licensed project rather than the alternative, which since a recent release ships only under a source-available licence with a capacity-limited free tier.
- 06 Where the data model is really relational — and it frequently is — collapse it into the database you already run. Fewer systems is worth more than a faithful reproduction of a data model nobody chose deliberately.
- 07 Dual-write, compare, then cut over with a thirty-minute window and keep the source for thirty days.

ROLLBACK

For thirty days the source table still exists and still holds the data, so reverting is repointing the repository interface. Anything written on the new side in the interim must be restored to the source by replaying it, which is why the dual-write runs for the whole window rather than only until the cutover. The point of no return is the deletion of the source table in Move 38, against a signed approval.

89 THE SERVERLESS ESTATE IN FOUR PILES

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
DATA	MANAGED FUNCTION-AS-A-SERVICE AND ITS SCHEDULERS	HIGH	30 DAYS

Triages the function estate into cron, provider-event glue, HTTP handlers that should have been Deployments, and the load-bearing few that need real design.

AWS	Lambda with EventBridge Scheduler	the largest of the three estates in practice, and the one with the most provider-event glue in it.
GOOGLE	Cloud Functions with Cloud Scheduler	which land on a serverless layer over Kubernetes rather than being rewritten.
AZURE	Azure Functions with Timer triggers	an open-source runtime that runs in a container, so this estate has a lift path the other two do not.

BEFORE YOU START

ACCESS

- An inventory of every function with its trigger, its invocation rate and its owner
- The event substrate from Moves 81 and 82, carrying the triggers that survived

SOFTWARE

- An event-driven autoscaler at a pinned version for the per-message pile
- A measured cold-start figure on your own fleet: poll interval plus image pull plus process start

PEOPLE

- The owner of each function, because a third of the estate is usually deletable and only they can say so

WHY THIS WORKS

A few hundred functions sounds like a migration and is mostly a triage. Four piles: things that run on a schedule, which become cron jobs; glue reacting to provider events, which Move 82 already answered or declared lost; HTTP handlers that should always have been ordinary services, which become deployments; and the load-bearing few that genuinely need per-message scaling. Only the fourth pile needs design, and it needs it here rather than earlier because scale-from-zero on a fixed fleet is a different problem from scale-from-zero on an autoscaler, and Move 69 is where the fleet became fixed.

SWAP

Where a function estate is small and genuinely bursty, keeping it rented is defensible and belongs on the Move 116 list of things you deliberately keep.

DO IT FASTER

Delete first. Most estates have functions that have not been invoked in a year, and deleting them is faster than migrating them.

WATCH OUT

The execution time limit enforced replay-safety. Once it is gone, a handler that runs for an hour and half-completes is a new class of failure, and the code was never written to survive it.

LEFTOVERS

Function versions, aliases and their logs keep billing after invocations stop, and log retention in particular runs for months.

\$1,720/mo	\$90/mo	95%	0 min	6 weeks	30 days
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Function versions, aliases and their schedulers after thirty days, and their log retention once it has been set down deliberately.

NEEDS 65 Argo CD, and the two infrastructure-as-code estates 66 Infrastructure code that only compiles for one cloud

67 The first stateless service, end to end 69 The fixed fleet: priority, preemption, NUMA and the GPUs

81 The queues come home, the fan-out does not 82 The event bus, and the control-plane events that stop arriving

84 Application buckets, and the object API your code assumes 88 Document and wide-column stores: the hardest exit

THE RUNBOOK

- 01 | Triage first and spend a week on it. Four piles, every function in exactly one, with its owner named. This week determines the size of the other five.
- 02 | Convert the scheduled pile to cron jobs. This is mechanical, it is usually a third of the estate, and it finishes in days.
- 03 | Convert the HTTP pile to ordinary deployments. Most of these were services all along and were written as functions because that was the platform, not because the shape fitted.
- 04 | Check the second pile against Move 82. The glue reacting to provider control-plane events either has a replacement there or was written down as lost, and either way it is not new work here.
- 05 | Design the fourth pile properly with the event-driven autoscaler, and measure the scale-from-zero figure honestly on your own fleet: poll interval, image pull and process start. It is slower than the managed platform and knowing the number is what lets somebody decide whether that matters.
- 06 | State what is lost rather than hiding it: per-invocation isolation, automatic retry with a dead-letter destination, and the wall-clock execution limit that quietly forced every handler to be safe to replay. That last one is the property most likely to be missed.
- 07 | Run both for thirty days with the new side handling a share of the traffic, then stop the old.

ROLLBACK

For thirty days both paths exist and traffic can be shifted back by changing the trigger, with the function versions still deployed. Anything the new side wrote in the interim has to be restored or reconciled by hand, which is why the shift is gradual rather than a switch. The point of no return is the deletion of the function versions, which is Part VII. Keep the triage document, because it is what Move 122 will check against.

90 ORCHESTRATED WORKFLOWS: THE DATED EXCEPTION

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
DATA	—	LOW	IMMEDIATELY

Argo Workflows and Temporal do not replace the provider service integrations, so low-frequency state machines keep running against a budgeted and dated rewrite.

AWS	Step Functions	a few dozen direct service integrations, which is the smallest of the three catalogues and still not nothing.
GOOGLE	Workflows	a connector library covering most of the provider's own services.
AZURE	Logic Apps	a connector catalogue in the hundreds, which is the single largest retention argument in this Part.

BEFORE YOU START

ACCESS

- Every state machine listed with its invocation frequency, its criticality and its owner
- The account-closure schedule from Move 122, ready to receive dated entries

SOFTWARE

- The two landings priced honestly: a workflow engine for graph-shaped batch, and a durable execution framework for long-running state
- A rewrite estimate per machine, in days, produced by somebody who has read the definition

PEOPLE

- The owner of each state machine and whoever holds the budget for the rewrite

WHY THIS WORKS

State machines are not the problem; the service integrations are. A workflow that calls twelve provider services directly, with retries and error handling expressed declaratively, has no equivalent because the twelve services are what it calls. So the honest answer is a dated exception: inventory the state machines, price the rewrite properly against the two open landings, and write the ones that stay into the account-closure schedule with a name and a date beside each. That gives Move 122 something concrete to discharge instead of a vague dependency it discovers at the end.

SWAP

Where a state machine calls only your own services, it converts almost directly and should be moved now rather than dated. The integrations are the whole difficulty.

DO IT FASTER

Delete the state machines that have not run in a year. There are always some, and they are on the list of things blocking account closure.

WATCH OUT

A retained state machine keeps an account alive, and an account that is alive has credentials, permissions and a bill. Move 122 has to know about every one of these.

LEFTOVERS

Workflow services bill per state transition, so a low-frequency machine costs almost nothing and is easy to leave running forever. That is the failure this Move exists to prevent.

\$290/mo	\$180/mo	38%	0 min	2 weeks	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The state machines that have not run in a year, today. The rest stay against a written date, and Move 122 collects them.

NEEDS 86 The batch estate: who schedules the jobs 89 The serverless estate in four piles

THE RUNBOOK

- 01 Inventory the state machines by frequency and criticality. Most estates find that a handful run constantly and the rest run monthly, and only the first group justifies a rewrite on cost.
- 02 Count the provider service integrations in each. This number, more than the size of the definition, predicts the rewrite cost, because each integration becomes code you write and maintain.
- 03 Price the two landings against the estate. A workflow engine suits graph-shaped batch work; a durable execution framework suits long-running business state. Both are openly licensed, which is why they are the landings.
- 04 Do not hand-roll a state machine on a queue. It is the obvious shortcut, it works for a fortnight, and then somebody has to reimplement retries, timeouts and idempotency badly.
- 05 Rewrite the ones that justify it, and leave the rest running with a named owner and a date. The date is the point of this Move.
- 06 Write the retained machines into the account-closure schedule so Move 122 has a concrete list with dates rather than an open dependency.
- 07 Review the list quarterly and shorten it, because the retention argument weakens every time one of the twelve services it calls is turned off elsewhere in the programme.

ROLLBACK

Nothing is migrated here that cannot be reverted by re-enabling the original state machine, which is still there. There is no point of no return in this Move at all; it is an inventory, a price and a set of dates. Keep the inventory in source control and re-run it quarterly, so the retained list can be restored, compared and shortened rather than quietly growing.

VI

EDGE

How traffic reaches you once the addresses are yours. Also the Part with the most honest advice to keep paying somebody else.

17	15	9	15
MOVES	ZERO DOWNTIME	HIGH RISK	MINUTES, TOTAL

IF YOU ONLY DO THREE

98

ENVOY GATEWAY, AFTER INGRESS-NGINX

Installs a Gateway API controller on an address of its own, so the layer-7 edge exists and is testable before a `…`

93

TRANSIT: TWO UPSTREAMS AND THE 95TH PERCENTILE

Buys transit from two upstreams on disjoint paths and reads a 95th-percentile invoice, where sustained rate rather `…`

91

YOUR OWN ASN, A /24 AND THE ADDRESS SPACE YOU ALREADY HAVE

Takes an ASN and provider-independent address space, and registers the IRR and RPKI objects that decide whether `…`

YOUR OWN ASN, A /24 AND THE ADDRESS SPACE YOU ALREADY HAVE · TURNING ON DUAL-STACK, AND THE CODE THAT PARSES AN ADDRESS · TRANSIT: TWO UPSTREAMS AND THE 95TH PERCENTILE · EGRESS OFF THE MANAGED NAT, AND THE ALLOWLISTS YOU RENEGOTIATE · THE FIRST VIP: L2, AND FAILOVER WHEN NOBODY WILL PEER · VIPS OVER BGP: ECMP, BFD AND THE REHASH · TLS OFF THE MANAGED CERTIFICATE SERVICE · ENVOY GATEWAY, AFTER INGRESS-NGINX · BALANCER RULES AND PROXY ANNOTATIONS INTO HTTPROUTE · THE MANAGED API FRONT DOOR: KEYS, QUOTAS AND WEBSOCKETS · EDGE AUTHENTICATION AFTER THE BALANCER DID IT · THE SHADOW EDGE: TWO GATEWAYS AND A WEIGHTED RECORD · GO-LIVE: THE SOAK, THE FREEZE, AND HOW YOU ABORT · AUTHORITATIVE DNS, AND THE REGISTRAR YOU FORGOT · CDN: CHANGE VENDOR, NOT MODEL · DDOS: THE CAPACITY YOU RENT FOREVER · EMAIL: INBOUND COMES HOME, OUTBOUND DOES NOT

■ 3 LOW RISK

■ 5 MEDIUM RISK

■ 9 HIGH RISK

91 YOUR OWN ASN, A /24 AND THE ADDRESS SPACE YOU ALREADY HAVE

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
☀️ EDGE	PROVIDER-ASSIGNED PUBLIC IP ADDRESSES AND BRING-YOUR-OWN-IP PROGRAMMES	🟡 MEDIUM	IMMEDIATELY

Takes an ASN and provider-independent address space, and registers the IRR and RPKI objects that decide whether anyone will accept the prefix.

AWS	Elastic IP addresses and the bring-your-own-IP programme	which requires a route origin authorisation naming the provider's own network as the origin.
GOOGLE	external addresses and bring-your-own-IP	the same requirement, expressed through its own onboarding flow.
AZURE	public IP addresses and custom IP prefixes	the same again, and the same trap when you publish your own authorisation too early.

BEFORE YOU START

ACCESS

- Registry membership under way, started in Move 39, with the legal entity documentation ready
- Any existing address space you already hold, and evidence of the right to use it

SOFTWARE

- The registry's portal for route objects and origin authorisations, and a validator to check what you publish
- A written plan for the transition period during which a provider still announces the prefix

PEOPLE

- Whoever can sign registry agreements, and whoever will be listed as the technical contact for the next decade

WHY THIS WORKS

Addresses you rent are addresses that end when the account does. Provider-independent space is yours, and taking it has three parts: membership of the registry for your region, an autonomous system number, and the address space itself. None of that is difficult and all of it is slow, which is why this Move opens the Part and why Move 39 told you to start it months ago. The part that is easy to get wrong is the routing security records, because a route origin authorisation naming your own network, published while a provider is still announcing the prefix on your behalf, invalidates the announcement carrying your production traffic.

SWAP

For a small estate, addresses from your transit provider are perfectly workable and cost nothing in paperwork. What you lose is the ability to change transit provider without renumbering, which is the whole argument for this Move.

DO IT FASTER

Start the registry membership on the day you read Move 39. Everything here is queue time and the queue does not care how urgent you are.

WATCH OUT

The technical contact on a registry record is a person who will be emailed about outages for years. Use a role address that survives them leaving.

LEFTOVERS

Provider-assigned addresses stay assigned and stay billing until Part VII, and releasing one early means it is gone for good.

—	—	—	0 min	4 days	6 to 12 weeks
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. Provider addresses carry every user you have until Move 103.

NEEDS 39 The address plan, and the three clocks you start today

91 YOUR OWN ASN, A /24 AND THE ADDRESS SPACE YOU ALREADY HAVE

7 STEPS · 0 MIN · MEDIUM RISK

THE RUNBOOK

- 01 Complete the registry membership and take an autonomous system number. This is paperwork against a wait measured in weeks, and nothing else in this Part can start without it.
- 02 Get the address space. An address allocation for a member is a fixed size and an assignment to an end user is a different, smaller one; the sizes are documented and worth reading rather than assuming. For legacy address space, the waiting list has a real eligibility condition — in one region only a member that has never held an allocation may join a queue currently measured in years — so the transfer market is usually the realistic path.
- 03 Publish route objects in the routing registry so that upstreams will build filters that accept your prefix. A prefix with no route object is a prefix most transit providers will drop.
- 04 Publish route origin authorisations, and choose the maximum length deliberately. Too tight and you cannot announce more specific prefixes during an attack; too loose and you have authorised a hijack.
- 05 Handle the transition carefully. While a provider is still announcing your prefix under its bring-your-own-IP programme, the authorisation must name that provider's network as the origin. Publishing your own too early invalidates the live announcement, which is an outage on production traffic caused by a routing record.
- 06 Validate what you published from outside your own network, using a public validator, before relying on any of it.
- 07 Announce nothing yet. This Move is registration only; the announcements are Move 95 and Move 96.

ROLLBACK

Nothing is announced and nothing is serving, so this Move is reversed by withdrawing registry objects, and membership can be relinquished. There is no point of no return here. The one genuinely dangerous action is publishing an origin authorisation that contradicts a live announcement, and that is reversed by restoring the previous record — which propagates in minutes but not instantly, so it is checked before publication rather than after.

92 TURNING ON DUAL-STACK, AND THE CODE THAT PARSES AN ADDRESS

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
☀️ EDGE	PROVIDER-MANAGED IPV6 SUBNETS AND TRANSLATION GATEWAYS	■ HIGH	NO

Turns on dual-stack Services and pod addressing, then audits the code that parses an address, because a v6 literal breaks a parser that never saw one.

AWS	VPC IPv6 subnets with an egress-only gateway	the IPv6 cluster mode there gives pods IPv6-only addresses with translation rather than dual-stack.
GOOGLE	dual-stack subnets	dual-stack is supported and IPv6-only is not, which is the more familiar starting point.
AZURE	dual-stack virtual networks	the same, with the same limitation and the same habits to unlearn.

BEFORE YOU START

ACCESS

- Address space for the second family, from Move 39 and Move 91, allocated rather than improvised
- A change window for the API server range change, which is not a live edit

SOFTWARE

- The network plug-in configured for both families with the order chosen deliberately
- A list of every place an address is parsed, stored or compared, which is the real deliverable

PEOPLE

- The owner of every application that logs, stores or filters on an address

WHY THIS WORKS

The cluster half of this Move is short: dual-stack pod and service addressing, an address family order, and a change to the service range on the API server. Existing services and pods stay single-stack until they are recreated, which is a rollout rather than a switch. The long half is the audit, because application code, log parsers, allowlists and database columns have spent years only ever seeing one address format, and the first time a longer literal with colons in it arrives, something that has worked for a decade produces a wrong answer rather than an error.

SWAP

Where nothing needs the second family yet, this Move can wait — but not indefinitely, because Move 94 uses public IPv6 to sidestep address translation limits and it is much easier to have it already.

DO IT FASTER

Audit the code before touching the cluster. The cluster change is an afternoon and the audit is the schedule.

WATCH OUT

Database columns sized for the shorter format silently truncate the longer one, which produces a stored address that is wrong rather than absent. That is the worst failure in this Move and it is found by reading schemas.

LEFTOVERS

Egress-only gateways and translation gateways in the cloud keep billing until Part VII and are not removed here.

—	—	—	0 min	2 weeks	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. The provider's egress-only and translation gateways stay until Move 94 has moved the traffic.

NEEDS 44 Cilium, with kube-proxy off 45 BGP to the top-of-rack, and the MTU that breaks five per cent

91 Your own ASN, a /24 and the address space you already have

92 TURNING ON DUAL-STACK, AND THE CODE THAT PARSES AN ADDRESS

7 STEPS · 0 MIN · HIGH RISK

THE RUNBOOK

- 01 | Allocate the second family properly from the plan in Move 39, rather than taking whatever the tooling suggests. This addressing is as permanent as the first.
- 02 | Change the service range on the API server and configure the network plug-in for both families, with the order set deliberately: the first family listed is what a single-stack client gets.
- 03 | Understand that existing services and pods stay on one family until they are recreated. This is a rolling change and the estate will be mixed for weeks, which is normal and should be expected rather than debugged.
- 04 | Audit the application code. Address parsing, string length assumptions, storage columns sized for the shorter format, comparison functions, and anything that splits on a colon. Two weeks of this Move is here.
- 05 | Audit the operational surfaces too: log parsers, dashboards that group by address, allowlists at partners, and any tooling that assumes an address matches a particular pattern.
- 06 | Roll out family by family and namespace by namespace, watching for the failure that looks like an application bug rather than a network change.
- 07 | Note the provider difference before assuming your starting point is the common one: one cloud's IPv6 cluster mode is not dual-stack at all but IPv6-only with translation, so those readers are changing model rather than adding a family.

ROLLBACK

This Move is irreversible at the cluster level: a cluster converted to dual-stack is not supported converting back, so the decision is taken once and taken here. The point of no return is the service range change on the API server. What remains reversible is the rollout — a workload can be recreated as single-stack — and the underlying data is untouched, so a truncated column is restored from backup rather than from a network change. Rehearse the whole thing on the second cluster from Move 51 first.

93 TRANSIT: TWO UPSTREAMS AND THE 95TH PERCENTILE

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
☀ EDGE	METERED DATA TRANSFER OUT AND DEDICATED INTERCONNECT	■ MEDIUM	30 DAYS

Buys transit from two upstreams on disjoint paths and reads a 95th-percentile invoice, where sustained rate rather than total bytes sets the bill.

AWS	metered data transfer out with Direct Connect	port hours plus a per-gigabyte rate, billed on total bytes rather than on rate.
GOOGLE	network service tiers with Interconnect	where the tier itself changes the egress rate, and an unlimited-data plan exists alongside the metered one.
AZURE	metered egress with ExpressRoute	port hours plus per-gigabyte, with an unlimited option on some circuit types.

BEFORE YOU START

ACCESS

- The cross-connect price list from Move 26 and the address space and network number from Move 91
- Traffic measurements at five-minute resolution, because that is the resolution the invoice uses

SOFTWARE

- Authorisation letters and cross-connect orders prepared, priced from the Move 26 list
- Shaping or scheduling for the backup and migration traffic, in place before the first invoice

PEOPLE

- Whoever will hold the transit contracts, because the commit level is a three-year decision

WHY THIS WORKS

Cloud egress is billed on total bytes, and transit is billed on sustained rate. That single difference reorganises everything: a nightly backup that costs a fortune as bytes costs nothing as rate if it runs off-peak, and a short daily spike that was invisible as bytes sets the price for the whole month. So this Move buys two upstreams on genuinely disjoint paths, sizes the commit against a percentile rather than a peak, and shapes the traffic that would otherwise set that percentile — the nightly backups and the migration pull from Move 53 — off the measured rate.

SWAP	DO IT FASTER	WATCH OUT	LEFTOVERS
A small reader should buy none of this. A blended-bandwidth port from the facility, with the facility handling the upstreams, is cheaper and adequate below a few hundred megabits, and this Move says so explicitly.	Order the circuits early, in parallel with everything else in this Part. Six to twelve weeks of lead time is normal and nothing depends on them until Move 103.	A single burst from a misconfigured job can set the percentile for a whole month. Alert on the rate approaching the commit, not on the monthly total.	The dedicated interconnect from Move 53 stays until Part VII, because the abort path in Move 121 needs it.

\$9,216/mo	\$1,800/mo	80%	0 min	2 weeks	6 to 12 weeks
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. Cloud egress carries production until Move 103 moves the traffic.

NEEDS 26 Carrier neutrality and the cross-connect price list 53 The links you build first: to the cloud, to the office

91 Your own ASN, a /24 and the address space you already have

THE RUNBOOK

- 01 Choose upstreams on genuinely disjoint paths, using the fibre route information gathered in Move 25. Two transit providers sharing a duct are one transit provider with two invoices.
- 02 Understand the billing before negotiating. The rate is measured in five-minute samples across the month, the top five per cent are discarded, and the highest remaining sample sets the bill. Total volume is almost irrelevant.
- 03 Size the commit against that measurement rather than against a peak. A commit that is too low costs a punitive overage rate; one that is too high is paid whether or not it is used.
- 04 Order the cross-connects with their authorisation paperwork, priced from Move 26's list, and remember the ninety-day validity on the cloud on-ramp paperwork noted there.
- 05 Add an exchange port if the traffic profile justifies it, and be honest about what peering saves: it is usually a meaningful share of traffic and a smaller share of cost, because the transit commit does not fall in proportion.
- 06 Shape the backup window and the migration pull from Move 53 off the measured rate. This is the single highest-value configuration change in this Move and it is frequently forgotten until the first invoice.
- 07 Compare the result against the cloud egress bill in your own units. One cloud sells an unlimited-data plan, one sells tiers that change the rate itself, and one charges port hours plus per gigabyte, so the comparison has to be re-expressed rather than read off.

ROLLBACK

Transit contracts have notice periods, so backing out costs the remainder of a term rather than being impossible, and for thirty days the cloud egress path is still carrying everything. The point of no return is the commit level on a signed contract, not the circuit itself. Keep the traffic measurements and the commit arithmetic in source control so the sizing can be restored and re-argued at renewal, when the invoice will look different from the model.

94 EGRESS OFF THE MANAGED NAT, AND THE ALLOWLISTS YOU RENEGOTIATE

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
☀️ EDGE	MANAGED NAT GATEWAYS AND PROVIDER-ASSIGNED EGRESS ADDRESSES	■ HIGH	30 DAYS

Sends pods out through node addresses, gives fixed-source workloads an explicit SNAT hop, and renegotiates every partner allowlist before the first packet moves.

AWS	NAT Gateway	pre-allocates translation ports per instance, so exhaustion is a limit you can compute in advance.
GOOGLE	Cloud NAT	a shared pool where one noisy workload can exhaust translation capacity for everything else behind it.
AZURE	NAT Gateway with public IP prefixes	a similar pooled model, with the same failure shape and the same need to reserve ports explicitly.

BEFORE YOU START

ACCESS

- The allowlist inventory started in Move 39, with a named contact for every entry
- Public address space from Move 91 and, for the egress-heavy workloads, the second family from Move 92

SOFTWARE

- An egress gateway or a dedicated translation hop, configured with a stable address
- Port budget arithmetic per workload, so exhaustion is computed rather than experienced

PEOPLE

- Whoever owns each partner relationship, because this is their change request, not yours

WHY THIS WORKS

Most pods do not need a fixed source address and can leave through the address of the machine they are on, which is simpler and free. A minority do need one, because a partner's firewall has their address written into it, and those get an explicit translation hop with a stable address. The hard part of this Move is not the networking at all. It is that every partner, webhook endpoint and payment gateway with your current address in an allowlist has to be renegotiated onto a hostname or a new range, and each of those sits on somebody else's change calendar.

SWAP

Where a single partner refuses to move off an address allowlist, keep one small cloud footprint with that address and route only that traffic through it. It is ugly and it is cheaper than losing the integration.

DO IT FASTER

Send the allowlist change requests on the day you start this Part, not when the engineering is ready. The engineering takes two weeks and the partners take a quarter.

WATCH OUT

Webhook senders often verify the source address as well as the signature. A webhook that stops arriving silently is the classic symptom, and nobody notices until a reconciliation runs.

LEFTOVERS

The managed translation gateway bills per hour and per gigabyte, and both continue until the last workload has moved.

\$1,340/mo	\$0/mo	100%	0 min	2 weeks	30 days
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The managed translation gateways, thirty days after the last workload moved and every partner has confirmed the new source.

NEEDS 39 The address plan, and the three clocks you start today 44 Cilium, with kube-proxy off 93 Transit: two upstreams and the 95th percentile

94 EGRESS OFF THE MANAGED NAT, AND THE ALLOWLISTS YOU RENEGOTIATE

7 STEPS · 0 MIN · HIGH RISK

THE RUNBOOK

- 01 Start the allowlist renegotiation first and treat it as the critical path. Assume a quarter for the long tail; some partners take six weeks to change a firewall rule and some require a contract amendment.
- 02 Ask every partner for a hostname rather than an address wherever they will accept one. It is the change that stops this Move recurring the next time anything moves.
- 03 Classify the workloads. Those that do not need a fixed source leave through node addresses, which is most of them and costs nothing.
- 04 Give the fixed-source workloads an explicit translation hop with a stable address from Move 91. This is a deliberate, small set rather than a default for everything.
- 05 Do the port arithmetic. Translation state is per source address and port, and a workload opening thousands of short-lived connections exhausts a pool. One provider pre-allocates per instance so the limit is computable; another shares a pool so a noisy neighbour exhausts it for everyone.
- 06 Where a workload is genuinely egress-heavy, give it public addresses in the second family from Move 92 instead of translating it at all. That removes the state entirely.
- 07 Expect established connections to reset when an assignment moves, because translation state does not migrate. That is the user-visible cost of this Move rather than a downtime window, and it is worth telling people about in advance.

ROLLBACK

For thirty days the managed path still exists and a workload is moved back by changing its egress policy, so this is reversible per workload rather than all at once. The point of no return is the release of the provider-assigned addresses, which cannot be reclaimed once released and which is deliberately deferred to Part VII. Keep the allowlist inventory current, because it is the document a restore of the old egress path depends on.

95 THE FIRST VIP: L2, AND FAILOVER WHEN NOBODY WILL PEER

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
☀️ EDGE	MANAGED LAYER-4 LOAD BALANCERS	■ MEDIUM	IMMEDIATELY

Produces a working LoadBalancer address in layer-2 mode, at the cost of one elected node carrying every packet destined for it.

AWS	Network Load Balancer	a layer-4 balancer with cross-zone distribution and its own health checking.
GOOGLE	the external passthrough load balancer	the same, implemented in the network rather than as an appliance.
AZURE	Standard Load Balancer	the same again, with its own health probe model and backend pool semantics.

BEFORE YOU START

ACCESS

- The address pool for services, from the plan in Move 39 and, for public ones, Move 91
- A layer-2 domain the nodes share, which is the requirement this mode has and Move 96 does not

SOFTWARE

- The load-balancer address management configured with a pool and an assignment mode
- A way to observe which node currently holds each address, because that is the thing that changes

PEOPLE

- Whoever will test failover, since the interesting behaviour is what neighbouring devices do

WHY THIS WORKS

The first service address on your own hardware should be the simplest thing that works, and that is layer-2 mode: one node claims the address, answers for it, and every packet destined for it arrives there. That is address failover rather than load balancing, and the cost is honest — one node carries all of it, and failover is measured in however long it takes neighbouring devices to update their address caches. It is enough to make a working address exist, and it stays in place afterwards to hold the address while Move 96 cuts distribution over pool by pool.

SWAP

Where the nodes do not share a layer-2 domain at all, skip this Move and go straight to Move 96. It exists as a stepping stone, not as a requirement.

DO IT FASTER

Use it for internal services first. They are lower stakes and they exercise the same mechanism.

WATCH OUT

A stale neighbour cache on a switch or a firewall will keep sending traffic to the old node after failover, and the symptom is that the address works from some places and not others. That is the classic layer-2 mode fault.

LEFTOVERS

The managed balancer keeps billing per hour and per capacity unit throughout, which is expected: this is the period where you pay for both.

—	—	—	0 min	3 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing. The managed balancer carries every user until Move 103 moves the traffic.

NEEDS 44 Cilium, with kube-proxy off 91 Your own ASN, a /24 and the address space you already have

95 THE FIRST VIP: L2, AND FAILOVER WHEN NOBODY WILL PEER

7 STEPS · 0 MIN · MEDIUM RISK

THE RUNBOOK

- 01 Define the address pool and assign the first address. Keep the pool small and documented; addresses assigned ad hoc become impossible to reconcile against the plan.
- 02 Bring up a service and confirm the address answers from outside the cluster. This is the moment the estate has its first address of its own.
- 03 Understand what you have built. One node holds the address and answers for it; the others do not. Traffic is not distributed, and the node holding it is a single point of failure for that address until Move 96.
- 04 Decide the external traffic policy deliberately. Sending traffic on to any node preserves reachability and hides the client address; keeping it local preserves the client address and requires the health-check port to work correctly.
- 05 Test failover by stopping the node that holds the address. Measure how long the address takes to answer again, which depends on how quickly neighbouring devices refresh their caches rather than on anything in the cluster.
- 06 On rented metal, note the constraint: address filtering on the provider's network usually prevents this mode entirely, and the answer is their own floating-address mechanism with a failover measured in tens of seconds.
- 07 Decommission nothing. The managed balancer keeps serving production until Move 103, which is why this Move is reversible immediately.

ROLLBACK

Nothing is serving production through this address, so the rollback is deleting the service and its address assignment. There is no point of no return in this Move at all. That is deliberate: it exists so that the first address is created in a situation where being wrong costs nothing, and so that the address itself survives into Move 96 rather than being created twice.

96 VIPS OVER BGP: ECMP, BFD AND THE REHASH

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
☀️ EDGE	MANAGED CROSS-ZONE LOAD BALANCING AND ANYCAST FRONT ENDS	■ MEDIUM	IMMEDIATELY

Advertises service addresses to the leaf switches so ECMP spreads them across nodes, with BFD for fast withdrawal and resilient hashing to protect live flows.

AWS	cross-zone load balancing on the Network Load Balancer	distribution across zones handled invisibly and billed as capacity units.
GOOGLE	the global external load balancer with anycast	a single address answered from many places, which is the property that does not come with you.
AZURE	Standard Load Balancer with zone redundancy	the same distribution, expressed as a zone property of the front end.

BEFORE YOU START

ACCESS

- The routing sessions from Move 45, already established and stable, on both leaf switches
- The redundancy answer from Move 12, because it decides how the two uplinks behave here

SOFTWARE

- Service address advertisement configured through the network plug-in, per pool
- Sub-second failure detection enabled on both ends, and resilient hashing confirmed on the switch

PEOPLE

- The switch administrator, because the hashing behaviour is configured on their side

WHY THIS WORKS

Move 95 made an address exist on one node. This Move makes it exist on all of them, by advertising it to both leaf switches so that the fabric spreads traffic across every node that is ready to receive it. Two details decide whether it behaves well. The default routing hold timers leave a dead node attracting traffic for tens of seconds, which is fixed by adding sub-second failure detection. And on many switches, changing the set of next hops rehashes every flow, so an ordinary node drain resets established connections unless the switch supports a hashing scheme designed to avoid exactly that.

SWAP

Where the fabric will not run a routing session with the hosts, the layer-2 mode from Move 95 remains a working answer for a small estate, with the single-node limitation stated.

DO IT FASTER

Cut the internal service pool first. It exercises everything and no external user is watching.

WATCH OUT

An anycast front end answering from many locations is a property of a global network, and you are not building one. What you have is one site with distributed reception, and Move 105 is where geography comes back.

LEFTOVERS

The managed balancer keeps billing throughout, which is expected until Move 103.

—	—	—	0 min	1 week	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing. The managed balancer keeps every user until Move 103.

NEEDS 12 Two switches, and the spine you do not need yet 45 BGP to the top-of-rack, and the MTU that breaks five per cent

95 The first VIP: L2, and failover when nobody will peer

THE RUNBOOK

- 01 | Configure advertisement of the service address pool through the network plug-in, alongside the pod prefixes already advertised in Move 45. Cut one pool at a time with the Move 95 path still holding the address.
- 02 | Confirm the addresses appear on both leaf switches with multiple next hops, and that traffic is genuinely spread rather than all arriving at one node.
- 03 | Fix the failure detection. Default hold timers mean a dead node keeps attracting traffic for tens of seconds; sub-second detection between the node and the switch reduces that to under a second, and it is the single most valuable setting in this Move.
- 04 | Check the rehash behaviour explicitly. Drain a node and watch whether established connections on the other nodes survive. On many switches they do not, because the next-hop set changed and every flow rehashed.
- 05 | Where the switch supports resilient hashing, turn it on and repeat the drain test. Where it does not, know that a drain resets flows and plan node maintenance in Move 109 accordingly.
- 06 | Spend the second half of the week testing failure rather than success: kill a node, kill a switch, drain a node gracefully, and watch each case.
- 07 | Leave the Move 95 configuration in place. It is the abort path, and reverting is one manifest change.

ROLLBACK

Reverting is one manifest change back to the Move 95 path, which is still configured and still able to hold the address. There is no point of no return here, and nothing is serving production through these addresses until Move 103. Keep the switch configuration and the advertisement resources in source control so a working routing state can be restored quickly, because a routing mistake here is fast and total rather than gradual.

97 TLS OFF THE MANAGED CERTIFICATE SERVICE

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
☼ EDGE	MANAGED CERTIFICATE SERVICES AND MANAGED PRIVATE CERTIFICATE AUTHORITIES	■ HIGH	30 DAYS

Issues and renews every certificate from cert-manager through a delegated challenge zone, and rebuilds the internal hierarchy the managed private CA was holding.

AWS	Certificate Manager and Private CA	public certificates are exportable for a per-name fee, and only when issued as exportable in the first place.
GOOGLE	Certificate Manager and Certificate Authority Service	where the private key of a managed public certificate is not handed over.
AZURE	App Service Certificates and Key Vault	where a certificate's key can be exported, which changes the custody argument for this reader.

BEFORE YOU START

ACCESS

- A delegated zone dedicated to challenge records, so the issuance credential is scoped to nothing else
- The root of trust from Move 57, which is where the internal authority's key material lives

SOFTWARE

- `cmctl` and the issuer configuration, with issuance rate limits understood as an availability dependency
- An offline root and an issuing authority for the internal hierarchy

PEOPLE

- Whoever owns the domain, because delegating a zone is a change on their side

WHY THIS WORKS

The case for leaving has to be made accurately, because the folklore is out of date: two of the three providers will now give you a public certificate's private key, one of them for a fee and only when the certificate was issued as exportable. So the argument is not that you are locked out of your own keys. It is renewal automation you control, cost, and the private hierarchy — which is the larger half of this Move. What replaces the managed service is automated issuance through a challenge delegated to a throwaway zone, plus an offline root and an issuing authority for everything internal.

SWAP

Where only a handful of public certificates exist, issuing them by hand and diarising renewals is workable and terrible. Automate the renewal even for three certificates; the failure mode is an outage on a Sunday.

DO IT FASTER

Do the public half first. It is a day and it removes the dependency on the managed service for anything new.

WATCH OUT

A certificate that renews correctly but is not reloaded by the process using it expires while a valid one sits on disk. Test the reload, not the renewal.

LEFTOVERS

The managed private authority bills a substantial monthly fee per authority plus a per-certificate charge, and both stop only when the authority is deleted, which has its own waiting period.

\$520/mo	\$0/mo	100%	0 min	2 weeks	30 days
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The managed private certificate authority after thirty days, and the public certificates once every listener has been serving from your own issuance for a full renewal cycle.

NEEDS 43 Three control-plane nodes, the API VIP and the cluster CA 57 The root of trust after managed key management

THE RUNBOOK

- 01 Delegate a dedicated zone for challenge records and scope the issuance credential to it alone. A credential that can edit your production zone in order to prove domain control is a much larger grant than the job needs.
- 02 Configure automated issuance for public names and confirm a renewal end to end, not just an issuance. The renewal is the part that fails silently ninety days later.
- 03 Treat issuance rate limits as an availability dependency. A cluster rebuild that requests two hundred certificates at once will hit them, and the failure arrives during a recovery.
- 04 Set expiry alerting as a fraction of remaining lifetime rather than as a fixed number of days, because the maximum lifetime of a public certificate is on a published downward schedule and a fixed threshold will stop making sense.
- 05 Build the internal hierarchy: an offline root, kept offline and sealed with the custody model from Move 57, and an online issuing authority for workload and service certificates. Most of this Move's two weeks is here.
- 06 Distribute the internal root to everything that has to trust it, and write down what that list is. A root nobody trusts is an outage waiting for the first internal call.
- 07 Name the case for keeping a managed private authority: where a regulator requires keys in a validated key store, buying that is the right answer and it belongs on Move 116's list of things you deliberately keep.

ROLLBACK

For thirty days the managed service still holds valid certificates, so reverting is repointing the listener at them. The point of no return is the deletion of the managed private authority, which has its own deletion window and after which certificates it issued cannot be reissued from it. Keep the internal root sealed and backed up in two places, because losing it means every internal certificate has to be reissued and every trust store updated.

98 ENVOY GATEWAY, AFTER INGRESS-NGINX

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
☀️ EDGE	MANAGED LAYER-7 LOAD BALANCERS AND IN-CLUSTER INGRESS CONTROLLERS	■ ■ ■ LOW	IMMEDIATELY

Installs a Gateway API controller on an address of its own, so the layer-7 edge exists and is testable before a single route moves.

AWS	Application Load Balancer with the load balancer controller	an appliance managed for you and configured from the cluster.
GOOGLE	the external application load balancer	the same shape driven by an ingress controller, and implemented in the network rather than as an appliance.
AZURE	Application Gateway with its ingress controller	the same again, with its own annotation vocabulary.

BEFORE YOU START

ACCESS

- An address from Move 95 or Move 96, and certificates issued by Move 97
- The link from Move 53, so backends still in the cloud are reachable from the new gateway

SOFTWARE

- The gateway controller at a pinned chart version, installed with `helm`
- A gateway class, a gateway and listeners defined as resources rather than configured by hand

PEOPLE

- Whoever owns the routes that will move in Move 99, so the structure suits them

WHY THIS WORKS

This Move builds the edge and moves nothing through it. A gateway controller, a gateway class, a gateway and its listeners, with certificates from Move 97 and an address from Move 95 or Move 96, fronting backends that are still running in the provider account across the link from Move 53. That last detail is what makes it low risk: the new edge can be exercised against real backends before any user is involved. Readers arriving from the in-cluster controller rather than a managed balancer have a deadline as well, because that project has been retired and no longer receives security fixes.

SWAP

Where an estate is already on a well-run alternative gateway implementation, keep it. The value here is the resource model, not the specific proxy.

DO IT FASTER

Stand it up on the second cluster from Move 51 first. It takes an hour and it removes every surprise from the production install.

WATCH OUT

Listener and certificate mismatches produce a gateway that reports ready and serves the wrong certificate. Check what is served from outside rather than what the resource says.

LEFTOVERS

The managed balancer keeps billing per hour and per capacity unit until Move 103, and the retired in-cluster controller keeps running until its routes move.

—	—	—	0 min	1 week	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing. Both edges are running and both are paid for until Move 103.

NEEDS 95 The first VIP: L2, and failover when nobody will peer 97 TLS off the managed certificate service

THE RUNBOOK

- 01 | Install the controller with `helm install` at a pinned chart version. The book names one implementation rather than surveying three, because a small team should run one edge and know it well.
- 02 | Define the gateway class, the gateway and its listeners. Terminate the certificates from Move 97 on the listener and confirm the chain is served correctly from outside.
- 03 | Bind the gateway to the address from Move 95 or Move 96. This is where the two halves of the Part meet, and it is worth confirming the address is reachable before adding any route.
- 04 | Point one throwaway route at a backend still running in the provider account, across the link from Move 53. It proves the whole path — address, certificate, gateway, link, backend — without touching production.
- 05 | Establish the delegation model: who may create a route, in which namespace, against which listener. Getting this right now avoids re-organising every route later.
- 06 | Note the deadline if you are leaving the in-cluster controller rather than a managed balancer. That project is retired and no longer receives security fixes, which makes this urgent rather than merely worthwhile.
- 07 | Cut nothing over. Route translation is Move 99 and traffic is Move 103.

ROLLBACK

Nothing is serving production, so this Move is undone by deleting the gateway resources. There is no point of no return. That is precisely the design: the edge exists, is tested and is boring long before Move 103 asks it to carry users. Keep the gateway and listener definitions in source control so a working edge can be restored quickly, and keep the throwaway route as a permanent smoke test.

99 BALANCER RULES AND PROXY ANNOTATIONS INTO HTTPROUTE

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
☀️ EDGE	MANAGED BALANCER LISTENER RULES AND INGRESS-CONTROLLER ANNOTATIONS	■ ■ ■ LOW	IMMEDIATELY

Translates every listener rule and controller annotation into HTTPRoute by hand, with a test case per rule, because only some have a standard equivalent.

AWS	Application Load Balancer listener rules	priority-ordered conditions with weighted target groups underneath them.
GOOGLE	URL maps	a different structure for the same job, with its own ordering semantics.
AZURE	Application Gateway routing rules	the same again, expressed through path maps and backend settings.

BEFORE YOU START

ACCESS

- Every listener rule and every annotation exported, from every balancer and every ingress resource
- The gateway from Move 98, ready to accept routes

SOFTWARE

- A conversion tool used for the first draft only, with its output reviewed line by line
- A test harness that can assert a request lands on the right backend with the right transformations

PEOPLE

- An owner named for every annotation that has no equivalent, because each is a decision rather than a translation

WHY THIS WORKS

Routing rules look mechanical and are not, because the ordering semantics differ between implementations and because a large share of real behaviour lives in annotations that have no standard equivalent at all. A conversion tool produces a useful first draft and a misleading final answer. So the deliverable here is not a set of resources; it is a table with one row per rule, its translation, a test case, and — for anything unmapped — a named owner and a decision. Nothing is cut over, which is why a Move that takes three weeks is Low risk.

SWAP

Where a rule exists only to work around something the application should do itself, delete it and change the application. Migrations are the only time anyone is allowed to do this.

DO IT FASTER

Run the conversion tool first and then review, rather than writing from scratch. The draft is worth an hour of typing and no more trust than that.

WATCH OUT

A rule that has been shadowed by another for years is dead, and translating it faithfully makes it live again. Test what the current balancer actually does, not what its configuration says.

LEFTOVERS

The old listener rules stay until Move 103 shifts traffic, and they remain the rollback for the whole Part.

—	—	—	0 min	3 weeks	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing. Both edges are configured and only one is serving.

NEEDS 98 Envoy Gateway, after ingress-nginx

99 BALANCER RULES AND PROXY ANNOTATIONS INTO HTTPROUTE

7 STEPS · 0 MIN · LOW RISK

THE RUNBOOK

- 01 | Export everything first: host and path conditions, priorities, header and query matching, redirects, rewrites, weights, session affinity and every annotation on every ingress resource.
- 02 | Translate the mechanical parts. Host and path matching, header and query matching, redirects and rewrites, and weighted target groups into backend references with weights.
- 03 | Handle ordering explicitly. Priority-ordered rules and specificity-ordered matching are not the same thing, and two rules that overlapped harmlessly under one model can shadow each other under the other.
- 04 | Deal with affinity and slow start deliberately. These are common, they matter to real applications, and their support differs by implementation, so each one is either configured, replaced or dropped with the owner's agreement.
- 05 | List the annotations with no equivalent and give each a named owner and a decision. This list is the honest output of the Move and it is what stops a surprise during Move 103.
- 06 | Write a test case per rule. A request that should land on backend A, one that should be redirected, one that should be rewritten. Run them against the gateway from Move 98 with backends still in the cloud.
- 07 | Budget about a day per twenty rules. A mid-sized estate is three weeks and it does not compress.

ROLLBACK

Nothing is serving through these routes, so the rollback is deleting them. There is no point of no return in this Move. The route table and the test cases are the durable output, and they should be in source control so that the mapping can be restored and re-checked when a rule behaves differently in Move 103 than it did in testing.

100 THE MANAGED API FRONT DOOR: KEYS, QUOTAS AND WEBSOCKETS

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
☀️ EDGE	MANAGED API GATEWAYS	📊 HIGH	30 DAYS

Rebuilds the managed API front door as keys, quotas, validation and throttling on the gateway, and gives long-lived sockets an answer of their own.

AWS	API Gateway	a connect-and-disconnect route model for sockets with a server-initiated callback API that has no standard equivalent and has to be rebuilt as a service you run.
GOOGLE	API Gateway and Apigee	no socket support in the lighter product at all, which pushes that traffic to the full platform.
AZURE	API Management	which proxies sockets through, making this the easiest of the three for that half.

BEFORE YOU START

ACCESS

- The usage plans, keys and quotas exported, and a list of every consumer holding a key
- The gateway and routes from Moves 98 and 99, already carrying test traffic

SOFTWARE

- Key authentication and rate-limiting policies on the gateway, with the rate-limit service and its cache deployed
- Request and schema validation moved to the gateway or explicitly to the application

PEOPLE

- Whoever can contact every external consumer, because keys you cannot reissue set this Move's timeline

WHY THIS WORKS

A managed API front door was doing more than proxying, and the extra is what this Move rebuilds: usage plans with per-consumer quotas, keys and their rotation, request and schema validation, throttling, and the authoriser functions that fronted code migrated in Move 89. On the gateway from Move 98, key authentication and global rate limiting are policy resources, and the rate limiting needs a small service and a small cache of its own on the storage from Move 49. The genuinely divergent part is long-lived sockets, where the three providers do three different things and one of them has no equivalent at all.

SWAP

Where the front door was doing nothing but proxying, delete it and route directly. A surprising number of managed gateways are an expensive layer with no policy in them.

DO IT FASTER

Start the consumer contact on the first day of the Part. The engineering is three weeks and the customers are a quarter.

WATCH OUT

Quota accounting resets differently between implementations — per calendar period against per rolling window — and a consumer at the edge of their plan will notice.

LEFTOVERS

The managed gateway bills per million requests plus its own caching tier, and the caching tier is billed per hour whether or not it is hit.

\$1,480/mo	\$80/mo	95%	0 min	3 weeks	30 days
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The managed gateway and its caching tier, thirty days after the last consumer moved and their old key stopped being used.

NEEDS 79 Redis: which half of it is actually a cache 89 The serverless estate in four piles

99 Balancer rules and proxy annotations into HTTPRoute

THE RUNBOOK

- 01 | Inventory the consumers and their keys first. Keys you can reissue on your own schedule are a week's work; keys held by external customers are a quarter, and that is what sets the timeline.
- 02 | Rebuild usage plans as rate-limiting policy with per-consumer quotas. Deploy the rate-limit service and its small cache on the storage from Move 49 rather than treating them as an afterthought.
- 03 | Rebuild key authentication as a security policy on the route, and design the rotation before issuing the first key. A key with no rotation path is a key you will be living with in five years.
- 04 | Move request and schema validation deliberately. Either it happens at the gateway or it happens in the application, and the failure mode of assuming the other one is doing it is an unvalidated payload reaching your code.
- 05 | Handle sockets according to which provider you are leaving. Where the managed gateway proxied them, the gateway here does too. Where the model was connect-and-disconnect routes with a server-initiated callback, that callback has no standard equivalent and is rebuilt as a service you run, which is a real piece of engineering rather than a configuration change.
- 06 | Run both front doors for thirty days with consumers migrating on their own schedule.
- 07 | Alert on quota rejections from day one, because a misconfigured quota looks to a customer like an outage and to you like nothing at all.

ROLLBACK

For thirty days both front doors work and a consumer is moved back by reissuing them the old key and endpoint. The point of no return is the deletion of the managed gateway with its keys, since the keys themselves cannot be restored afterwards and every consumer would have to be reissued at once. Keep the exported usage plans and the key inventory in source control so the configuration can be restored onto a rebuilt gateway if the migration is aborted.

101 EDGE AUTHENTICATION AFTER THE BALANCER DID IT

CUTOVER

5 MIN



LAYER	LEAVING	RISK	REVERSIBLE
☀️ EDGE	LOAD-BALANCER AUTHENTICATION, MANAGED USER DIRECTORIES AND CLIENT VPN	■ HIGH	7 DAYS

Rebuilds the authentication the balancer was performing, before internal tools that never had a login of their own become publicly reachable.

AWS	Application Load Balancer authentication	with Cognito behind it and Client VPN beside it, the balancer signed an identity header the application trusted.
GOOGLE	Identity-Aware Proxy	the same pattern, with its own header name and its own signature scheme.
AZURE	App Service authentication with Entra ID	again the same, with a third header name, and a point-to-site VPN beside it.

BEFORE YOU START

ACCESS

- The identity provider from Move 50, with an application registration for the proxy
- A list of every service currently relying on a balancer-injected identity header

SOFTWARE

- An authentication proxy wired to the gateway's external authorisation filter, at a pinned version
- A session store and a cookie domain chosen deliberately, because the defaults are wrong for a multi-service estate

PEOPLE

- The owner of every internal tool, because their users will all be signed out at once

WHY THIS WORKS

A number of internal tools have no login of their own, because the balancer in front of them did the authenticating and passed a signed header the application trusted. Move those tools without rebuilding that, and they become publicly reachable with no authentication at all — which is the most dangerous single failure available in this Part. So this Move rebuilds it: an authentication proxy behind the gateway's external authorisation filter, against the identity provider from Move 50, with the cookie domain and the session store chosen deliberately rather than defaulted, and the downstream code updated for a header with a different name.

SWAP

Where an application already has its own login, do not put a proxy in front of it. Two authentication layers is a support burden and a source of confusing failures.

DO IT FASTER

Move the least-used internal tool first and use it as the pattern. It will find the cookie domain problem for you at no cost.

WATCH OUT

An application that trusts a header will trust that header from anyone if the proxy is bypassed. Network policy from Move 60 must ensure the only path to the application is through the gateway.

LEFTOVERS

The managed user directory bills per monthly active user, and the client VPN bills per endpoint-hour plus per connection-hour, both until deleted.

\$720/mo	\$30/mo	96%	5 min	2 weeks	7 days
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The client VPN endpoints today, and the managed user directory after seven days and a verified export.

NEEDS 50 Humans on the API server: OIDC, after break-glass 98 Envoy Gateway, after ingress-nginx

THE RUNBOOK

- 01 | List every service that trusts an injected identity header. The dangerous ones are the tools nobody thinks of as applications: a dashboard, an admin panel, a metrics interface.
- 02 | Deploy the authentication proxy and wire it to the gateway's external authorisation filter, against the identity provider from Move 50.
- 03 | Choose the cookie domain deliberately. A cookie scoped too broadly is a session shared between services that should not share one; scoped too narrowly and users sign in repeatedly. Neither default is right.
- 04 | Choose the session store deliberately too. An in-memory store means every proxy restart signs everybody out, which is fine for a tool and unacceptable for anything customer-facing.
- 05 | Update the downstream code for the new header name and signature. Each provider injected a differently named header and applications parsed it directly; that parsing has to change in every one.
- 06 | Retire the client VPN once the internal tools are behind this, which removes a per-hour endpoint charge and a client nobody enjoyed.
- 07 | Cut over in a single change and accept that live sessions are invalidated. Five minutes of everybody signing in again, nothing down, and it is visible enough to warn people about in advance.

ROLLBACK

For seven days the balancer-based authentication still exists and a service is moved back by repointing its route. The point of no return is the deletion of the managed user directory, because the accounts in it cannot be restored afterwards and any user identity that lived only there is gone. Export the directory before deleting it, and confirm the export can be read, not merely that it completed.



102 THE SHADOW EDGE: TWO GATEWAYS AND A WEIGHTED RECORD

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
 EDGE	—	 LOW	IMMEDIATELY

Runs the new gateway on a shadow hostname and compares it route by route under mirrored traffic, then restores the drain the managed balancer was providing.

AWS	Application Load Balancer deregistration delay	connection draining you never configured because it was on by default.
GOOGLE	backend service connection draining	the same property under a different name and a different default.
AZURE	Application Gateway connection draining	the same again, enabled per backend pool.

BEFORE YOU START

ACCESS

- A shadow hostname with its own certificate, resolving to the new edge
- Mirroring or replay of production requests, at a rate that does not affect the real backends

SOFTWARE

- Per-route status-code and latency comparison, on the metrics stack from Move 62
- A load test that can hold connections open while a pod is deleted, to verify draining rather than assert it

PEOPLE

- The owner of each route, to say whether a difference is a defect or an improvement

WHY THIS WORKS

Before shifting a single user, run the new edge on a hostname nobody uses and compare it route by route with mirrored or replayed traffic. Status codes and latency per route, against the metrics from Move 62, for long enough that the awkward traffic shapes have all appeared — which is usually a fortnight rather than a day. The second half of this Move is the thing that gets forgotten: the managed balancer was quietly draining connections when a backend went away, and on your own edge that is a stop delay and a termination grace period that have to be configured and then verified with load actually in flight.

SWAP

Where mirroring is not practical, replaying a day of access logs is a decent substitute for everything except concurrency behaviour.

DO IT FASTER

Start the comparison on the day Move 99 produces its first routes, and let it run while the remaining routes are translated.

WATCH OUT

A difference in latency at the ninety-ninth percentile with identical medians usually means a connection-reuse difference rather than a slow backend. Look at connection behaviour before blaming the application.

LEFTOVERS

The shadow hostname and its certificate are worth keeping permanently as a place to test edge changes.

—	—	—	0 min	2 weeks	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing. This is the last Move before the traffic moves, and both edges are paid for.

NEEDS 62 Metrics off the managed monitor, and the cardinality budget 99 Balancer rules and proxy annotations into HTTPRoute

100 The managed API front door: keys, quotas and WebSockets 101 Edge authentication after the balancer did it

102 THE SHADOW EDGE: TWO GATEWAYS AND A WEIGHTED RECORD

7 STEPS · 0 MIN · LOW RISK

THE RUNBOOK

- 01 Create the shadow hostname and its certificate, and point it at the gateway from Move 98 with the routes from Move 99. Nothing external resolves to it.
- 02 Mirror or replay production requests at the shadow edge. Mirroring is better because it captures the real distribution; replay is easier and misses the unusual shapes.
- 03 Compare per route rather than in aggregate. Status code distribution and latency percentiles, route by route, because one broken route out of two hundred disappears in an aggregate and does not disappear from a customer's day.
- 04 Run it for a fortnight. Most of that time is waiting for the awkward traffic to appear: the large upload, the slow client, the request with the unusual header, the monthly report.
- 05 Configure connection draining explicitly. A stop delay long enough for the load balancer to notice, and a termination grace period long enough for in-flight requests to finish.
- 06 Verify the draining under load. Delete a pod while a load test holds connections open and count the errors. A manifest that claims a grace period and an application that ignores the signal are indistinguishable until this test.
- 07 Shift nothing. Move 103 does that, and this Move exists so that Move 103 is boring.

ROLLBACK

Nothing is shifted and nothing is decommissioned, so there is nothing to reverse and no point of no return. The comparison data is the output, and it should be kept so that a difference observed during Move 103 can be checked against what the shadow edge showed. Keep the drain configuration in source control, because it is the setting most likely to be lost in a later refactor and its absence is invisible until a deploy drops requests.

103 GO-LIVE: THE SOAK, THE FREEZE, AND HOW YOU ABORT

CUTOVER

10 MIN



LAYER	LEAVING	RISK	REVERSIBLE
☀️ EDGE	THE PARALLEL MANAGED EDGE: LAYER-7 BALANCERS AND WEIGHTED DNS RECORDS	■ HIGH	7 DAYS

Shifts production traffic to your own edge a percentage at a time under a freeze, against abort criteria written before the week began.

AWS	Route 53 weighted records	sitting in front of the Application Load Balancer, and the mechanism that both shifts the traffic and takes it back.
GOOGLE	Cloud DNS weighted routing	the same shape, in front of the external application load balancer.
AZURE	Traffic Manager weighted routing	the same again in front of Application Gateway, with its own probe behaviour.

BEFORE YOU START

ACCESS

- The record lifetime lowered a week in advance, so a rollback propagates in minutes rather than hours
- A freeze on unrelated changes for the duration, agreed with everyone who deploys

SOFTWARE

- A synthetic peak that reproduces real traffic shapes at above real volume
- A migration dashboard everybody watches, with the abort criteria as visible thresholds on it

PEOPLE

- A named go and no-go owner, and the on-call rota briefed on what abort looks like

WHY THIS WORKS

This is the largest cutover in the book and it is done as a weighted shift rather than a switch, because the way back has to be one record change at every point. Two halves have to be right. The technical half is a synthetic peak that reproduces the failures that only appear under load: connection-tracking exhaustion, ephemeral port limits, name-resolution replica counts and the flow rehash from Move 96. The business half is the one engineers skip: a freeze window, customer notice, a status-page entry, a named person who says go or no-go, and a lowered record lifetime set a week in advance.

SWAP

Where a customer contract forbids a freeze window, shift during their quiet period at a lower percentage over a longer period. It takes a fortnight instead of a night and it is entirely legitimate.

DO IT FASTER

Do the synthetic peak two weeks early. It is the step that finds real problems and it is the one that gets compressed when the date slips.

WATCH OUT

A minority of clients cache records far beyond their stated lifetime, including some corporate resolvers and some mobile stacks. The old edge must keep working during the whole window for exactly this reason.

LEFTOVERS

Both edges bill for the full seven days, and that overlap is a real line in Move 120's cost model rather than a rounding error.

\$2,410/mo	\$0/mo	100%	10 min	1 week	7 days
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The managed layer-7 balancer and its listener rules, seven days after full weight, once the record lifetime has been restored to a normal value.

NEEDS 94 Egress off the managed NAT, and the allowlists you renegotiate 96 VIPs over BGP: ECMP, BFD and the rehash

102 The shadow edge: two gateways and a weighted record

THE RUNBOOK

- 01 Run the synthetic peak against the new edge at above real volume. Look specifically for connection-tracking table exhaustion, ephemeral port limits, name-resolution replica counts under load and the flow rehash from Move 96. These do not appear at any lower volume.
- 02 Lower the record lifetime a week in advance and confirm it has actually propagated by checking from several networks. A rollback is only as fast as the lifetime clients cached.
- 03 Write the abort criteria before the week starts, as numbers on the dashboard: error rate above a figure, latency above a figure, any data-path error at all. Somebody who is tired at midnight should be able to read them and act.
- 04 Open the freeze window, publish the customer notice and the status-page entry, and name the go and no-go owner for the session.
- 05 Shift a small percentage and hold. Then a larger one, and hold. The holds are the point; a shift with no hold is a switch with extra steps.
- 06 Expect the ten minutes. There is no write stop, but connections reset across the final weight shifts and a minority of clients ignore the record lifetime entirely. Printing zero here would make the book's largest cutover read as safer than a database audit, which it is not.
- 07 Hold at full weight for seven days with the managed edge still configured and still paid for, and keep the cost of running both in the programme's numbers rather than hiding it.

ROLLBACK

Setting the weight back to zero returns every new connection to the managed edge within the record lifetime, which is why it was lowered a week in advance. For seven days that path stays fully configured. The point of no return is the deletion of the managed balancer and its rules, which is Part VII. If anything in the data path failed rather than the traffic path, the way back is the restore from Move 74 rather than a weight change, and the abort criteria should say so explicitly.

104 AUTHORITY DNS, AND THE REGISTRAR YOU FORGOT

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
☀️ EDGE	MANAGED AUTHORITY DNS AND PROVIDER-INTEGRATED REGISTRARS	■ HIGH	7 DAYS

Moves the zone to a managed anycast provider and dual-serves for the parent TLD's NS TTL, because the recovery path needs DNS too.

AWS	Route 53 with its own registrar	the provider is also the registrar, so both have to be considered together.
GOOGLE	Cloud DNS	the provider exited domain registration and handed its customers to a third party, so the registrar is already somewhere else.
AZURE	Azure DNS with App Service domains	a partner registrar fronted under the provider's own name, which is the case people forget entirely.

BEFORE YOU START

ACCESS

- Administrative access to the registrar, which is frequently held by one person who left
- A zone export from the current provider, including the records created by automation

SOFTWARE

- A managed anycast DNS provider, with the zone imported and verified record by record
- The cluster's record publisher repointed, so automation keeps writing to the right place

PEOPLE

- Whoever holds the registrar account, because changing nameservers happens there and nowhere else

WHY THIS WORKS

Name resolution is the one service that has to work when everything else does not, which is why this Move moves the zone to a managed anycast provider rather than to nameservers you run. Running your own is refused here, and deliberately: a recovery path that depends on your own infrastructure resolving your own names has a circular dependency in exactly the situation you built it for. The transition is a dual-serving window at least as long as the parent zone's nameserver lifetime, during which every edit is applied to both sides — a discipline that is simple to state and easy to break.

SWAP

Two managed anycast providers serving the same zone is a genuinely good permanent arrangement for anyone whose availability requirement justifies it, and it costs very little.

DO IT FASTER

Import the zone and dual-serve weeks before you need to, then take your time verifying. Nothing forces the removal of the old provider on a schedule.

WATCH OUT

An edit applied to one side during the dual-serving window produces answers that differ depending on which nameserver a resolver picked, which is the hardest class of fault to diagnose because it is intermittent by design.

LEFTOVERS

The old hosted zone bills per zone and per million queries, and queries continue for a long time after the nameservers change.

\$180/mo	\$30/mo	83%	0 min	1 week	7 days
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The old hosted zone, seven days after the nameservers changed and the query count there has fallen to nothing.

NEEDS 97 TLS off the managed certificate service 103 Go-live: the soak, the freeze, and how you abort

THE RUNBOOK

- 01 Find the registrar before anything else. One provider is its own registrar, one exited registration and handed customers to a third party, and one fronts a partner under its own name. Establish which you have and who can log in.
- 02 Export the zone and import it, then compare record by record rather than by count. Records created by automation are the ones that go missing.
- 03 Deal with the apex record. A name at the zone apex cannot be a normal alias, so providers implement flattening differently, and the behaviour you rely on may not exist on the other side.
- 04 Reproduce or retire the routing features. Latency and geography routing keyed to cloud regions is keyed to regions that will not exist much longer, and health-check-driven failover has to be rebuilt against endpoints you now own.
- 05 Repoint the cluster's record publisher at the new provider so automation keeps working, and confirm it can create and delete records.
- 06 Add the new nameservers at the registrar and dual-serve for at least the parent zone's nameserver lifetime, which is typically a day or two. Every edit during the window is applied to both sides, without exception.
- 07 Move the certificate challenge delegation from Move 97 with the zone, or issuance stops working the moment the old provider is removed.

ROLLBACK

For seven days both providers serve the zone, so reverting is removing the new nameservers at the registrar. The point of no return is the deletion of the old hosted zone, after which the record set exists only in your export — so keep that export in source control and confirm it can be restored into a new zone before deleting anything. The registration itself stays where it is; this Move does not move it.

105 CDN: CHANGE VENDOR, NOT MODEL

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
☀️ EDGE	PROVIDER-BUNDLED CDN, EDGE FUNCTIONS AND MANAGED WEB APPLICATION FIREWALL	■ HIGH	7 DAYS

Changes CDN vendor rather than insourcing one, because the points of presence stay rented either way and only the invoice changes.

AWS	CloudFront with Lambda@Edge and the managed firewall	edge compute that is a rewrite rather than a migration.
GOOGLE	Cloud CDN with Cloud Armor	the same bundle, with the firewall rules expressed differently.
AZURE	Azure Front Door	where a previous CDN tier has already been retired and its customers moved once, which is the argument against treating any of this as permanent.

BEFORE YOU START

ACCESS

- The current cache configuration exported: cache keys, header handling, TTLs and behaviours per path
- Signed link and cookie configuration, which does not port between vendors

SOFTWARE

- A new vendor with origin shielding configured, so a cache miss storm does not reach your origin directly
- The firewall running in detection-only mode, with the trusted-hop count set explicitly

PEOPLE

- The owner of any edge compute, because it is a rewrite rather than a configuration move

WHY THIS WORKS

There is no version of this where you own the points of presence. A content network is a few hundred locations you rent, and repatriating it means building a few hundred locations, which nobody in this book's audience is doing. So this is a keep-renting verdict with a vendor change attached, and the vendor change is worth making because it decouples the edge from the cloud account you are closing. What makes it a High-risk Move is not the switch; it is the semantics that differ between vendors in ways that can serve one authenticated user's page to another.

SWAP

Where the content is small and mostly dynamic, no content network at all is a legitimate answer and removes a vendor. Measure the cache hit ratio before assuming one is needed.

DO IT FASTER

Move the static asset hostnames first. They are low risk, they are most of the volume, and they prove the configuration.

WATCH OUT

A cache key that omits a header the application varies on will serve the wrong response to somebody. Test with authenticated requests specifically, not only with anonymous ones.

LEFTOVERS

The bundled network bills per gigabyte and per request, and the firewall bills per rule and per million requests, both until the last hostname moves.

\$3,180/mo	\$1,900/mo	40%	0 min	2 weeks	7 days
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The bundled distribution and its firewall rules, seven days after the last hostname moved. The vendor bill does not go away; it changes name.

NEEDS 103 Go-live: the soak, the freeze, and how you abort 104 Authoritative DNS, and the registrar you forgot

THE RUNBOOK

- 01 | Export the cache configuration and read it for the semantics rather than the values. Cache key composition and header variation are where a subtle difference serves the wrong user's content, and that is the worst failure available in this Part.
- 02 | Configure origin shielding at the new vendor. Without it, a cache flush or a cold start sends every point of presence to your origin at once, and your origin is now a rack rather than a fleet.
- 03 | Rebuild signed links and signed cookies. These do not port between vendors and the format differs, so this is a code change in whatever generates them.
- 04 | Treat edge compute as a rewrite and price it as one. The runtimes differ, the limits differ, and a mechanical port produces something that runs and behaves differently under load.
- 05 | Decide where the firewall sits, and run it in detection-only mode for a fortnight before enforcing anything. A firewall enforcing on day one blocks a legitimate customer within a week.
- 06 | Set the trusted-hop count explicitly so the client address is taken from the right position in the forwarded header. Taking it from anywhere a client can set is how a rate limiter is defeated and how an audit log becomes fiction.
- 07 | Cut over one hostname at a time with the old vendor still configured, and keep it for seven days.

ROLLBACK

For seven days both vendors are configured and a hostname is moved back by changing one record. The point of no return is the deletion of the old distribution and its firewall rules, which is Part VII. Keep the exported cache configuration and firewall rules in source control so the previous behaviour can be restored and compared when a cache-related fault appears weeks later.



106 DDOS: THE CAPACITY YOU RENT FOREVER

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
☀️ EDGE	MANAGED DDOS PROTECTION SUBSCRIPTIONS	■ HIGH	NO

Rents mitigation capacity, because a volumetric attack saturates transit upstream of every device you own.

AWS	Shield Advanced	an annual commitment with a response team attached.
GOOGLE	Cloud Armor with its managed protection tier	the same shape, sold as a subscription.
AZURE	DDoS Protection	the same again, priced per protected plan with an annual term.

BEFORE YOU START

ACCESS

- The address space and network number from Move 91, since prefix-level protection uses both
- The transit contracts from Move 93, because inline scrubbing may be available there

SOFTWARE

- An origin access list, refreshed as vendor ranges change, replacing origin secrecy entirely
- A diversion mechanism configured and tested, whether tunnelled or announced

PEOPLE

- Whoever can authorise diversion at three in the morning, with a deputy and a telephone number

WHY THIS WORKS

A volumetric attack does not reach your firewall. It saturates your transit, upstream of every device you own, which means no equipment you buy can mitigate it. The capacity to absorb it is the product, and it is rented. This Move therefore prices four options with time-to-mitigate against each, kills origin secrecy as a strategy, and reaches a keep-paying verdict. The reason it is irreversible is commercial rather than technical: announcements and tunnels can be withdrawn in an afternoon, but diversion authority is a signed contract with a minimum term and a notice period.

SWAP

For a small estate already behind an always-on proxying network from Move 105, that may be the whole answer and no separate subscription is needed. Confirm what the proxy's own terms cover.

DO IT FASTER

Get the tunnels or the announcements configured and tested before you need them. Configuring a diversion path during an attack is not a plan.

WATCH OUT

Prefix-level protection depends on your announcements being accepted, which depends on the routing records from Move 91 being correct. Validate them before relying on this path.

LEFTOVERS

The provider subscription runs to the end of its annual term whether or not anything is behind it, which is a line Move 120 has to carry.

\$3,000/mo	\$2,400/mo	20%	0 min	1 week	4 weeks
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The provider subscription at the end of its annual term, and nothing before it. This is a cost you keep paying, deliberately.

NEEDS 91 Your own ASN, a /24 and the address space you already have 93 Transit: two upstreams and the 95th percentile

105 CDN: change vendor, not model

THE RUNBOOK

- 01 Price the four options with time-to-mitigate against each: transit with inline scrubbing, a scrubbing provider reached over a tunnel or a private interconnect with on-demand diversion, always-on proxying behind the network from Move 105, and prefix-level protection using your own address space and routing.
- 02 Kill origin secrecy as a strategy. It fails to the first certificate transparency log entry, the first misconfigured subdomain and the first email header. Replace it with an origin access list that only accepts the mitigation vendor's ranges, and a process to refresh those ranges when they change.
- 03 Choose the diversion mechanism and test it during business hours. A diversion that has never been performed takes an hour longer than it should, and the hour is during an attack.
- 04 Establish who can authorise diversion out of hours, with a deputy, a telephone number and an agreed form of words. Vendors will not divert on the word of somebody they cannot identify.
- 05 Read the commitment terms. All three cloud subscriptions are annual, and what replaces them has the same shape, so the exit cost is a notice period rather than a switch.
- 06 Set the detection thresholds and rehearse the runbook once, end to end, including the return to normal routing afterwards, which is the half people forget.
- 07 Put the recurring cost in Move 116's list of things you deliberately keep, and in Move 120's model as a permanent line.

ROLLBACK

The technical parts are reversible in an afternoon, but this Move is irreversible in the way that counts: diversion authority is a signed contract with a minimum term and a notice period, so leaving costs the remainder of the term. The point of no return is that signature. Nothing here holds data, so there is nothing to restore; what has to be kept is the origin access list, which is the thing that must be current before an attack rather than during one.

107 EMAIL: INBOUND COMES HOME, OUTBOUND DOES NOT

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
☀️ EDGE	MANAGED INBOUND EMAIL RECEIVING AND ITS OBJECT-STORE PIPELINES	■ MEDIUM	7 DAYS

Brings inbound mail home to a receiver you run and keeps outbound on a relay, because sender reputation cannot be rebuilt on demand.

AWS	SES inbound receiving with its rule sets	the only first-party inbound receiving of the three, writing to a bucket or invoking a function.
GOOGLE	no first-party inbound receiving	so these readers are leaving a third party rather than their cloud.
AZURE	Communication Services email	outbound only, so again the inbound half is somebody else's product.

BEFORE YOU START

ACCESS

- The zone from Move 104, because every record this Move touches lives there
- The current inbound rules exported, including the ones that invoke code

SOFTWARE

- A receiver you run with a real spam pipeline in front of it, not a bare mail server
- Storage of received mail on the object store from Move 49, under Move 54's durability terms

PEOPLE

- The owner of every transactional sender and every mail-to-webhook integration, named individually

WHY THIS WORKS

The two halves of email have opposite answers. Inbound is a service you can run: a receiver, a spam pipeline, and storage of what arrives, and it is not difficult. Outbound is not, because sending mail that reaches inboxes depends on a sender reputation that takes months to build and can be destroyed in a day, and on alignment records, reverse name records, list-unsubscribe handling and a failure mode that is invisible — your mail is accepted and filed in a spam folder, and nothing tells you. So this Move brings inbound home and refuses the outbound half explicitly, which closes the Part on a deliberate no.

SWAP

Where inbound volume is trivial, a hosted mailbox provider with forwarding is simpler than running a receiver and costs almost nothing.

DO IT FASTER

Run both destinations at equal preference for a fortnight rather than cutting. There is no cost to the overlap and it removes the timing risk entirely.

WATCH OUT

A mail-to-webhook integration that reports a bounce to a customer looks like your outage even though it is their retry policy. Warn those senders individually before the record changes.

LEFTOVERS

The managed receiving rules and the object-store pipeline behind them bill per message received and per bucket write, and both stop when the rules are deleted.

\$260/mo	\$40/mo	85%	0 min	2 weeks	4 weeks
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The managed inbound receiving rules and their pipelines, a week after the new destination is the only one receiving. The outbound relay stays bought.

NEEDS 49 Ceph under Rook: block, shared filesystem and the object store 104 Authoritative DNS, and the registrar you forgot

THE RUNBOOK

- 01 List the senders that will be affected, individually. Standards-compliant senders retry for four to five days and will survive any cutover; transactional senders and mail-to-webhook integrations frequently give up in minutes and report a bounce, and those are the ones that need warning.
- 02 Stand up the receiver with a genuine spam pipeline in front of it. A bare mail server on the internet receives mostly abuse, and filtering is the product rather than an add-on.
- 03 Store received mail on the object store from Move 49, under the durability terms Move 54 established. Mail is data and it inherits the same rules as everything else.
- 04 Reproduce the rules that invoked code. Where an inbound rule triggered a function, that becomes a service or a job on the substrate from Move 81, and it is ordinary work.
- 05 Cut the mail exchanger record over with both destinations live and equal preference, so mail arrives at either while the change propagates. For compliant senders this is lossless.
- 06 Watch both destinations for a week and confirm the old one goes quiet before removing it.
- 07 Refuse the outbound half in writing, with the reasons: alignment, reverse name records, warmed reputation, one-click unsubscribe handling and an invisible failure mode. Keep the relay bought and put it on Move 116's list.

ROLLBACK

For seven days both destinations are live, so reverting is removing the new record and leaving the old one. Nothing is lost for compliant senders because they retry. The point of no return is deleting the managed receiving rules, after which mail addressed under them is rejected rather than queued — so keep them until the new destination has been the only one receiving for a full week, and confirm the stored mail can be restored and read before removing anything.

VII

WATCH

Running it once it is yours. Every repatriation article stops before this Part, and every repatriation that regrets itself failed inside it.

15	14	8	10
MOVES	ZERO DOWNTIME	HIGH RISK	MINUTES, TOTAL

IF YOU ONLY DO THREE

110

THE INVENTORY, THE RMA LOOP AND WHICH RACK UNIT

Holds serial, rack unit, bay and node name in one inventory with a funded return loop, because hardware nobody can …

116

CAPACITY PLANNING, AND THE CLOUD FOOTPRINT YOU KEEP

Counts headroom per failure domain so it holds with the largest node gone, and keeps a tested cloud overflow where …

108

THE UPGRADE CALENDAR AND THE CONTROL-PLANE UPGRADE

Upstream maintains a minor for about fourteen months and the extra year sold by the hour is gone, so the treadmill …

THE UPGRADE CALENDAR AND THE CONTROL-PLANE UPGRADE · THE FLEET ROLL WITH NO THIRTEENTH MACHINE · THE INVENTORY, THE RMA LOOP AND WHICH RACK UNIT · THE 03:00 DISK · THE SUPPORT CONTRACTS YOU BUY FOR OPEN SOURCE · THE ROTA · ALERTS THAT SURVIVE THE CLUSTER, AND THE PAGER YOU DROP · VELERO, THE OFF-SITE COPY AND THE REBUILD DRILL · CAPACITY PLANNING, AND THE CLOUD FOOTPRINT YOU KEEP · THE PATCH SLA AFTER THE VULNERABILITY FEED STOPPED · AUDIT LOGS AFTER THE PROVIDER'S · THE AUDITOR'S CHECKLIST, AND THE CONTRACTS NOBODY READ · WHAT IT ACTUALLY COSTS · GOING BACK: HOW THE PROGRAMME IS ABORTED · THE LAST ACCOUNT

■ 3 LOW RISK	■ 4 MEDIUM RISK	■ 8 HIGH RISK
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108 THE UPGRADE CALENDAR AND THE CONTROL-PLANE UPGRADE

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
 WATCH	MANAGED CONTROL-PLANE UPGRADES, PURCHASABLE EXTENDED SUPPORT FOR END-OF-LIFE KUBERNETES MINORS, AND A MANAGED ADD-ON LIFECYCLE	 HIGH	NO

Upstream maintains a minor for about fourteen months and the extra year sold by the hour is gone, so the treadmill becomes a dated calendar with a named owner.

AWS	EKS extended support	a year beyond upstream, billed at a higher hourly rate, which is the option that disappears here.
GOOGLE	GKE release channels with extended support	the same idea sold as a channel rather than as a surcharge.
AZURE	AKS long-term support	the same again, with its own eligibility rules and its own end dates.

BEFORE YOU START

ACCESS

- The second cluster from Move 51, matching production's versions, for every rehearsal
- A verified snapshot from Move 52's procedure before each step, not merely a scheduled one

SOFTWARE

- A written calendar with dates and a named owner, published where the team will see it
- The ownership matrix: every chart, operator and driver, its current version, its cadence and its owner

PEOPLE

- One named owner for the calendar, because a shared responsibility for upgrades is nobody's

WHY THIS WORKS

Upstream maintains a minor release for roughly fourteen months, and the extra year every provider sold you by the hour does not exist on your own cluster. That is not a crisis; it is a calendar. What replaces the managed lifecycle is a written schedule with a named owner, a rehearsal on the second cluster before every step, and — the larger half — an ownership matrix for the forty-odd charts, operators and drivers that each move on their own cadence. On this host operating system the image and the Kubernetes version move on separate commands, which is a feature: two smaller changes rather than one large one.

SWAP

Where the estate genuinely cannot upgrade on schedule, a support subscription from Move 112 buys advice rather than time. Nothing buys back an unmaintained minor.

DO IT FASTER

Upgrade the second cluster a fortnight ahead of production, always. It converts an upgrade into a rehearsal that has already been performed.

WATCH OUT

Add-ons that were managed for you are now yours, and they have their own end-of-support dates that do not align with the cluster's. The matrix is what stops that being a surprise.

LEFTOVERS

Extended support surcharges on the old cluster stop when it does, and until then they are a real and growing line.

\$210/mo	\$0/mo	100%	0 min	1 week	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Extended support surcharges on the old cluster, once it is gone. On your own cluster there is nothing to buy and nothing to cancel.

NEEDS 43 Three control-plane nodes, the API VIP and the cluster CA 51 The second cluster, and what it shares 52 The etcd restore drill

62 Metrics off the managed monitor, and the cardinality budget 65 Argo CD, and the two infrastructure-as-code estates

THE RUNBOOK

- 01 | Write the calendar first, with real dates derived from upstream's support window rather than from when it feels convenient. Publish it and name its owner.
- 02 | Build the ownership matrix. Forty-odd components on independent cadences is normal and it is the part that actually consumes time; without the matrix, an upgrade stalls on a chart nobody knows the owner of.
- 03 | Take and verify a snapshot before every step, using the procedure proved in Move 52. Verified means restored somewhere, not merely written.
- 04 | State the version skew rules per component and check them before each step. The control plane, the kubelets and the client tools have separate tolerances and exceeding one produces failures that do not name the cause.
- 05 | Check for removed APIs and for your own admission webhooks. A webhook configured to fail closed during a control-plane upgrade can block the upgrade itself, and that is a memorable afternoon.
- 06 | Upgrade sequentially, never skipping a minor, rehearsing each step on the second cluster first. On this host operating system the image and the Kubernetes version are separate commands, so take them separately.
- 07 | Record how long each step took, because that number is the input to next quarter's calendar.

ROLLBACK

A control-plane upgrade is not reversible: there is no way back from a completed minor upgrade except restoring the snapshot taken before it, which loses everything written since. The point of no return is the first control-plane component upgraded. That is why the snapshot is verified rather than assumed and why every step is rehearsed on the second cluster first. Restoring is a real procedure from Move 52 with a measured time, not a theoretical option.

109 THE FLEET ROLL WITH NO THIRTEENTH MACHINE

CUTOVER

10 MIN



LAYER	LEAVING	RISK	REVERSIBLE
WATCH	MANAGED NODE GROUPS, IMAGE-BASED NODE ROLLOUTS, AUTOSCALER SURGE CAPACITY AND MANAGED NODE AUTO-REPAIR	HIGH	IMMEDIATELY

Rolls the fleet one failure domain at a time, because the surge node the provider added before taking one away does not exist on metal you already own.

AWS	managed node group updates	a surge node added before one is drained, which is capacity you were renting by the minute.
GOOGLE	surge upgrades on node pools	the same mechanism, with the surge count configurable.
AZURE	node image upgrades with max surge	the same again, and the same assumption of spare capacity on demand.

BEFORE YOU START

ACCESS

- The headroom from Move 11, confirmed to still exist after Move 69's scheduling work
- The failure-domain labels from Move 42, because the roll is ordered by them

SOFTWARE

- A disruption budget audit: every workload, its budget, and whether that budget is achievable
- Reboot orchestration that reads the rack, feed and switch labels and respects a concurrency limit

PEOPLE

- The owner of any workload pinned to a local volume, because theirs is the one that sees a window

WHY THIS WORKS

Every managed rollout worked the same way: add a machine, drain one, remove it. On hardware you own there is no thirteenth machine, so the roll has to work inside the headroom that Move 11 bought deliberately. That makes three things matter: disruption budgets that are honest, an inventory of pods that cannot be evicted, and awareness that a local volume from Move 47 pins its pod to one machine. The saving grace is that upgrades here are image-based with a fallback, so a bad image is a reboot back rather than a rebuild, which is what makes this reversible immediately.

SWAP

Where headroom is genuinely insufficient, the cold spare machine from Move 19 can be racked temporarily as the surge node. It is exactly what it is for.

DO IT FASTER

Roll the non-production cluster first, always. It finds the disruption-budget problems at no cost.

WATCH OUT

A drain that hangs is almost always a disruption budget that cannot be satisfied, and the temptation is to force it. Forcing a drain kills the pod the budget was protecting.

LEFTOVERS

Node auto-repair and the surge capacity settings stay configured on the old cluster and stop billing when it does.

—	—	—	10 min	1 week	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing here. This Move replaces a capability rather than a service, and its cost is headroom you already bought.

NEEDS 42 Netboot, node identity and failure-domain labels 47 Local NVMe, and the default StorageClass nobody gave you

108 The upgrade calendar and the control-plane upgrade

THE RUNBOOK

- 01 | Audit the disruption budgets before rolling anything. A budget that permits zero disruption blocks the drain forever, and a budget nobody set permits everything. Both are common.
- 02 | List the pods that cannot be evicted: single-replica workloads, pods with local volumes from Move 47, and anything without a controller. Each is a decision rather than an obstacle.
- 03 | Order the roll by failure domain using the labels from Move 42. One rack at a time, or one feed at a time, so that a mistake affects a domain the estate is designed to lose.
- 04 | Set a concurrency budget and hold to it. The temptation on a slow roll is to widen it, and widening it is how a fleet roll becomes an outage.
- 05 | Roll with the image-based upgrade and its fallback. A machine that does not come back healthy is rebooted into the previous image, which is a minutes-long recovery rather than a reprovision.
- 06 | Accept the ten minutes where it applies. A workload pinned to a local volume is unavailable while its machine reboots; anything with a replica in another rack sees nothing at all.
- 07 | Time the first full roll and record it. That number is the input to Move 108's calendar and to Move 116's capacity planning.

ROLLBACK

An image-based upgrade with a fallback is reverted by rebooting into the previous image, so a bad roll is undone per machine within minutes. There is no point of no return in this Move. Where a workload with a local volume did not come back, the way back is the restore path built in Move 74 or Move 115 rather than anything in this Move, which is why those exist before this one.



WATCH · 10 MIN · HIGH RISK

Nothing here. This Move replaces a capability rather than a service, and its cost is headroom you already bought.

110 THE INVENTORY, THE RMA LOOP AND WHICH RACK UNIT

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
WATCH	THE PROVIDER'S CAPACITY POOL AND ITS HARDWARE-RETIREMENT NOTICES	LOW	IMMEDIATELY

Holds serial, rack unit, bay and node name in one inventory with a funded return loop, because hardware nobody can find is hardware nobody can replace.

AWS	EC2 instance retirement notices	a failing machine was replaced by somebody else and announced to you as an event.
GOOGLE	Compute Engine host maintenance events	the same, mostly handled by migration before you noticed.
AZURE	Virtual Machines planned and unplanned maintenance events	again a notification rather than a part in your hand.

BEFORE YOU START

ACCESS

- The as-built from Move 37 and the bill of materials from Move 20, both current
- The support contracts from Move 17, with their return terms read rather than assumed

SOFTWARE

- An inventory system that holds physical position as a first-class field, not a tag on a virtual resource
- A seeding script, because typing three hundred serial numbers by hand produces three hundred opportunities for error

PEOPLE

- A named contact for returns at every supplier, and a named owner for the inventory

WHY THIS WORKS

This Move is logistics rather than engineering, and it is the difference between a two-hour repair and a two-week one. One inventory holds serial number, rack unit, bay and node name for every machine and every device, seeded from the as-built in Move 37 and the bill of materials in Move 20. Behind it sits a return loop that is funded and has a named contact, because a warranty claim with no owner is a warranty claim nobody files. Diagnosis is Move 111; the spares themselves were budgeted in Move 19. What is new here is being able to find the thing.

SWAP

For a single rack, a well-maintained document from Move 37 plus a spares list is enough. The inventory system earns its place at the second rack or the second site.

DO IT FASTER

Seed it during the build in Move 36, when the serial numbers are visible and the machines are open. Reading a serial number off an installed machine is a torch-and-mirror job.

WATCH OUT

A device replaced under warranty comes back with a different serial number. An inventory that is not updated at replacement time drifts immediately and silently.

LEFTOVERS

The provider's retirement notification wiring — the event rules and the notifications behind it — can be removed when the old estate goes.

—	—	—	0 min	1 week	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The provider's retirement notification wiring, once the old estate is gone. This Move adds a small cost and removes a category of two-week repairs.

NEEDS 19 The shelf, and the failure rates you stock against 20 The bill of materials 37 Labels, port maps and the as-built

THE RUNBOOK

- 01 Choose an inventory system that models racks, rack units, bays and devices properly. A spreadsheet works for a single rack and stops working at the second.
- 02 Seed it with a script from the as-built and the bill of materials. Serial number, position, node name, and the acceptance test result from Move 18 for each machine.
- 03 Record the firmware baseline per machine while you are there, because Move 111 treats firmware as a data-loss concern and needs a starting point.
- 04 Build the return loop: who raises a claim, with which supplier, under which contract, and what advance replacement does to your holdings. Advance replacement is worth paying for and it obliges you to return the failed part or be billed for it.
- 05 Read the warranty terms for rented dedicated machines separately. They differ sharply from owned hardware, and the provider replaces the machine rather than the part, which changes what your inventory has to track.
- 06 Establish a parts service level with a named contact and a telephone number, and test it once with a trivial claim while nothing is broken.
- 07 Reconcile the inventory against reality monthly. Half a day a month keeps it true; a year of neglect makes it worse than nothing, because people trust it.

ROLLBACK

An inventory is corrected by correcting it and there is nothing here to reverse. The point of no return is the moment it stops matching reality, which happens quietly and is why the monthly reconciliation exists. Keep the inventory's data exported to source control periodically so a corrupted or lost system can be restored, and so that the physical estate can be reconstructed on paper if the system is unavailable during an incident.



WATCH · 0 MIN · LOW RISK

The provider's retirement notification wiring, once the old estate is gone. This Move adds a small cost and removes a category of two-week repairs.

111 THE 03:00 DISK

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
WATCH	TRANSPARENT BLOCK-DEVICE REPLACEMENT AND DEGRADED-HARDWARE NOTICES	HIGH	IMMEDIATELY

Wires per-device latency, SMART attributes and storage slow-op alerts, because a dying drive rarely fails cleanly and leaves every node still reporting Ready.

AWS	EBS with transparent device replacement	a failing device was replaced beneath you and you were never told.
GOOGLE	Persistent Disk with the same transparency	again handled entirely by the provider.
AZURE	Managed Disks with the same again	which is why nobody arriving from a cloud has ever watched a drive die slowly.

BEFORE YOU START

ACCESS

- Per-device metrics reaching the store from Move 62, including wear and error counters
- The inventory from Move 110, so a device identifier maps to a rack unit and a bay

SOFTWARE

- Alerts on device latency, wear attributes and the storage cluster's own slow-operation warnings
- A firmware baseline per component class, treated as a data-loss concern rather than a chore

PEOPLE

- Whoever will do the physical replacement, or the remote hands relationship from Move 35

WHY THIS WORKS

A drive that fails cleanly is the easy case: the storage cluster notices, marks it out, and rebuilds. The hard case is the drive that is dying rather than dead — latency climbing, errors being corrected, throughput halving — while every machine still reports itself healthy and the cluster still reports itself available. Nothing above the device notices, and the symptom is an application that is slow for reasons nobody can find. So this Move instruments the device layer directly, and then rehearses the physical replacement against the inventory from Move 110, which is genuinely new work for anyone arriving from a cloud.

SWAP

Where the estate is small, replacing a suspect device rather than diagnosing it is often cheaper than the engineering time to be certain. The spares from Move 19 exist for this.

DO IT FASTER

Label the bays physically and in the inventory using the same identifier the operating system reports. Translating between two naming schemes at three in the morning is how the wrong drive gets pulled.

WATCH OUT

Pulling the wrong device from a degraded pool can lose data. The ticket template from Move 35 carries a photograph and a bay number precisely because a technician follows it exactly.

LEFTOVERS

Nothing to remove. This is a capability the provider supplied invisibly and you now supply deliberately.

—	—	—	0 min	1 week	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing. This Move replaces something the provider did silently, and it costs instrumentation rather than money.

NEEDS 49 Ceph under Rook: block, shared filesystem and the object store 62 Metrics off the managed monitor, and the cardinality budget

110 The inventory, the RMA loop and which rack unit

WATCH

THE RUNBOOK

- 01 Collect per-device metrics properly: latency percentiles, wear indicators, available spare against its own threshold, media and integrity errors, and unsafe shutdown counts.
- 02 Alert on the trends rather than the thresholds. A wear indicator that moves faster than its historical rate is a better signal than one that crosses a line, because the line is usually crossed long after the drive started misbehaving.
- 03 Wire the storage cluster's slow-operation warnings into the same alerting path. Those warnings are the earliest reliable signal that one device is dragging a pool, and they are frequently ignored because they are not errors.
- 04 Model the rebuild. When a device is marked out, the cluster backfills, and that backfill uses the fabric sized in Move 12 for workloads rather than for recovery. Measure what a rebuild does to application latency before you need to know.
- 05 Set the hot-spare policy deliberately. Immediate automatic replacement shortens the degraded window and starts a rebuild at a time nobody chose; a delay lets a human decide. Both are defensible and one has to be written down.
- 06 Treat firmware as a data-loss concern. Baselines for the machine firmware, the management controller, the network card, the storage adapter and the drives, with a controlled update path, because specific firmware revisions have lost data in real fleets.
- 07 Drill the replacement end to end: identify the device, map it to a bay through the inventory from Move 110, raise a ticket using the template from Move 35, and confirm the cluster rebuilds. Time it.

ROLLBACK

Nothing here changes the running estate, so the instrumentation is reversible by removing it. The point of no return is the physical act of pulling a device, which is why the drill and the ticket template matter more than the alerting does. If the wrong device is pulled, the way back is the cluster's own recovery and, failing that, the restore path from Move 115 — which is the reason that Move exists before this one.

112 THE SUPPORT CONTRACTS YOU BUY FOR OPEN SOURCE

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
WATCH	PAID CLOUD SUPPORT PLANS PRICED AGAINST SPEND	LOW	365 DAYS

Prices support subscriptions for the host OS, Ceph, Postgres and the switch OS, because the answer at 03:00 on a corruption bug is not always someone you employ.

AWS	Business and Enterprise Support	priced as a percentage of spend, so the saving here is largest where the bill was largest.
GOOGLE	Standard and Enhanced Support	the same percentage-of-spend model with its own thresholds.
AZURE	Standard and Professional Direct	flat-rate tiers, so a small estate saves almost nothing by cancelling.

BEFORE YOU START

ACCESS

- The current support plan's price and what it actually covers, read rather than remembered
- A list of the components you would not want to debug alone at three in the morning

SOFTWARE

- Quotes from vendors for each candidate subscription, on the versions you actually run
- The escalation path each one offers, in writing, with a response commitment attached

PEOPLE

- Whoever holds the budget, because these are annual commitments rather than monthly ones

WHY THIS WORKS

Cancelling the cloud support plan is a real saving on two of the three clouds, because it was priced as a share of a bill that is about to shrink. What replaces it is not nothing. There are four components where a subscription genuinely earns its money — the host operating system, the storage cluster, the database and its operator, and the switch operating system — and what such a contract buys is a named escalation path and a fix in a release, not an engineer on site. Knowing that distinction is what stops the subscription being either skipped or over-bought.

SWAP

Where the team genuinely has deep expertise in a component, skip its subscription and put the money into the rota from Move 113. Support contracts and headcount are the same budget.

DO IT FASTER

Buy the storage subscription first. It is the component where an unsolved problem is most likely to be a data problem.

WATCH OUT

A support subscription is not a substitute for the restore drills in Moves 52, 74 and 115. Vendors help you diagnose; they do not restore your data.

LEFTOVERS

The cloud support plan is billed monthly and cancels on notice, so it can be dropped as soon as the estate that justified it has moved.

\$4,100/mo	\$1,600/mo	61%	0 min	1 week	4 weeks
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The cloud support plan, once the estate it covered has moved. What you buy instead is smaller and it is not nothing.

NEEDS 13 The switch, and the software running on it 41 Talos as the host OS, and where it cannot go

49 Ceph under Rook: block, shared filesystem and the object store 73 Postgres onto your own disks

THE RUNBOOK

- 01 | Price what you are cancelling. On two of the three clouds support was a percentage of spend, so the saving falls with the bill and is largest at the start; on the third it was a flat tier and cancelling saves very little.
- 02 | Shortlist the components that warrant a subscription. The host operating system from its vendor, the storage cluster from a company that employs its maintainers, the database and its operator from a database company, and the switch operating system from whoever sold it.
- 03 | Read what each contract buys. A named escalation path and a fix in a future release is the realistic answer. An engineer at your site is not, and a subscription bought expecting one will disappoint.
- 04 | Check the versions covered. A subscription that only supports versions you are not running is a subscription you cannot use, and this is the most common mismatch.
- 05 | Get quotes early. Procurement here is a quarter, mostly spent waiting, so start it when Part VII starts rather than when the cloud plan lapses.
- 06 | Note the commitment shape. Most of these are annual and cancellable only at renewal, which is what the reversibility figure records, and that is a reason to buy fewer of them rather than more.
- 07 | Test each escalation path once with a real, low-severity question. A path nobody has used is a path with an unknown response time.

ROLLBACK

Subscriptions are annual and cancellable only at renewal, so leaving one costs the remainder of a year. The point of no return, such as it is, is the purchase order. Cancelling the cloud plan is reversible at any time by re-subscribing, which is worth knowing before Part VII closes the account entirely. Keep the escalation paths and contract terms in source control with the runbooks, so the right number can be found at three in the morning rather than searched for. No vendor on this list will restore your data for you; the drills in Moves 52, 74 and 115 do that.

113 THE ROTA

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
WATCH	—	MEDIUM	IMMEDIATELY

Builds the rota from the engineers who can actually be called, because one week in four needs a bench that leave and resignation do not empty.

AWS	the shared responsibility model	the provider carried the hardware layer and its on-call for it, invisibly and permanently.
GOOGLE	its shared responsibility documentation	the same split, described in a whitepaper you never had to staff against.
AZURE	its shared responsibility guidance	the same again, and the reason the hardware rota is a new line in the budget.

BEFORE YOU START

ACCESS

- The staffing conclusion from Move 22, and the budget it implied
- The alert inventory from Move 114, so the rota is designed against real volume rather than hope

SOFTWARE

- A rota schedule with escalation, published where everyone can see who is on
- A handover practice: what is written down, when, and by whom

PEOPLE

- Enough engineers that one week in four is achievable with someone on leave

WHY THIS WORKS

A rota is arithmetic before it is a policy. One week in four needs four people who can genuinely be called; one in three needs three and burns them out; one in two is not a rota, it is two people waiting to resign. Whether the team is large enough at all was settled back in Move 22, against each landing option, at the point where the reader could still act on the answer. What happens here is building it, funding it and writing the handover practice — and being honest that the alert volume will include signals the managed monitor never surfaced, because nobody was ever paged about an inlet temperature before.

SWAP

Where the team is genuinely too small, the honest answers are the rented-metal option from Move 22 or a managed service partner, not a rota of two people. Both are cheaper than losing them.

DO IT FASTER

Start the rota before go-live, on the non-production estate, so that the first real page is not the first page anybody has received.

WATCH OUT

Alert fatigue is the failure mode, and it arrives about six weeks in. Move 114's tuning is not optional; it is what keeps the rota staffed.

LEFTOVERS

The rented paging service, if you keep one, bills per user per month, and it is one of the lines Move 114 examines.

—	—	—	0 min	1 week	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing. This Move adds a standing cost, and pretending otherwise is how the comparison in Move 120 goes wrong.

NEEDS 22 The four places the machines could live 62 Metrics off the managed monitor, and the cardinality budget 111 The 03:00 disk

THE RUNBOOK

- 01

Do the arithmetic and publish it. Depth of four is the minimum for a sustainable weekly rota, and depth of three works only with an explicit plan for leave and an accepted risk during it.
- 02

Settle compensation and rest. Standby that materially constrains someone's free time counts as working time under several jurisdictions' rules, and rest entitlements after a disturbed night are a legal obligation rather than a courtesy.
- 03

Estimate the alert volume honestly. Power distribution, inlet temperature, fans, memory errors and disks are all new sources that the managed monitor never surfaced, and they arrive at night by their nature.
- 04

Set the escalation path with names rather than roles, and a second name behind each. A rota that escalates to a role nobody occupies at three in the morning escalates to nobody.
- 05

Write the handover practice down: what the outgoing person records, what the incoming person reads, and where it lives. This single practice decides whether year two is calm.
- 06

Rehearse a handover during a real incident review rather than only in theory, and adjust the template based on what was actually missing.
- 07

Price it and give the number to Move 120, because a rota is a standing cost and the comparison in that Move is dishonest without it.

ROLLBACK

A rota is changed by changing it, and there is nothing here to reverse technically. The point of no return does not exist; what does exist is a human cost that is slow to appear and slow to recover from, which is why the depth arithmetic is done honestly rather than optimistically. Keep the rota, the compensation policy and the handover template in source control so the agreed practice can be restored when it drifts.

114 ALERTS THAT SURVIVE THE CLUSTER, AND THE PAGER YOU DROP

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
WATCH	MANAGED ALARMS, MANAGED SYNTHETIC CHECKS, A RENTED PAGING AND INCIDENT SERVICE AND A HOSTED STATUS PAGE	HIGH	IMMEDIATELY

Places the prober, the heartbeat and the status page outside the cluster they watch, then routes to an on-call estate you run rather than services you rent.

AWS	CloudWatch alarms with Synthetics canaries	evaluation happened outside your blast radius, which is the property that matters.
GOOGLE	Cloud Monitoring alerting with uptime checks	the same property, provided invisibly.
AZURE	Azure Monitor alerts with availability tests	the same again, and the same thing to rebuild.

BEFORE YOU START

ACCESS

- A small footprint outside the cluster, in a second location, that does not depend on it for identity or name resolution
- The alert inventory from Move 62 and the rota from Move 113

SOFTWARE

- Out-of-band probing, a dead-man's-switch heartbeat, and a status page not hosted on what it reports
- An on-call estate: routing, escalation, incidents and postmortems

PEOPLE

- The rota from Move 113, because alert routing without a rota is a mailbox

WHY THIS WORKS

The managed alarm service was quietly supplying one thing that is easy to miss: it evaluated your alerts somewhere that was not affected by your outage. Rebuild that badly — alert evaluation inside the cluster it watches — and the first real failure is also the failure of the thing that would have told you about it. So this Move puts the prober, the heartbeat and the status page on a small footprint outside the cluster, in a second location, and then routes alerts to an on-call estate. That constraint matters more than the choice of software.

SWAP

A small rented paging service plus a hosted status page is a perfectly good answer and costs little. The requirement is independence from the cluster, not ownership of the software.

DO IT FASTER

Move the probes and the heartbeat first, before the alert routing. They are the part that protects you and they take a day.

WATCH OUT

Alert fatigue kills a rota. Review the alerts that fired every month, and delete or tune anything that fired and needed no action.

LEFTOVERS

The managed alarms, canaries, rented pager seats and hosted status page all bill monthly and can be dropped once the replacement has run through a real incident.

\$1,240/mo	\$110/mo	91%	0 min	2 weeks	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Managed alarms, synthetic checks, rented pager seats and the hosted status page, once the replacement has carried a real incident end to end.

NEEDS 49 Ceph under Rook: block, shared filesystem and the object store 50 Humans on the API server: OIDC, after break-glass

62 Metrics off the managed monitor, and the cardinality budget 63 Logs and traces, brought home 113 The rota

114 ALERTS THAT SURVIVE THE CLUSTER, AND THE PAGER YOU DROP

7 STEPS · 0 MIN · HIGH RISK

THE RUNBOOK

- 01 Place the probe outside the cluster and in a second location. A check that runs inside what it checks tells you nothing on the day it matters.
- 02 Add a dead-man's-switch heartbeat. Something outside must expect a regular signal and page when it stops, because the failure where nothing alerts is the failure that lasts until a customer calls.
- 03 Host the status page somewhere independent. A status page on the estate it reports on is a status page that goes down with it, which is exactly when people look at it.
- 04 Choose the on-call estate. This book recommends OneUptime, which is openly licensed and self-hostable and puts uptime and synthetic monitoring, alert routing, the on-call rota and escalation, incidents and postmortems and the status page into one estate rather than four. The honest alternative is the alert manager you already run with a rented pager and a hosted status page, which is the smaller change for a team already using one — though that market has moved under its users more than once, with one widely used on-call product discontinued and one open-source one archived. The author of this book founded OneUptime, and the copyright page says so.
- 05 Route every alert to a person through the rota from Move 113, and delete every alert that has no action attached. An alert nobody acts on trains people to ignore the ones that matter.
- 06 Write the ordered cold-start list: from nothing running, in what order does the estate come back, and what does each step depend on. This list is what somebody follows at four in the morning.
- 07 Drill it with the identity provider deliberately unreachable, which is the break-glass case from Move 40 and the one that catches circular dependencies.

ROLLBACK

Everything here can be reverted by re-enabling the managed alarms, which stay configured until you delete them. There is no point of no return. The genuine risk is subtler: a monitoring estate that has never seen a real incident is untested, so keep both running until one has, and keep the cold-start list in source control so the recovery order can be restored rather than reconstructed under pressure.



WATCH · 0 MIN · HIGH RISK

Managed alarms, synthetic checks, rented pager seats and the hosted status page, once the replacement has carried a real incident end to end.

115 VELERO, THE OFF-SITE COPY AND THE REBUILD DRILL

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
WATCH	MANAGED BACKUP, SNAPSHOT LIFECYCLE POLICIES, CROSS-REGION COPY AND MANAGED FAULT INJECTION	HIGH	IMMEDIATELY

Backs up cluster state with Velero and volumes with CSI snapshots to another credential domain, and publishes an RTO measured by an actual rebuild.

AWS	Backup with cross-region copy and Fault Injection Service	a policy engine and a game-day tool, both of which stop with the account.
GOOGLE	Backup for GKE with cross-region storage	the only one of the three that sold a cluster-workload backup product.
AZURE	Backup for AKS with geo-redundant vaults	which exists and is frequently overlooked, so this reader may have more to replace than they expect.

BEFORE YOU START

ACCESS

- The object store from Move 49, made durable by Move 54, with an off-site target in a second credential domain
- Two drill days a year in the calendar, funded and defended

SOFTWARE

- veLero at a pinned version for cluster state, and volume snapshots configured through the storage driver
- Immutability on the off-site target, so a compromised credential cannot delete both copies

PEOPLE

- Whoever will run the game day, and whoever will act on what it finds

WHY THIS WORKS

Two things need backing up and they are different: the cluster's own state, which is a set of objects, and the volumes, which are data. Velero handles the first and snapshots handle the second, and both go to the object store from Move 49 under the durability terms Move 54 established. The rule that makes them backups rather than copies is where the second copy sits: a different building, a different credential domain and, where it matters, a different jurisdiction, under immutability. Database backups stay with the repository from Move 74. And the whole thing ends with a measured rebuild, because an untested recovery time is a guess.

SWAP

Where a full rebuild drill is too disruptive, rebuild into the second cluster from Move 51 instead. It measures the same thing at a fraction of the risk.

DO IT FASTER

Automate the rebuild as a script rather than a document, and run it every game day. A script that has been executed is worth more than a runbook that has been read.

WATCH OUT

A backup of cluster objects without the volume data restores an estate that starts and has nothing in it, which is a specific and demoralising kind of failure. Test both together.

LEFTOVERS

Managed backup vaults keep billing for their retention after the resources are gone, and vault lock can prevent deletion entirely until the retention expires.

\$1,480/mo	\$120/mo	92%	0 min	2 weeks	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Managed backup vaults, snapshot lifecycle policies and cross-region copy, once a rebuild has been performed from your own backups and timed.

NEEDS 49 Ceph under Rook: block, shared filesystem and the object store 54 The object store you now own, and how it loses data

65 Argo CD, and the two infrastructure-as-code estates 74 Point-in-time recovery with pgBackRest

114 Alerts that survive the cluster, and the pager you drop

THE RUNBOOK

- 01 | Back up cluster state with `velero` on a schedule, and confirm what it captures and what it does not. Objects are captured; the data behind a volume is captured only if snapshots are configured.
- 02 | Configure volume snapshots through the storage driver and confirm one can be restored into a new claim before trusting the schedule.
- 03 | Send the off-site copy to a different building, a different credential domain and, where regulation requires it, a different jurisdiction, under immutability. A copy the cluster can delete with its own identity is not an off-site copy.
- 04 | Leave the database backups where they are. The repository from Move 74 is purpose-built for point-in-time recovery and this Move does not duplicate it.
- 05 | Schedule a game day twice a year to replace whatever fault-injection service the provider offered. Break something real, on purpose, in daylight, with people watching.
- 06 | Do the rebuild drill: from the repository in Move 65 and these backups, rebuild a working cluster and restore a service into it. Measure the wall-clock time.
- 07 | Publish that number as the recovery time objective, and use it to settle the second-site question that Move 24 deferred. If the measured rebuild is faster than the business can tolerate, the deferral was correct; if it is not, the trigger for a second site has been met.

ROLLBACK

Nothing here changes the running estate, so it is reversible by removing the schedules. There is no point of no return. The purpose of this Move is the opposite of risk: it is what makes every rollback paragraph elsewhere in the book true, because a restore that has been performed and timed is the only kind that counts. Treat a failed drill as an incident, with a postmortem, rather than as an inconvenience to be repeated later.



116 CAPACITY PLANNING, AND THE CLOUD FOOTPRINT YOU KEEP

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
WATCH	ON-DEMAND ELASTICITY AND MANAGED CLUSTER AUTOSCALING (KEEPING A MINIMAL TESTED OVERFLOW)	MEDIUM	IMMEDIATELY

Counts headroom per failure domain so it holds with the largest node gone, and keeps a tested cloud overflow where owning the peak costs more than renting it.

AWS	on-demand instances and Auto Scaling	capacity in ninety seconds, which is what you are choosing to keep a small amount of.
GOOGLE	on-demand instances with managed instance groups	the same, and the least conditional exit terms if you do leave entirely.
AZURE	on-demand instances with scale sets	the same again, with exit terms that require the subscription to close.

BEFORE YOU START

ACCESS

- A year of real load on your own estate, because a capacity model built on three months of migration traffic is fiction
- The procurement lead times from Move 17, which turn adding capacity into a quarterly decision

SOFTWARE

- A headroom model per failure domain, tested with the largest node removed
- A pre-built overflow target in the cloud, deployable and tested rather than described

PEOPLE

- Whoever approves capital, because the model's output is a purchase order placed a quarter early

WHY THIS WORKS

On your own hardware, headroom is a purchase made a quarter in advance, so it has to be counted properly: per failure domain, and held so that it still stands with the largest machine removed. That is a stricter test than an average utilisation figure and it is the one that matters during a rack failure. The second half of this Move is deliberate retention. A tested overflow target, occasional accelerators and the appliances that did not become workloads in Move 70 are all legitimately kept — and keeping them costs more than the invoice, because the free-egress-on-exit programmes generally require a full departure.

SWAP

Where the load is genuinely predictable, retaining nothing is the cheapest and cleanest answer, and it makes Move 122 much simpler.

DO IT FASTER

Build the model after a year of real load, not before. Anything earlier is measuring the migration rather than the business.

WATCH OUT

An overflow target that has never been used at scale will fail at scale. Quarterly testing means running real traffic through it, not confirming it deploys.

LEFTOVERS

The retained footprint keeps a whole account alive, with credentials, permissions and an audit obligation attached. Move 119 and Move 122 both inherit that.

\$56,513/mo	\$1,900/mo	97%	0 min	1 week	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Everything except the retained list, and the retained list is short, written down and tested quarterly.

NEEDS 11 Headroom, and the number of machines you order 20 The bill of materials

62 Metrics off the managed monitor, and the cardinality budget 69 The fixed fleet: priority, preemption, NUMA and the GPUs

70 The workloads that must stay virtual machines 103 Go-live: the soak, the freeze, and how you abort

116 CAPACITY PLANNING, AND THE CLOUD FOOTPRINT YOU KEEP

7 STEPS • 0 MIN • MEDIUM RISK

THE RUNBOOK

- 01 | Build the headroom model against real failure domains from Move 42 rather than against the fleet total. A cluster with twenty per cent headroom spread evenly has none in the rack that just failed.
- 02 | Test the model by removing the largest node on paper. If the remaining capacity cannot carry production, the headroom figure is wrong regardless of what the average says.
- 03 | Add the procurement lead time. A model that says you need a machine in six weeks and a supplier who needs twelve is a model that produces an outage on schedule.
- 04 | Decide the retained cloud footprint deliberately and list it: a pre-built overflow target, accelerators used intermittently, and the appliances from Move 70 that stayed rented.
- 05 | Test the overflow quarterly or delete it honestly. An untested overflow is a line on an invoice and a false sense of safety, and the second is worse than the first.
- 06 | Resolve the egress conflict now rather than in Move 122. The free-egress-on-exit programmes generally require full departure, so a retained footprint may cost you the credit — and that trade should be made deliberately with the numbers in front of you.
- 07 | Refresh the model annually and review it quarterly. A day a quarter keeps it honest.

ROLLBACK

The model is revised by revising it, and the retained footprint is deleted whenever you decide to. There is no point of no return in this Move. What is not reversible is under-buying against a twelve-week lead time, which is why the model is built early and reviewed often. Keep it in source control so a previous year's assumptions can be restored and compared against what actually happened.

 WATCH • 0 MIN • MEDIUM RISK

Everything except the retained list, and the retained list is short, written down and tested quarterly.

117 THE PATCH SLA AFTER THE VULNERABILITY FEED STOPPED

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
WATCH	MANAGED VULNERABILITY SCANNING, REGISTRY IMAGE SCANNING AND MANAGED PATCH-COMPLIANCE REPORTING	MEDIUM	IMMEDIATELY

Sets patch SLAs by severity and triages from the exploited-vulnerability catalogue and vendor advisories, because the feed no longer enriches every CVE.

AWS	Inspector with Systems Manager Patch Manager	scanning and a compliance report, both tied to the account.
GOOGLE	Security Command Center with VM Manager	the same pair under different names.
AZURE	Microsoft Defender for Cloud with Update Manager	the same again, with the findings landing in a portal you are leaving.

BEFORE YOU START

ACCESS

- The scanning output from Move 59, in the registry and in the pipeline
- The asset register and firmware baselines from Move 110

SOFTWARE

- A written patch policy with service levels by severity, for nodes, images, firmware and management controllers
- A suppression mechanism for findings that are genuinely unreachable, recorded rather than deleted

PEOPLE

- One named owner for triage, because a shared triage queue is an unread queue

WHY THIS WORKS

Two things changed and only one of them is about your migration. Your provider's scanning and compliance reporting go away with the account, and Move 59 already rebuilt the scanning half. The other change is external: the public vulnerability feed no longer enriches every identifier promptly, so a triage process that waits for a severity score to appear will wait indefinitely. What replaces it is triage from the catalogue of vulnerabilities known to be exploited, vendor advisories and the newer regional database, with unreachable findings suppressed deliberately rather than ignored quietly.

SWAP

Where the estate is small, a weekly review of the exploited-vulnerability catalogue against your own component list is most of the value for a fraction of the tooling.

DO IT FASTER

Patch by rebuilding rather than by updating. An image roll is a well-tested path in Move 109 and it applies every fix at once.

WATCH OUT

A service level that is written but not achievable with the mechanism you have is worse than none, because it will be reported as breached every month until somebody stops reading the report.

LEFTOVERS

The managed scanning and compliance services bill per resource and per scan, and can be dropped once your own triage has run for a month.

\$680/mo	\$0/mo	100%	0 min	1 week	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The managed scanning and patch-compliance services, once your own triage has produced a month of reports somebody has read.

NEEDS 59 cosign, Trivy and an admission policy 109 The fleet roll with no thirteenth machine

110 The inventory, the RMA loop and which rack unit

117 THE PATCH SLA AFTER THE VULNERABILITY FEED STOPPED

7 STEPS · 0 MIN · MEDIUM RISK

THE RUNBOOK

- 01 Write the service levels by severity and by asset class: node images, container images, firmware and management controllers. Four classes, three severities, twelve numbers, published.
- 02 Write the node service level against the mechanism you actually have. There is no in-place package update on this host operating system, so a node patch is an image roll through Move 109, and the service level has to be achievable with that mechanism rather than with a package manager.
- 03 Triage from sources that are actually current: the catalogue of vulnerabilities known to be exploited, vendor advisories for the components you run, and the newer regional database. Do not wait for enrichment that may not arrive.
- 04 Suppress what is genuinely unreachable, using a documented exchange format rather than by deleting findings. A suppression with a reason and a date is a decision; a deleted finding is a gap.
- 05 Treat firmware as a patch class with change management behind it. Baselines and the asset register from Move 110 are the input, and a firmware update is a planned change rather than a background task.
- 06 Report against the service levels monthly, and treat a breach as a finding rather than as a metric. Move 119's auditor will ask for exactly this report.
- 07 Budget the running cost honestly: about a day a week, ongoing, for somebody who is not doing it in the gaps between other work.

ROLLBACK

Nothing here changes the running estate; it changes what somebody does every week. There is no point of no return. Keep the policy, the suppressions and the monthly reports in source control, so the position at any past date can be restored and shown — which is precisely what an auditor asks for, and precisely what nobody can produce when the record lived in a console that has been closed.

118 AUDIT LOGS AFTER THE PROVIDER'S

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
WATCH	THE PROVIDER'S CONTROL-PLANE AUDIT TRAIL, ITS CONFIGURATION-COMPLIANCE RULES AND ITS POSTURE-MANAGEMENT SERVICE	MEDIUM	IMMEDIATELY

Writes the audit policy by hand and ships it to an append-only store in another credential domain, because Kubernetes audit logging is off by default.

AWS	CloudTrail with Config and Security Hub	the management-event trail could not be switched off and was retained for you.
GOOGLE	Cloud Audit Logs with Security Command Center	admin activity logs retained without charge, and the rest a resource you paid for.
AZURE	Activity Log with Policy and Defender for Cloud	the same split again, with its own retention defaults.

BEFORE YOU START

ACCESS

- A destination in a separate credential domain, under immutability, that the cluster cannot delete from
- The reconciler from Move 65, because drift detection against the repository is half the control

SOFTWARE

- A hand-written audit policy, with each level's storage cost estimated before it is enabled
- Scheduled benchmark runs against the cluster, so configuration compliance is measured rather than assumed

PEOPLE

- Whoever will answer the auditor in Move 119, so the evidence collected is the evidence they need

WHY THIS WORKS

Two of the three providers retained a control-plane audit trail whether or not you asked, which means most readers have never had to think about audit logging existing. On your own cluster it is off by default, the policy is written by hand, and each level of detail has a storage cost you are choosing. It also has a structural weakness the provider's did not: the log is written by the thing being audited. That is why the destination is an append-only store in a separate credential domain, and why the preventive and detective controls from Moves 59 and 65 matter alongside it.

SWAP

Where the volume is genuinely unaffordable, log metadata for everything and bodies only for secret and role changes. That covers most of what is ever asked for.

DO IT FASTER

Start from the upstream example policy and prune. Writing one from nothing takes a day and produces the same result.

WATCH OUT

An audit log that is never read is not a control. Sample it monthly, and put one query in front of a human, or the first time anybody looks at it will be during an incident.

LEFTOVERS

The provider's trail, configuration rules and posture service bill per event and per rule evaluation, and the trail in particular keeps billing after everything else has gone.

\$940/mo	\$90/mo	90%	0 min	1 week	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The provider's configuration rules and posture service once your own benchmarks are running. The trail itself goes with the account in Move 122.

NEEDS 50 Humans on the API server: OIDC, after break-glass 54 The object store you now own, and how it loses data

59 cosign, Trivy and an admission policy 63 Logs and traces, brought home 65 Argo CD, and the two infrastructure-as-code estates

THE RUNBOOK

- 01
- Write the audit policy by hand, rule by rule. Full request and response bodies for sensitive resources, metadata only for the noisy ones, and nothing at all for the health checks that would otherwise dominate the volume.
- 02
- Estimate the storage before enabling it. Audit logging at a high level on a busy cluster produces a surprising volume, and the estimate is what stops the first month's bill being the reason it gets turned off.
- 03
- Ship it to an append-only store in a separate credential domain, under immutability. The cluster writes the log about itself, so the destination has to be somewhere the cluster's own identity cannot delete from.
- 04
- Set retention deliberately and budget it. Retention that nobody chose is retention that either costs too much or is too short when it is asked for.
- 05
- Rebuild the continuous control monitoring the provider's configuration service was doing: scheduled benchmark runs against the cluster and its nodes, on a schedule, with the results stored.
- 06
- Name the two halves of control explicitly. The admission policy from Move 59 is the preventive control; drift detection against the repository from Move 65 is the detective one. An auditor will ask which is which.
- 07
- Test the trail by doing something notable and finding it in the store afterwards, from the auditor's side rather than from the cluster's.

ROLLBACK

The policy is reverted through the machine configuration and the shipping is stopped by removing it, so this is reversible in minutes. There is no point of no return. What cannot be recovered is a period during which auditing was off, which is why it is enabled early rather than before an audit. Keep the policy in source control so the exact configuration in force at any past date can be restored and evidenced.

119 THE AUDITOR'S CHECKLIST, AND THE CONTRACTS NOBODY READ

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
 WATCH	INHERITED COMPLIANCE ATTESTATIONS, THE PROVIDER'S AUDIT-EVIDENCE PORTAL AND ITS COMMITTED-SPEND AGREEMENT	 HIGH	NO

Rebuilds the evidence pack from the facility's report, the asset register and the sanitisation certificates, because physical controls stop being inherited.

AWS	Artifact	a portal where compliance reports were downloaded and inherited without further thought.
GOOGLE	Compliance Reports Manager	the same, with the same effect on how nobody learned what the reports covered.
AZURE	the Service Trust Portal	the same again, and the same set of controls that quietly stop being somebody else's.

BEFORE YOU START

ACCESS

- The facility's own audit report, including its complementary user entity controls section
- Every customer contract, data-processing agreement and security questionnaire that names a provider

SOFTWARE

- An evidence pack assembled as documents rather than as links into consoles you are closing
- The committed-spend agreement read for what happens when spending stops

PEOPLE

- Counsel, the auditor and whoever owns customer contracts, all engaged a quarter before the audit window

WHY THIS WORKS

A large part of your compliance posture was inherited from a provider and evidenced by a report you downloaded. Moving into a facility changes who owns which controls, and the ones that stop being inherited are physical: access, environmental, media handling and disposal. So the evidence pack is rebuilt from what you now have — the facility's own report and, critically, the complementary controls it expects you to operate — plus the sanitisation certificates from Move 38, the asset register from Move 110 and the trail from Move 118. The legal half is the part that blocks migrations hardest and is started a quarter early.

SWAP

Where an attestation genuinely cannot be reproduced in time, a scoped exception agreed with the auditor in advance is far better than discovering the gap during the audit.

DO IT FASTER

Collect evidence continuously rather than in a scramble. Every Move in this book that says keep this in source control is feeding this pack.

WATCH OUT

A data-processing agreement naming a specific provider is a contractual commitment, not a description. Changing where the data lives may require customer notice or consent.

LEFTOVERS

The committed-spend agreement runs to its end date whether or not anything runs on it, and Move 120 has to carry that as a real line.

—	—	—	0 min	3 weeks	8 to 12 weeks
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing. This Move adds work that the provider was doing invisibly, and it is the work that keeps the contracts valid.

NEEDS 30 The notice period and the door out 38 The drives that leave, and the snapshots that stay

110 The inventory, the RMA loop and which rack unit 118 Audit logs after the provider's

119 THE AUDITOR'S CHECKLIST, AND THE CONTRACTS
NOBODY READ

7 STEPS · 0 MIN · HIGH RISK

THE RUNBOOK

- 01

Read the facility's report properly, including the complementary controls section. That section lists what the facility assumes you are doing, and every item in it is now yours to evidence.
- 02

List the controls that stop being inherited: physical access, environmental, media handling and media disposal. Map each to evidence you actually hold.
- 03

Assemble the evidence: sanitisation certificates and chain of custody from Move 38, the asset register from Move 110, the audit trail from Move 118, the access list from Move 35 and the acceptance results from Move 18.
- 04

Start the legal work a quarter early, because it lives in other people's calendars. Sub-processor notices, data-processing agreements that name the old provider explicitly, and customer security questionnaires whose answers were written against cloud controls.
- 05

Read the committed-spend agreement for what happens when spending stops. On all three providers the commitment does not shrink because you stopped using the service, so the remaining term is a cost regardless.
- 06

Rewrite the questionnaire answers before a customer asks. An answer that says a control is inherited from a provider you have left is a false statement in a contractual document.
- 07

Rehearse the audit conversation once with somebody playing the auditor. It finds the missing evidence while there is still time to produce it.

ROLLBACK

This Move is irreversible in the sense that matters: statements made to auditors and customers cannot be unmade, and a commitment surrendered or a notice served cannot be recalled. The point of no return is the first formal notice to a customer or a regulator. Keep every version of the evidence pack, so the state of the controls at any past date can be restored and shown, which is what an auditor is actually asking for.

 WATCH · 0 MIN · HIGH RISK

Nothing. This Move adds work that the provider was doing invisibly, and it is the work that keeps the contracts valid.

120 WHAT IT ACTUALLY COSTS

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
WATCH	THE PROVIDER'S COST EXPLORER, BILLING EXPORTS, BUDGETS AND COST-ADVISORY SERVICE	LOW	IMMEDIATELY

Prices the cluster from real invoices, counting power, transit burst, spares, support contracts, audit hours, the rota and the months you paid twice.

AWS	Cost Explorer with Cost and Usage Reports and Trusted Advisor	a cost model somebody else maintained for you.
GOOGLE	Cloud Billing reports with the Recommender	the same, with recommendations attached.
AZURE	Cost Management with Advisor	the same again, and the same absence once the account closes.

BEFORE YOU START

ACCESS

- Real invoices for a full quarter, from every supplier, not estimates
- The discounted rate you actually paid on the cloud, not list, so the comparison is fair

SOFTWARE

- A custom pricing file if a cost tool is used, because the defaults are placeholders
- A showback model against scarce capacity rather than against money

PEOPLE

- Somebody from finance to check the invoices with you, because this number will be quoted in board papers

WHY THIS WORKS

Every repatriation comparison you have read omits the same lines, and this Move is where the honest number gets built from real invoices rather than from a model. Power at the committed figure from Move 28, transit at the percentile from Move 93, the spares holding from Move 19, the support subscriptions from Move 112, audit hours from Move 119, the rota from Move 113, and the months where both estates were paid for at once. The last of those is the one that makes a good migration look bad in month six and is always left out. If a cost tool is used, the default on-premises prices in it are placeholders and have to be replaced.

SWAP

Where the estate is small, a spreadsheet updated quarterly beats any tool. The discipline is counting everything, not the software you count it in.

DO IT FASTER

Build it once properly and then update it quarterly. The first build is two weeks and each update is an afternoon.

WATCH OUT

Amortisation choices change the answer substantially. Pick a life, state it, and use the same one on both sides of the comparison.

LEFTOVERS

The provider's billing exports keep writing to a bucket after everything else has gone, and that bucket bills.

\$74,013/mo	\$38,337/mo	48%	0 min	2 weeks	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The cost-advisory services and, eventually, the billing exports — but not until Move 122 has taken the billing history you will need for the comparison.

NEEDS 19 The shelf, and the failure rates you stock against 20 The bill of materials 28 Committed kilowatts, metered kilowatts and the derate
93 Transit: two upstreams and the 95th percentile 112 The support contracts you buy for open source 113 The rota
116 Capacity planning, and the cloud footprint you keep

THE RUNBOOK

- 01
- Build the cost model from invoices. Every supplier, a full quarter, actual amounts. Estimates in this Move undermine every other number in the book.
- 02
- Replace the default prices in any cost tool. On-premises prices in these tools are placeholders and using them produces a confident wrong answer.
- 03
- Count the lines that get omitted: committed power from Move 28, transit burst at the percentile from Move 93, the spares holding from Move 19, support subscriptions from Move 112, audit hours from Move 119 and the rota from Move 113.
- 04
- Count the double-running months explicitly. Both estates were paid for during Parts V and VI, and that is real money that belongs in the programme's total rather than outside it.
- 05
- Compare against the discounted rate you actually paid rather than list price. A comparison against list flatters the result and will be dismantled by the first person who checks it.
- 06
- Do showback against scarce capacity rather than against money. On a fixed fleet the constraint is cores, memory and storage, and charging teams for those changes behaviour in a way that charging them for currency does not.
- 07
- Write the refresh policy: what is replaced when, staggered so that year five is a cadence rather than a cliff, with the money set aside annually rather than found in a crisis.

ROLLBACK

A cost model is corrected by correcting it and there is nothing to reverse. The point of no return does not exist here, though the number this Move produces will be quoted for years, which is an argument for building it carefully rather than quickly. Keep every quarterly version in source control so the model at any past date can be restored and compared, which is how you answer the question of whether the programme delivered what it promised.

121 GOING BACK: HOW THE PROGRAMME IS ABORTED

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
WATCH	—	HIGH	NO

Writes the abort plan while going back is still cheap: what returns, what the surrendered commitments cost to restart, and who is allowed to make the call.

AWS	Savings Plans and Reserved Instances	surrendered on the way out and repriced at current rates on the way back.
GOOGLE	Committed Use Discounts	the same, with the commitment tied to a project or organisation you may have closed.
AZURE	reservations and savings plans	the same again, and the one where a closed subscription complicates the restart most.

BEFORE YOU START

ACCESS

- The cost model from Move 120, so the comparison is against a real number rather than a fear
- The notice dates from Move 30 and the disposal plan from Move 38

SOFTWARE

- Each Move's own rollback paragraph, collected into one ordered document
- A current list of which managed services still exist in the form you left them in

PEOPLE

- A named decision owner with the authority to stop the programme, and their deputy

WHY THIS WORKS

Somewhere around month nine a board asks what happens if this does not work, and the answer should already be written. It is answerable now, because Move 120 knows the real number and Moves 52, 74 and 115 have proved the restores. The abort plan says what would actually return, what it costs to restart the discounts that were surrendered, which managed services no longer exist in the version you left, who is allowed to make the call, and what would trigger it. It also names the genuine point of no return, which is not a technical step at all.

SWAP

Where the programme is genuinely in trouble, a partial abort is usually the right answer: keep the platform and return one workload. The plan should support that rather than treating this as all or nothing.

DO IT FASTER

Write it at the first programme gate rather than at the ninth month. It takes a week and it is the document that keeps a nervous board calm.

WATCH OUT

The way back depends on the interconnect from Move 53 and the cloud accounts staying open. Move 122 must not close them before this plan is retired.

LEFTOVERS

The old estate's configuration, launch templates and node groups are the way back, and they cost almost nothing to leave in place. Leave them.

—	—	—	0 min	1 week	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing. This Move exists to keep options open, and every option it protects costs a little to keep.

NEEDS 53 The links you build first: to the cloud, to the office 103 Go-live: the soak, the freeze, and how you abort

116 Capacity planning, and the cloud footprint you keep 120 What it actually costs

- 01 Collect every Move's rollback into one ordered document. The abort inherits each Move's own cutover window in reverse rather than adding one of its own, which is why the individual paragraphs matter.
- 02 Price the restart. Reserved capacity and committed spend surrendered on the way out are repriced at current rates on the way back, and current rates are not the rates you left.
- 03 Check what still exists. Managed services change, are retired and are renamed; a service you left eighteen months ago may not be available in the form your code expects.
- 04 Name the decision owner and the deputy. An abort decision with no owner is an abort decision that gets made too late by a committee.
- 05 Write the trigger criteria in advance and in numbers, the same way Move 103 wrote its abort criteria. What would have to be true for this to be the right call.
- 06 Name the real point of no return, which is commercial rather than technical: the notice served on the facility in Move 30 and the hardware disposal in Move 38. Everything before those is recoverable at a price.
- 07 Print the egress and switching-charge position with its source and a review date, because it is regulated, it is dated, and it moves.

ROLLBACK

This is the rollback, and it is irreversible in the sense that the decisions it describes cannot be taken twice: notice served on a facility cannot be recalled, and hardware disposed of under Move 38 cannot be restored. The point of no return for the whole programme is those two events, not any technical cutover. Review this document at every programme gate, and keep each version, so the board can see how the answer changed as the evidence did.

122 THE LAST ACCOUNT

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
WATCH	THE PROVIDER'S ORGANISATION AND IDENTITY CENTRE, AUDIT TRAIL, KEY MANAGEMENT, REGISTRY, MARKETPLACE SUBSCRIPTIONS AND SUPPORT PLAN	HIGH	30 DAYS

Extracts the reports, the history and the key material, discharges the dated exceptions, then closes all but what four earlier Moves deliberately kept.

AWS	Organizations with IAM Identity Center	an organisation closes only with its management account, and a closed account is recoverable for about ninety days.
GOOGLE	the organisation with Cloud Identity	a deleted project is recoverable for about thirty days, which is the shortest window and therefore the one this Move quotes.
AZURE	the tenant with Entra ID	a cancelled subscription is recoverable for about ninety days, with its own reactivation process.

BEFORE YOU START

ACCESS

- The management account or root project, because the organisation cannot be closed from anywhere else
- Every account and subscription enumerated, including the ones nobody has opened this year

SOFTWARE

- An extraction list: compliance reports, billing history, audit trail and key material, with a destination for each
- The retained list from Moves 85, 88, 90 and 116, consolidated with dates and owners

PEOPLE

- Whoever signed the committed-spend agreement, since that spend does not end when the usage does

WHY THIS WORKS

This is the last Move and it is mostly extraction and waiting. What has to come out before anything closes: the compliance reports, the billing history that Move 120's comparison depends on, the audit trail from Move 118 and the key material. What trips people is the mechanics — key deletion has a waiting period, the audit trail keeps billing after everything else has gone, reserved capacity cannot be cancelled, and an organisation closes only with its management account. And what stays open is deliberate rather than accidental: four earlier Moves each kept something, and each said so in writing.

SWAP

Where a single retained resource is keeping a large account alive, moving it to a small separate account first makes the closure much simpler and cheaper.

DO IT FASTER

Start the key deletion waiting periods first, in parallel with everything else. They are the longest pole and nothing depends on them.

WATCH OUT

Closing an account does not cancel a committed-spend agreement. It continues to bill against the organisation, and Move 119 should already have established what that costs.

LEFTOVERS

Billing exports, audit trails and marketplace subscriptions all survive the deletion of what they were attached to. Each needs closing individually.

\$2,180/mo	\$1,900/mo	13%	0 min	2 weeks	30 days
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Everything except the four retained items, and those are written down with owners and review dates rather than forgotten.

NEEDS 55 Workload identity without the cloud's 57 The root of trust after managed key management

85 The data that stays: deep archive and the warehouse 88 Document and wide-column stores: the hardest exit

90 Orchestrated workflows: the dated exception 116 Capacity planning, and the cloud footprint you keep

119 The auditor's checklist, and the contracts nobody read 121 Going back: how the programme is aborted

THE RUNBOOK

- 01 | Extract before closing anything: compliance reports for the audit in Move 119, the full billing history for the model in Move 120, the audit trail from Move 118, and any key material still needed to read archived data.
- 02 | Consolidate the retained list. The archive from Move 85, the store held below its threshold in Move 88, the workflows dated in Move 90 and the tested overflow from Move 116. Each stays, each has an owner and each has a review date.
- 03 | Resolve the egress credit conflict. The free-egress-on-exit programmes are time-boxed and at least one requires actually leaving, which conflicts with the footprint Move 116 keeps. Decide which is worth more, with the numbers, before closing anything.
- 04 | Order the identity removal correctly. The federated identity provider registered in Move 55 is removed after the workloads that used it, not before, or the last workloads lose their access on the way out.
- 05 | Work through the mechanics that delay closure: key deletion waiting periods, an audit trail that keeps billing, a committed-spend agreement and reserved capacity that cannot be cancelled, and marketplace subscriptions with their own terms.
- 06 | Close accounts and subscriptions from the leaves inward, and close the organisation last, from its management account.
- 07 | Wait out the recovery windows before declaring it finished. They differ by provider and the shortest is about thirty days, which is what this Move's reversibility figure records.

ROLLBACK

For about thirty days a closed account or a deleted project can be recovered, which is the window this Move quotes because it is the shortest of the three. After that, the extraction you took is all that remains, so verify that every extracted artefact can be restored and read before the window expires rather than after. The point of no return is the end of the shortest recovery window, and it is the last one in the book.

A FEW THINGS WORTH KNOWING

None of these is a Move. They are the ten things that make the other 122 work, and most of them are the things people learn in the second year rather than the first.

THE BILL LAGS THE CHANGE BY A MONTH

A resource stopped on the third still bills for the first two days, and a reservation you no longer use bills for a year. Judge a Move by the invoice after next, never by the console.

TWO PEOPLE KNOW, OR NOBODY DOES

The single most common failure mode of a self-run platform is not hardware. It is one engineer who understands the network, and a holiday.

KEEP ONE FOOT IN THE CLOUD ON PURPOSE

An account with a working credential, a Terraform state and a warm standby is not a failure of nerve. It is the cheapest disaster recovery site you will ever have.

WRITE THE RUNBOOK BEFORE YOU NEED IT

Every Move in this book is a runbook because that is the artefact that survives the person who wrote it. A migration that leaves no runbooks behind has to be done again by whoever comes next.

CAPACITY YOU OWN IS CAPACITY YOU HAVE

The cloud taught a generation to treat headroom as waste. On your own metal, idle capacity is the thing that absorbs a bad deploy at four in the afternoon. Buy thirty per cent more than the model says.

HARDWARE FAILS PREDICTABLY AND BORINGLY

Disks, fans, PSUs, optics, in that order, at a rate you can put in a spreadsheet. It is not the drama people expect. The drama is in the change you made on Friday.

THE SAVING IS REAL; THE FREEDOM IS THE POINT

Most teams start this for the bill and finish it for the other thing: being able to answer a question about your own system without opening a support case.

YOU CAN GO BACK

The industry talks about repatriation as though it were one-way. It is not. 13 of the 122 Moves here are genuinely irreversible and the other 109 are not, and knowing which is which is most of the skill.

The best infrastructure decision you make this year will probably be unglamorous, take a fortnight, save a five-figure sum and never be written up anywhere. That is rather the point.

THE BOOK

122 Moves across five Parts. 110 cost no downtime at all, 13 cannot be undone, and the whole book executed end to end costs 202 minutes of user-visible outage.

THE TYPE

Set in Archivo, drawn by Omnibus-Type, and JetBrains Mono, drawn by Philipp Nurullin and Konstantin Bulenkov. Both are open source. Every number in this book is set in the mono.

THE NUMBERS

Every price is a public list rate observed while writing, before any discount, and every saving is illustrative of a shape rather than a quotation. Check the arithmetic against your own invoice.

Disclosure: the author founded OneUptime, which this book recommends for alerting, on-call, incidents and status pages. It is recommended because it is Apache-2.0 licensed, self-hostable, and does in one place what that Part would otherwise ask you to assemble from four separate tools. The alternatives named beside it are real ones, and a reader who prefers them loses nothing else in this book.